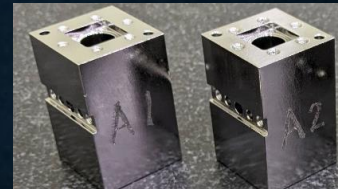
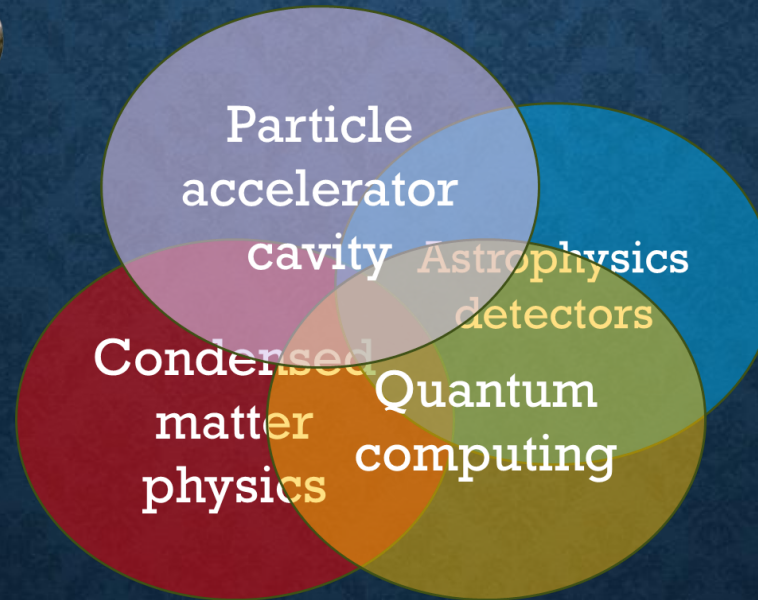
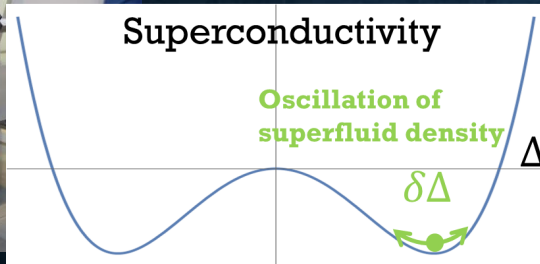
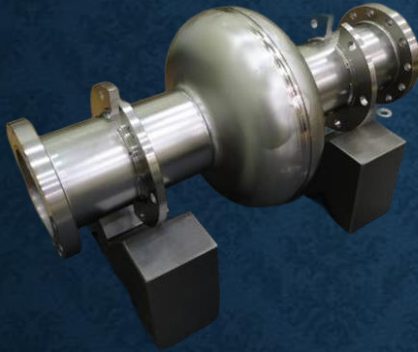
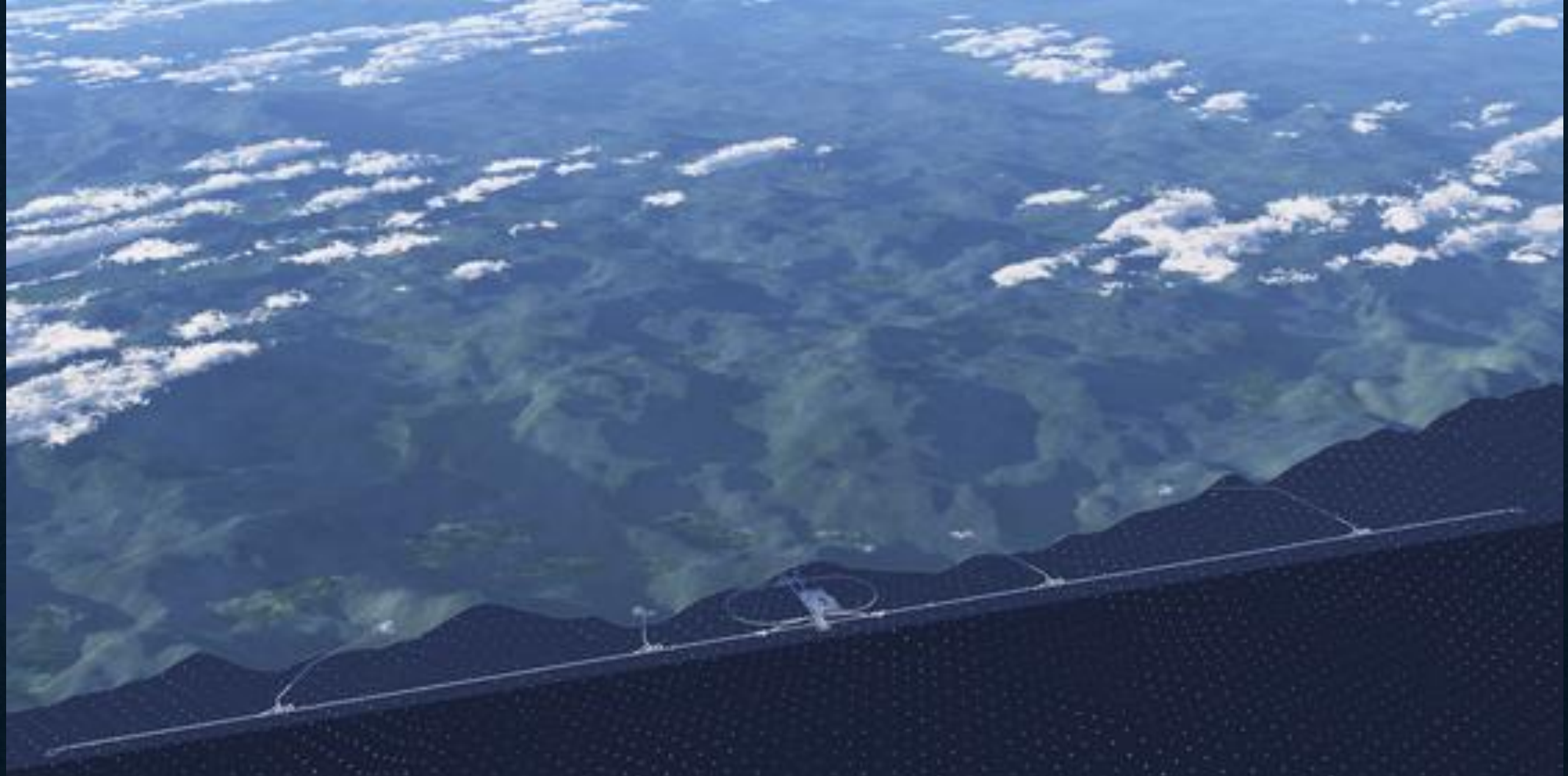


Superconducting Resonators: a Crossroads of High-Energy Accelerators, Astrophysical Detectors, Condensed Matter Theory, and Quantum Computing

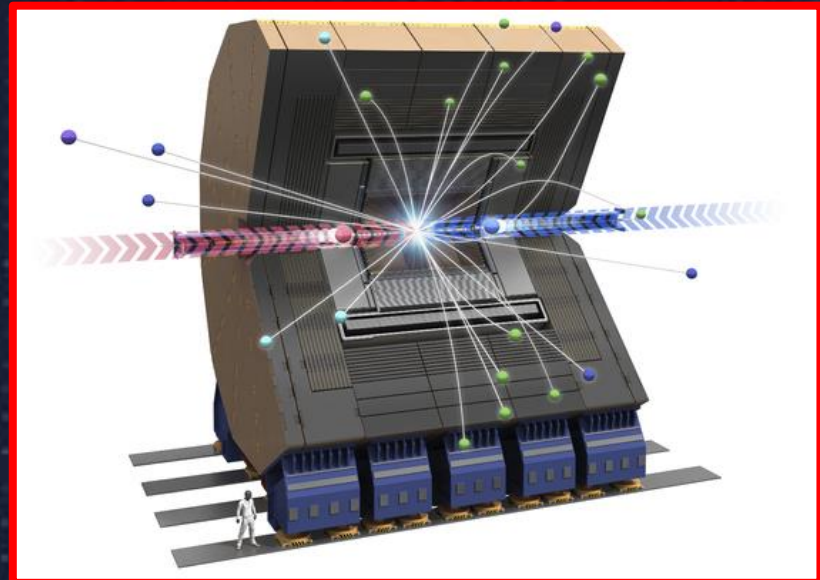
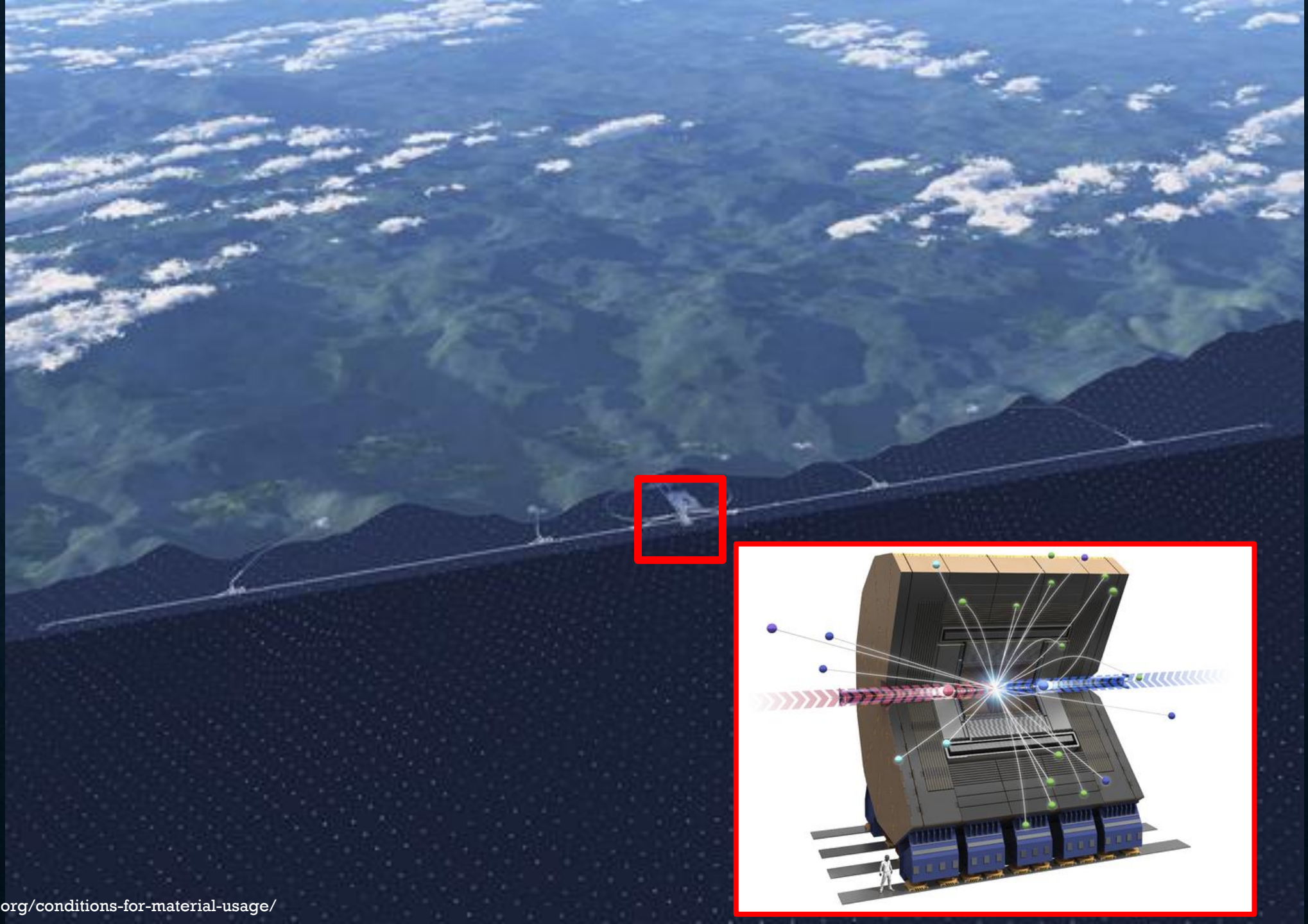


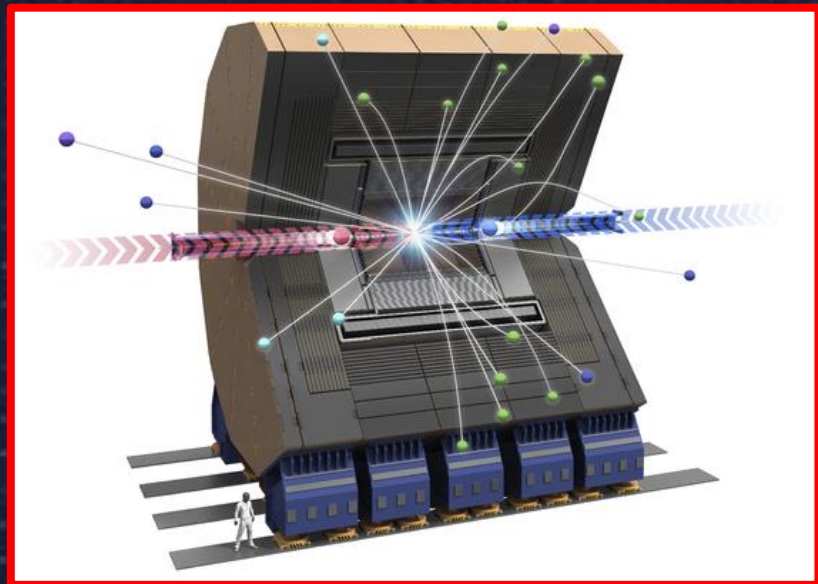
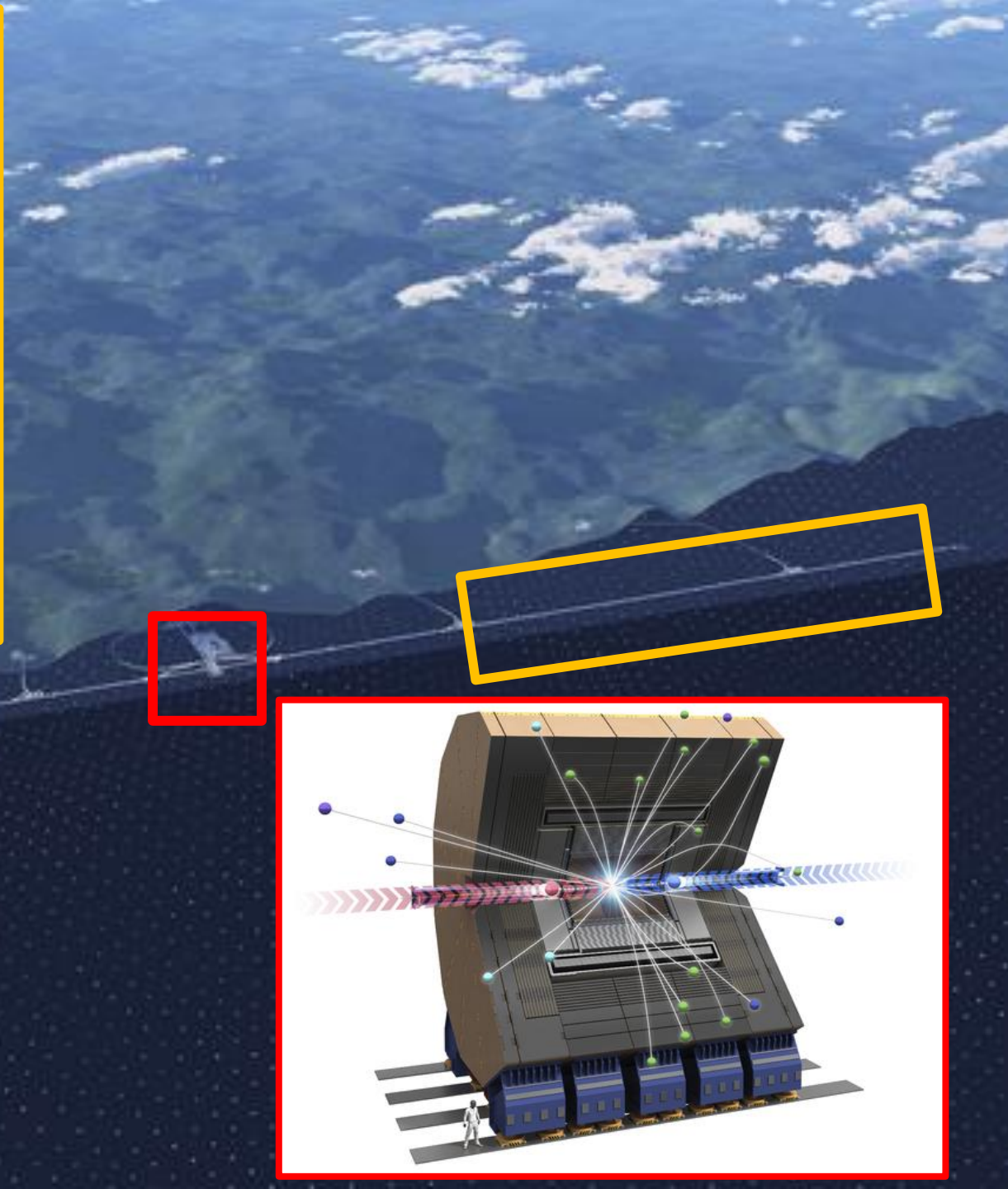
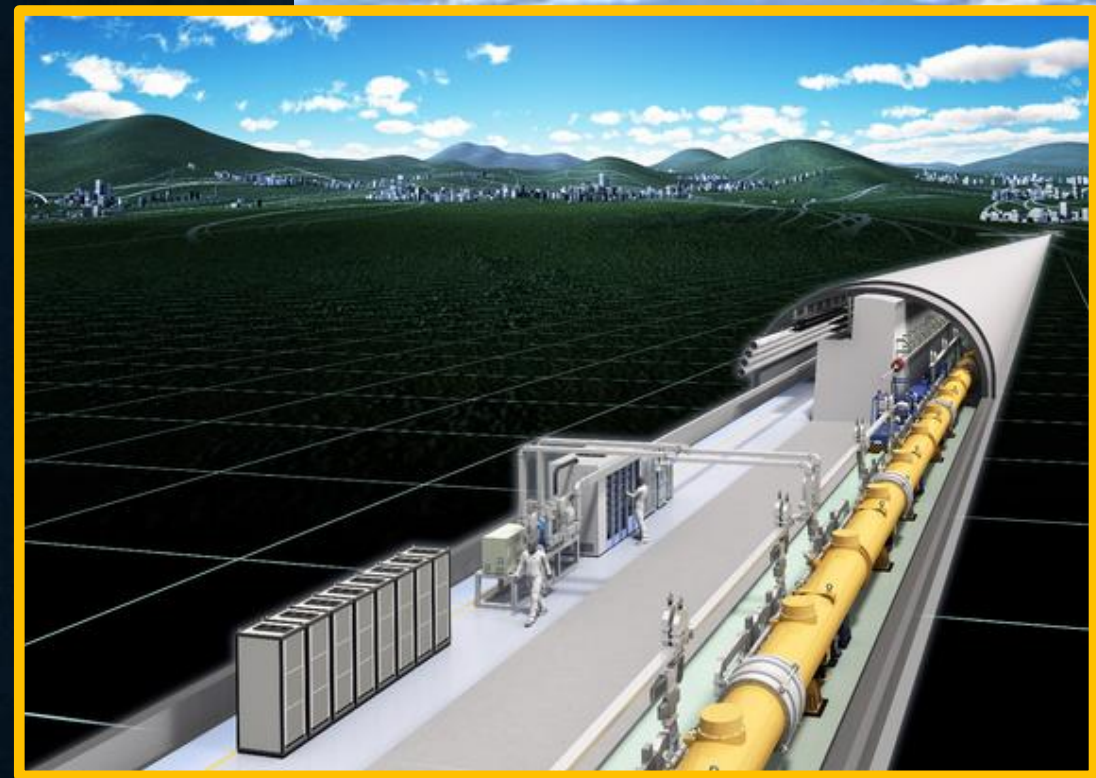
High energy Accelerator Research Organization (KEK)
Innovation Center for Applied Superconducting Accelerators (iCASA)

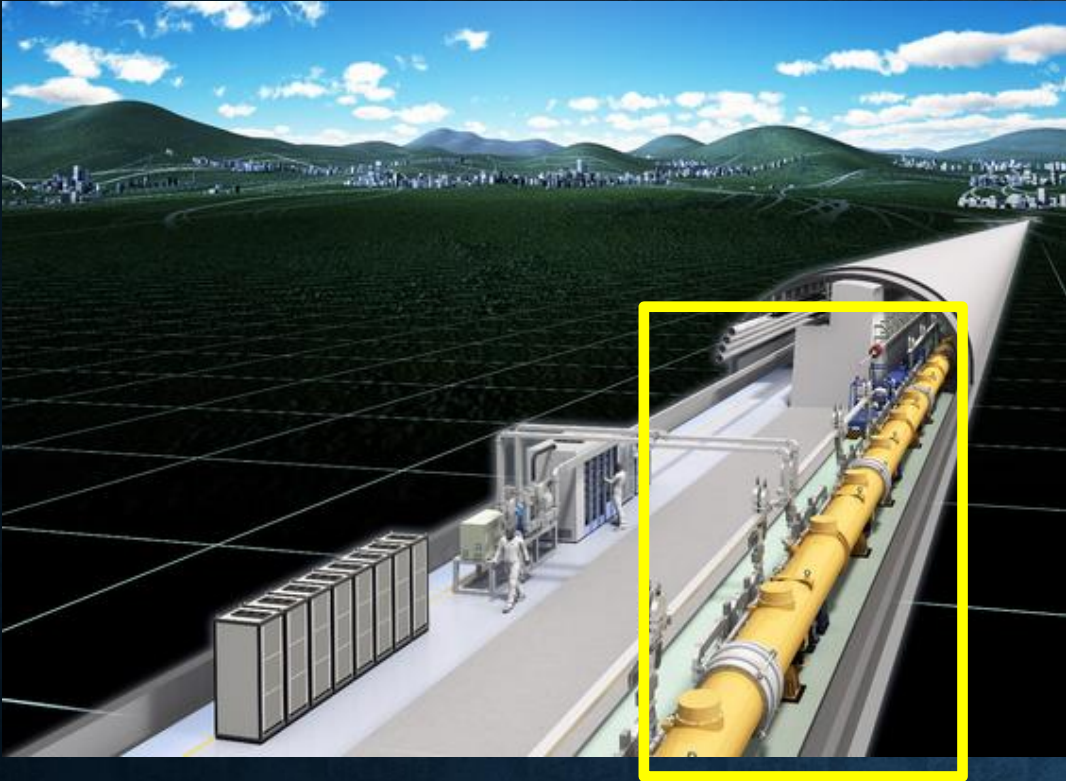
Takayuki Kubo



Here we use the International Linear Collider (ILC), an electron–positron Higgs factory, **as an example**, but the European X-ray Free-Electron Laser (European XFEL) in Germany and the Linac Coherent Light Source II (LCLS-II) at SLAC in California are also based on similar superconducting RF linac technology.

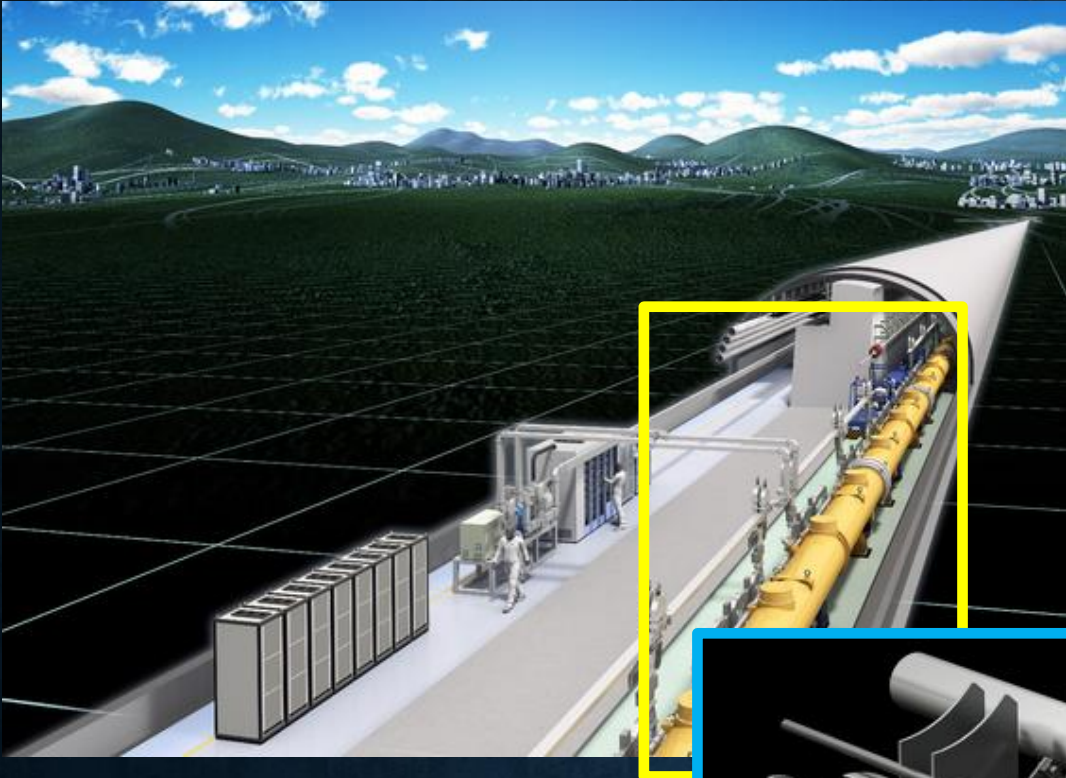




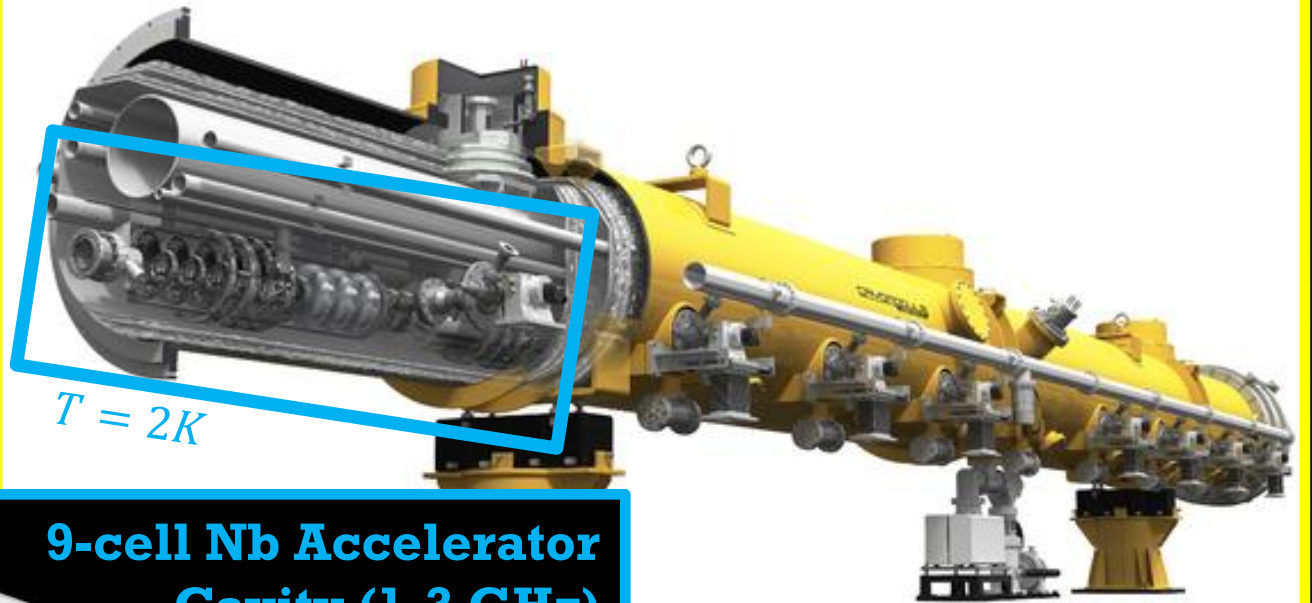


Cryomodule



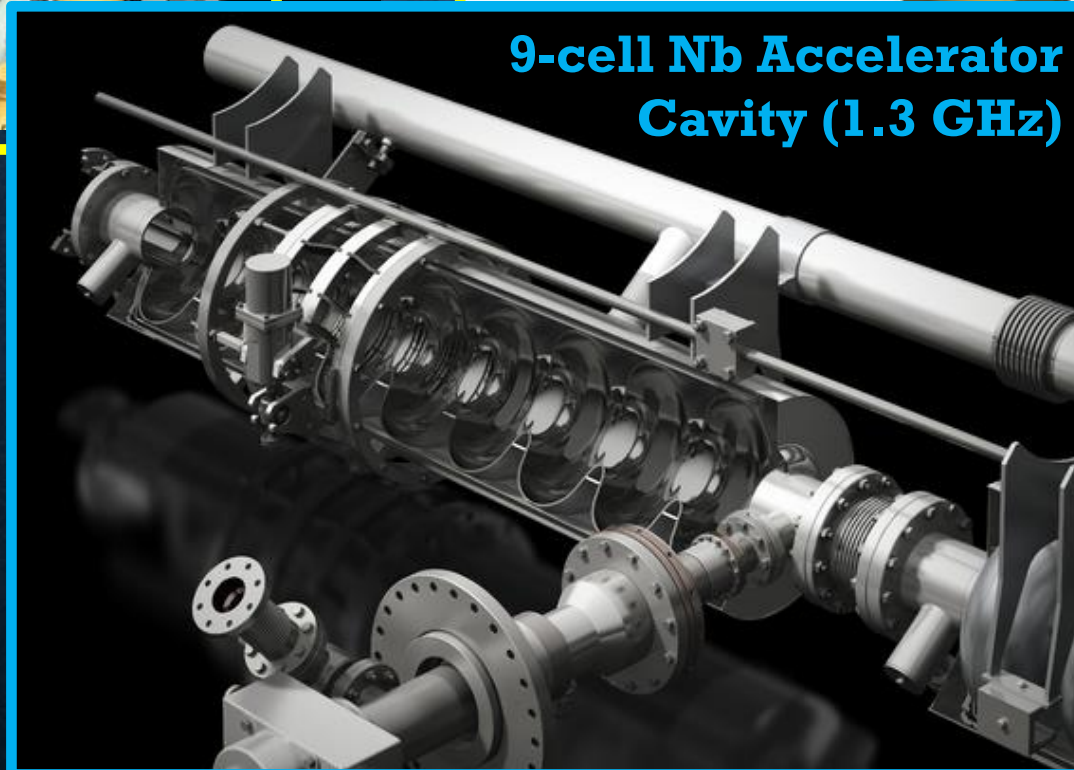


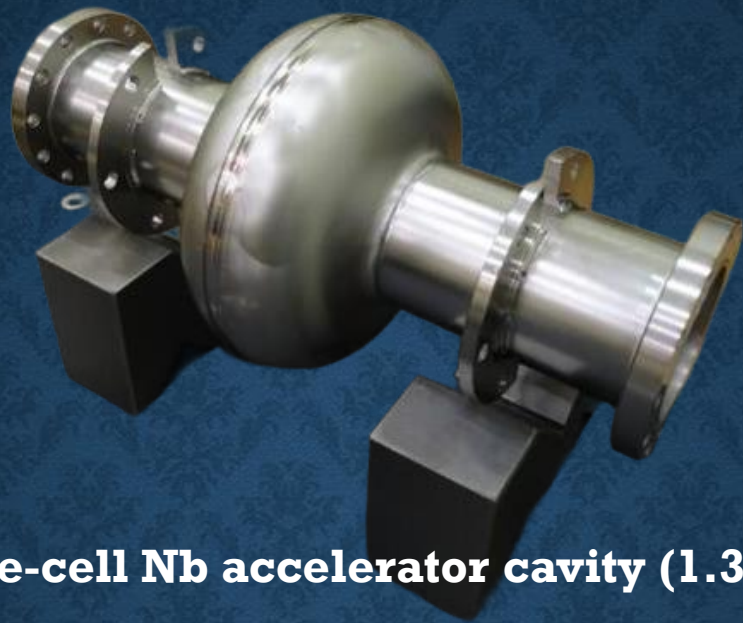
Cryomodule



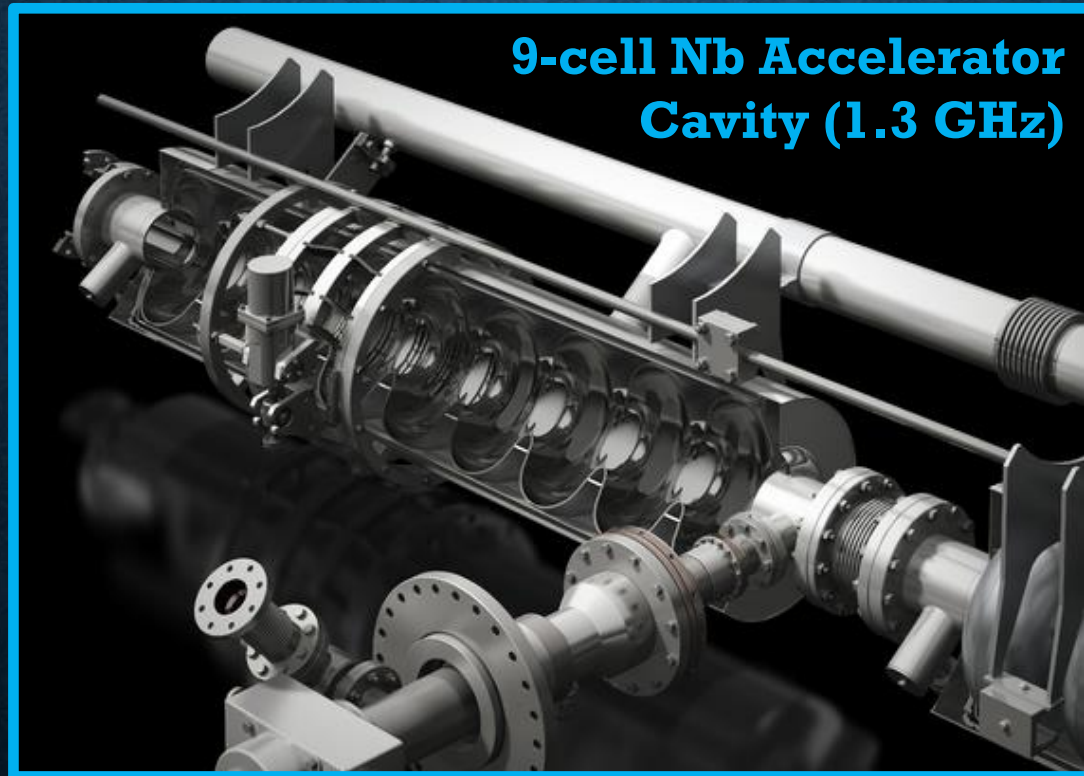
$T = 2K$

**9-cell Nb Accelerator
Cavity (1.3 GHz)**

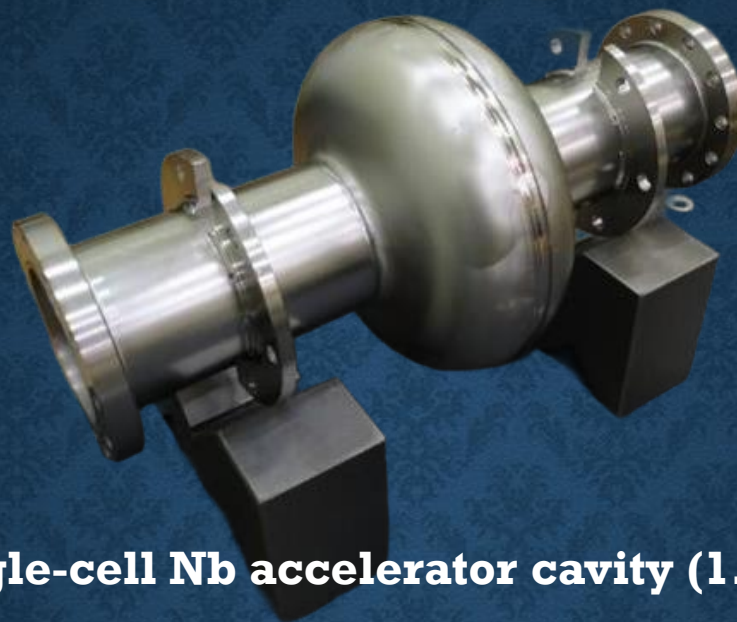
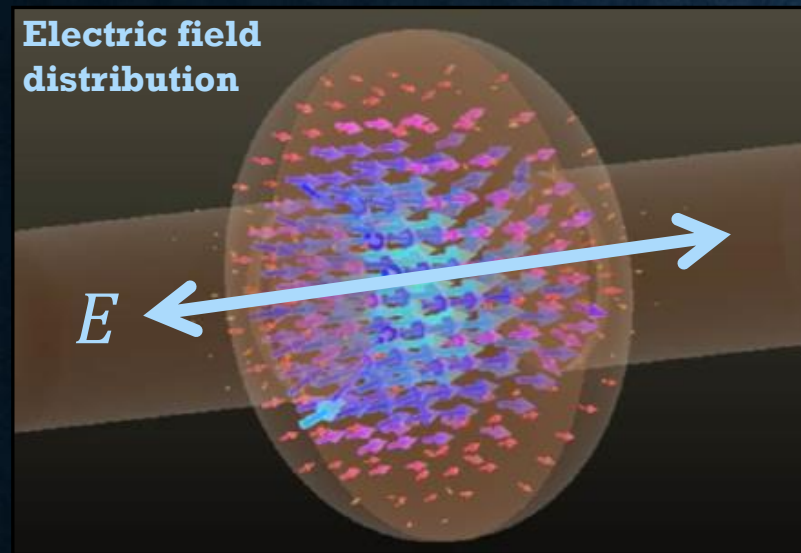




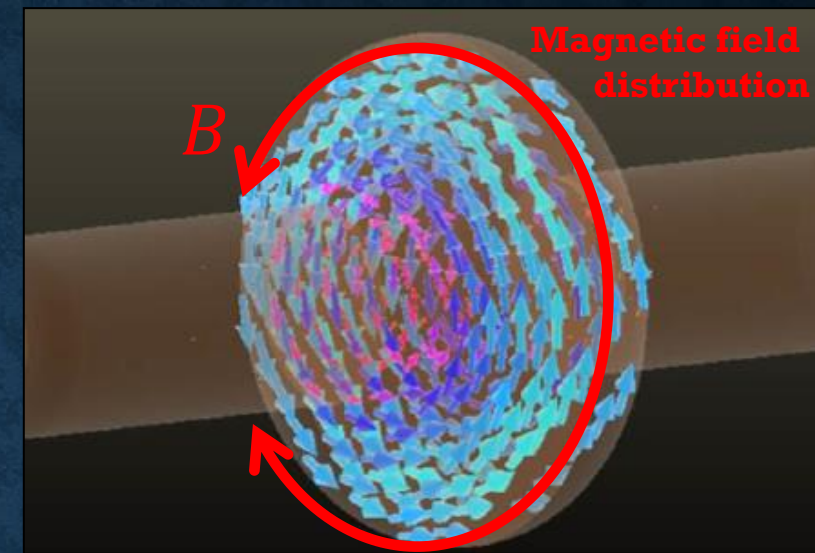
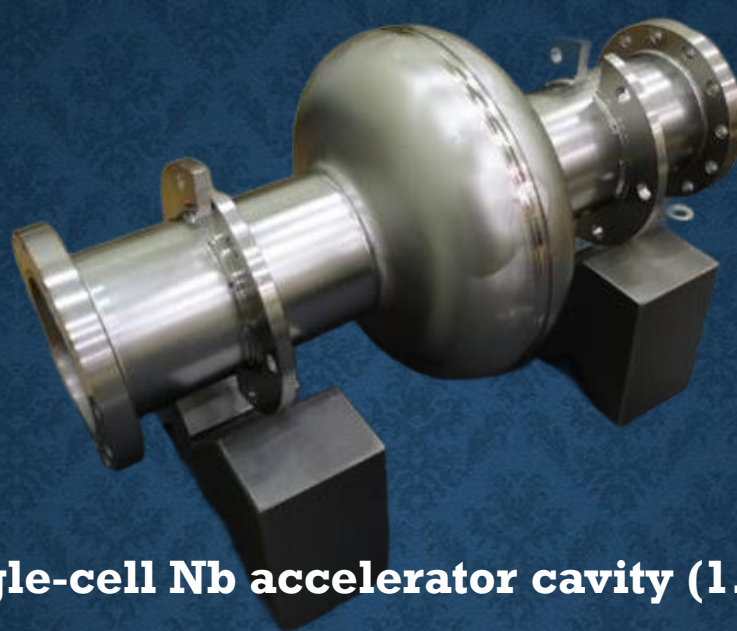
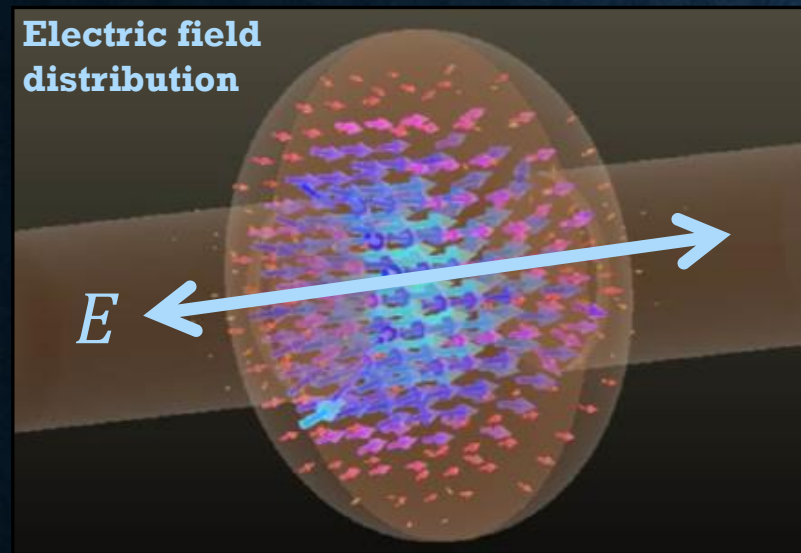
Single-cell Nb accelerator cavity (1.3 GHz)



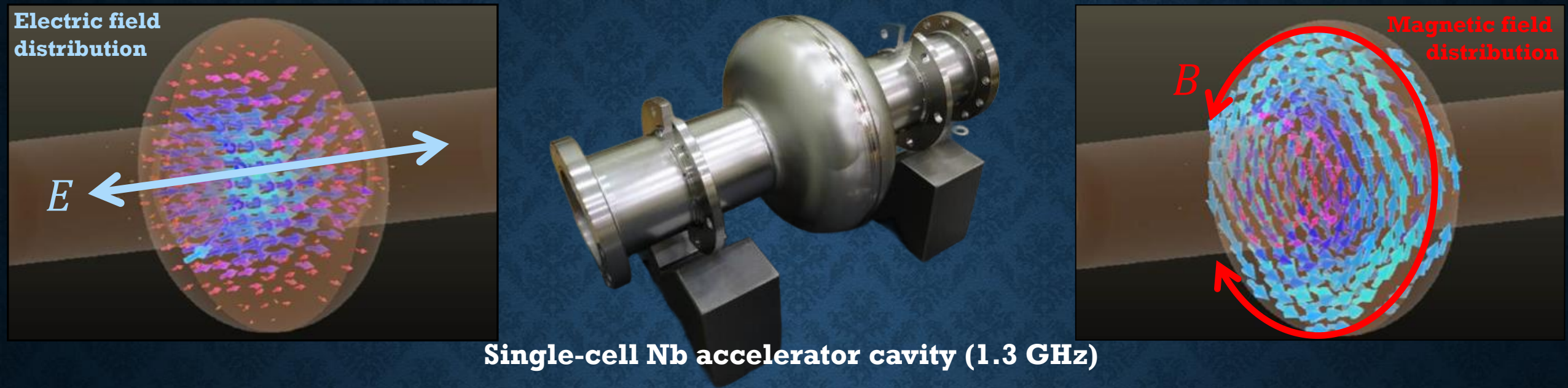
**9-cell Nb Accelerator
Cavity (1.3 GHz)**



Single-cell Nb accelerator cavity (1.3 GHz)

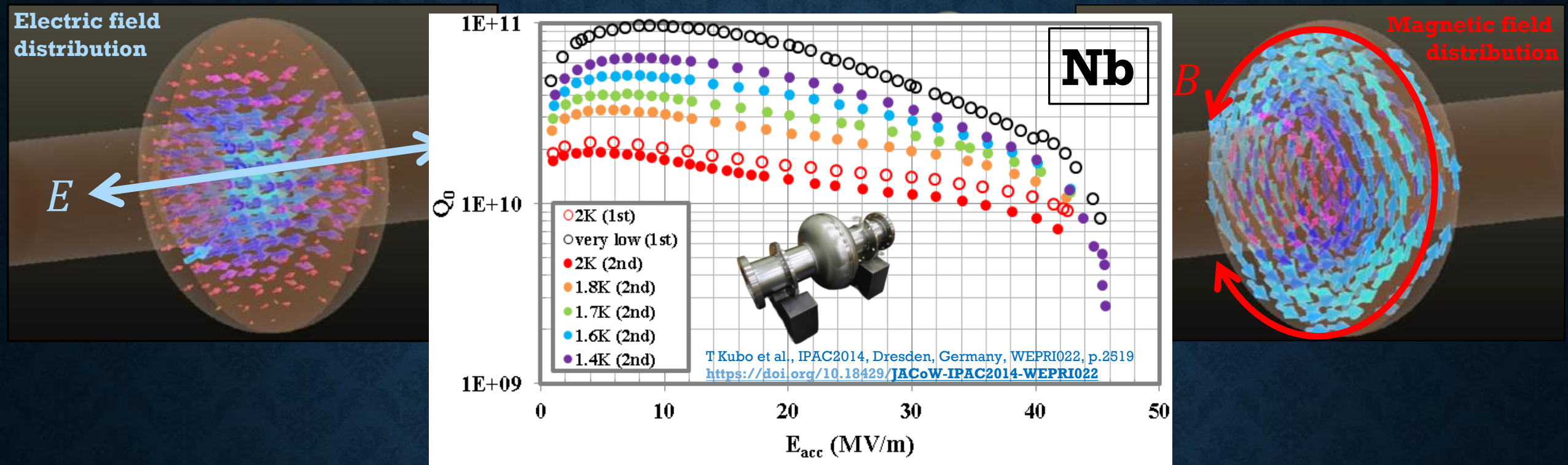


Single-cell Nb accelerator cavity (1.3 GHz)



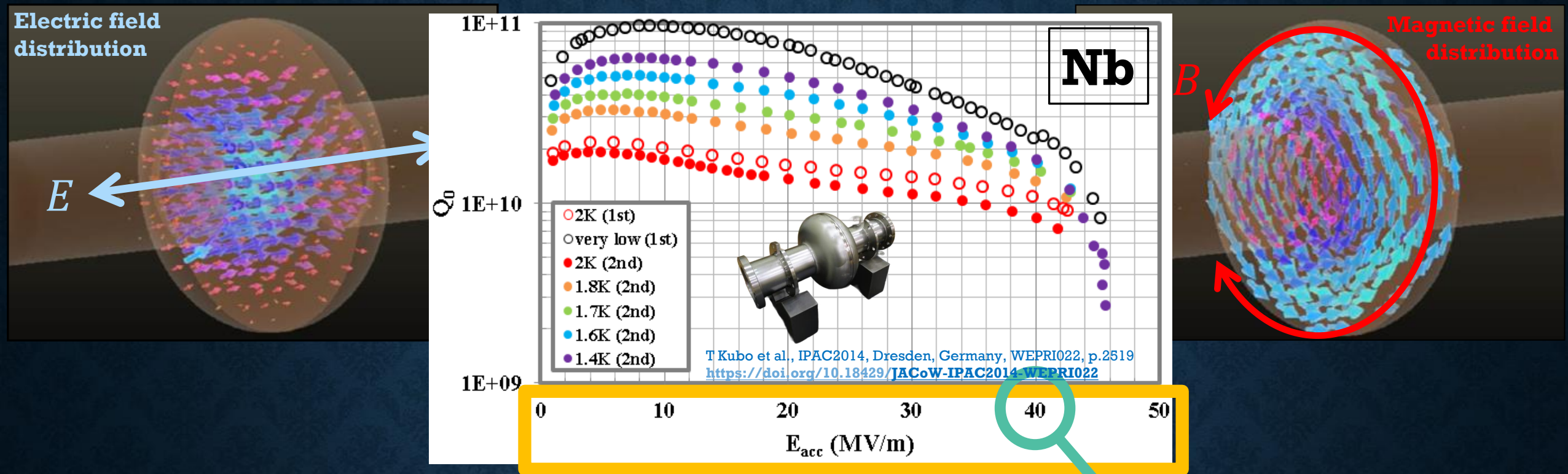
There are two key performance metrics

- (1) Accelerating electric field E_{acc}
- (2) Quality factor Q_0 (smallness of dissipation)



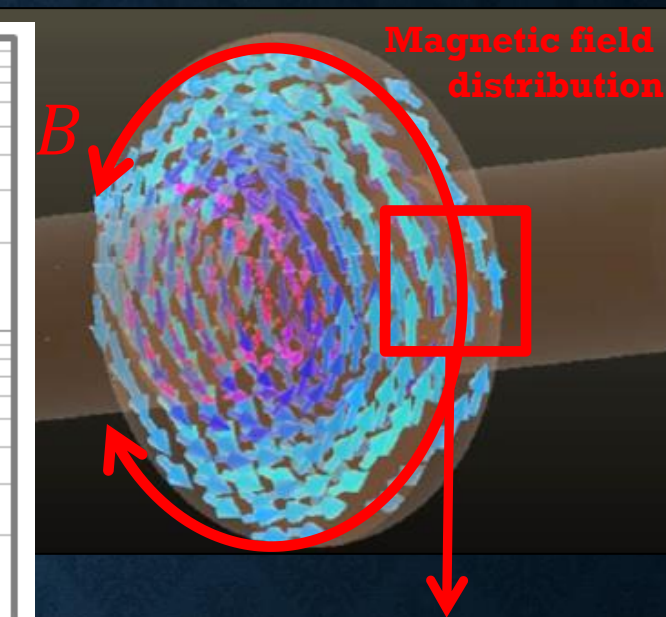
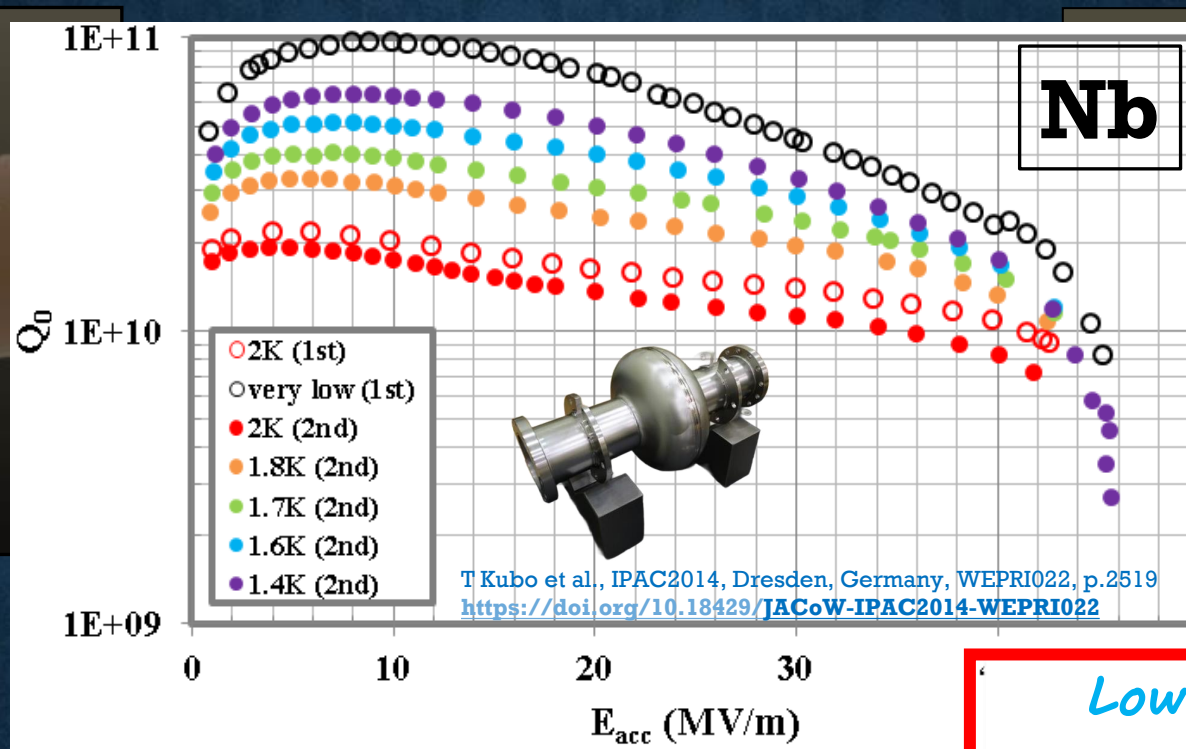
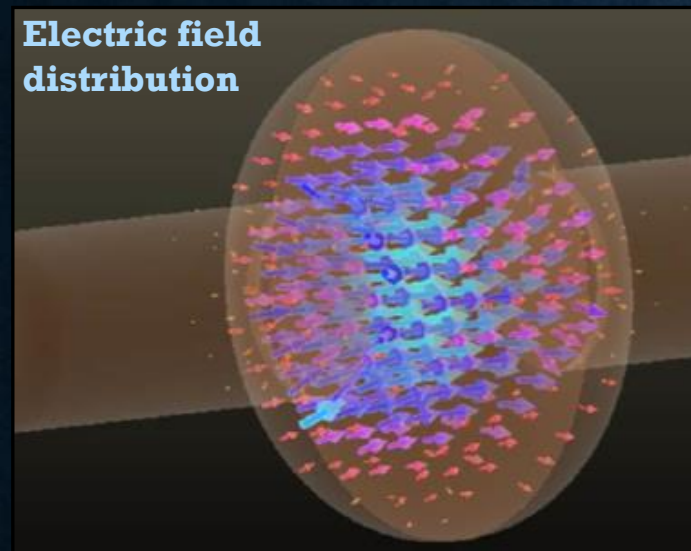
There are two key performance metrics

- (1) Accelerating electric field E_{acc}
- (2) Quality factor Q_0 (smallness of dissipation)

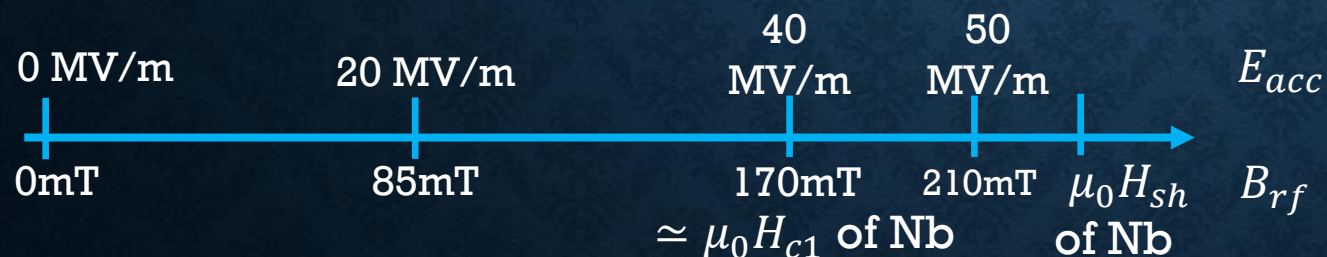


There are two key performance metrics

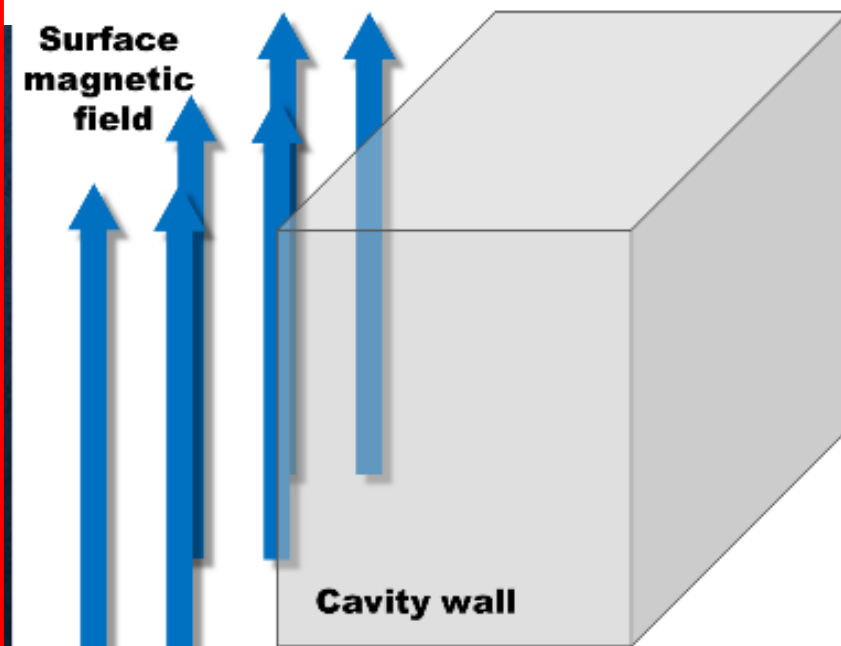
- (1) **Accelerating electric field E_{acc}**
 - (2) **Quality factor Q_0 (smallness of dissipation)**
- At 40MV/m, microwave B field reaches $B \simeq B_{c1} \simeq 170$ mT



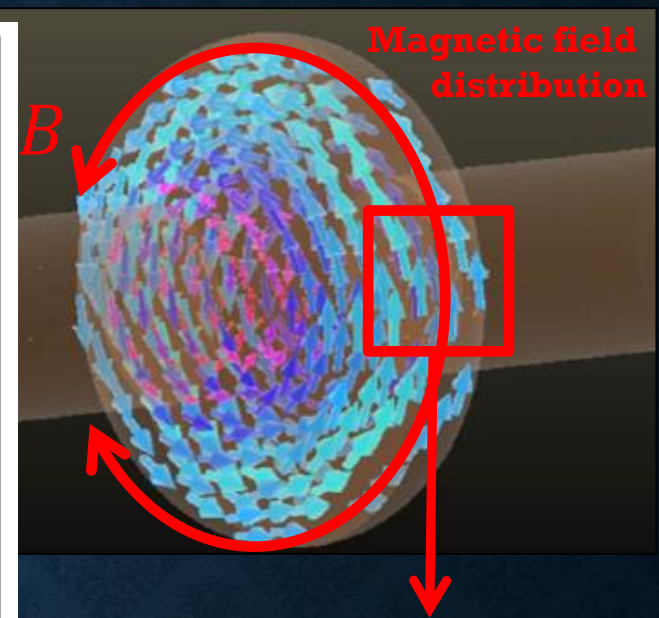
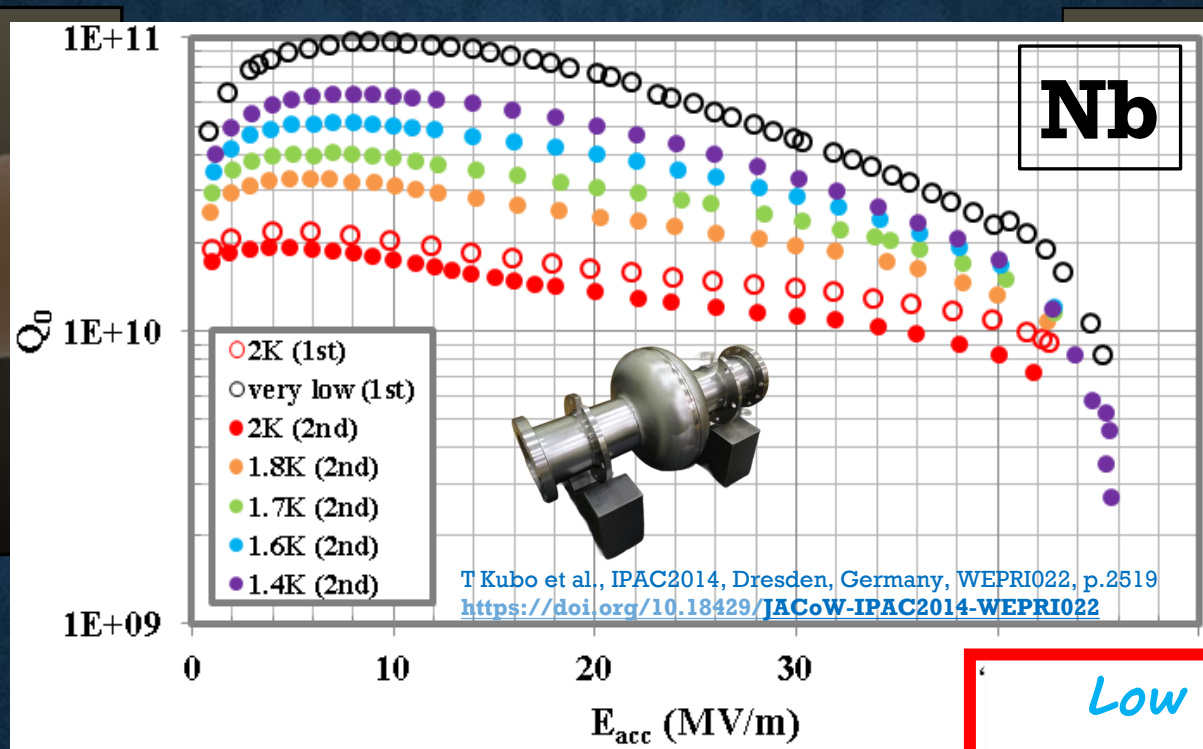
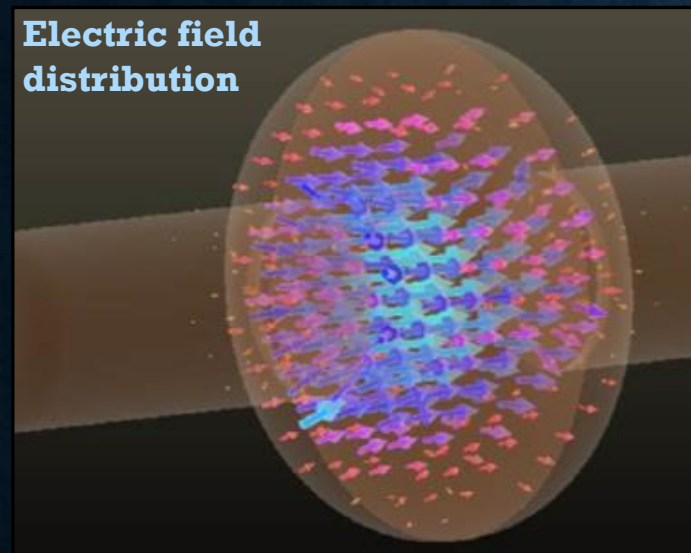
Note: Cavities are operated under the **Meissner state**



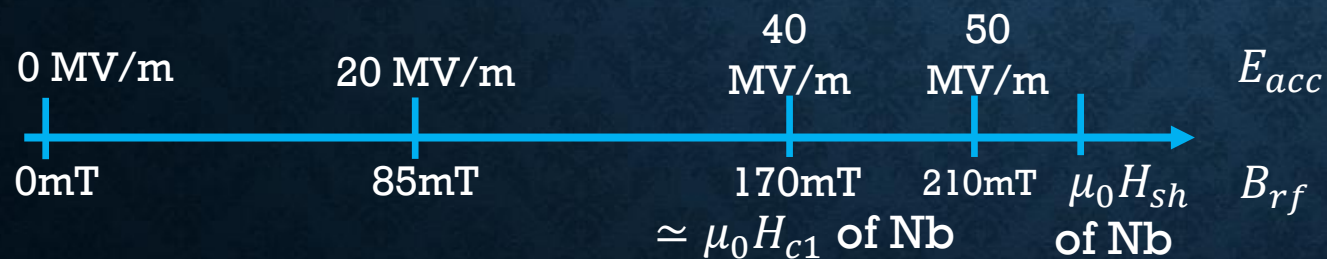
Low field (Meissner state)



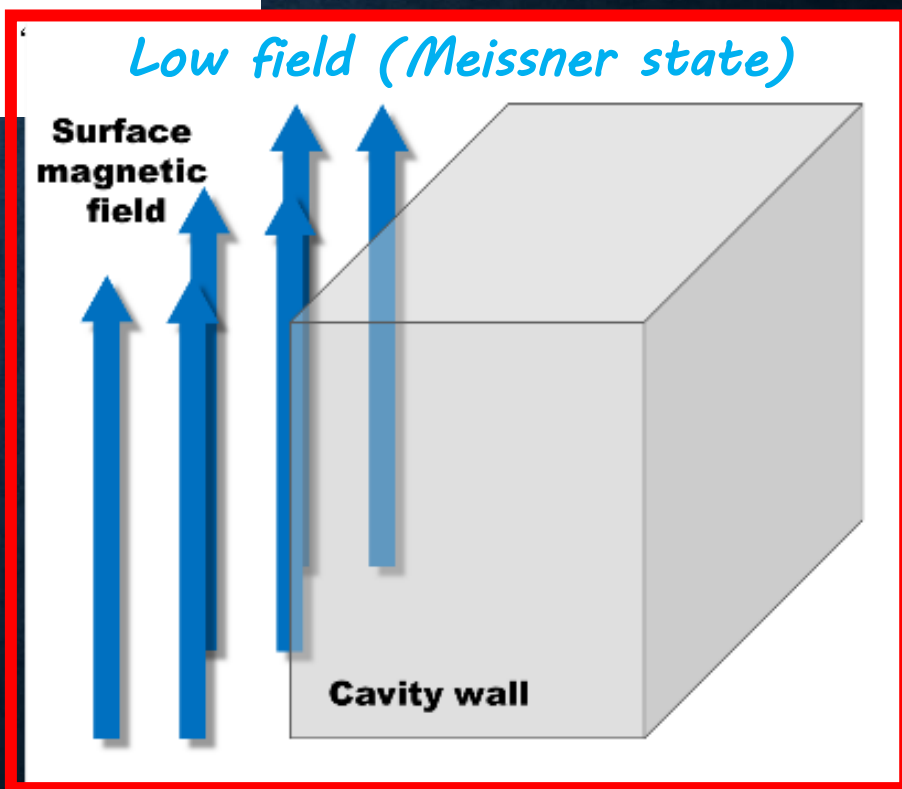
(Γ $\square$ $\square$) good

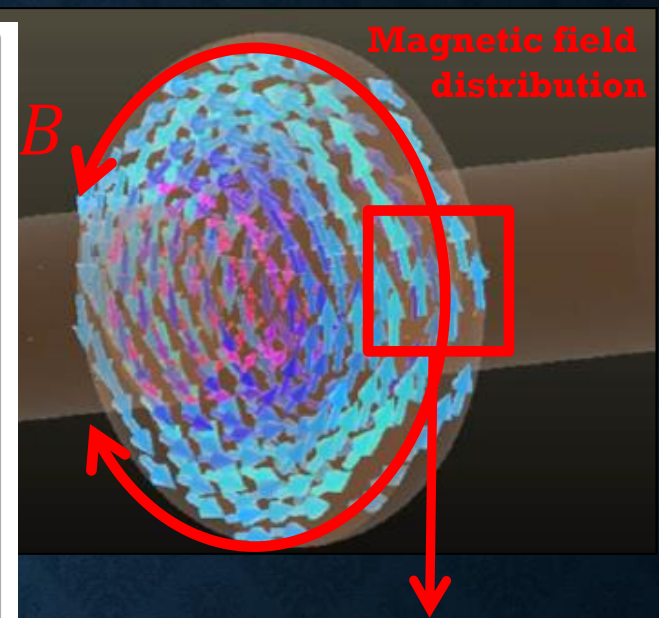
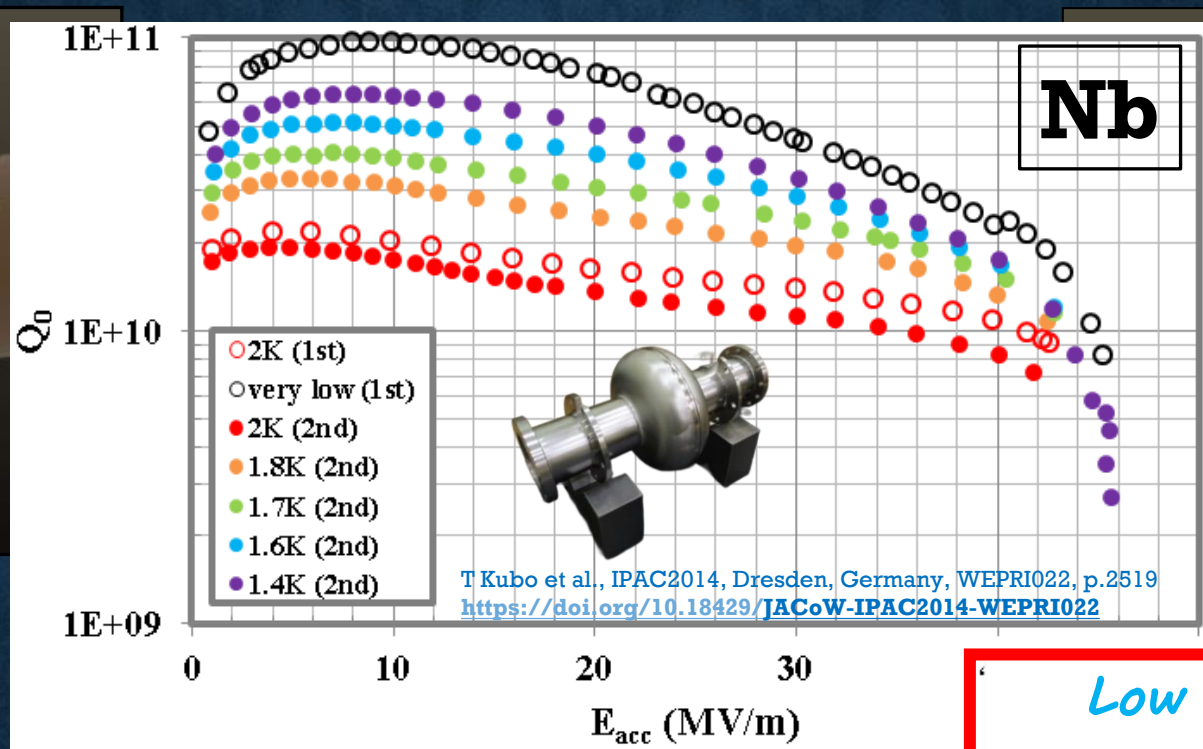
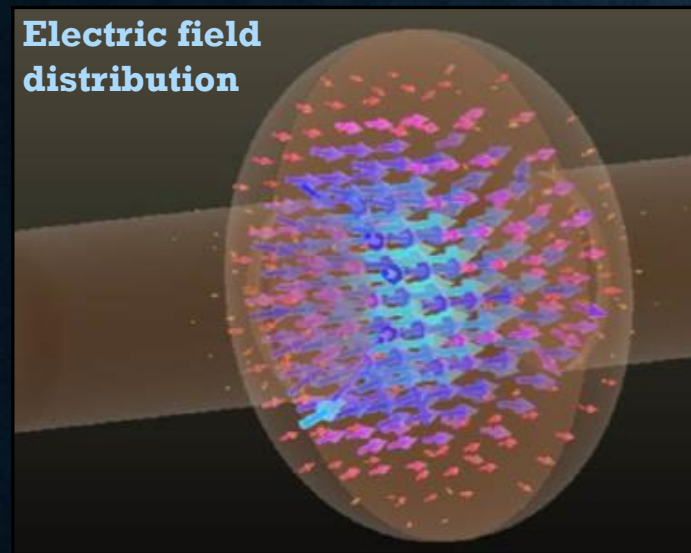


Note: Cavities are operated under the **Meissner state**

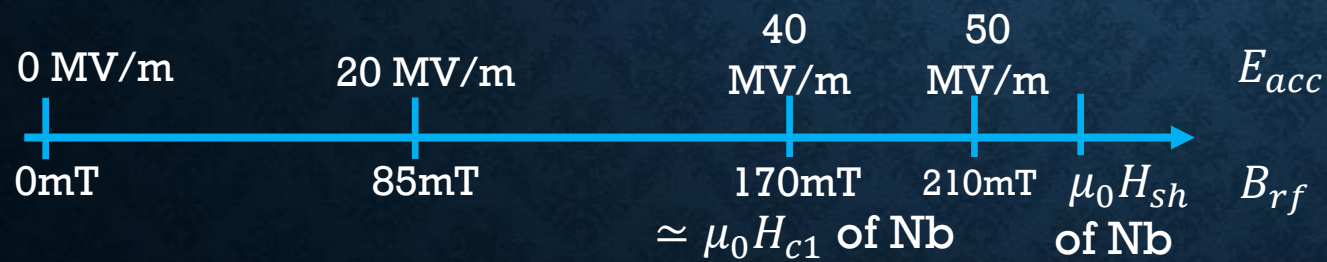


(Γ $\square$ $\square$) good

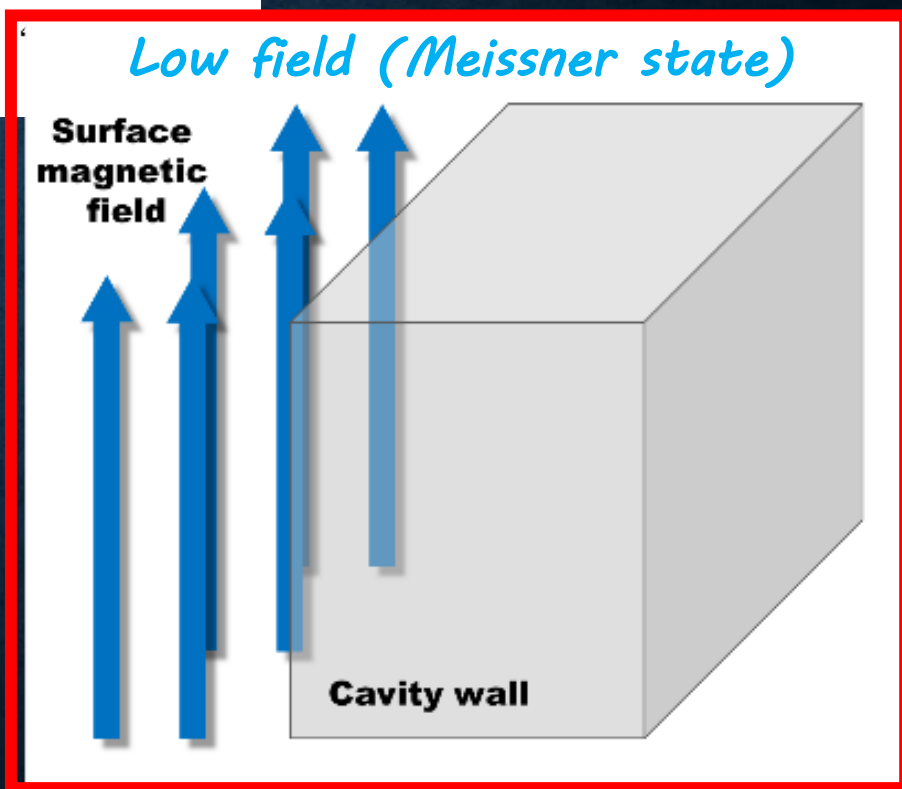


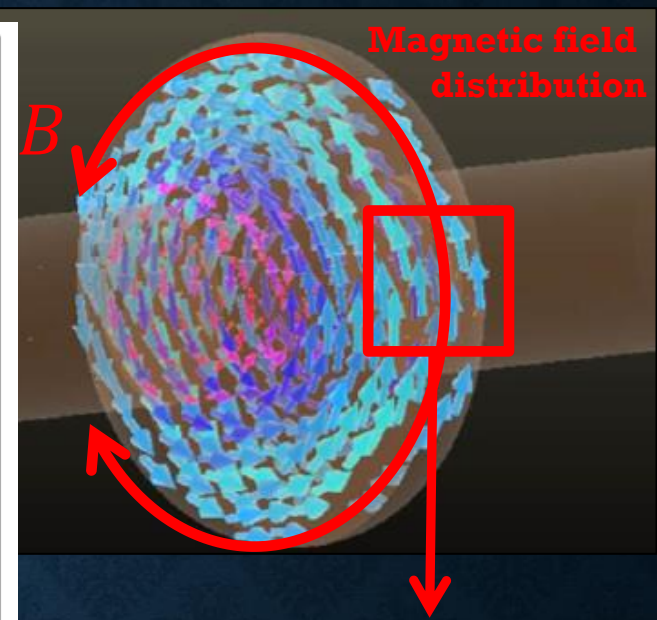
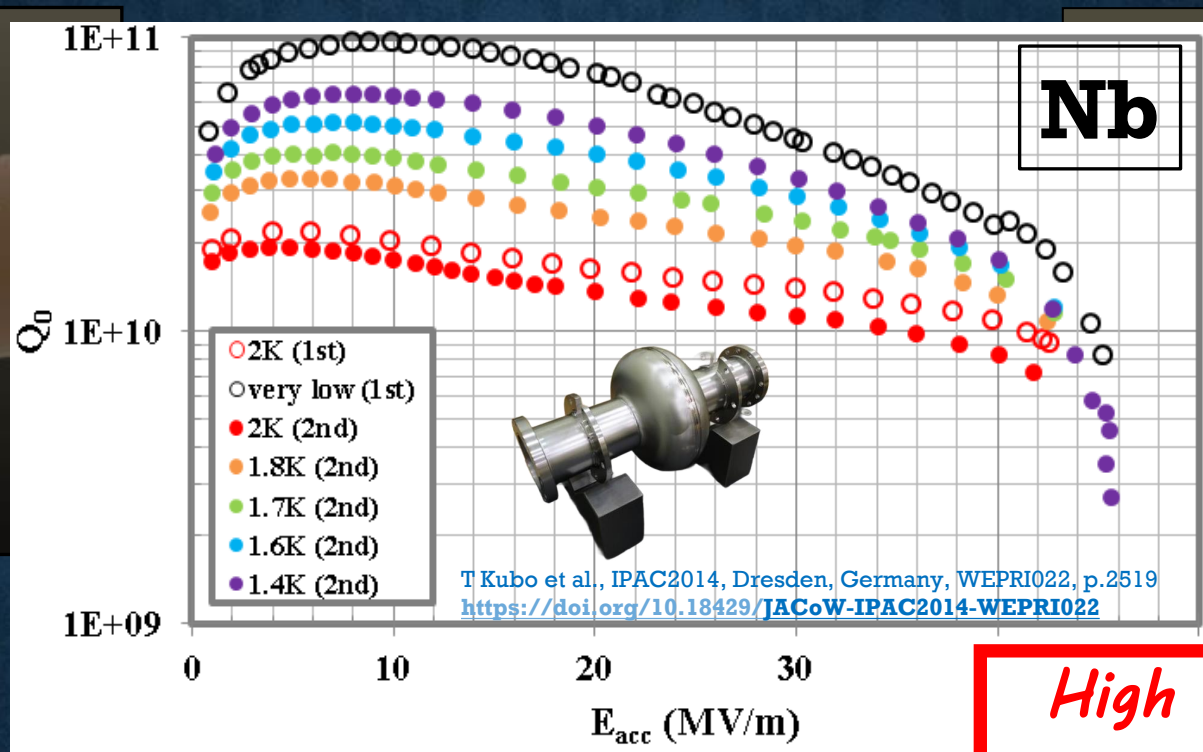
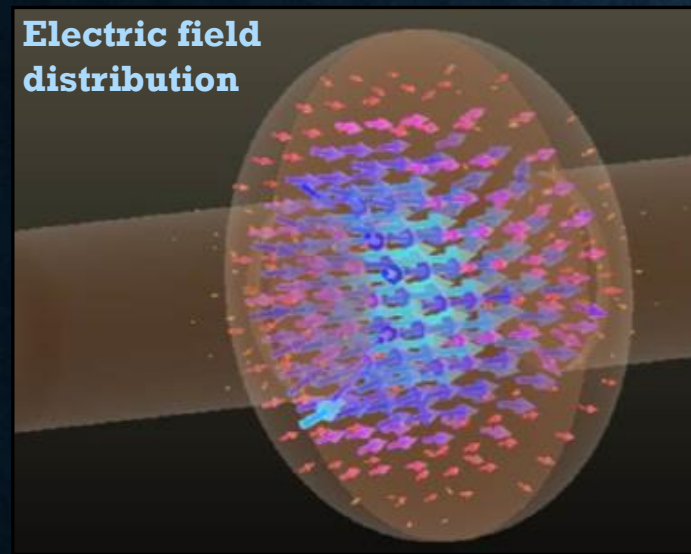


Note: Cavities are operated under the **Meissner state**

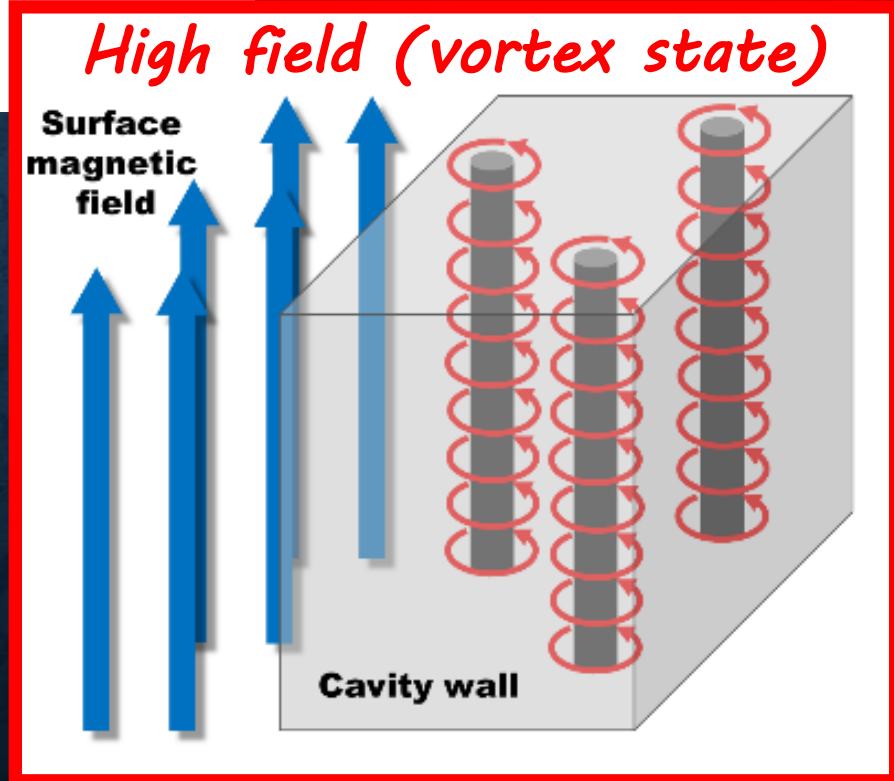
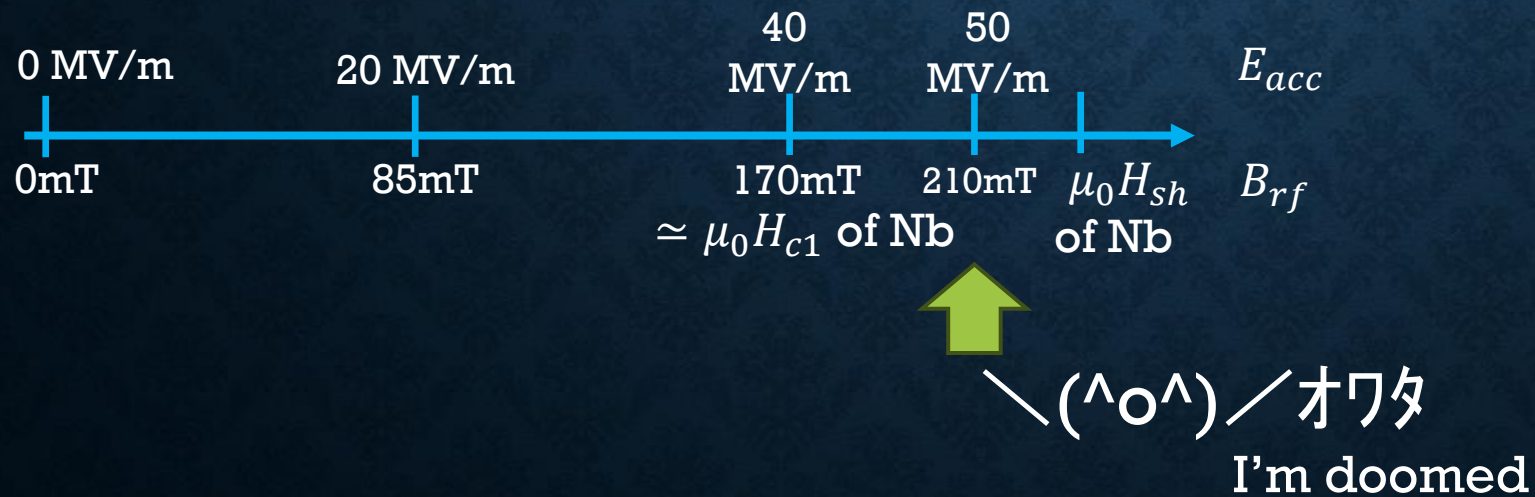


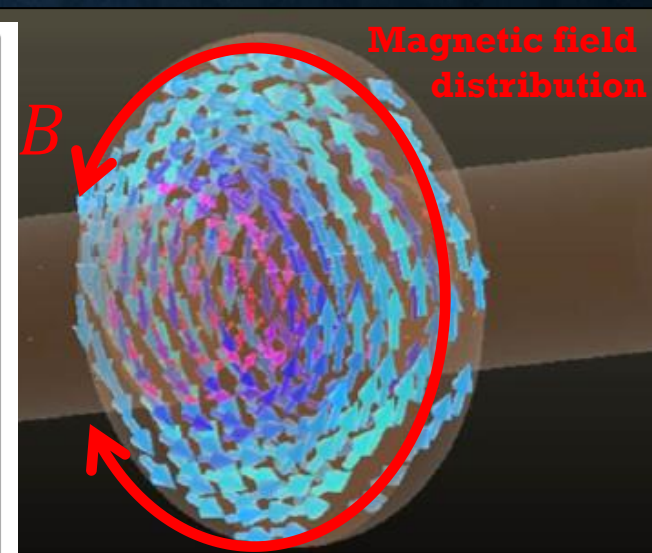
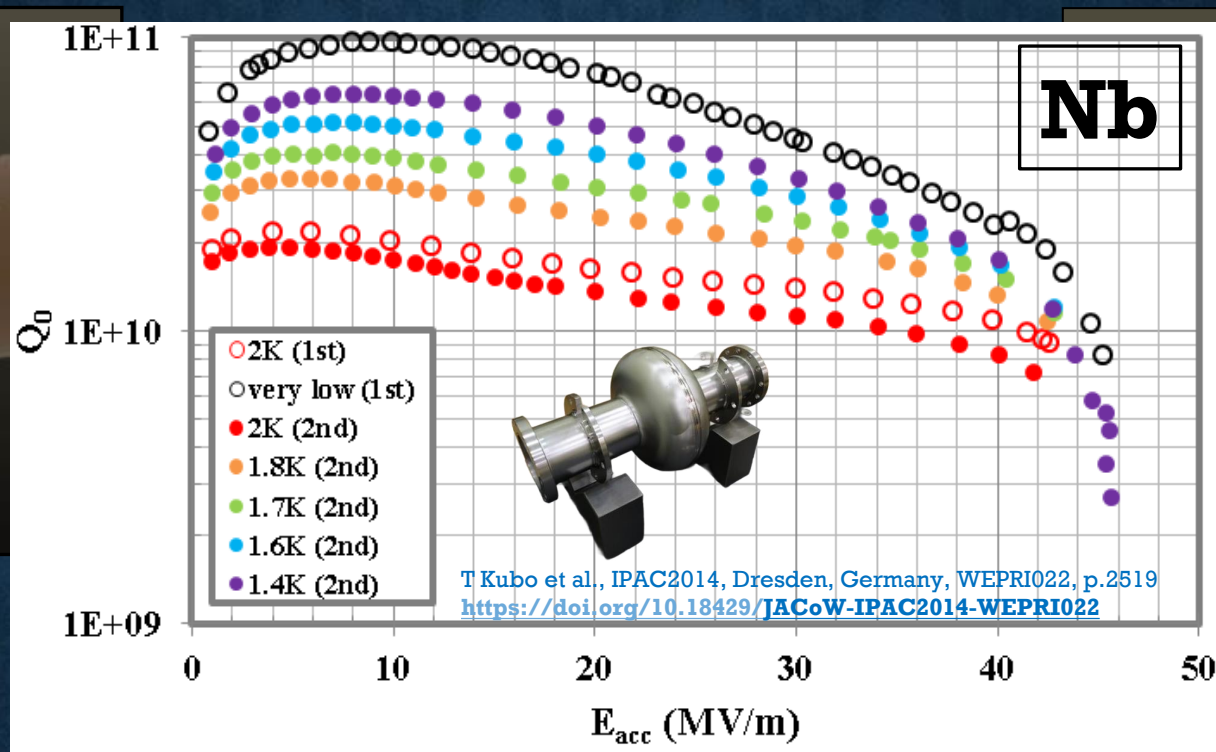
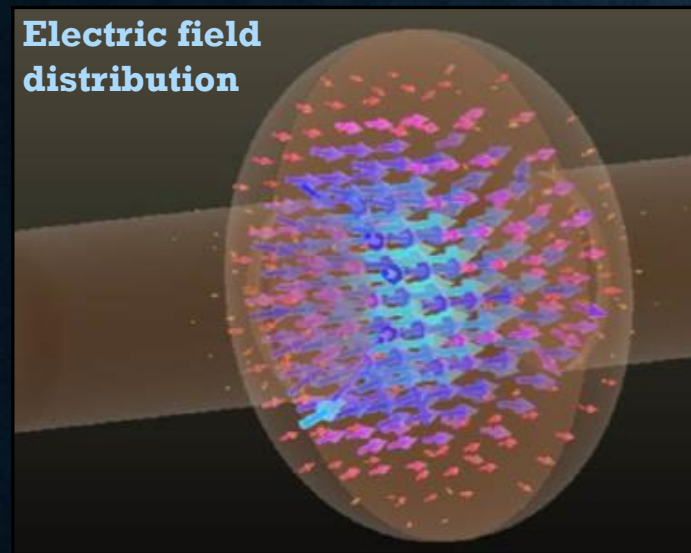
($\square$ $\square$) good



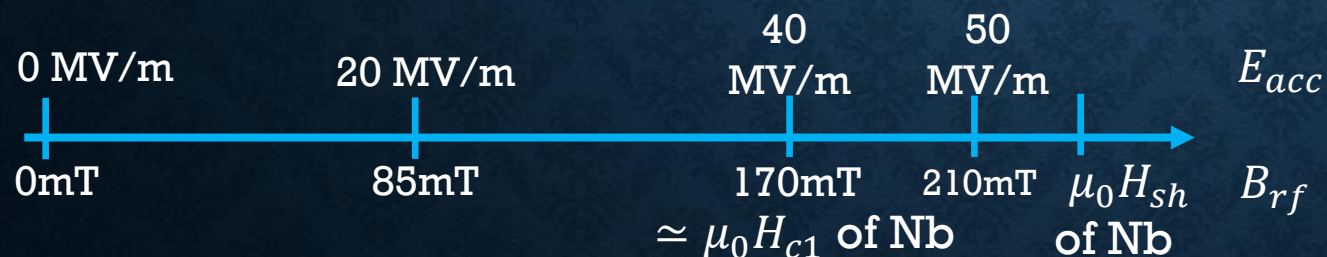


Note: Cavities are operated under the **Meissner state**





Note: Cavities are operated under the **Meissner state**

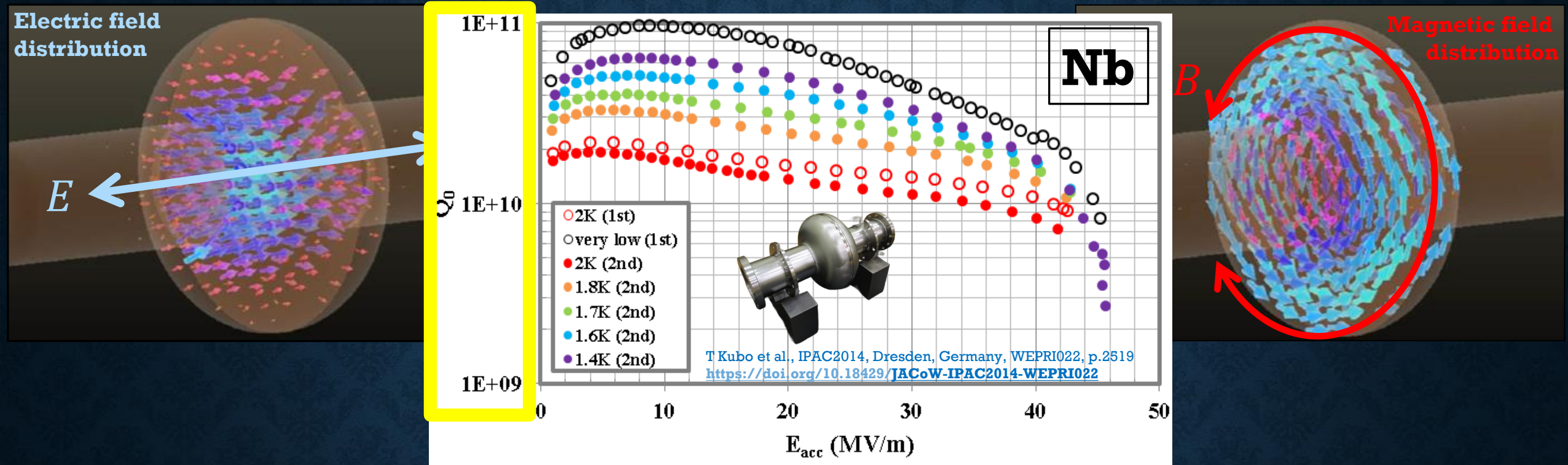


B_{c1} : the lower critical field, above which the Meissner state becomes energetically unfavorable.

B_{sh} : the superheating field, which represents the stability limit of the Meissner state. **Ultimate limit for the ideal material.**

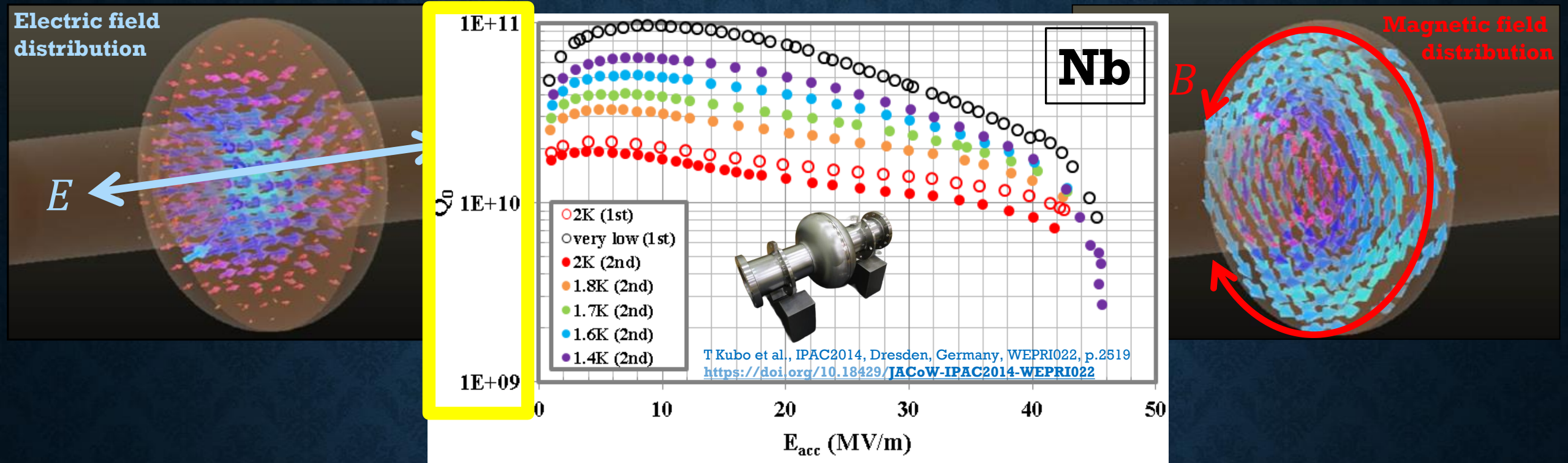
$$B_{c1} \leq B_0^{max} < B_{sh}$$

- G. Catelani and J. P. Sethna, Phys. Rev. B **78**, 224509 (2008)
- F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)
- T. Kubo, Phys. Rev. Research **2**, 033203 (2020)
- T. Kubo, Supercond.Sci. Technol. **34**, 045006 (2021)



There are two key performance metrics

- (1) Accelerating electric field E_{acc}
- (2) **Quality factor Q_0** (smallness of dissipation)



There are two key

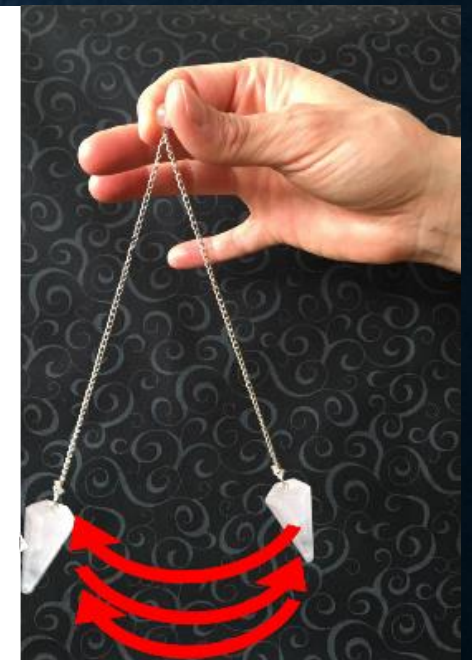
- (1) Accelerating electric field
- (2) **Quality factor Q_0**

This is the key parameter common to all superconducting resonators.

$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1} \propto \frac{1}{R_s}$$

P_1 : dissipation per RF cycle

Q_0 : how many cycles it takes for the stored energy to completely dissipate

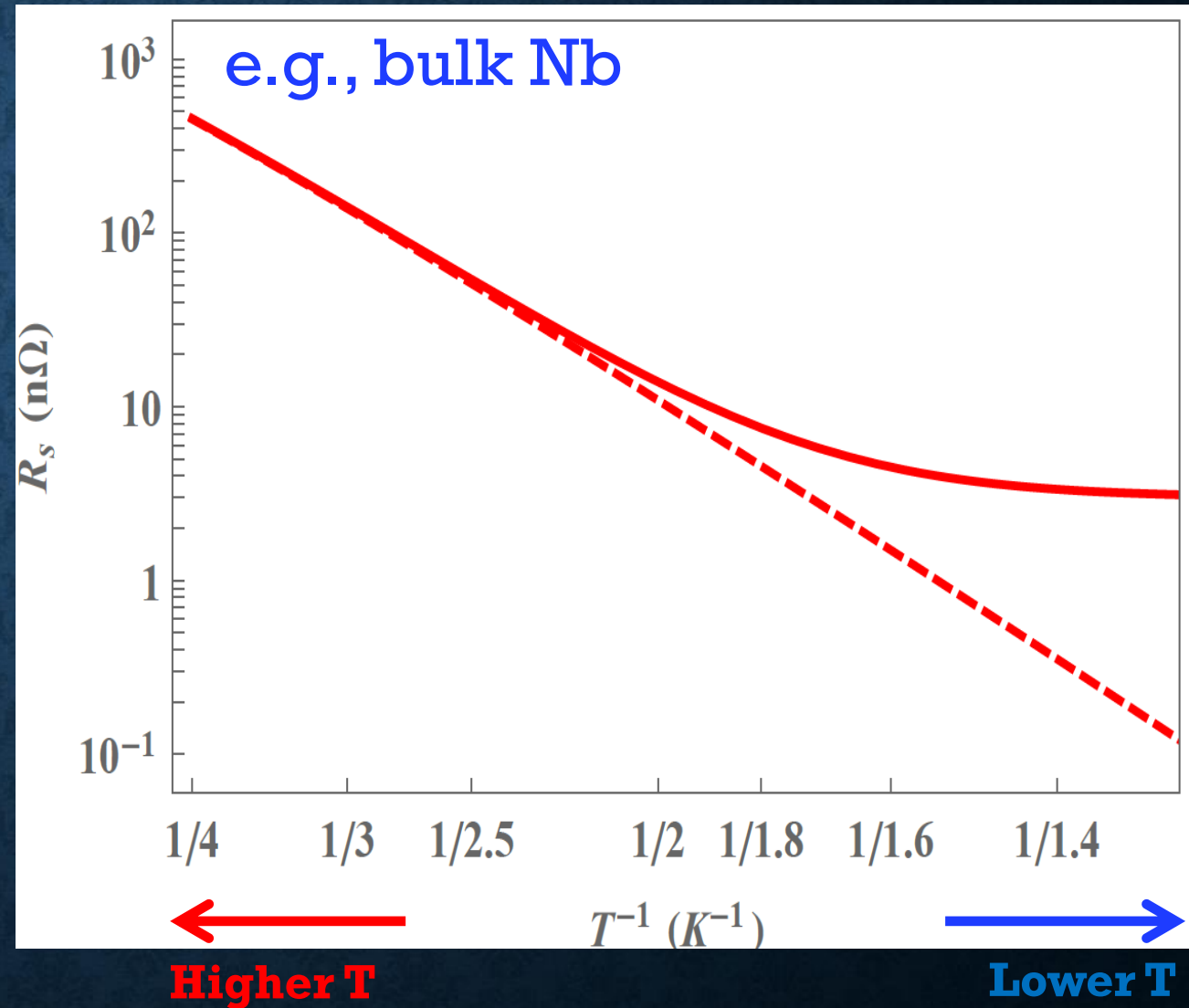


Overview of energy dissipation mechanisms in superconducting resonators

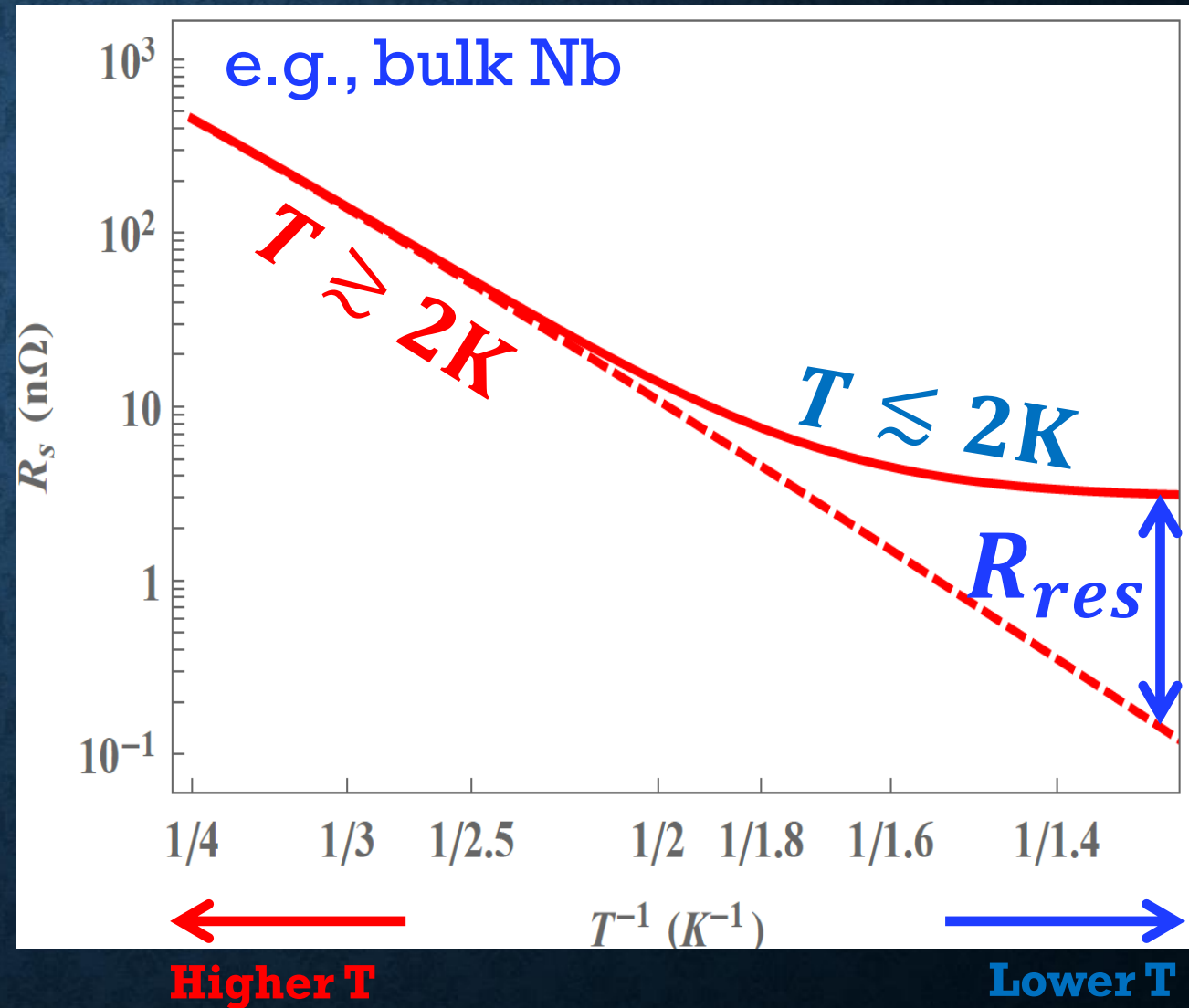
$$Q_0 = \frac{\omega U}{\boxed{P}} \sim \frac{U}{\boxed{P_1}} \propto \frac{1}{\boxed{R_s}}$$


What are the causes of dissipation?

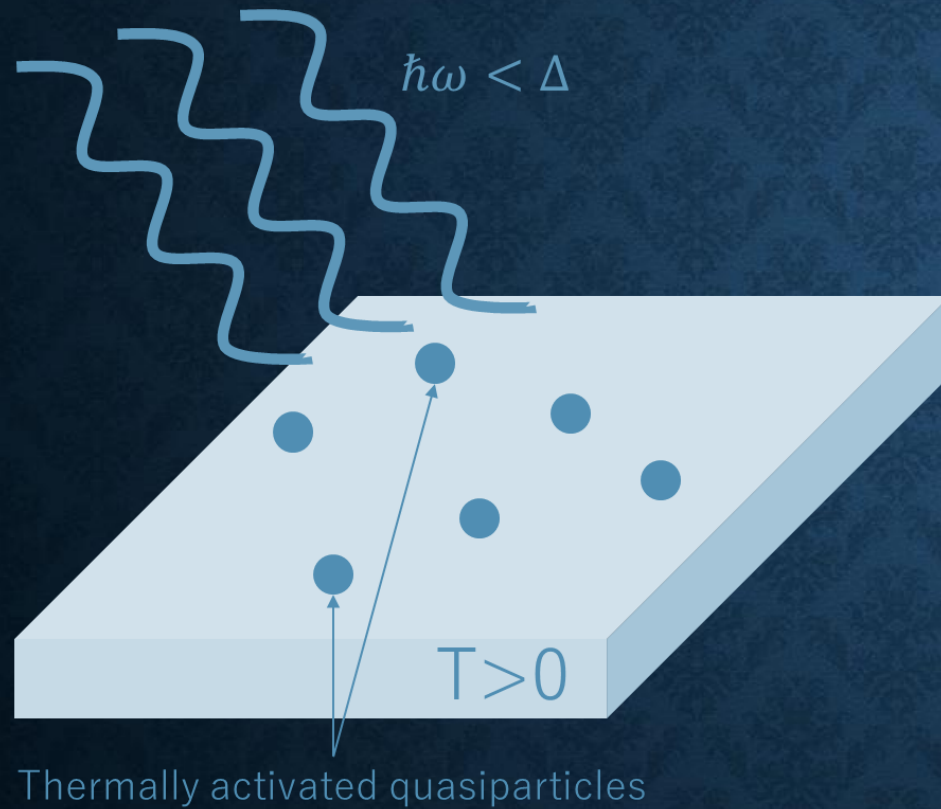
Surface resistance $\propto \frac{1}{Q_0}$



Surface resistance $\propto \frac{1}{Q_0}$

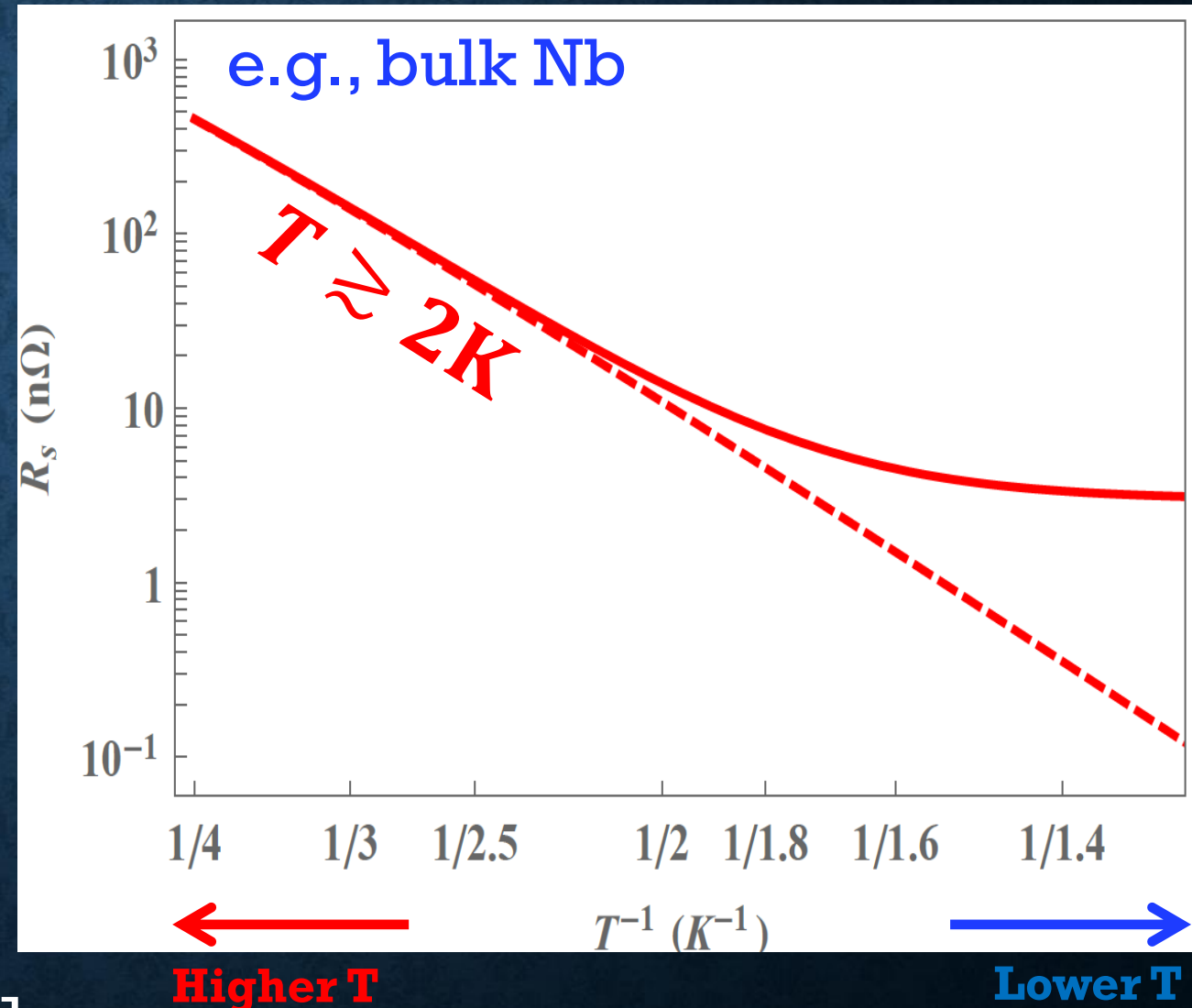


At a higher T ($\gtrsim 2\text{K}$)



The contribution from **thermally activated quasiparticles** decreases exponentially [$\propto \exp(-\Delta/kT)$].

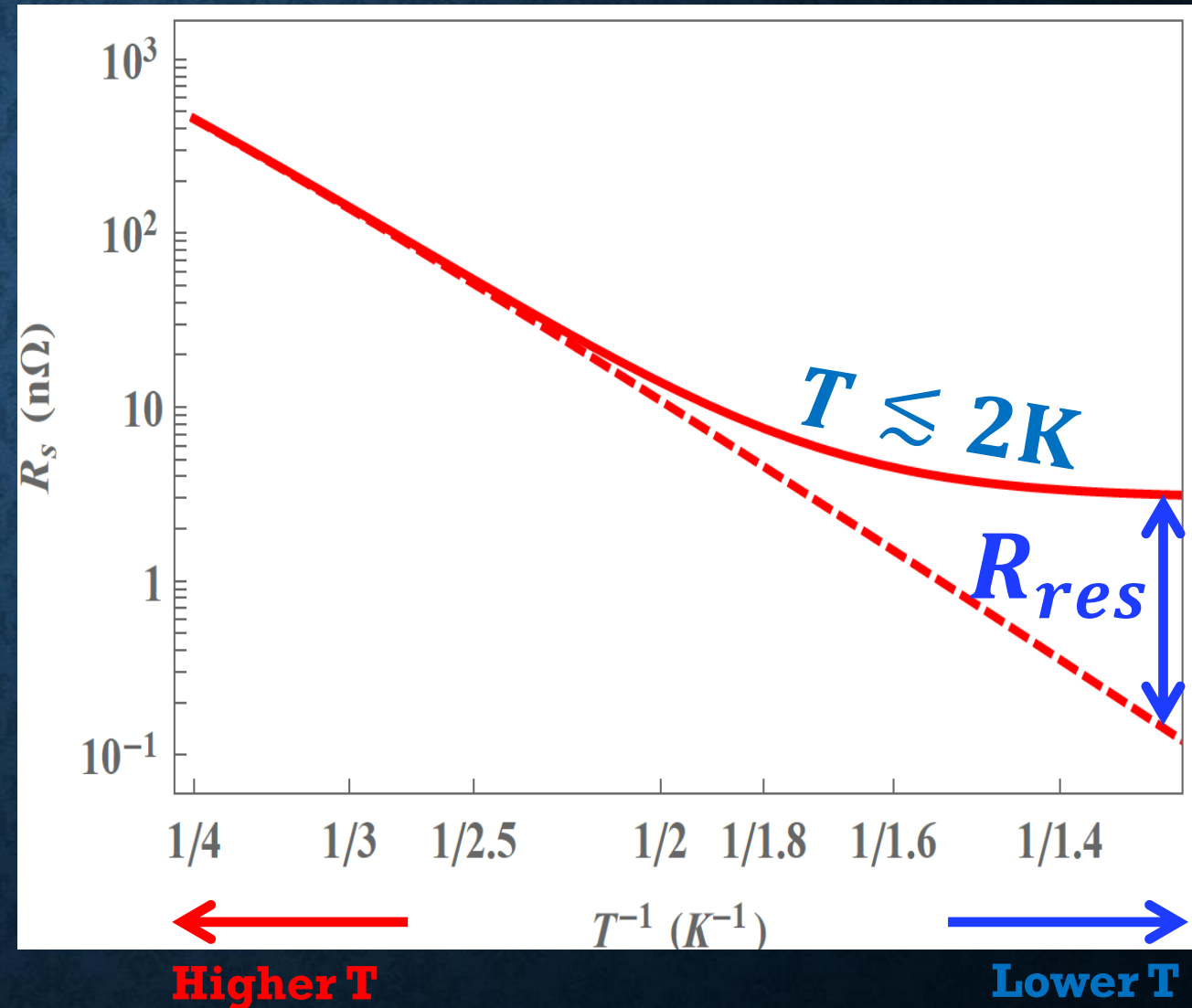
Surface resistance $\propto \frac{1}{Q_0}$



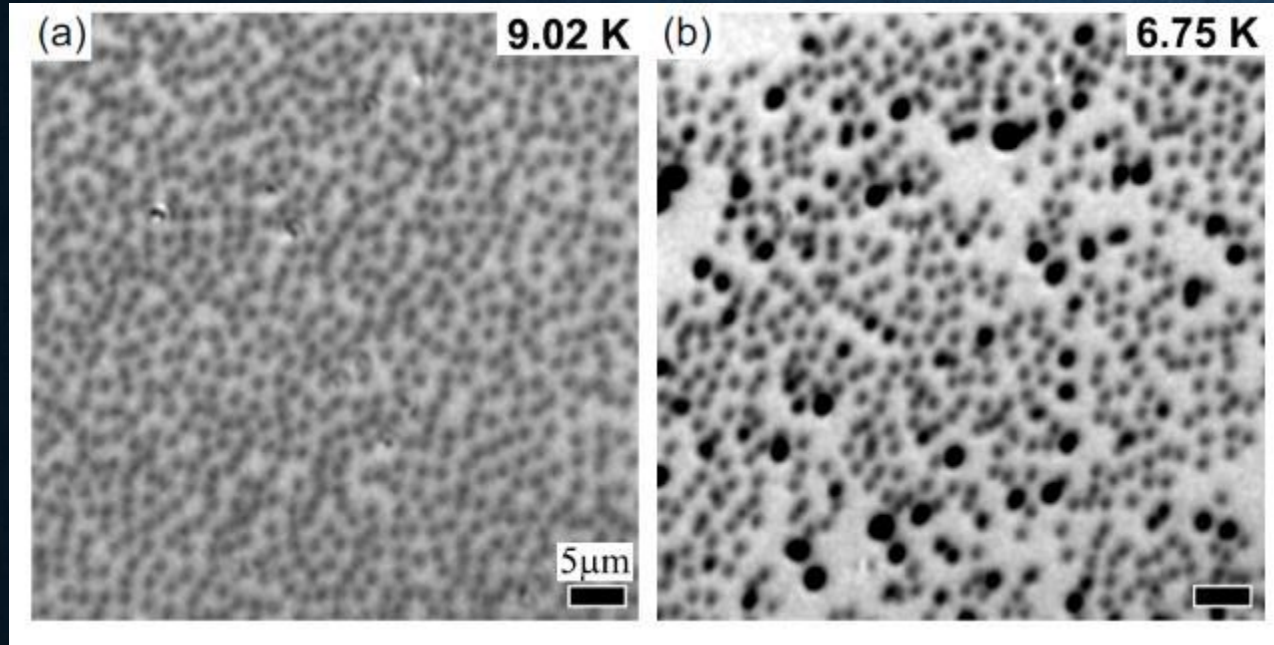
At a lower T ($\lesssim 2\text{K}$)

The quasiparticle contribution $R_s \sim e^{-\frac{\Delta}{k_B T}}$ is exponentially suppressed (dashed curve), but **other mechanisms can dominate at low temperatures.**

Surface resistance $\propto \frac{1}{Q_0}$



Dominant cause of R_{res} in SRF cavities is Vortices trapped during cool down

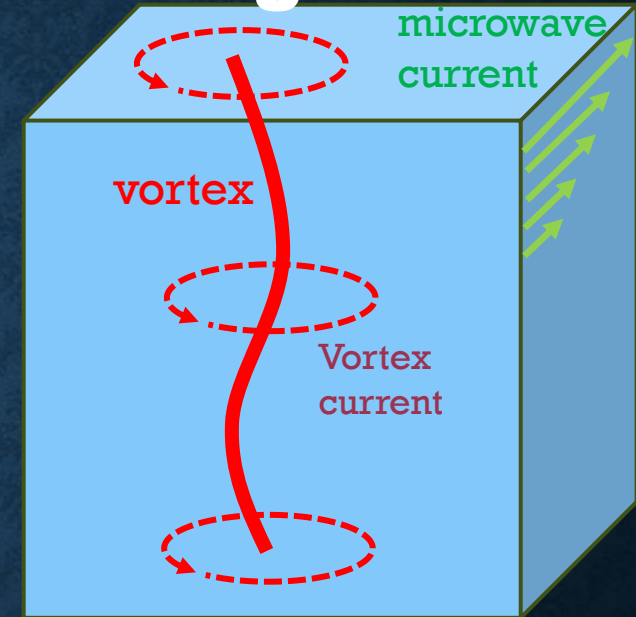


Figures adapted from Ooi et al., PRB **111**, 094519 (2025)

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Videos of vortex dynamics in **cavity-grade Nb (RRR>300)** can be found in Ooi et al., PRB **111**, 094519 (2025)

Microwave-driven vortex dynamics cause dissipation, which is only weakly dependent on temperature, leading to an elevated surface resistance: R_{res}



See also,

A. Gurevich and G. Ciovati, PRB **87**, 054502 (2013)

A. Romanenko et al., Appl. Phys. Lett. **105**, 234103 (2014).

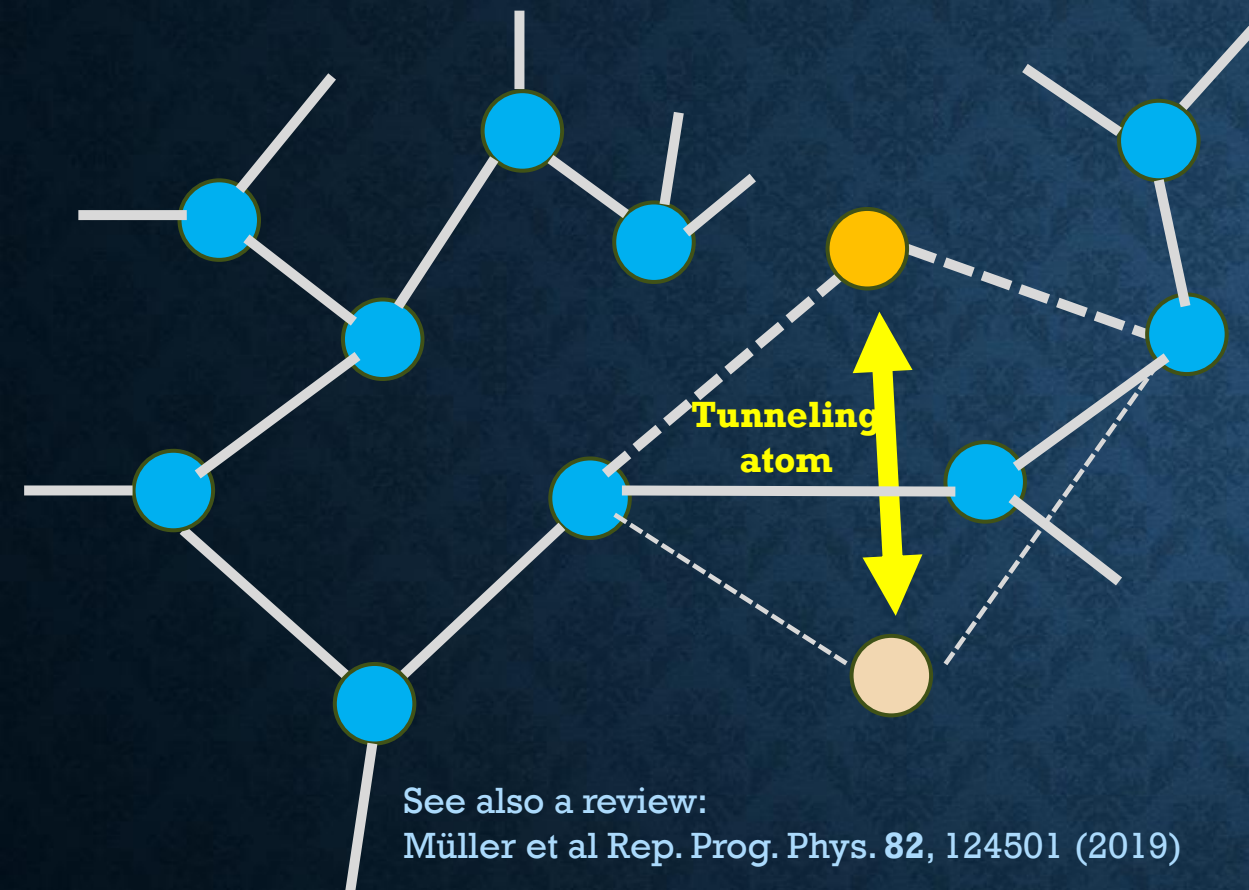
S. Huang et al., Phys. Rev. Accelerators and Beams **19**, 082001 (2016).

S. Posen et al., J. Appl. Phys. **119**, 213903 (2016).

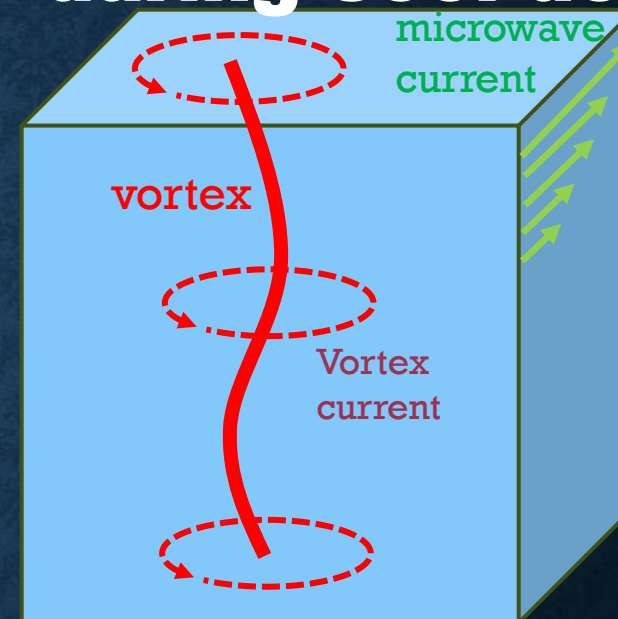
S. Ooi et al., Physical Review B **104**, 064504 (2021).

S. Ooi et al., Physical Review B **111**, 094519 (2025)

Two-level system defects



Vortices trapped during cool down



See also,

A. Gurevich and G. Ciovati, PRB **87**, 054502 (2013)

A. Romanenko et al., Appl. Phys. Lett. **105**, 234103 (2014).

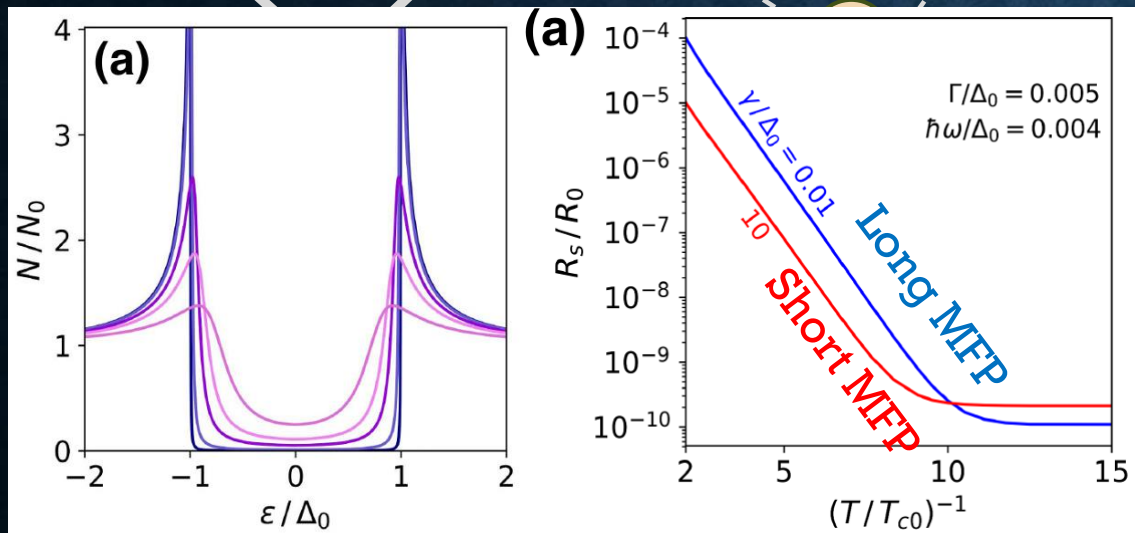
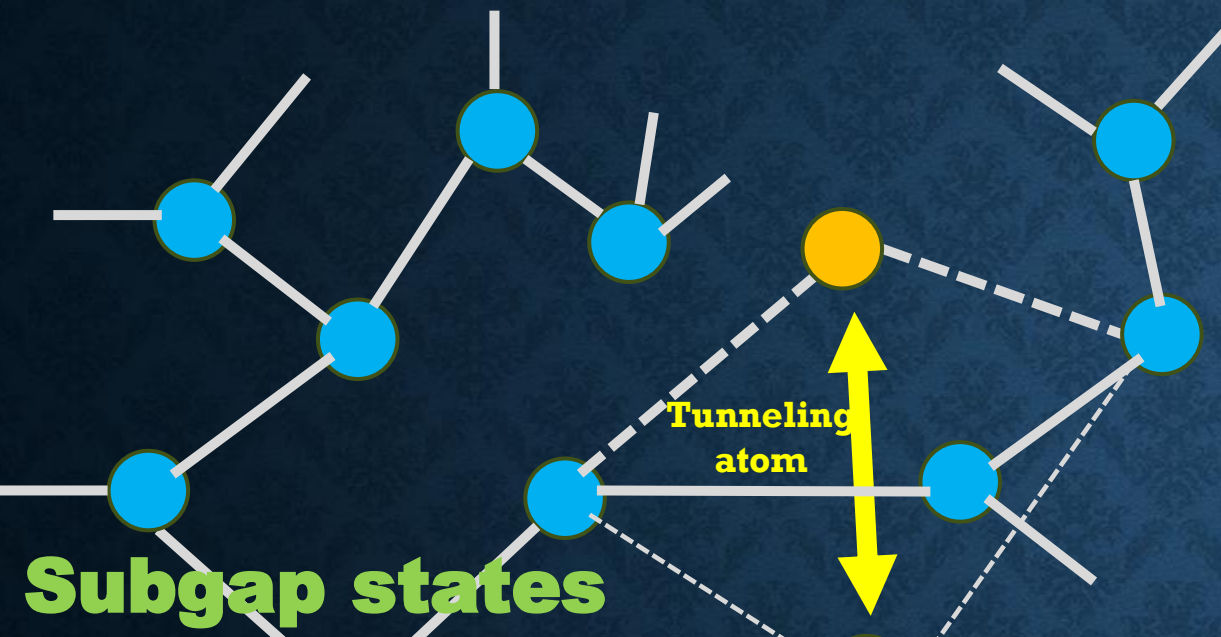
S. Huang et al., Phys. Rev. Accelerators and Beams **19**, 082001 (2016).

S. Posen et al., J. Appl. Phys. **119**, 213903 (2016).

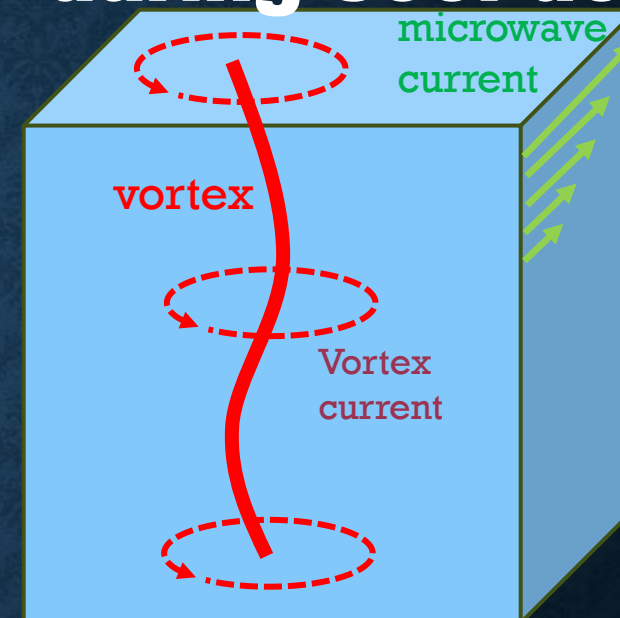
S. Ooi et al., Physical Review B **104**, 064504 (2021).

S. Ooi et al., Physical Review B **111**, 094519 (2025)

Two-level system defects



Vortices trapped during cool down



See also,

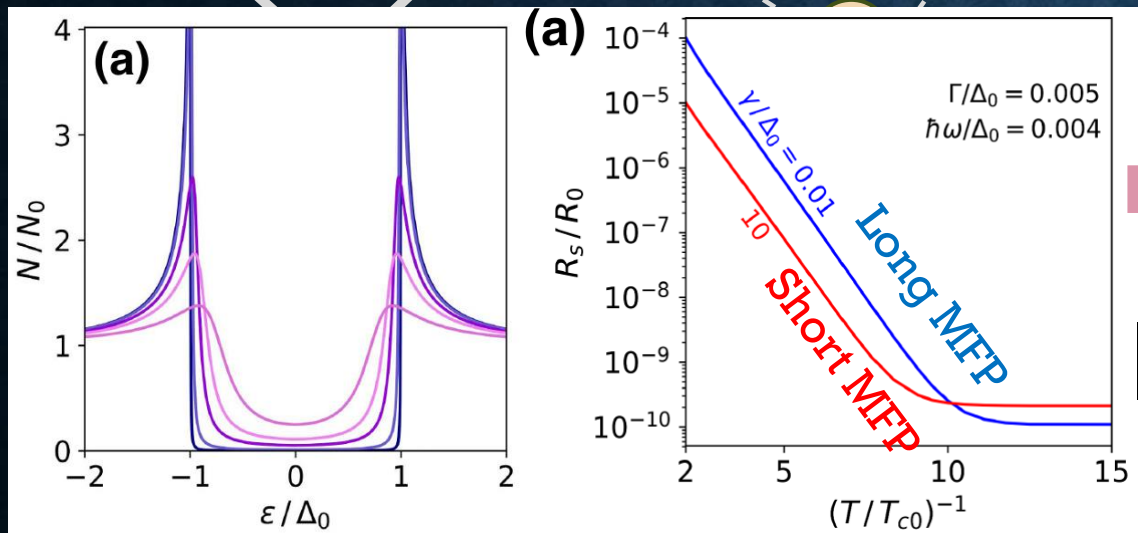
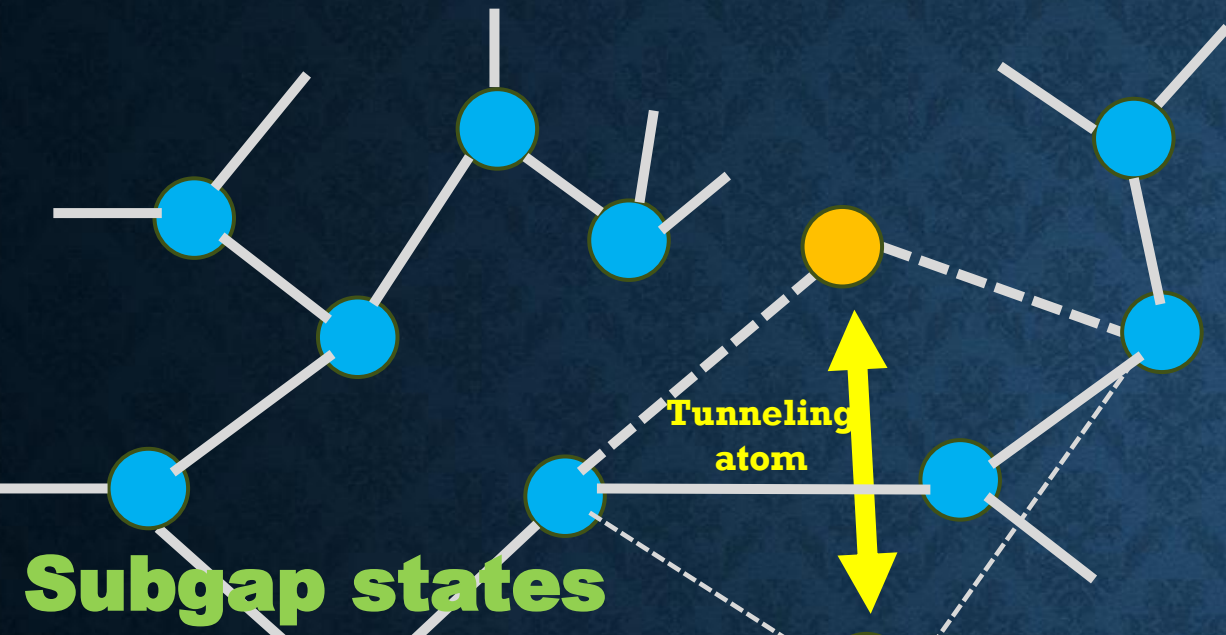
- A. Gurevich and G. Ciovati, PRB **87**, 054502 (2013)
- A. Romanenko et al., Appl. Phys. Lett. **105**, 234103 (2014).
- S. Huang et al., Phys. Rev. Accelerators and Beams **19**, 082001 (2016).
- S. Posen et al., J. Appl. Phys. **119**, 213903 (2016).
- S. Ooi et al., Physical Review B **104**, 064504 (2021).
- S. Ooi et al., Physical Review B **111**, 094519 (2025)

Figures adapted from T. Kubo, Physical Review Applied **17**, 014018 (2022)

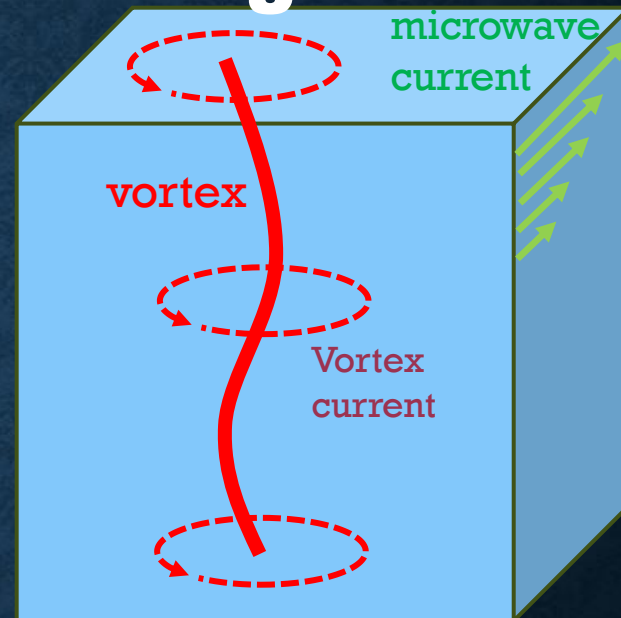
© American Physical Society.

See also, A Gurevich and T Kubo, Physical Review B **96**, 184515 (2017)

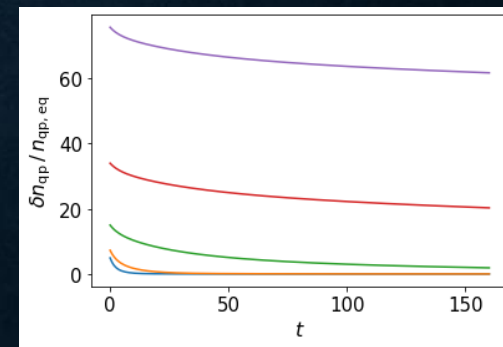
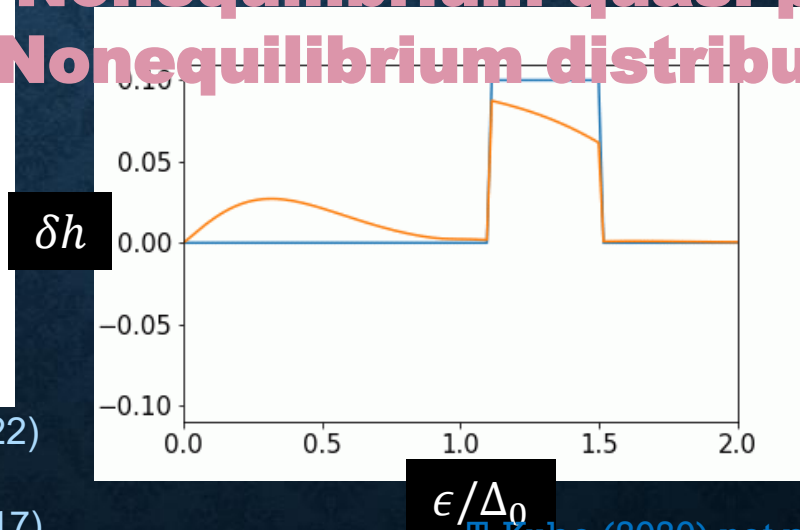
Two-level system defects



Vortices trapped during cool down



Nonequilibrium quasi particles and Nonequilibrium distribution function



Figures adapted from T. Kubo, Physical Review Applied 17, 014018 (2022)

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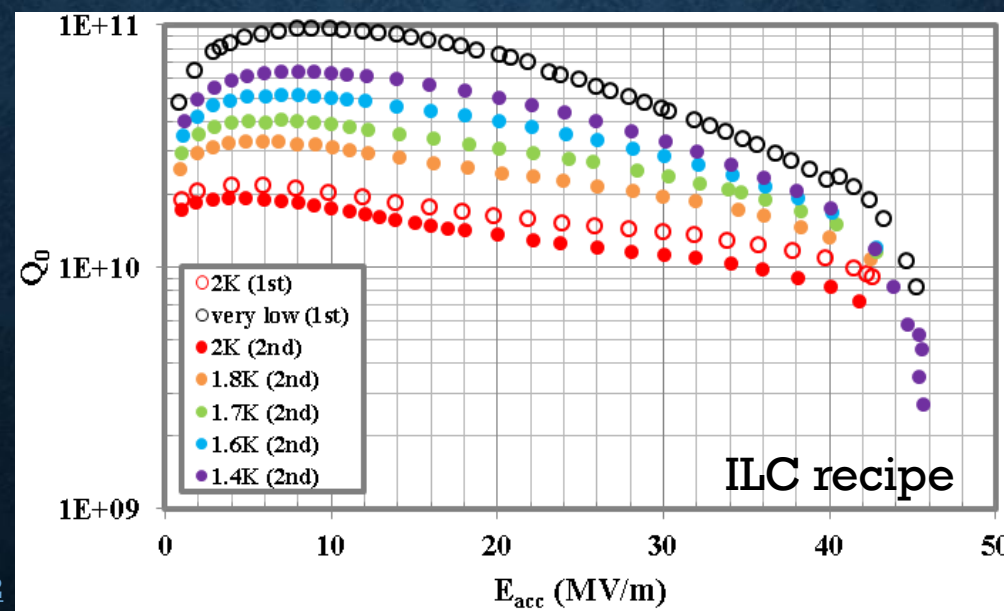
See also, A Gurevich and T Kubo, Physical Review B 96, 184515 (2017)

T. Kubo (2020) not published

t/τ ($\tau \sim 0.2\text{ns}$ for Nb)

In such a seemingly complex situation, how do accelerator researchers extract high cavity performance?

The accelerator community can build
the best cavities on the planet — achieving $Q \sim 10^{11}$.



← This is not special but
typical in accelerator
experiments



Nb cavity technology for accelerators:

Classical ILC recipe as a representative example

ILC recipe

Materials

Fabrication

Electropolishing(1)
(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity

RRR>300
270 mm

High purity Nb

ILC recipe

Materials

Fabrication

Electropolishing(1)
(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

Electropolishing(2)
(several μ m)

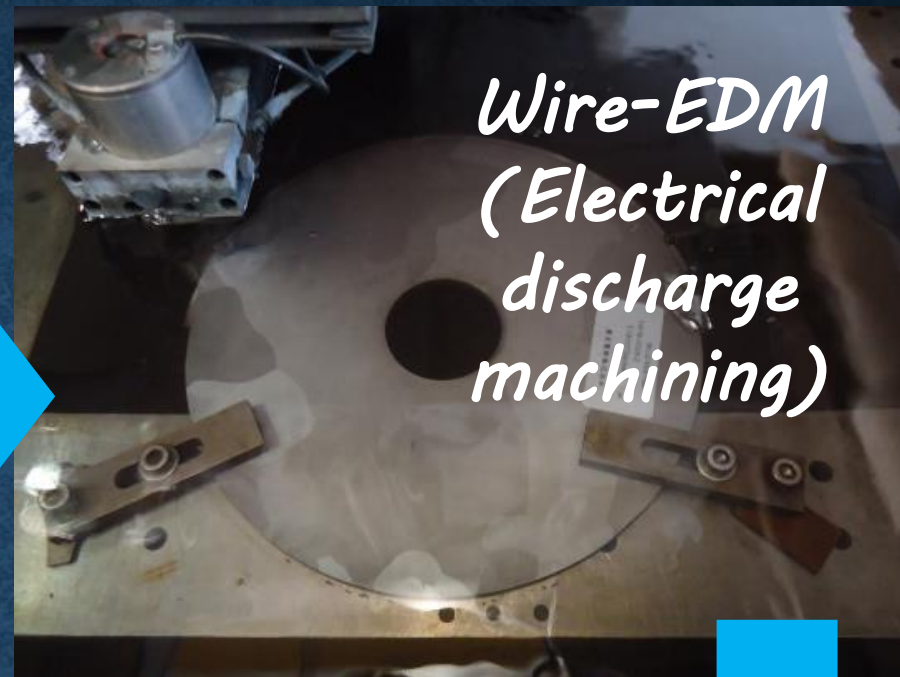
High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity

*Let's start
fabrication.*



ILC recipe

Materials

Fabrication

Electropolishing(1)
(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity



Half cells

Electron beam welding
Cavity material quality is maintained



Half cell + beam pipe

ILC recipe

Materials

Fabrication

Electropolishing(1)
(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

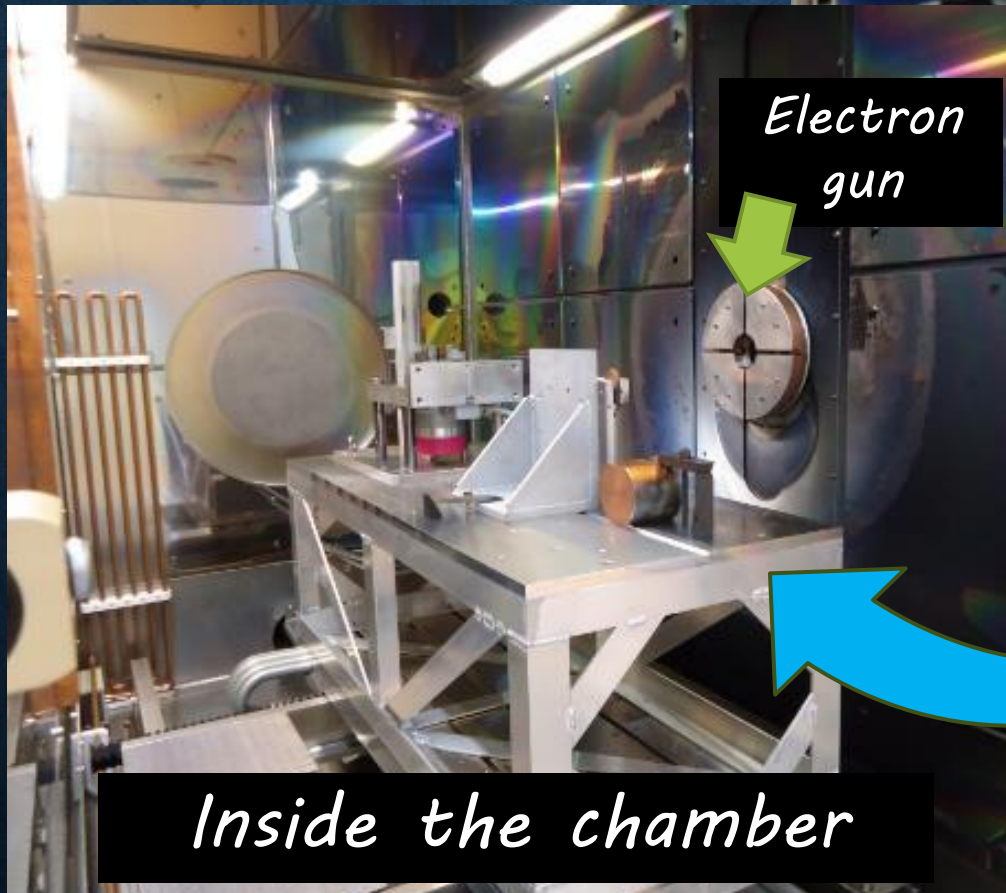
Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity



ILC recipe

Materials

Fabrication

Electropolishing(1)
(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

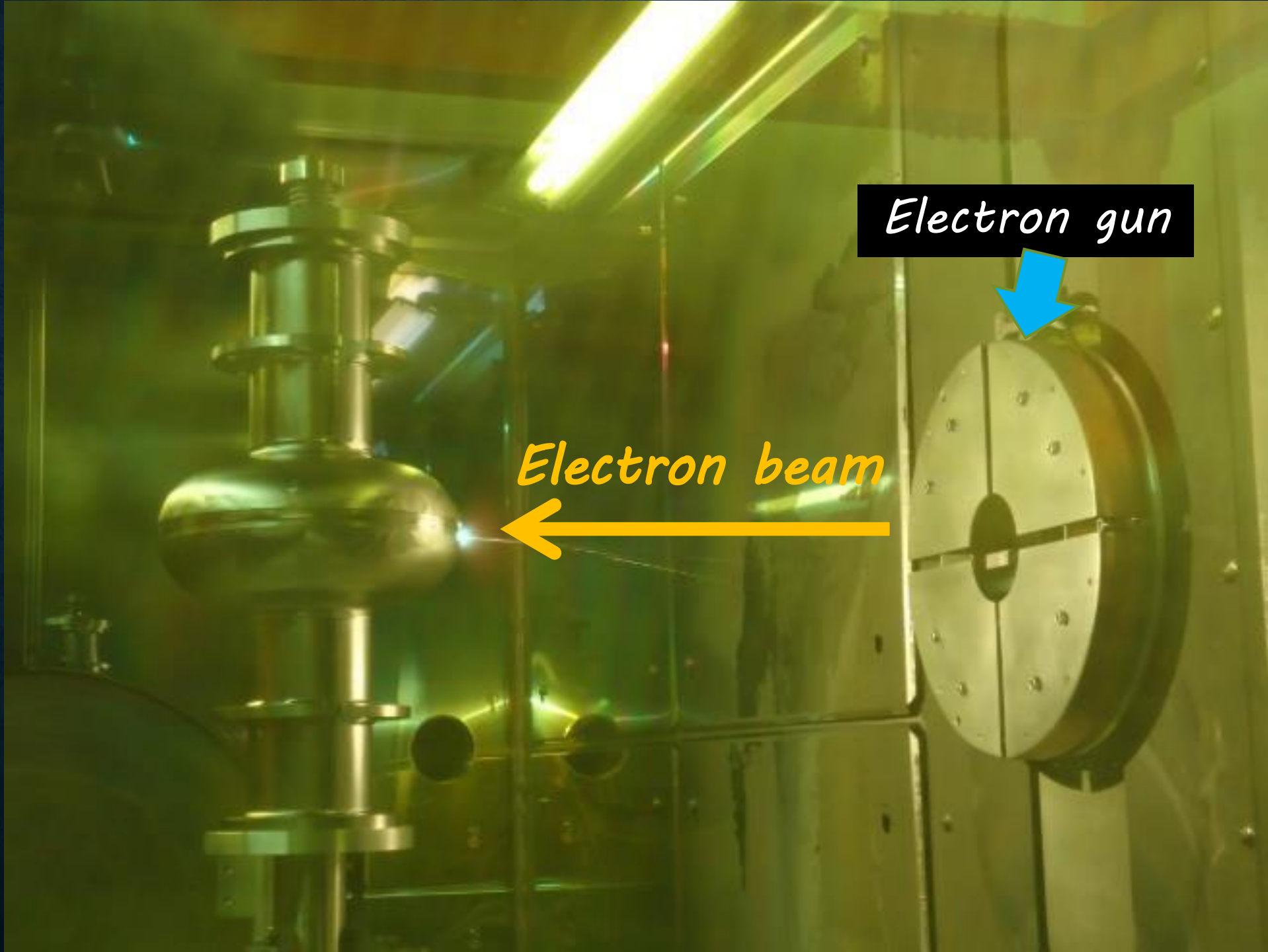
Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity



Electron gun

Electron beam

ILC recipe

Materials

Fabrication

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(100-200 μ m)

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(750-900°C, 2-3hours)

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(100-140°C, 48hours)

Test cavity



ILC recipe

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(100-200 μ m)

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(750-900°C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity

Electropolishing (EP)

- The electrolyte is a mixture of (9:1) sulfuric acid and hydrofluoric acid (HF).
- A DC voltage is applied with Nb as the anode and aluminum as the cathode.
- Niobium is oxidized to form Nb_2O_5 .
- This oxide layer is dissolved and polished away by HF present in the electrolyte.



ILC recipe

Materials

Fabrication

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(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

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(several μ m)

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ILC recipe

Materials

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(100-200 μ m)

Annealing
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(several μ m)

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rinsing

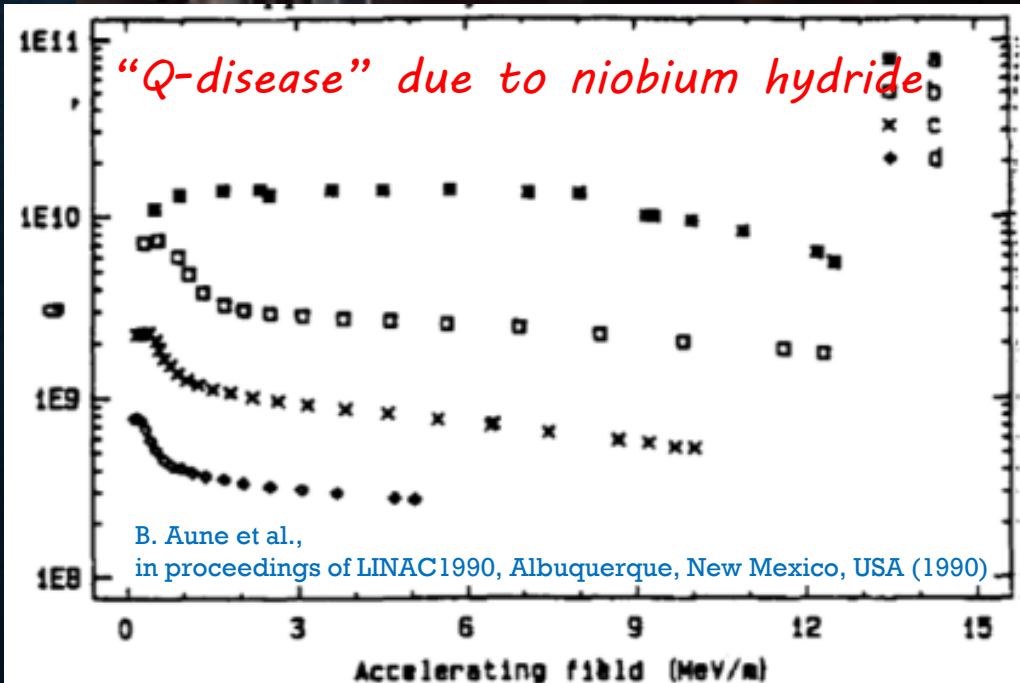
Assembling

Low-temperature
baking
(100-140 $^{\circ}$ C, 48hours)

Test cavity

Annealing

- To degas hydrogen — prevents hydrogen Q-disease (requires $>700^{\circ}\text{C}$)



ILC recipe

Materials

Fabrication

Electropolishing(1)
(100-200 μ m)

Annealing
(750-900 $^{\circ}$ C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

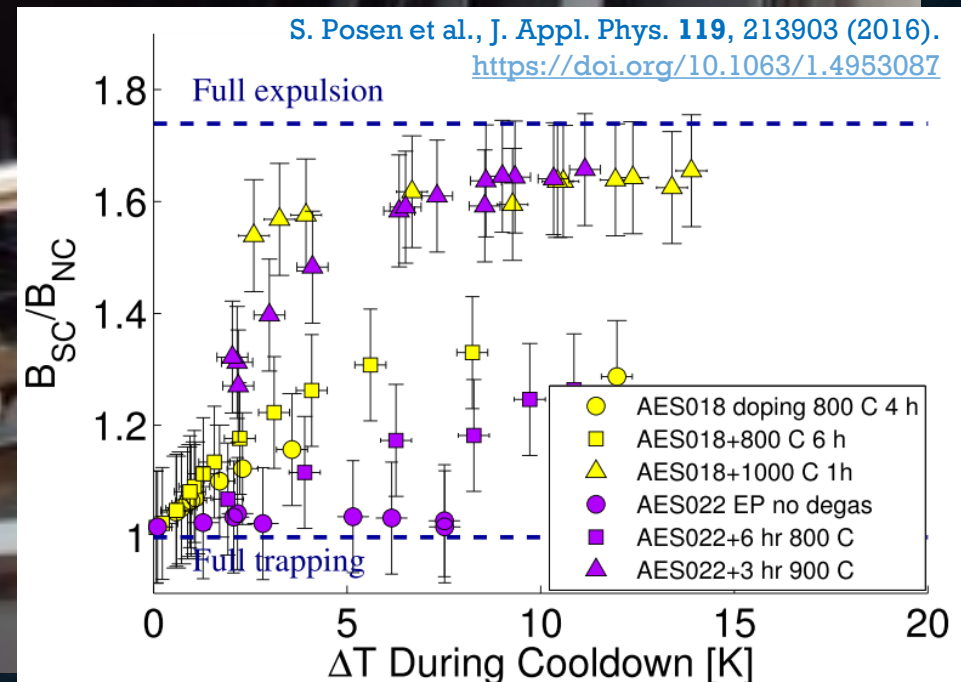
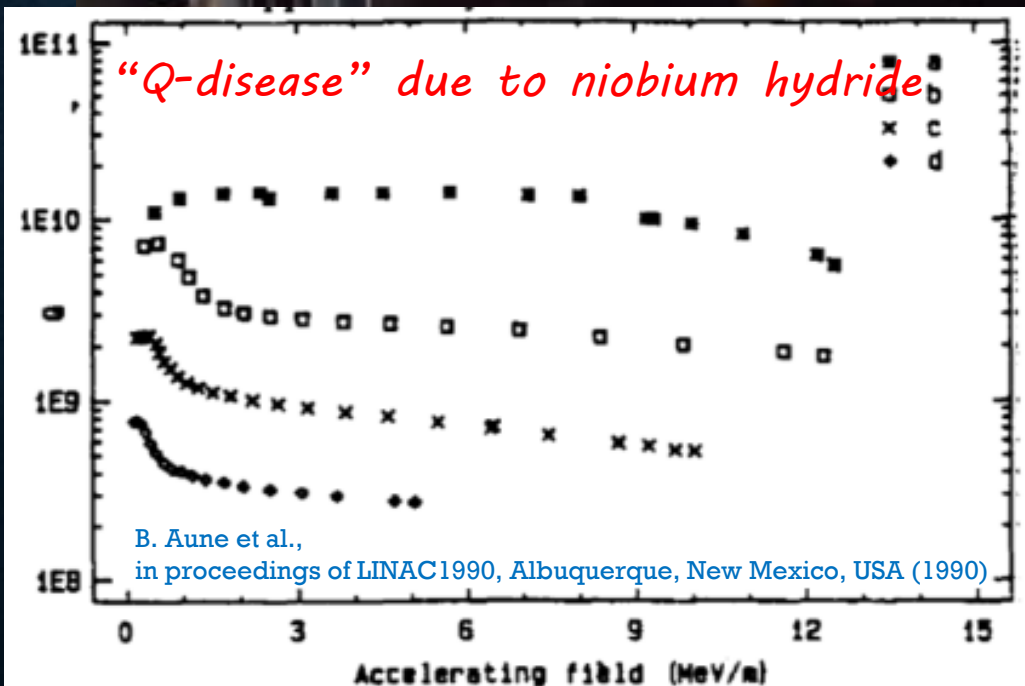
Assembling

Low-temperature
baking
(100-140 $^{\circ}$ C, 48hours)

Test cavity

Annealing

- To degas hydrogen — prevents hydrogen Q-disease (requires $>700^{\circ}\text{C}$)
- To reduce pinning centers — minimizes trapped flux (requires $\sim 900^{\circ}\text{C}$)



ILC recipe

Materials

Fabrication

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(100-200 μ m)

Annealing
(750-900°C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

Low-temperature
baking
(100-140°C, 48hours)

Test cavity

A **second EP** step, removing only a few microns.



ILC recipe

Materials

Fabrication

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(100-200 μ m)

Annealing
(750-900 $^{\circ}$ C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

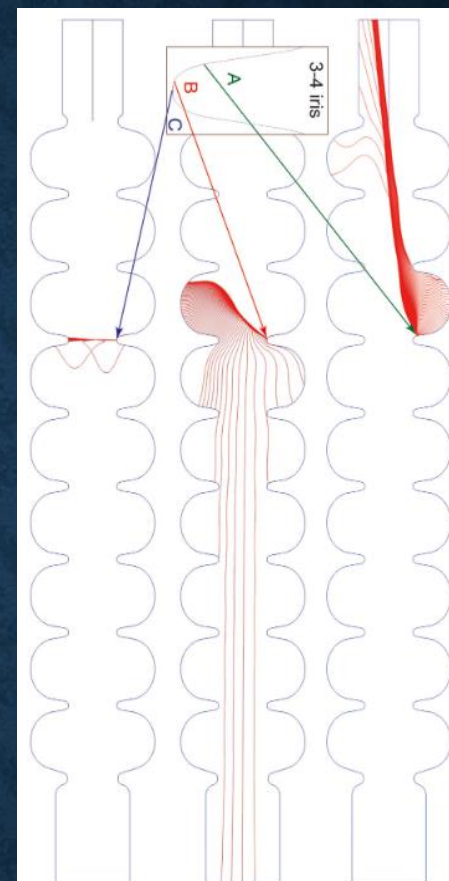
Assembling

Low-temperature
baking
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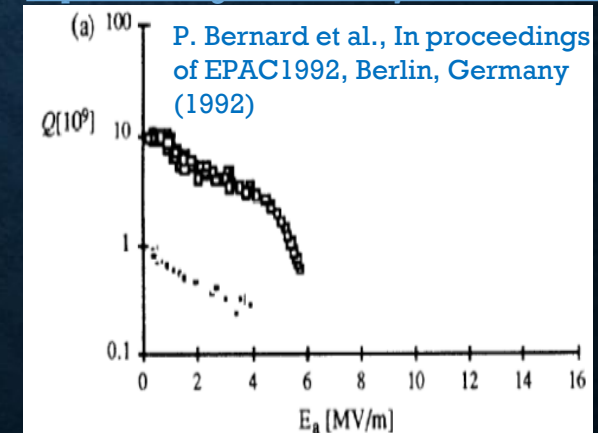
Test cavity



Courtesy of Mathieu Omet (KEK)



H. Sakai et al.,
Phys. Rev. Accel. Beams **22**,022002 (2019)
<https://doi.org/10.1103/PhysRevAccelBeams.22.022002>



ILC recipe

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Fabrication

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(100-200 μ m)

Annealing
(750-900 $^{\circ}$ C, 2-3hours)

Electropolishing(2)
(several μ m)

High pressure
rinsing

Assembling

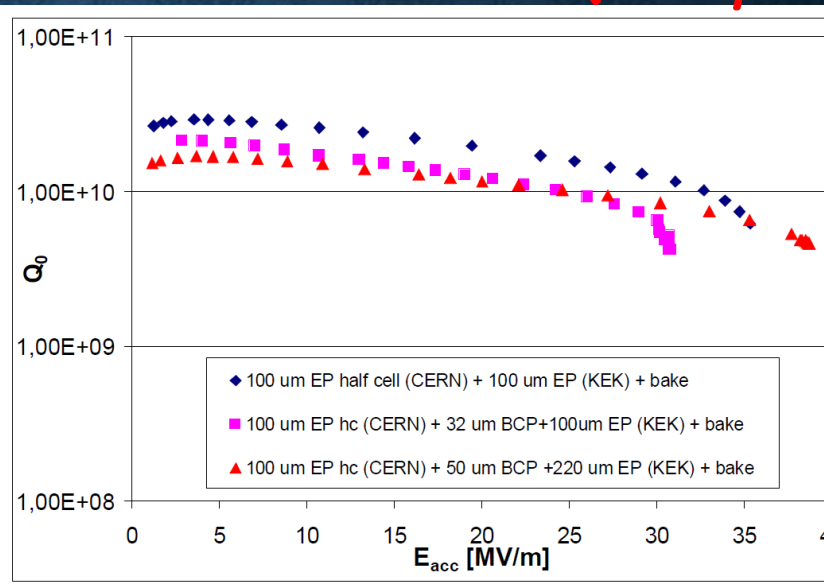
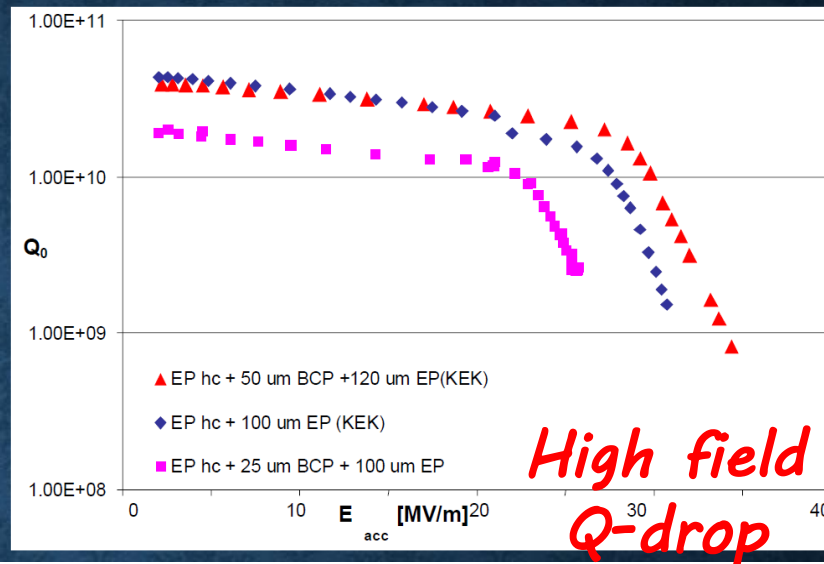
Low-temperature
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Test cavity

120 $^{\circ}$ C—48h bake



L. Lilje et al., in Proceedings of SRF1999, Santa Fe, New Mexico, USA (1999), p. 74, TUA001.



Spatial variation of the mean free path on the scale of the penetration depth suppresses the surface current, thereby increasing the maximum achievable field.

T. Kubo, Supercond. Sci. and Technol. **30**, 023001 (2017); Sci. and Technol. **34**, 045006 (2021)

V. Ngampruetikorn and J. A. Sauls, Phys. Rev. Research **1**, 012015(R) (2019)

ILC recipe

Materials

Fabrication

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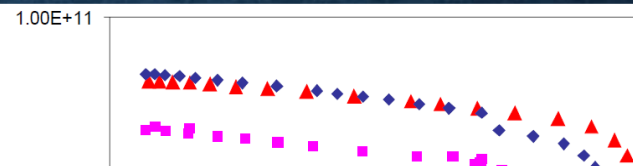
Test cavity

120 $^{\circ}$ C—48h bake

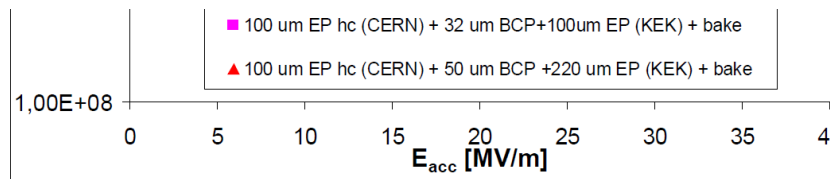
Labs in Japan and the US traditionally applied a 100 $^{\circ}$ C bake to remove residual water after high-pressure rinsing. In contrast, labs in Germany and France did not adopt this step — and continued to suffer from the so-called Q-drop.

At the time, the reason was unknown, and the phenomenon was even nicknamed the "European disease." It wasn't until the late 1990s that the community realized the low-temperature bake was the key. This is a prime example of the serendipitous discoveries that have accumulated over the past few decades.

L. Lilje et al., in *Proceedings of SRF1999*, Santa Fe, New Mexico, USA (1999), p. 74, TUA001.



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ILC recipe

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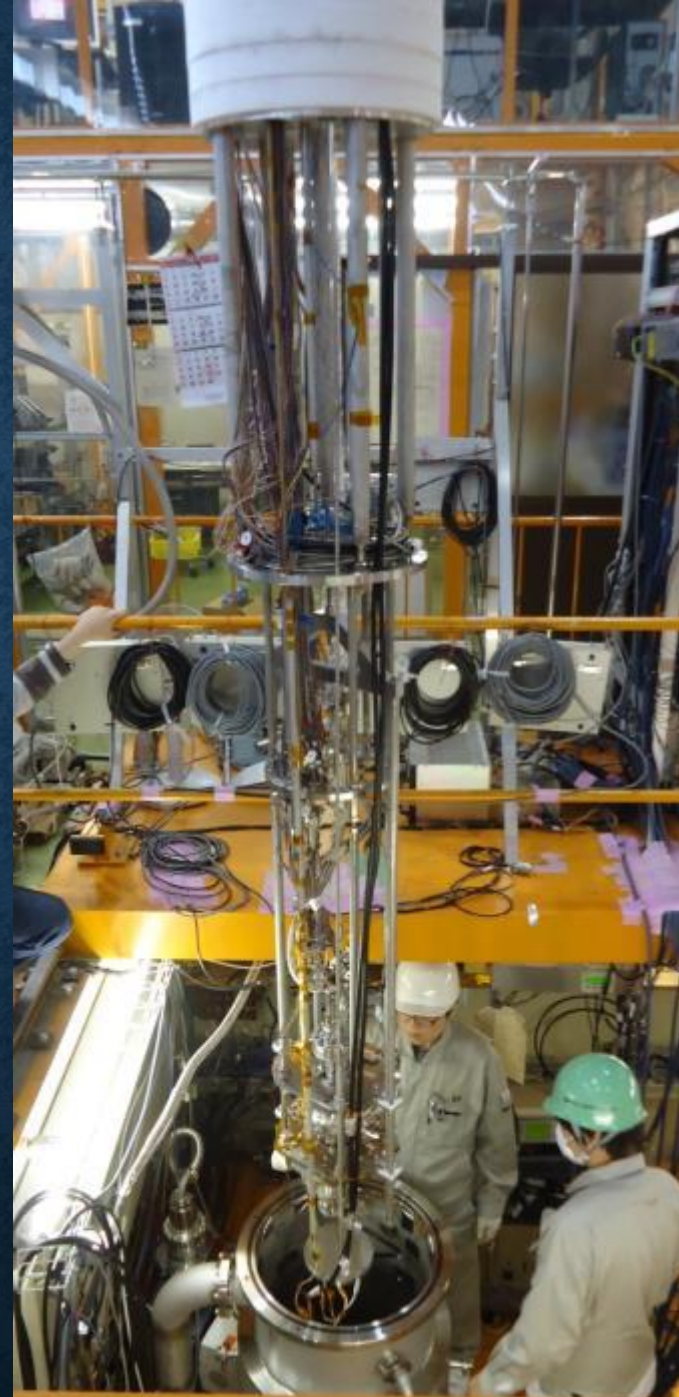
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ILC recipe

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(750-900 $^{\circ}$ C, 2-3hours)

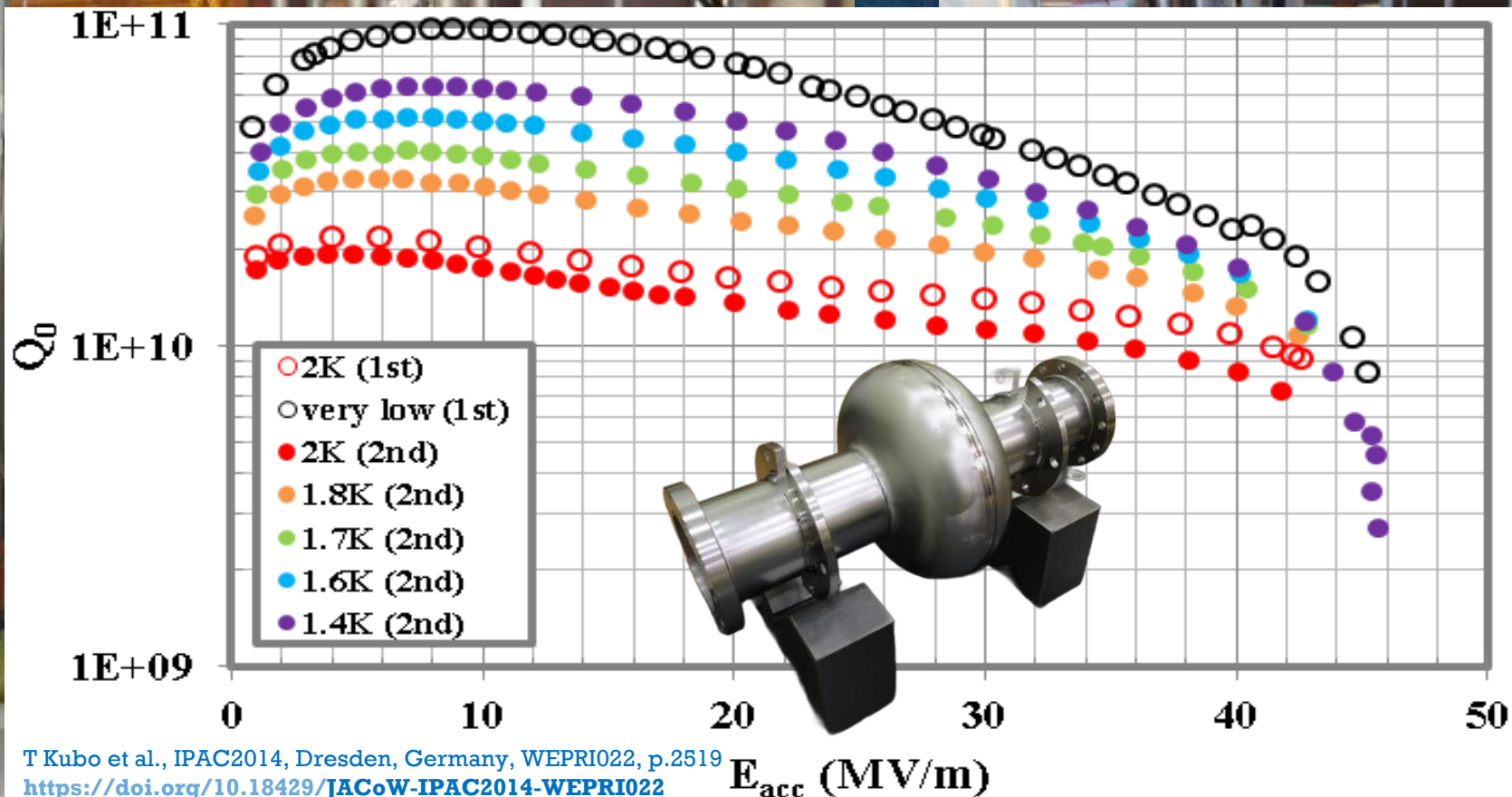
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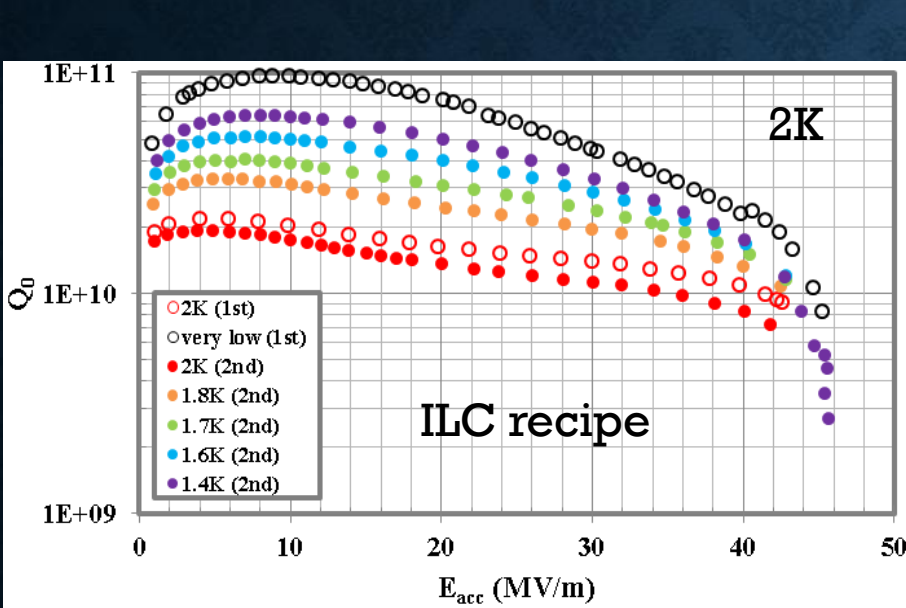
Assembling

Low-temperature
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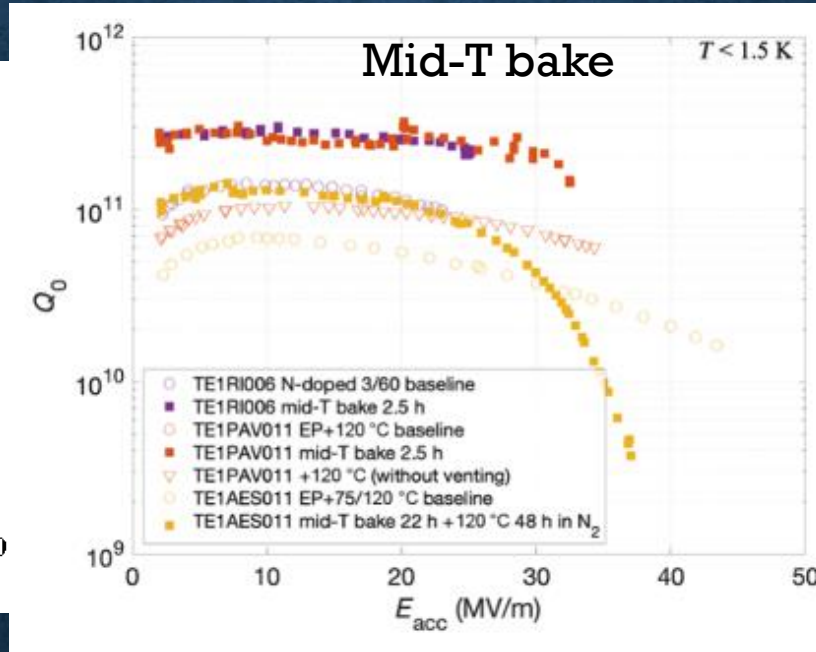
Test cavity



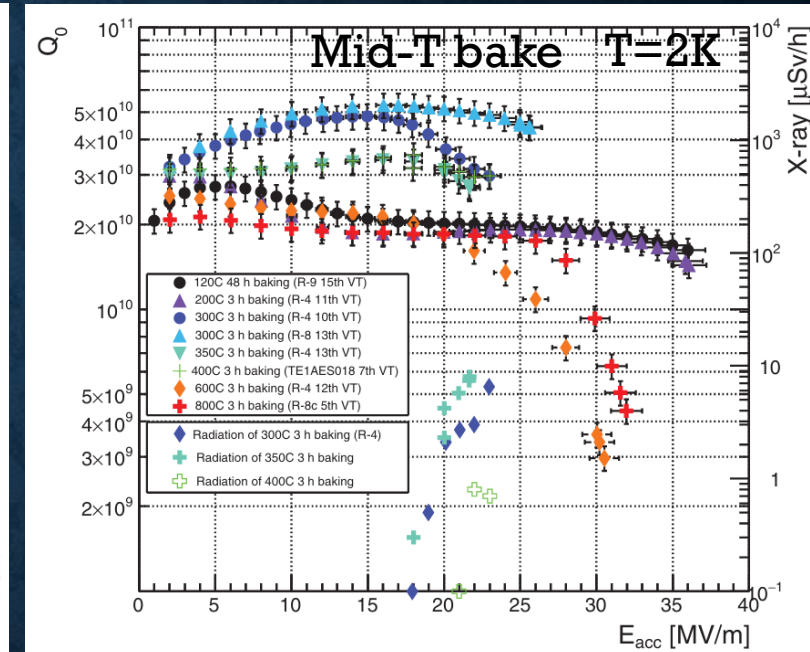
Many modern recipes exist: N-doping, N-infusion, two-step bake, mid-T bakes, etc. **They are all variants of the classical recipe**, with some steps replaced or modified.



T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
<https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022>

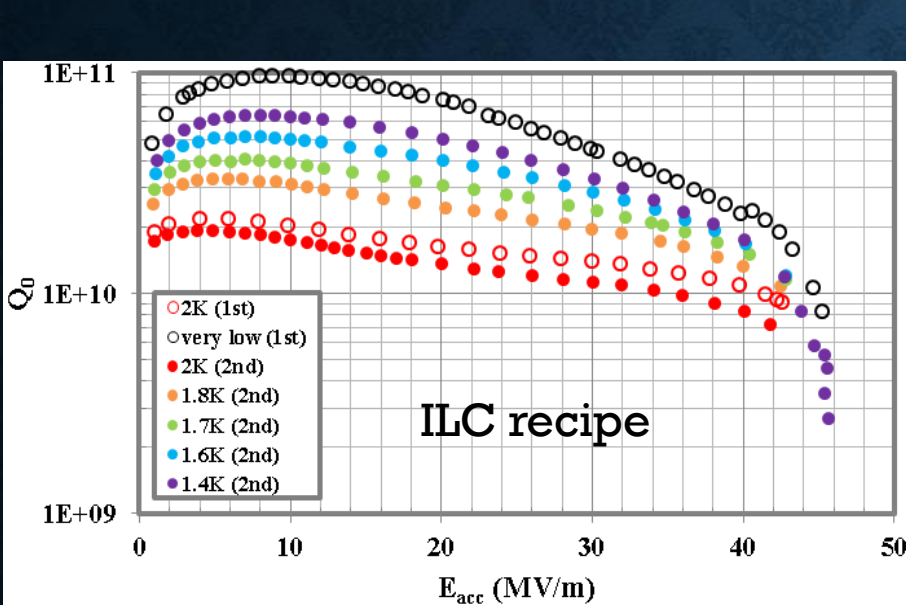


S. Posen et al., Ultralow Surface Resistance via Vacuum Heat Treatment of Superconducting Radio-Frequency Cavities, Phys. Rev. Applied **13**, 014024 (2020)
<https://doi.org/10.1103/PhysRevApplied.13.014024>

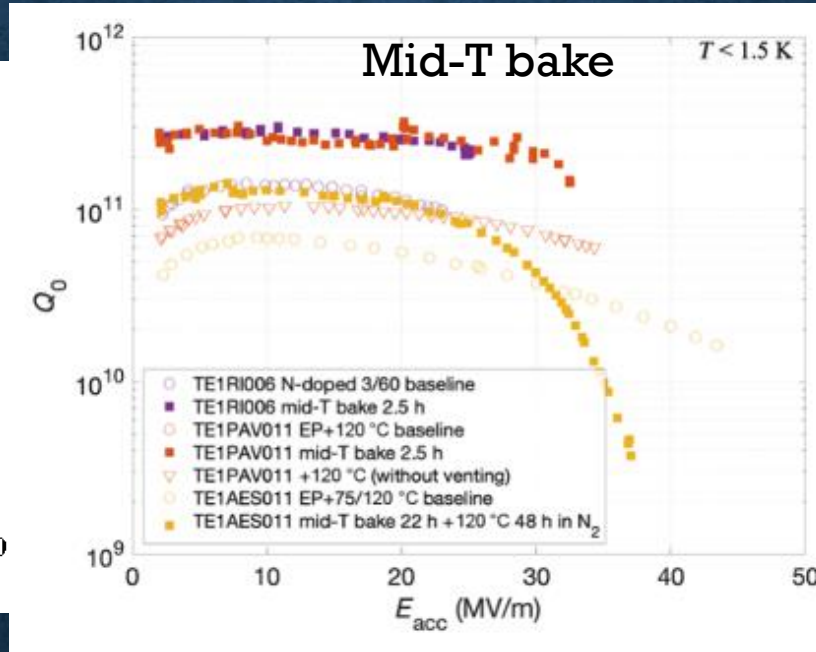


H. Ito et al., Influence of furnace baking on Q-E behavior of superconducting accelerating cavities, Prog. Theor. Exp. Phys. **2021**, 071G01 (2021)
<https://doi.org/10.1093/ptep/ptab056>

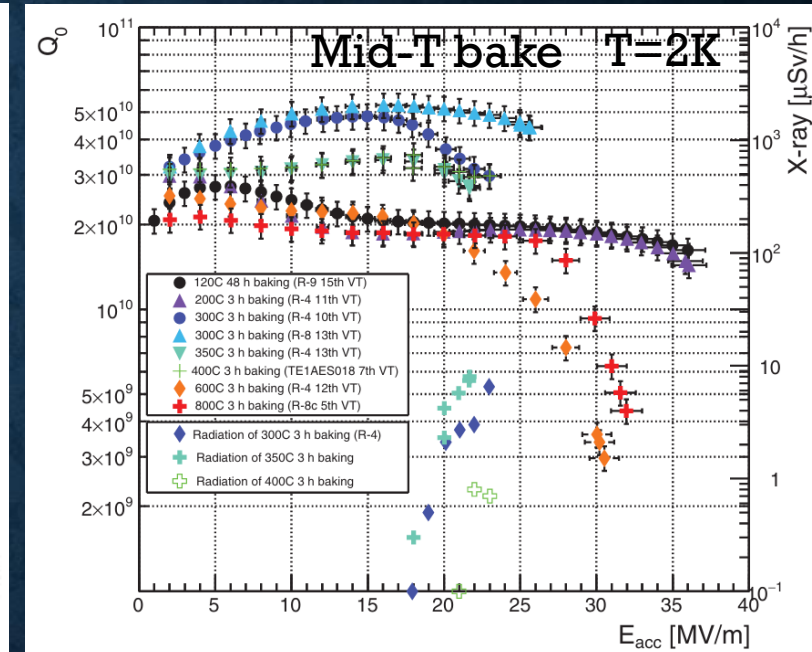
The accelerator community can build the best cavities on the planet — achieving $Q > 10^{11}$.



T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
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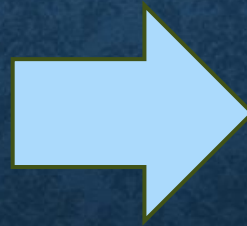
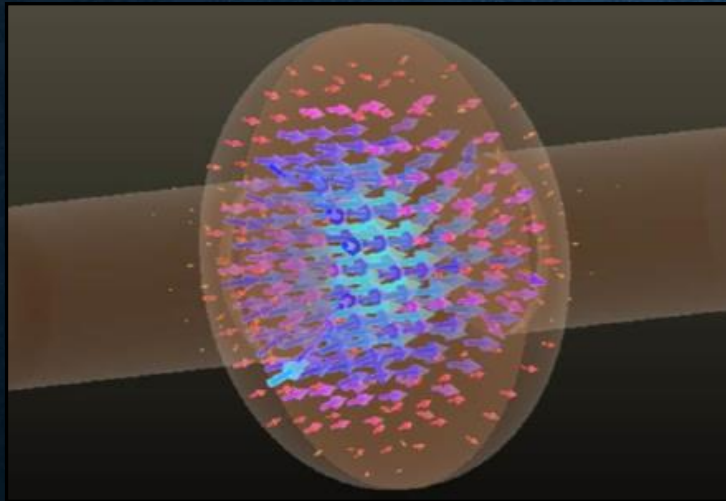


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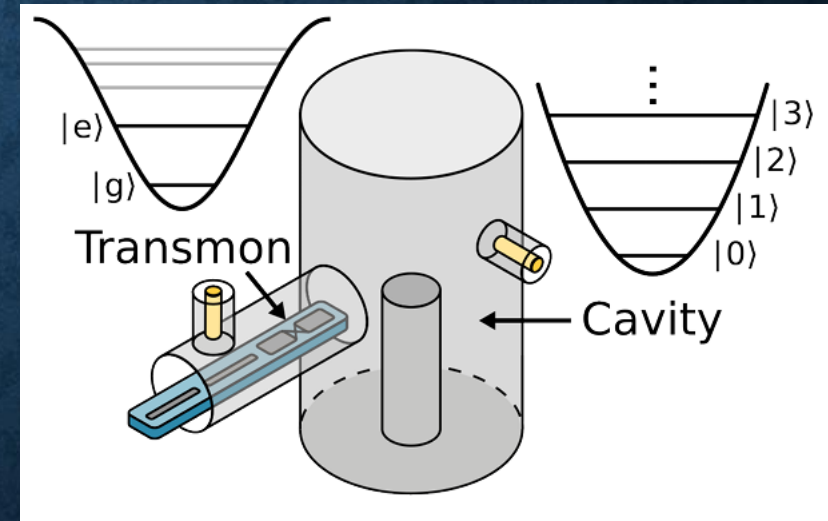
So why not make use of this technology in other applications?

Quantum applications

Accelerator cavity



Quantum memory cavity
for quantum computing



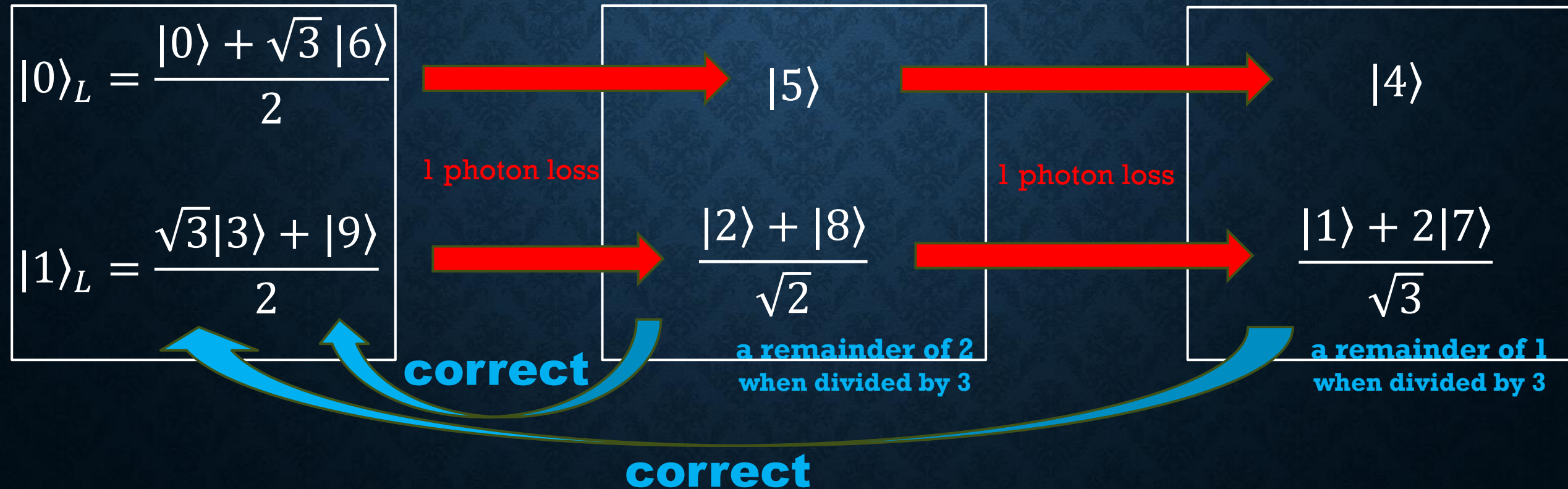
Brock et al., Nature **641**, 612 (2025)

<https://doi.org/10.1038/s41586-025-08899-y>

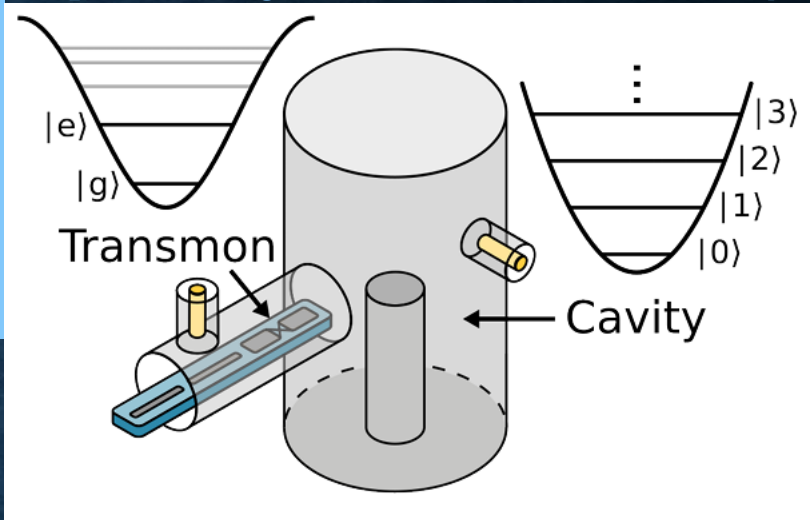
High-Q superconducting cavities allow us to store quantum information redundantly, enabling bosonic error correction.

Known to reduce the required number of qubits by one to two orders of magnitude.

e.g. binomial code



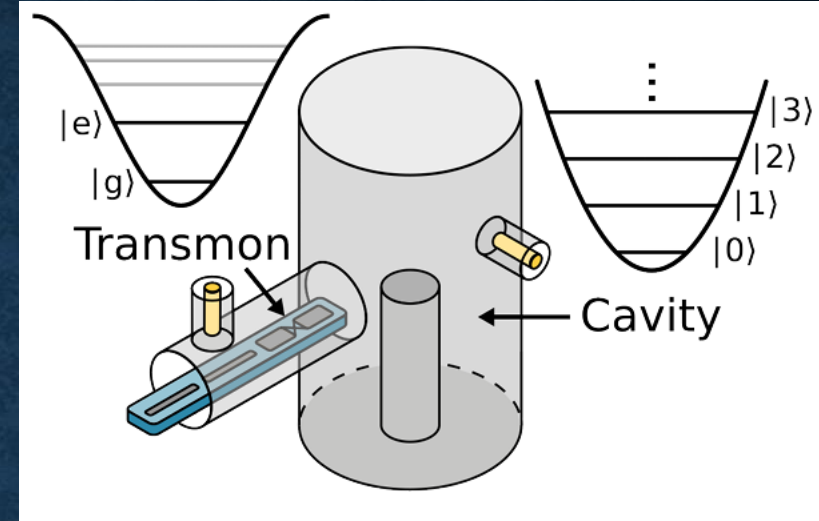
Brock et al., Nature **641**, 612 (2025)
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3D coaxial cavity progress

- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
Aluminum,
 $Q = 7 \times 10^7$ (single photon level)

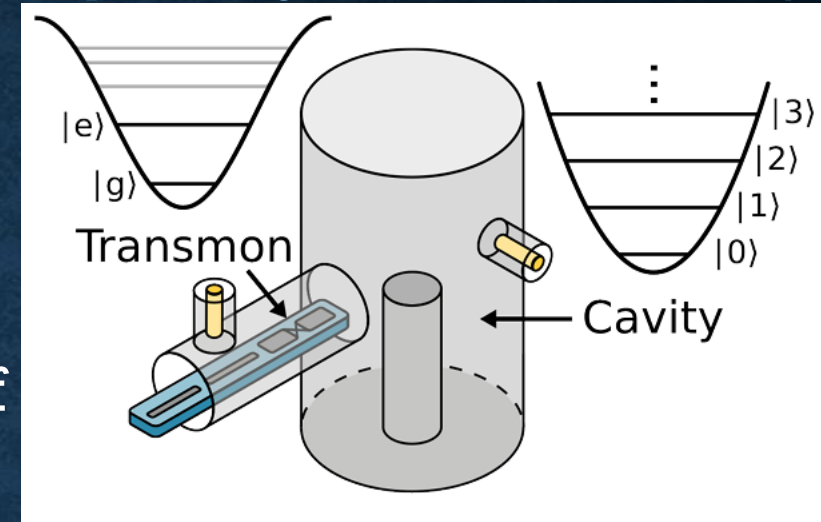
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Niobium (1:1:1 buffered chemical polishing at 0-5°C), Innsbruck
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Brock et al., Nature **641**, 612 (2025)
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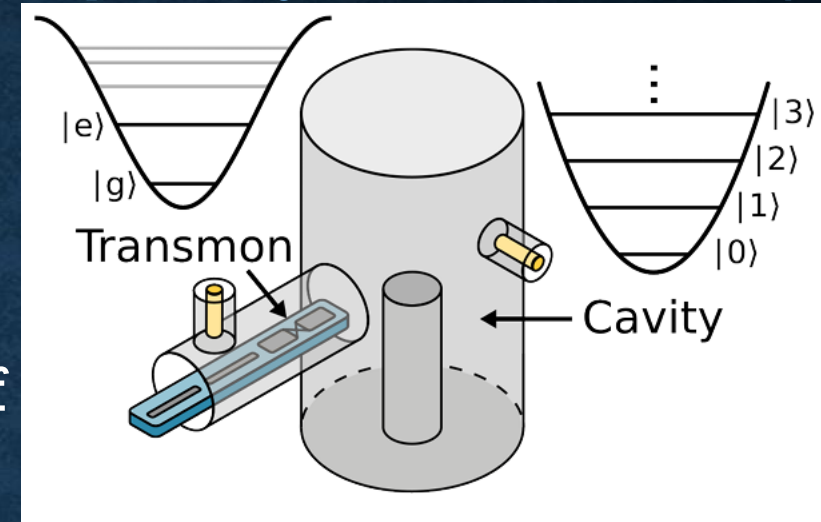
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- Oriani et al., Phys. Rev. Applied **24**, 044080 (2025) University of Chicago
Niobium (1:1:2 water buffered chemical polishing below 10°C),
 $Q = 1.4 \times 10^9$ (single photon level)



The world record (as of 2024)



Why don't we give it a shot?



Why don't we give it a shot?

T. Takenaka, T. Kubo et al.,
arXiv:2510.01819.

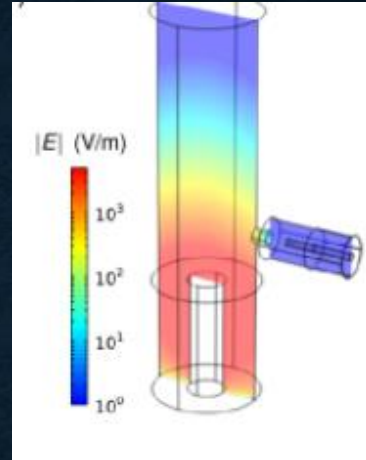
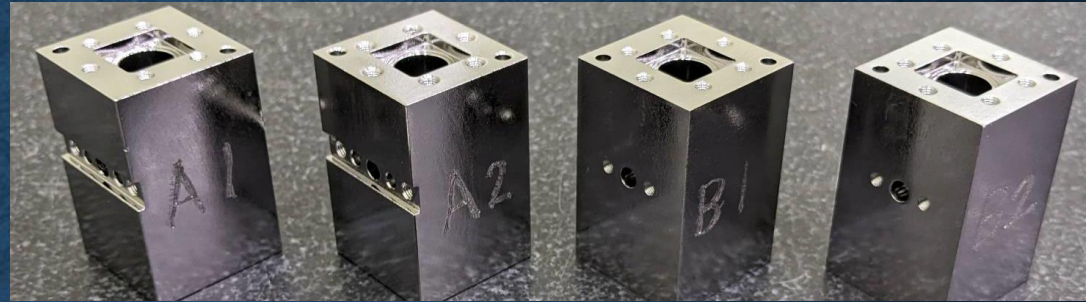


cavity→

T. Kubo



Nb coaxial cavity of **KEK-NTT** collaboration



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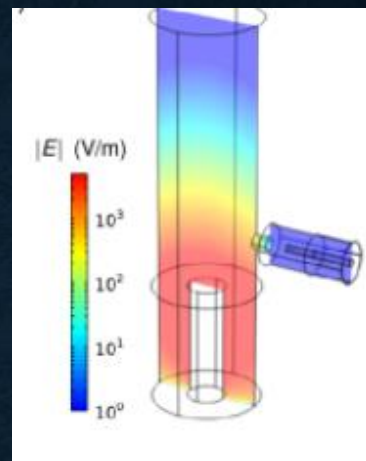
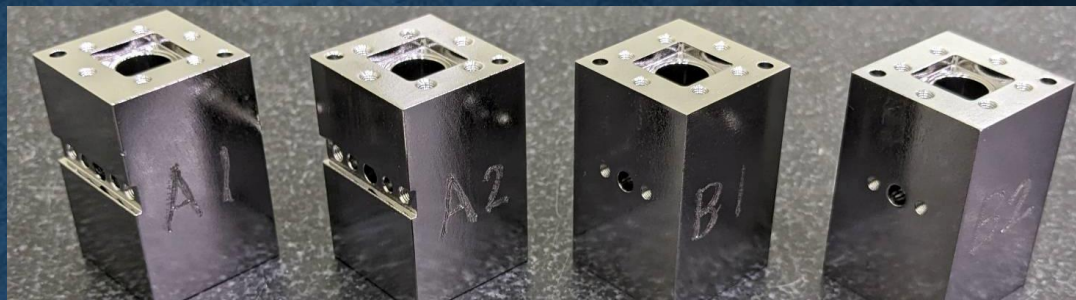
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cavity→

T. Kubo



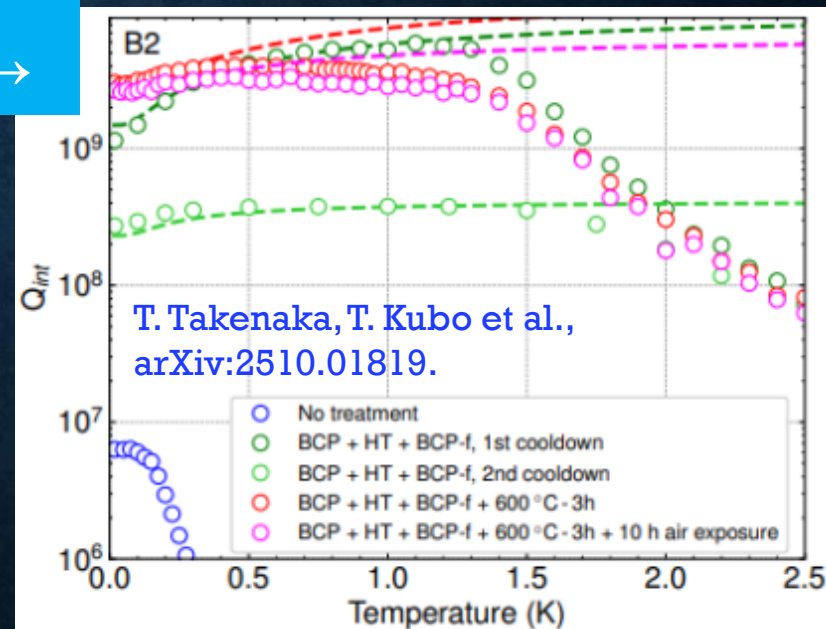
Nb coaxial cavity of **KEK-NTT** collaboration



We broke
the world record!!!→

Corresponding to
photon lifetime of
~0.1 sec.

20mK & Single photon level

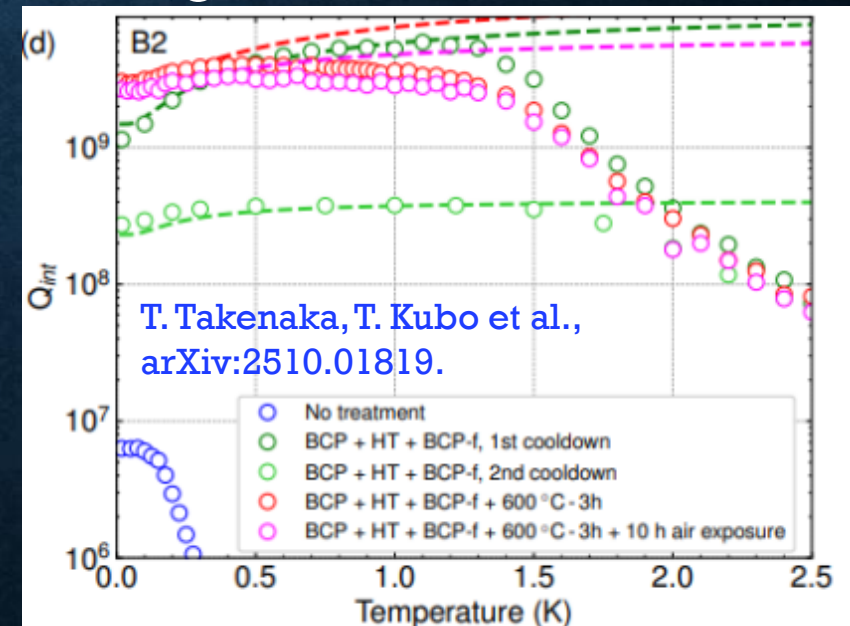
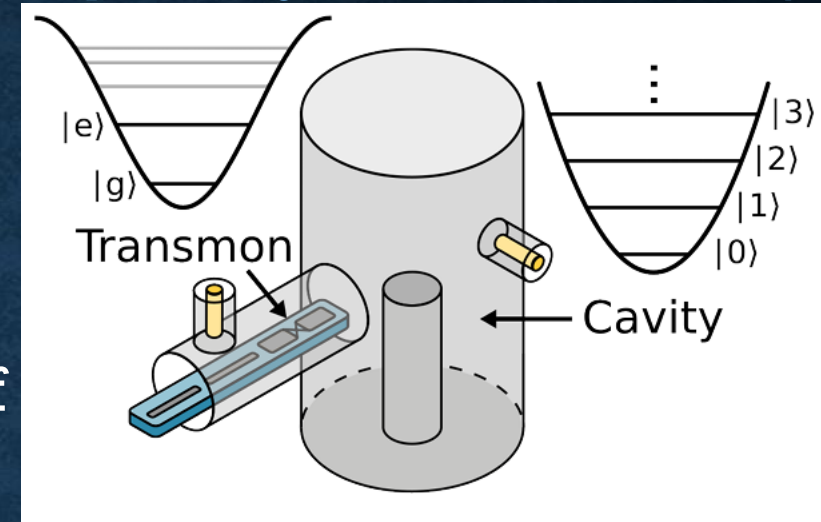


3D coaxial cavity progress

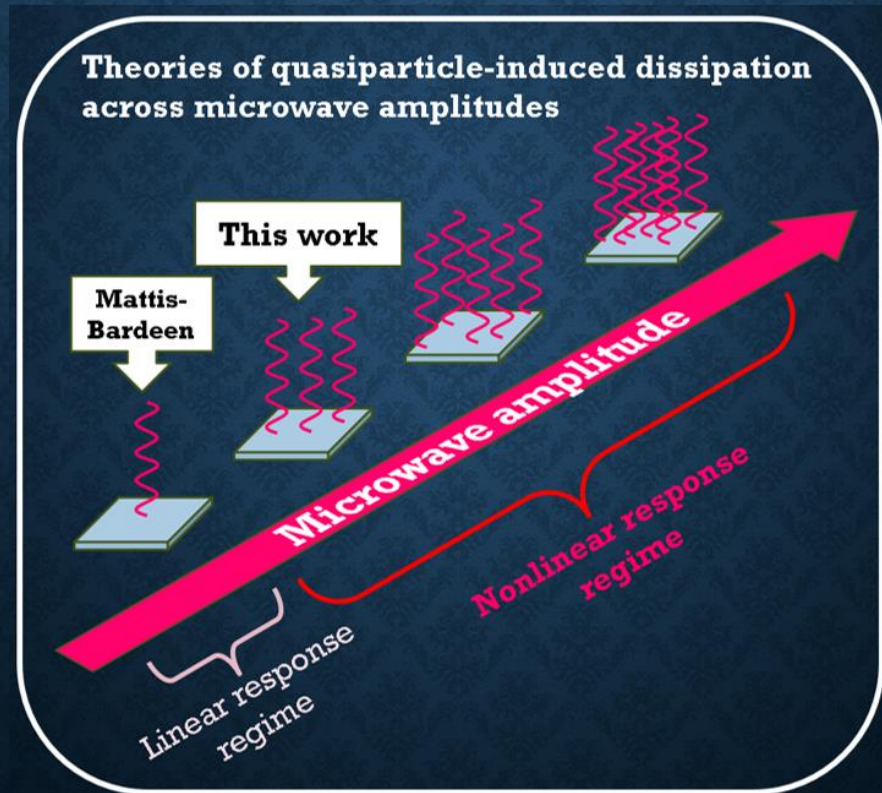
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**Niobium (1:1:1 BCP at 25°C + 900°C anneal
+ 1:1:1 BCP flush at 25°C + 600°C anneal)**
 $Q=3 \times 10^9$ (single photon level)

The world record

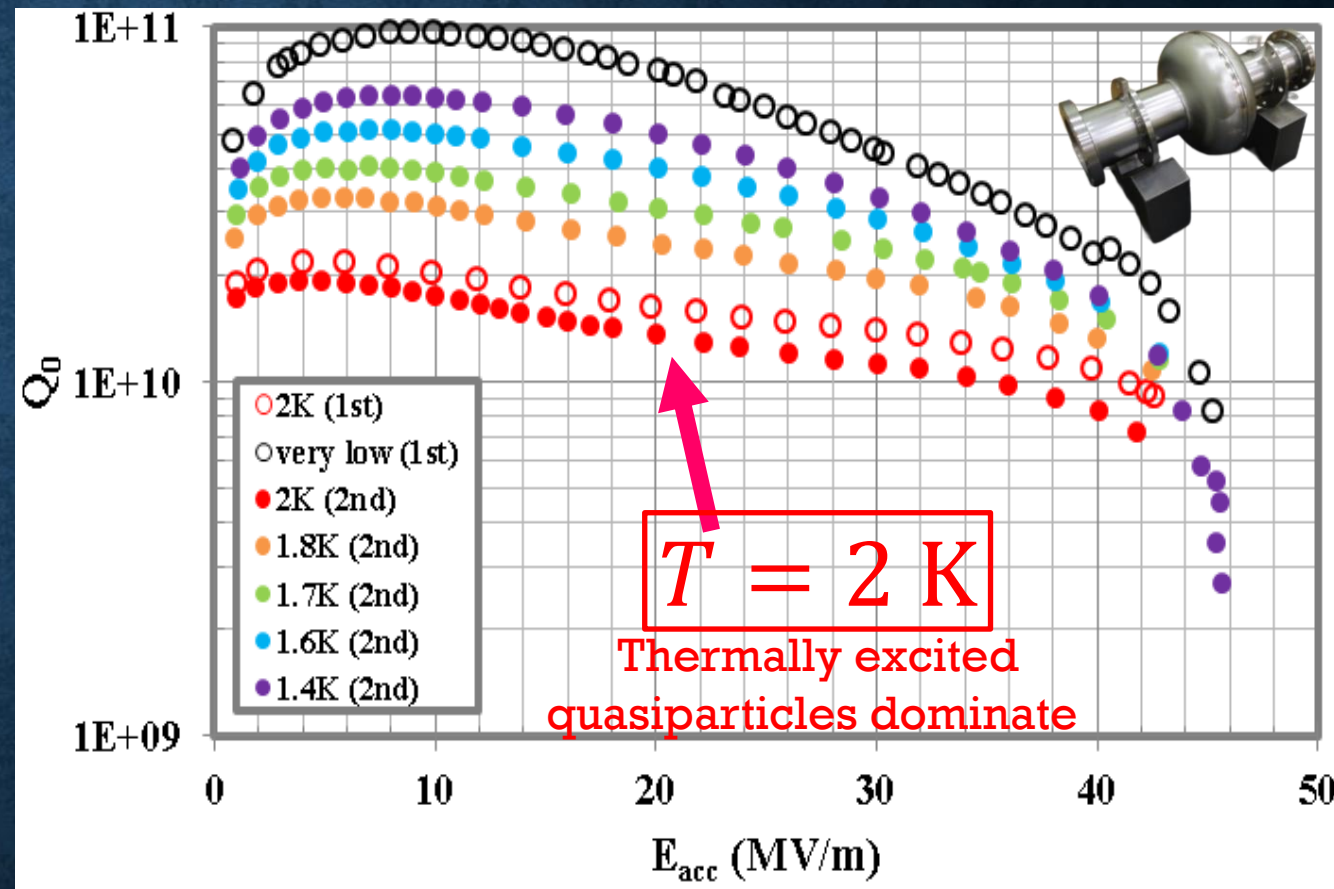
Brock et al., Nature **641**, 612 (2025)
<https://doi.org/10.1038/s41586-025-08899-y>



Understanding more about physics of Q factor

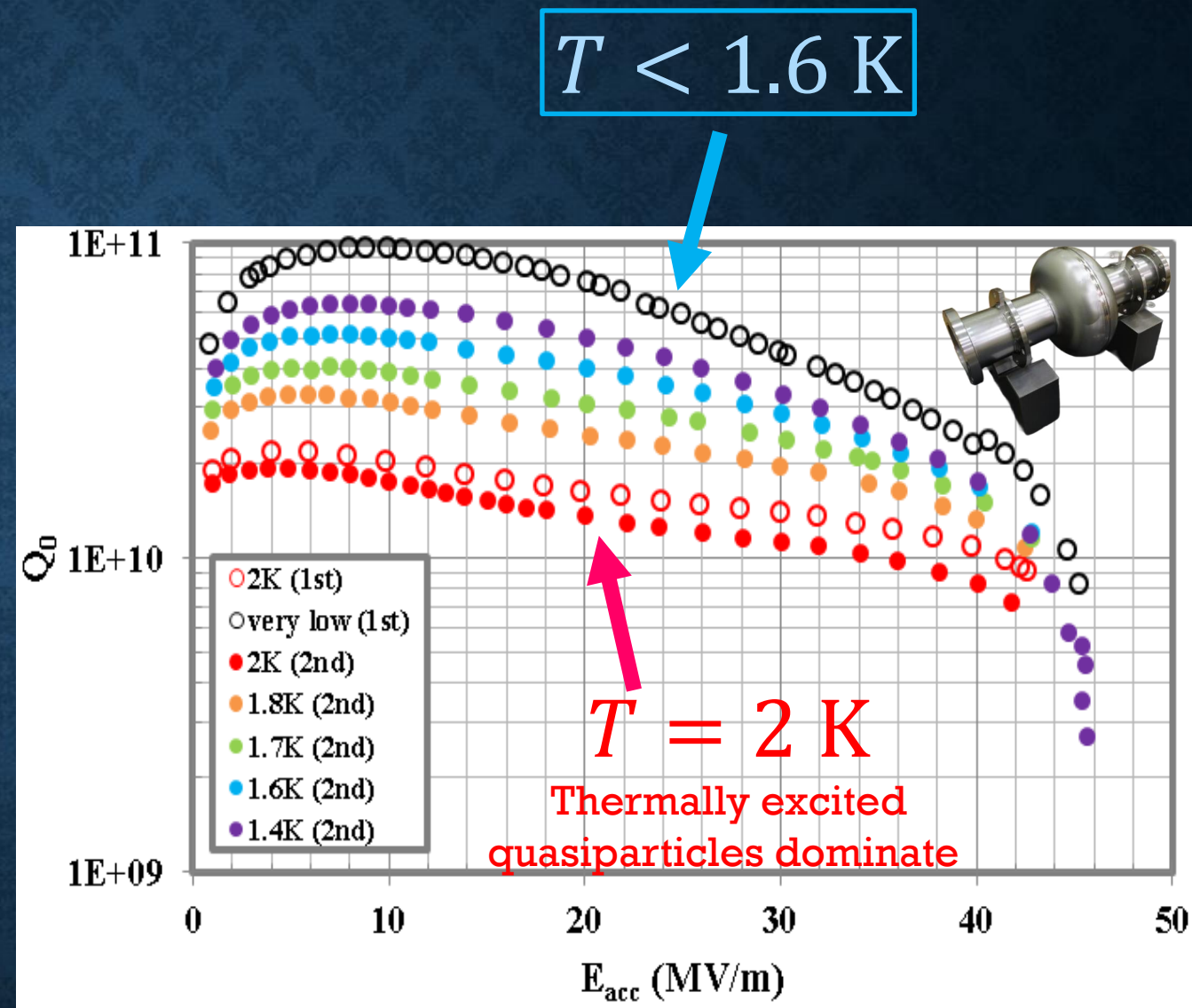


Dissipation mainly comes from
thermally excited
quasiparticles at $T \gtrsim 2$ K



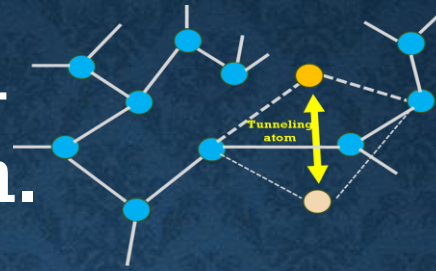
T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
<https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022>

Dissipation mainly comes from
**thermally excited
quasiparticles at $T \gtrsim 2$ K**
and
some others at $T \lesssim 2$ K

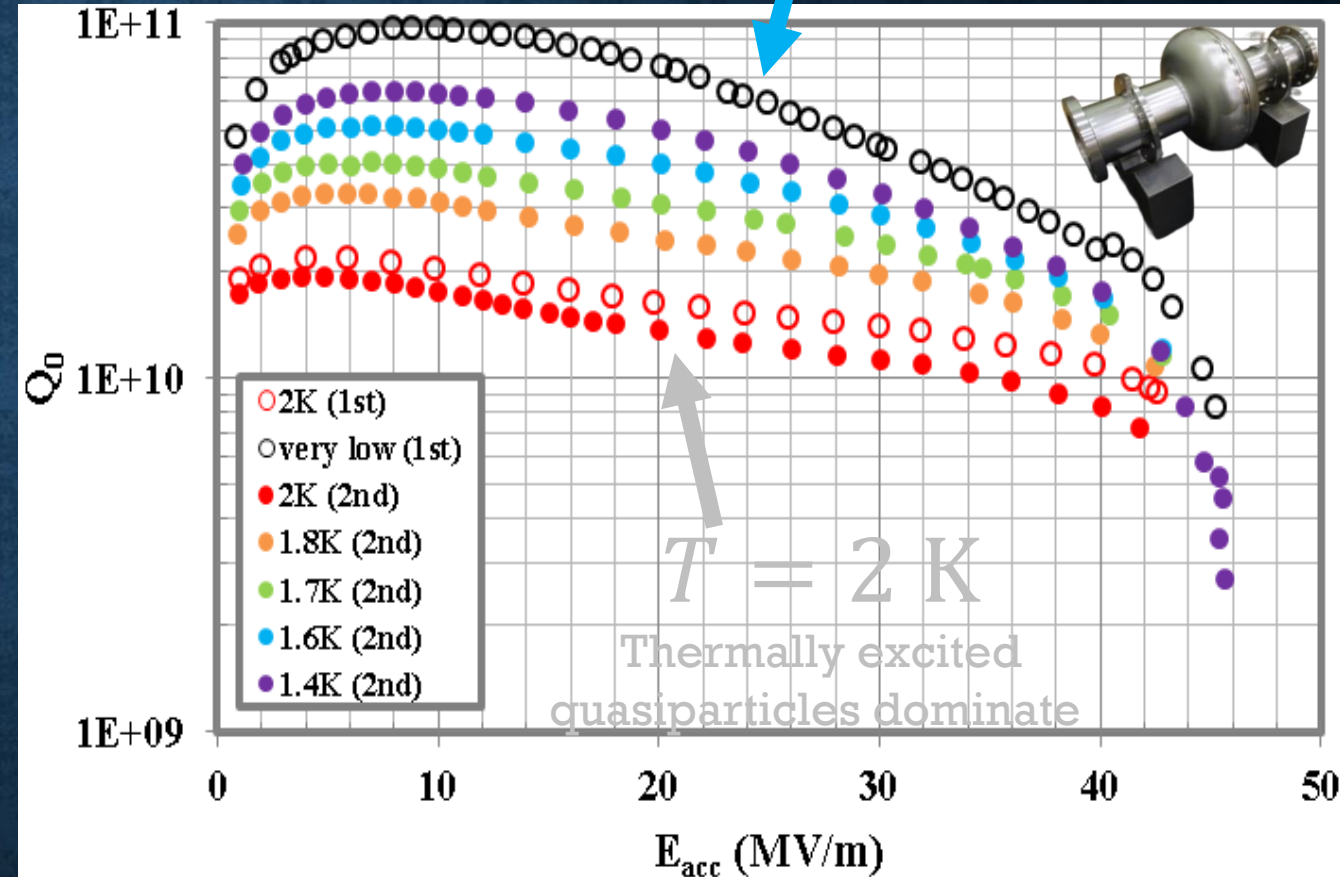
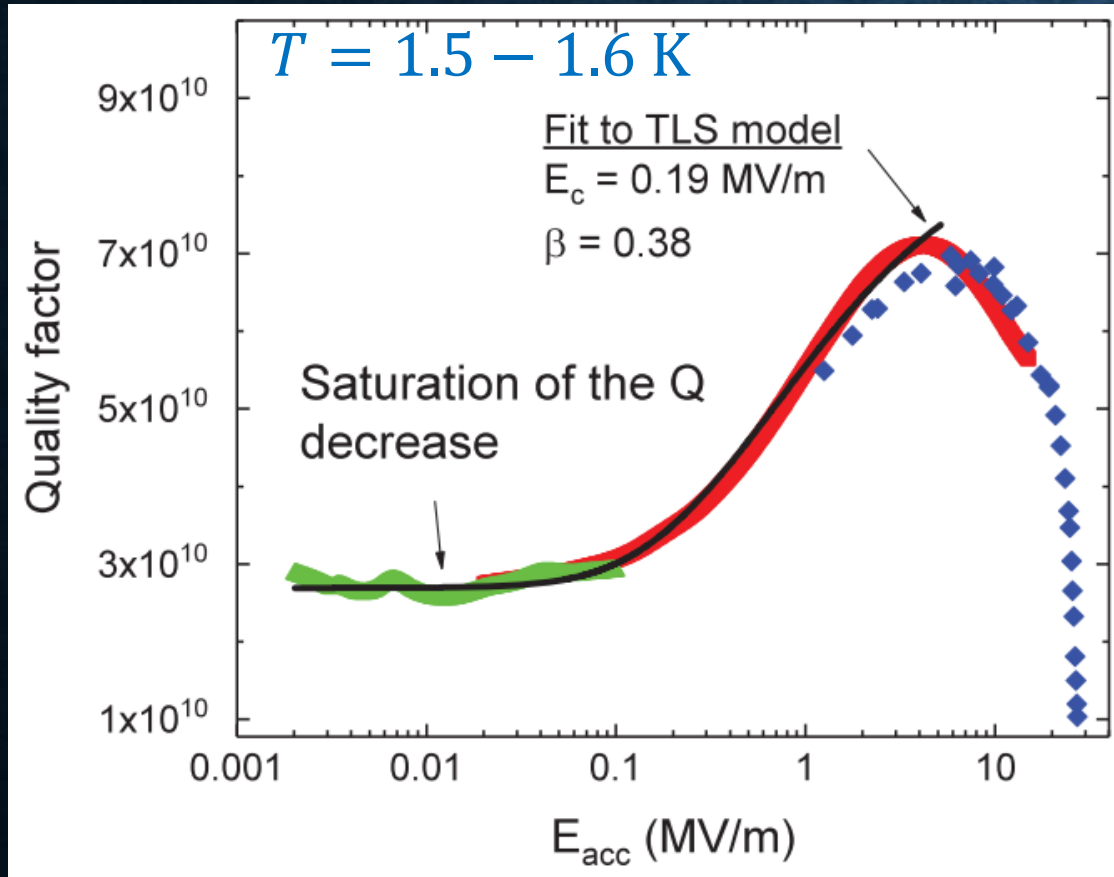


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TLS model explains low-T and low- E_{acc} nonlinear Q evolution.



$$T < 1.6 \text{ K}$$

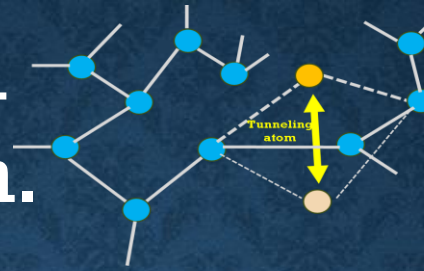


A. Romanenko and D. I. Schuster, Understanding Quality Factor Degradation in Superconducting Niobium Cavities at Low Microwave Field Amplitudes, Phys. Rev. Lett. **119**, 264801 (2017)
<https://doi.org/10.1103/PhysRevLett.119.264801>

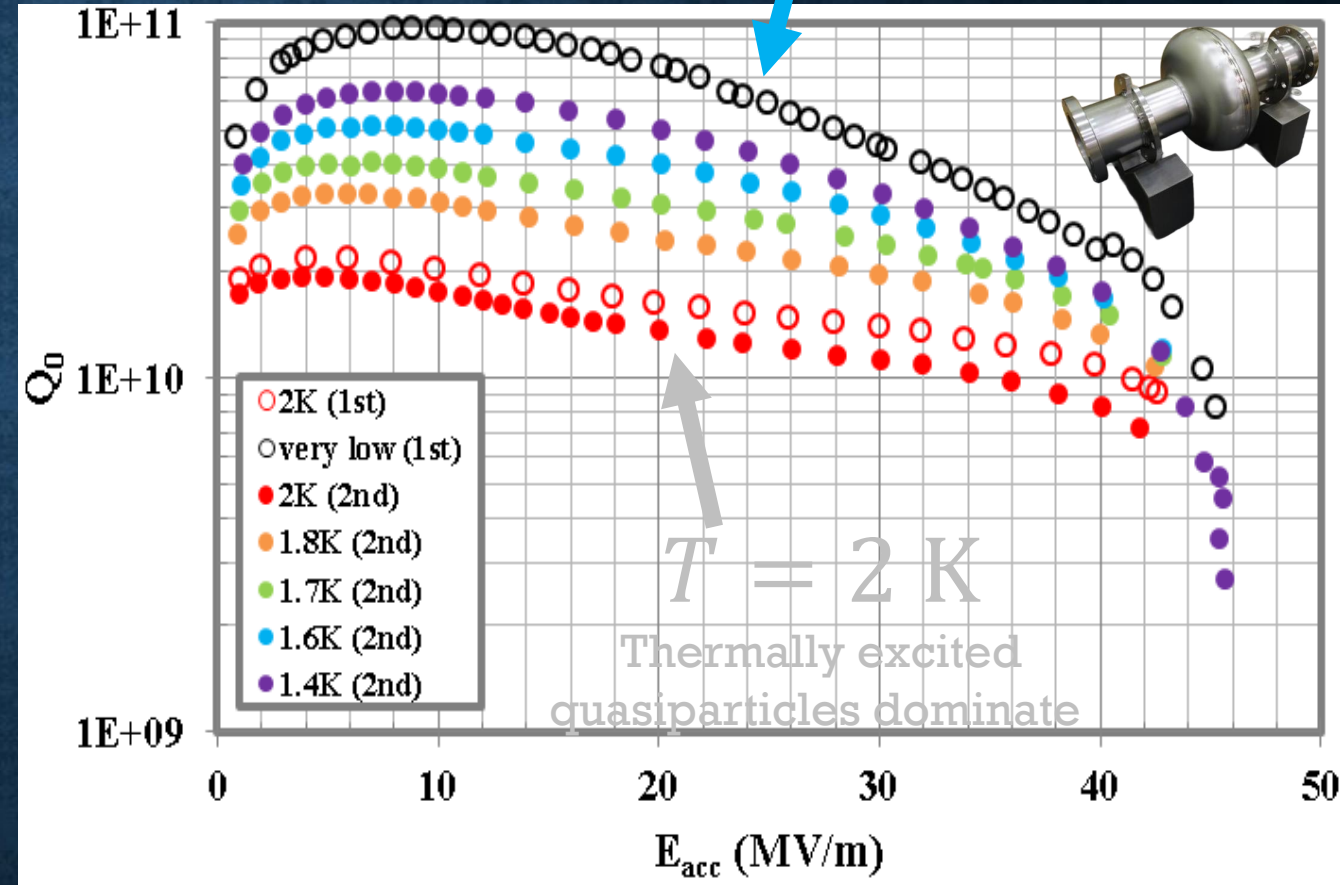
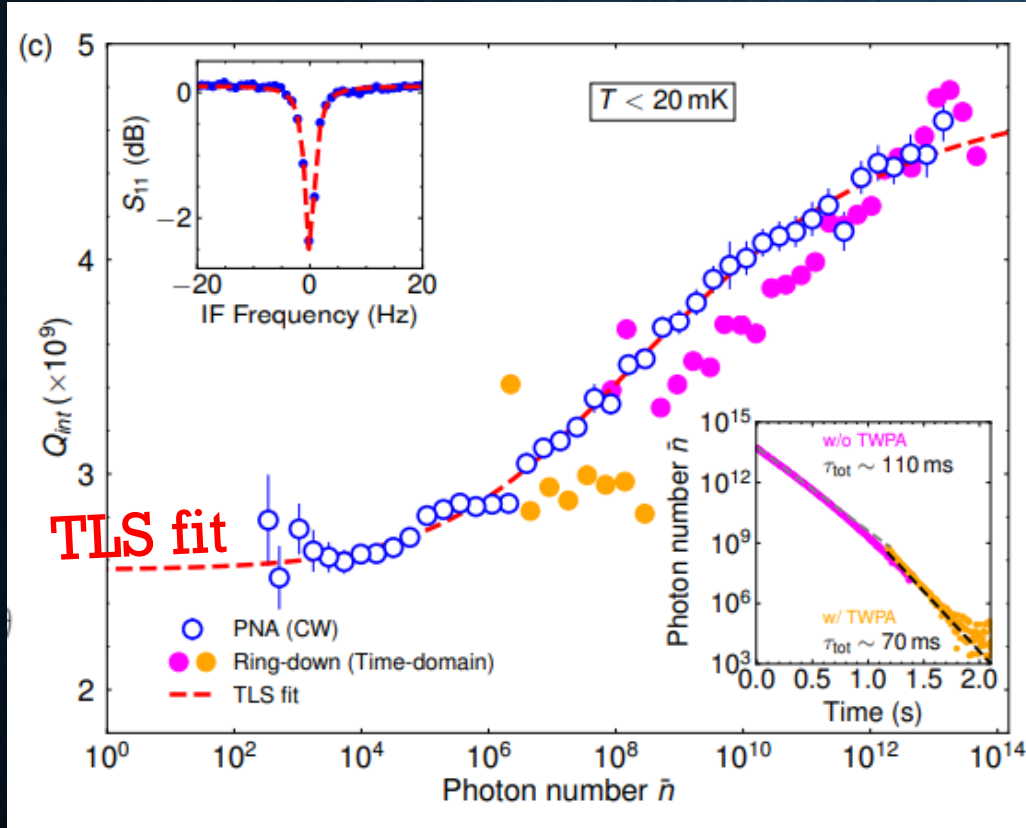
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TLS model explains low-T and low- E_{acc} nonlinear Q evolution.

The same is true for the quantum regime ($T \simeq 20$ mK and a tiny photon number)



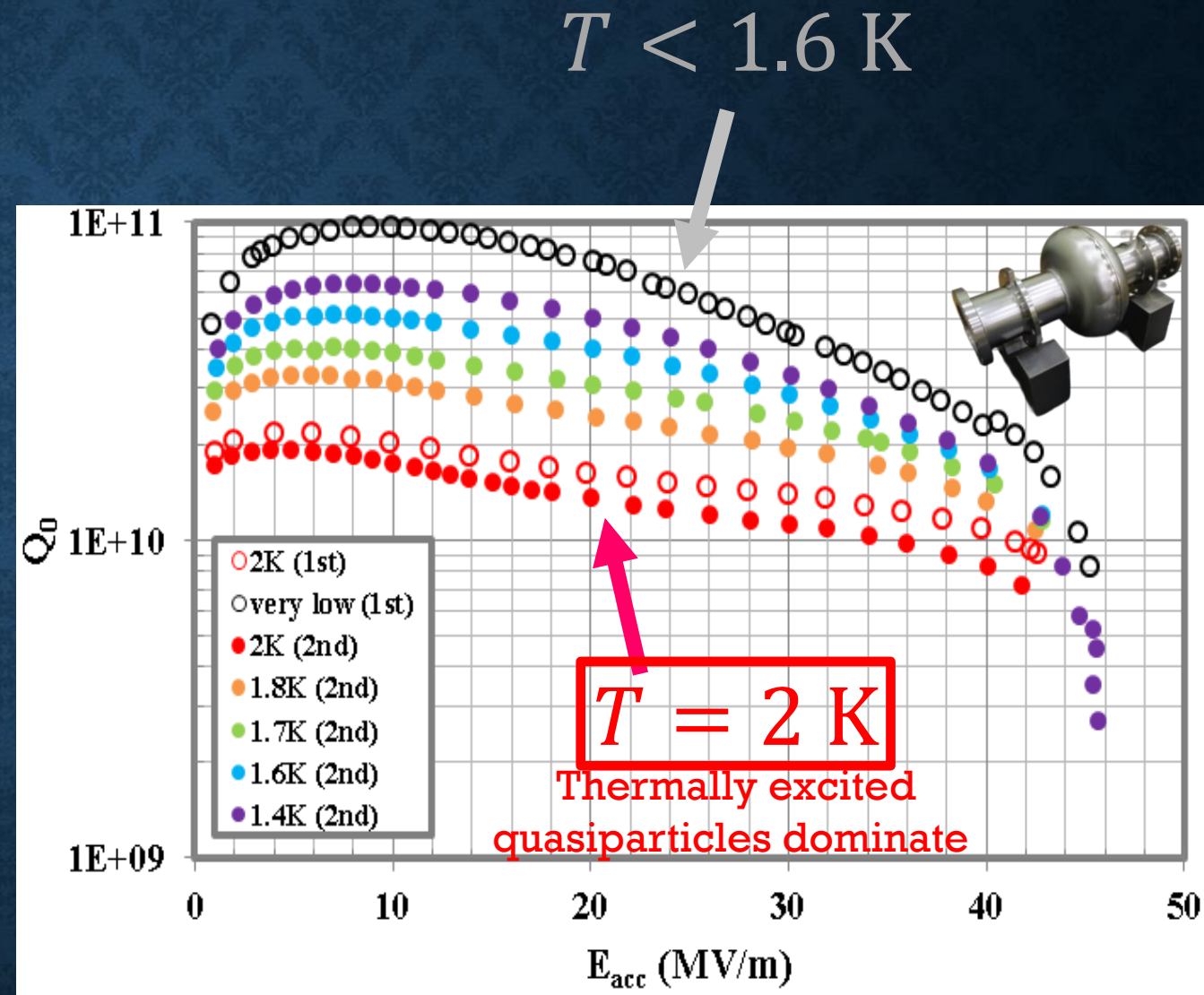
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T. Takenaka, T. Kubo et al., arXiv:2510.01819.
<https://doi.org/10.48550/arXiv.2510.01819>

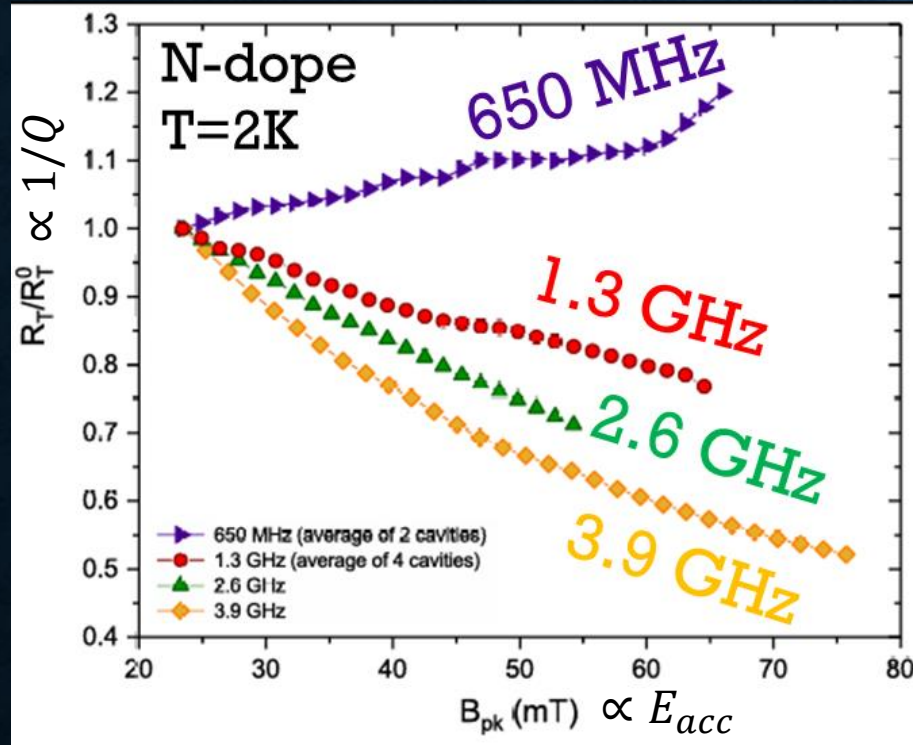
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<https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022>

What about the dissipation mechanisms inherent to superconductors?

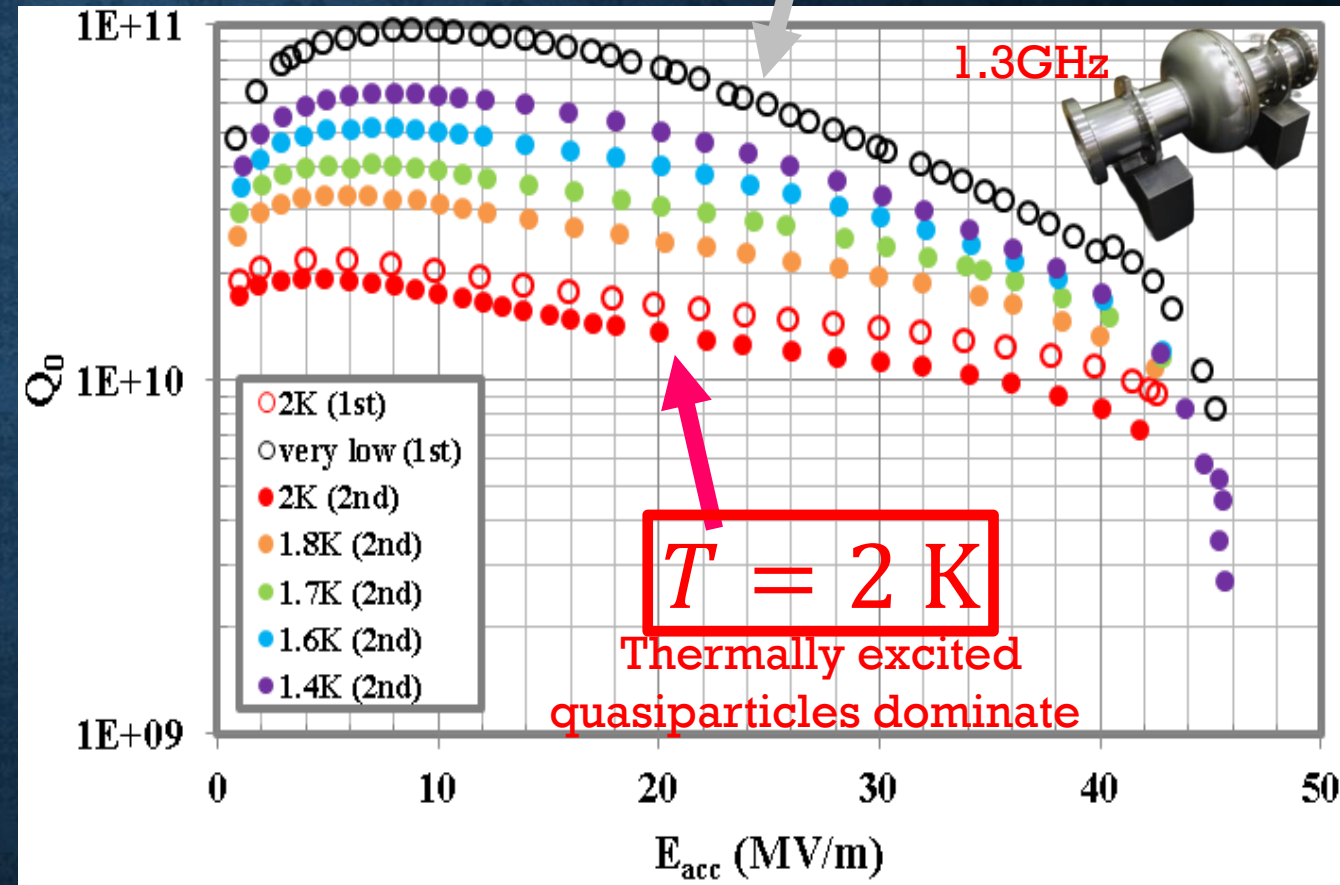


T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
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M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: <https://doi.org/10.1103/PhysRevLett.121.224801>



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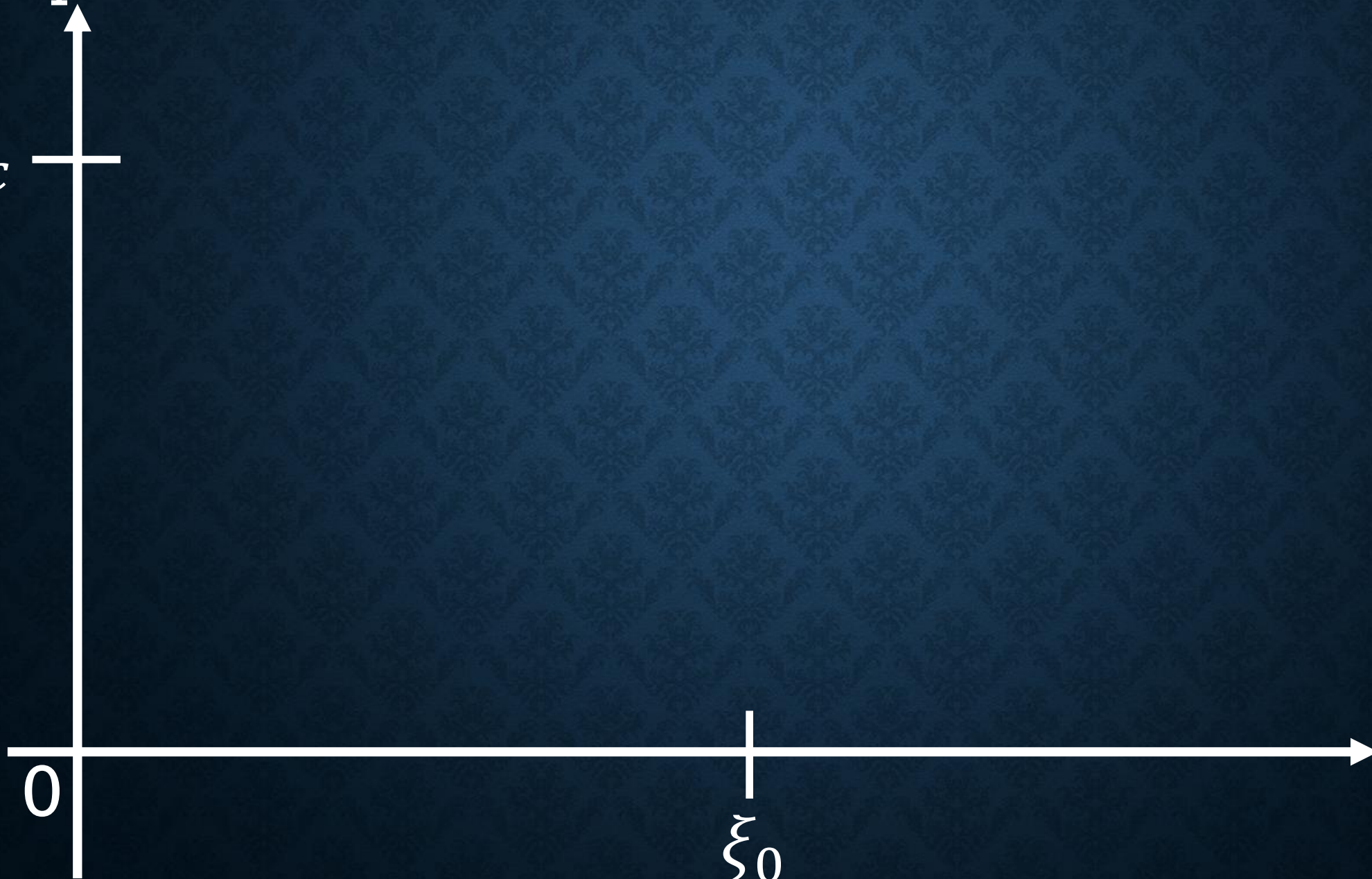
Temperature

T_c

0

ξ_0

Mean
free
path



Temperature

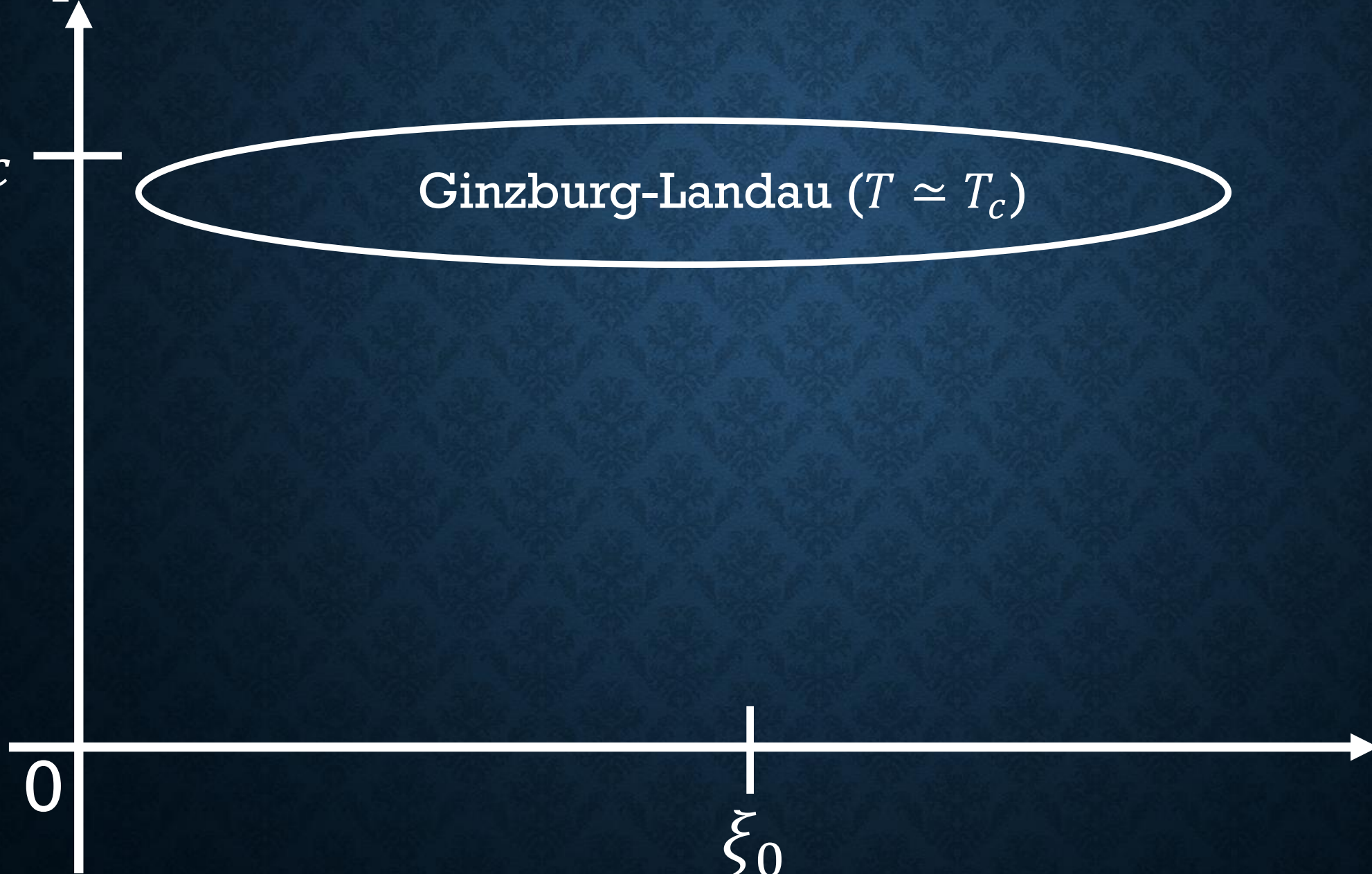
T_c

Ginzburg-Landau ($T \simeq T_c$)

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Temperature

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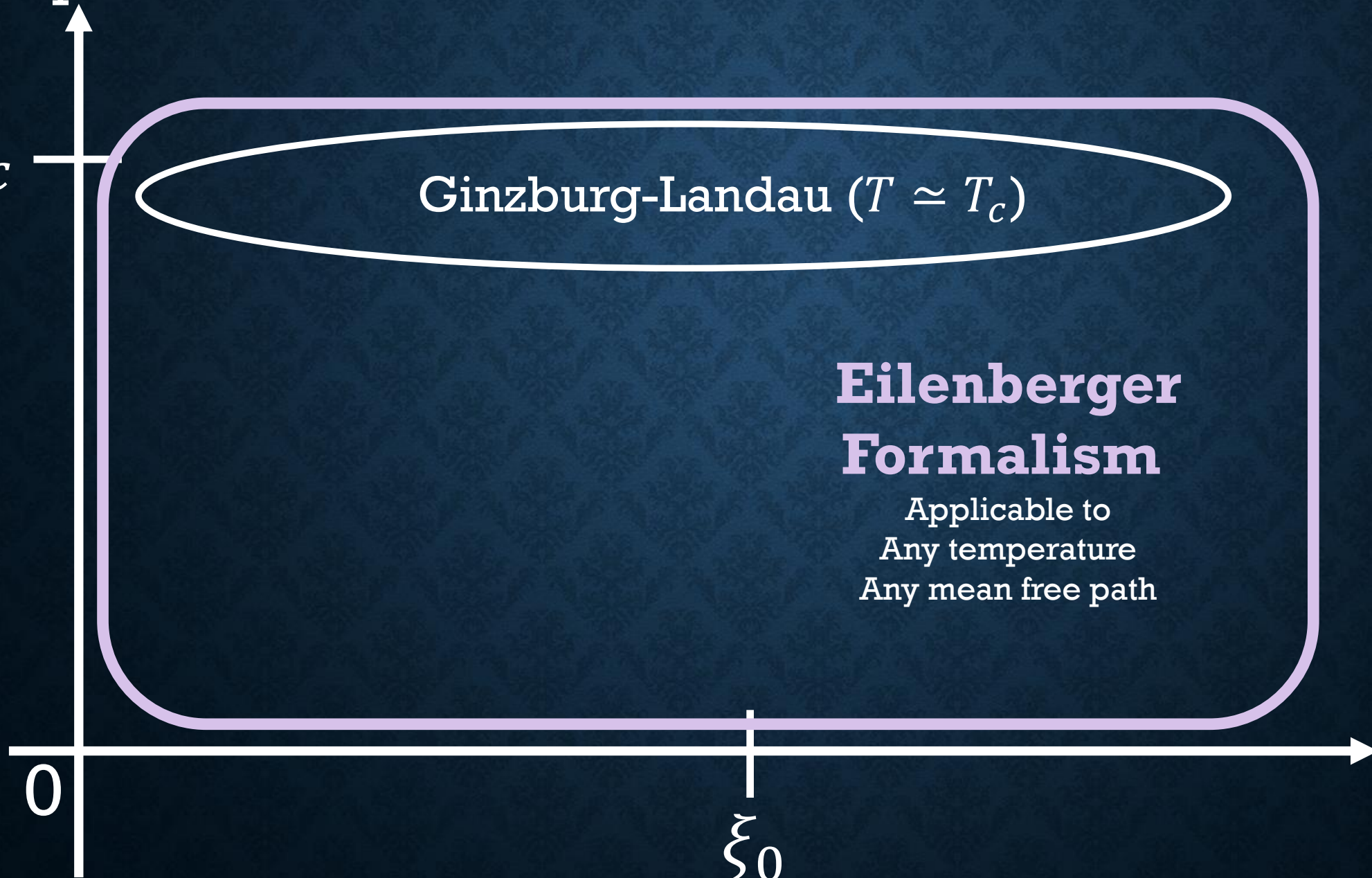
**Eilenberger
Formalism**

Applicable to
Any temperature
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$$mfp < \xi_0$$

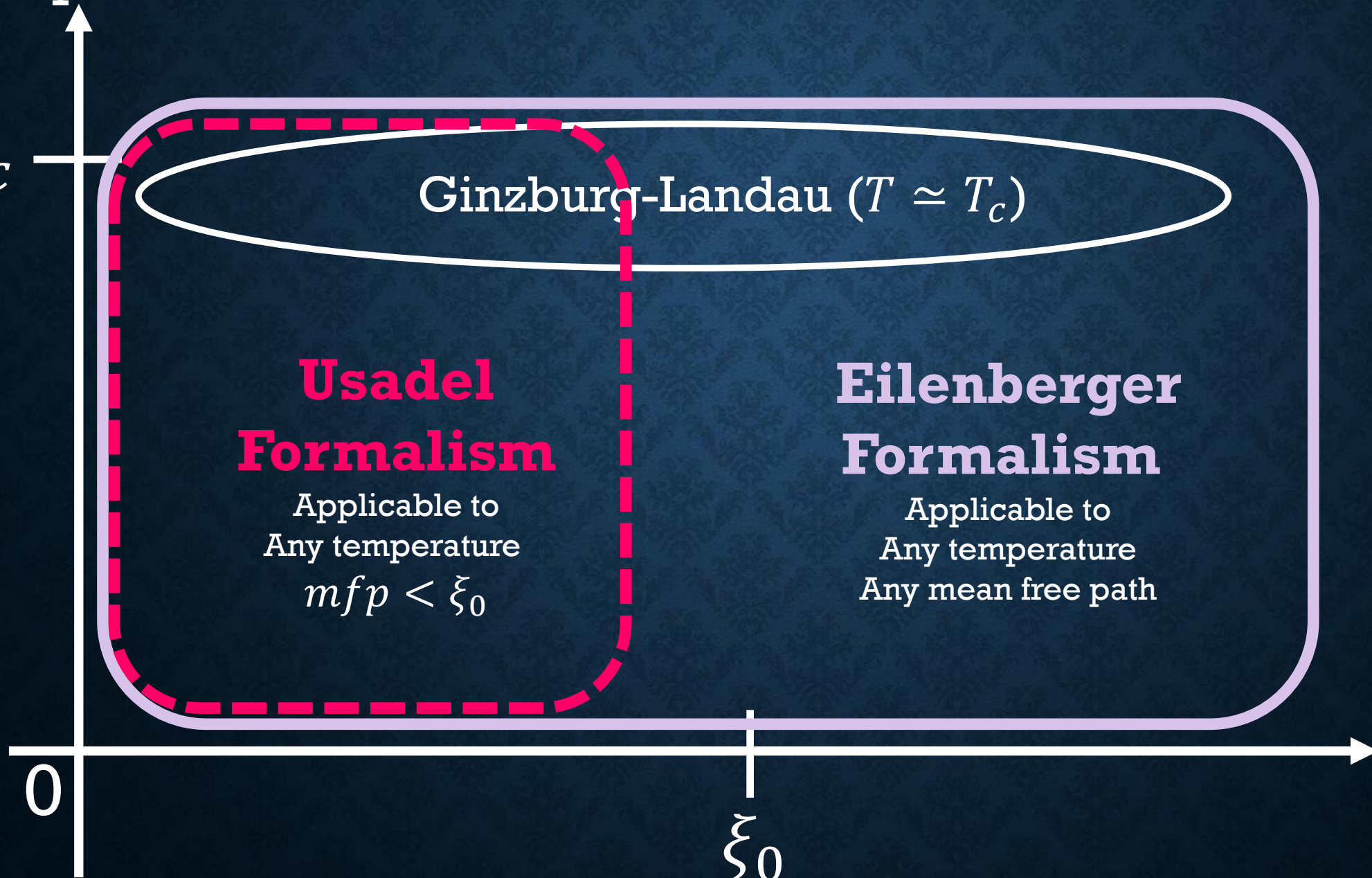
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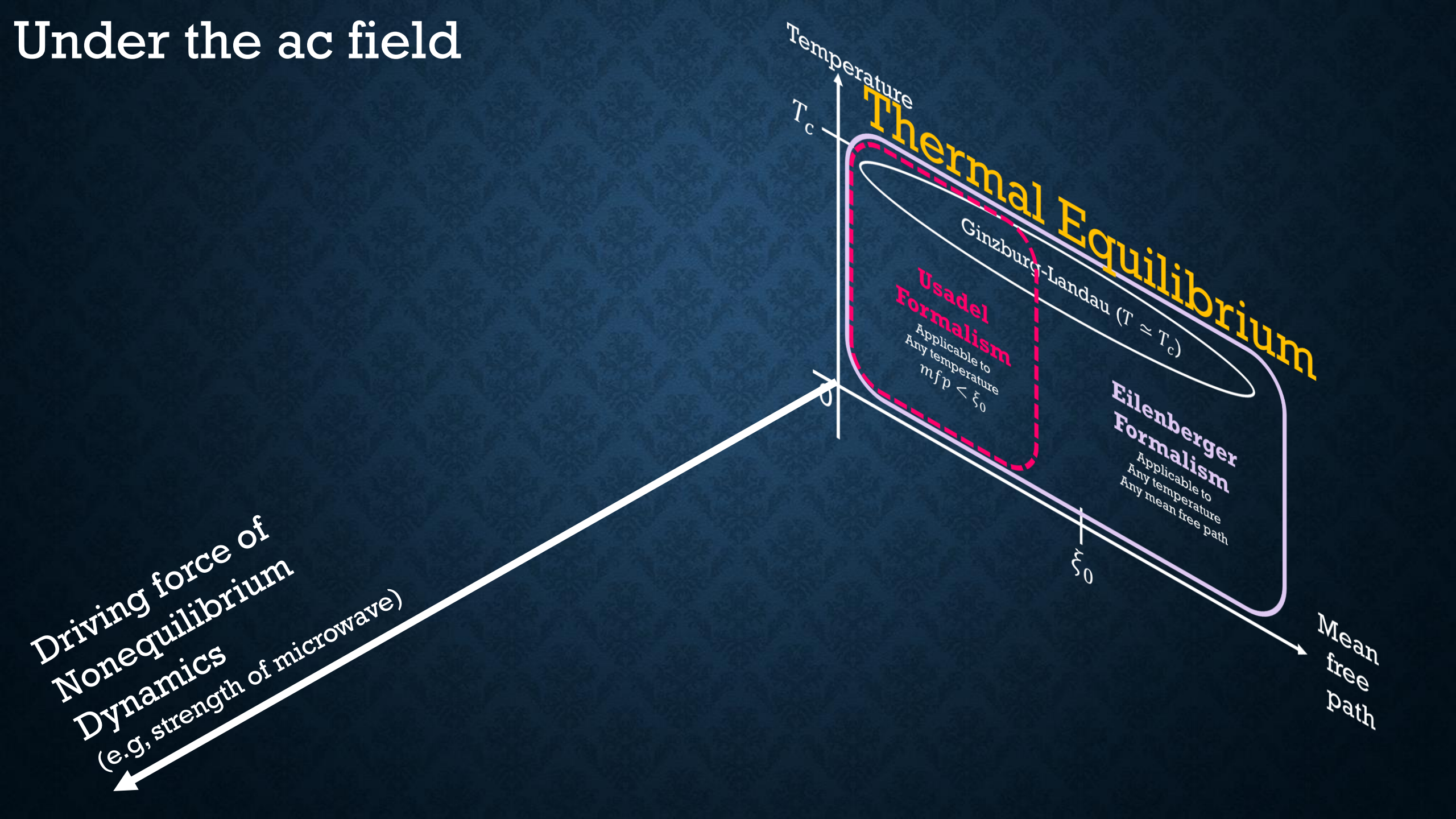
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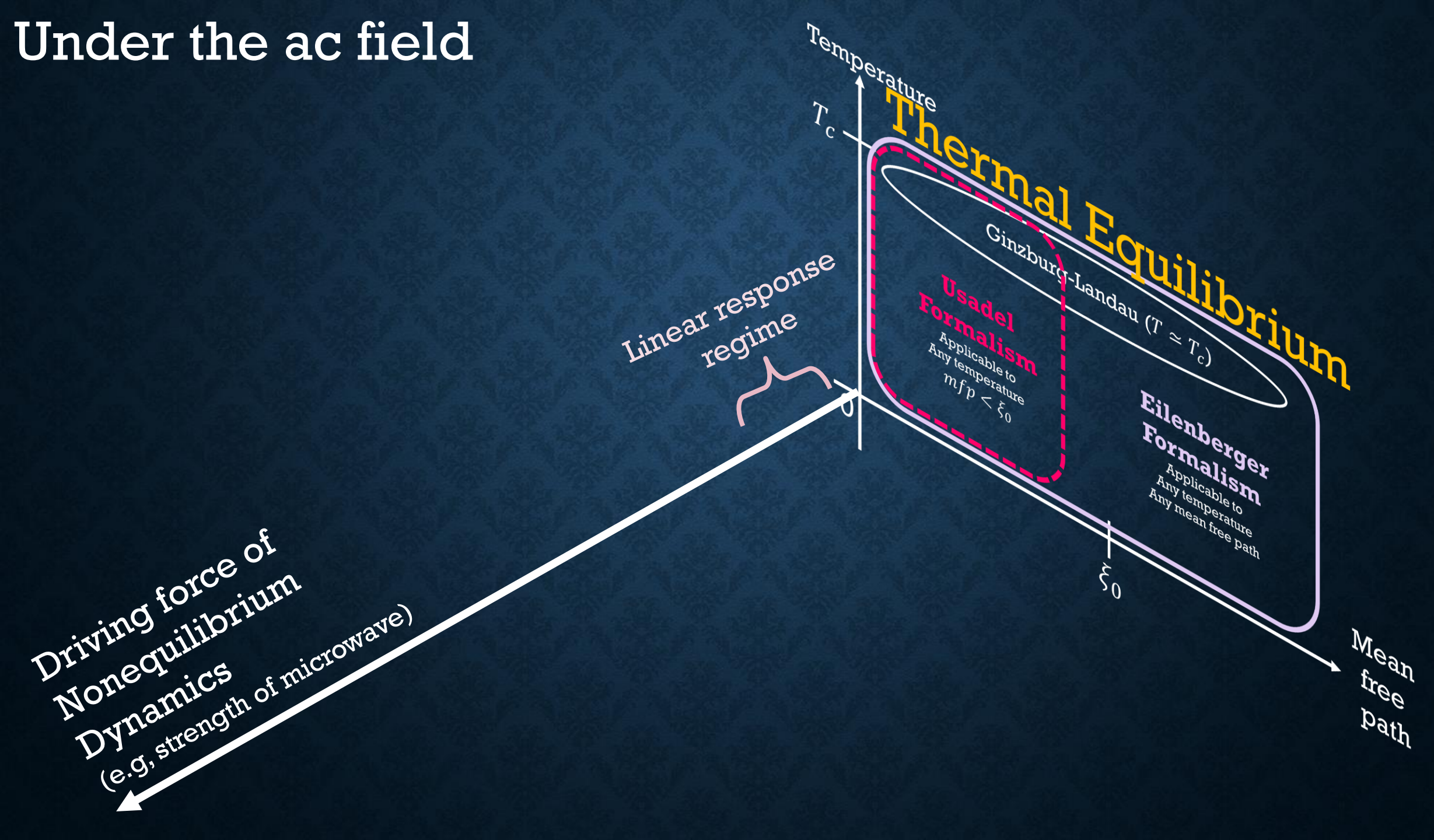
Mean
free
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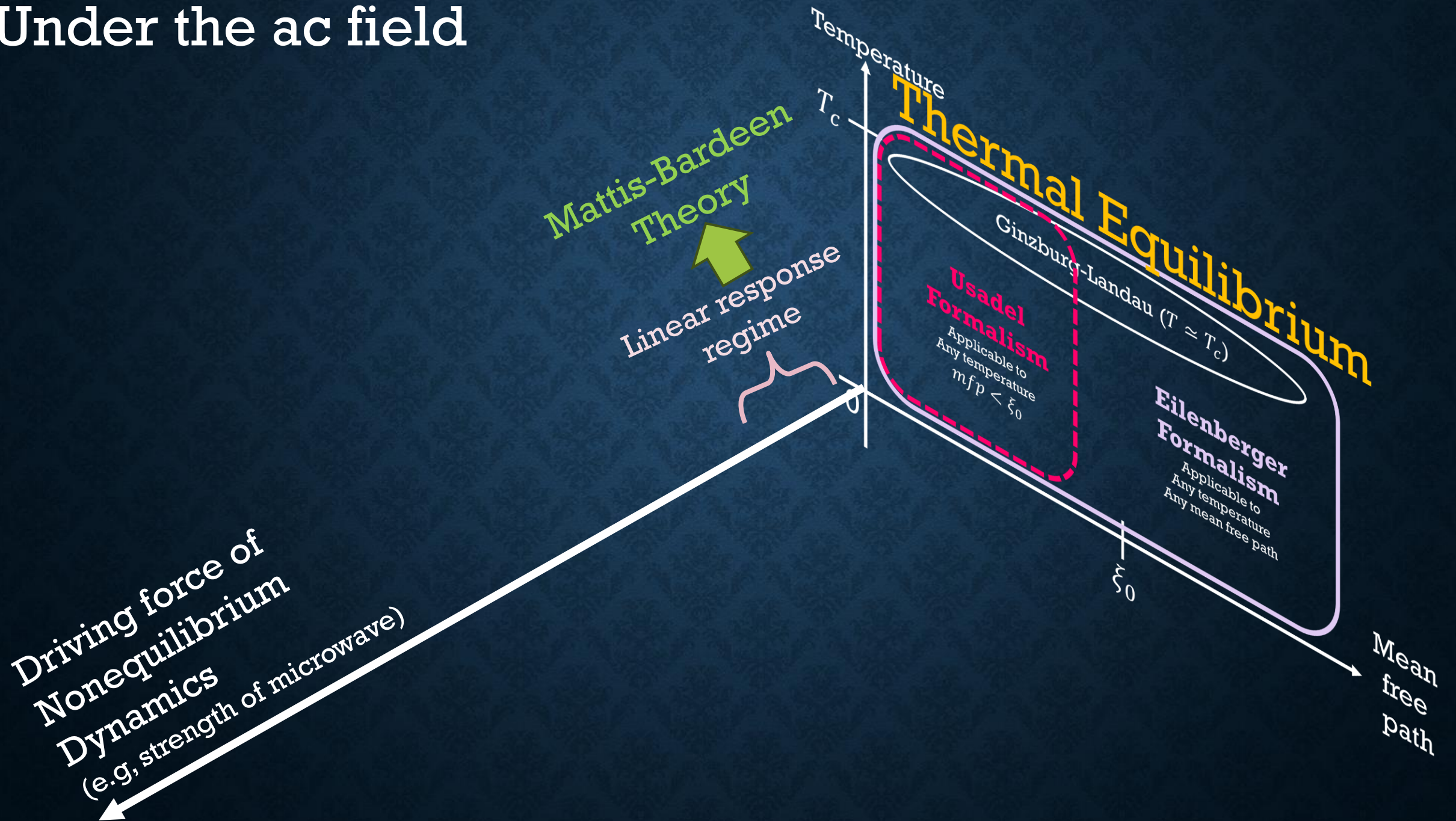
Under the ac field



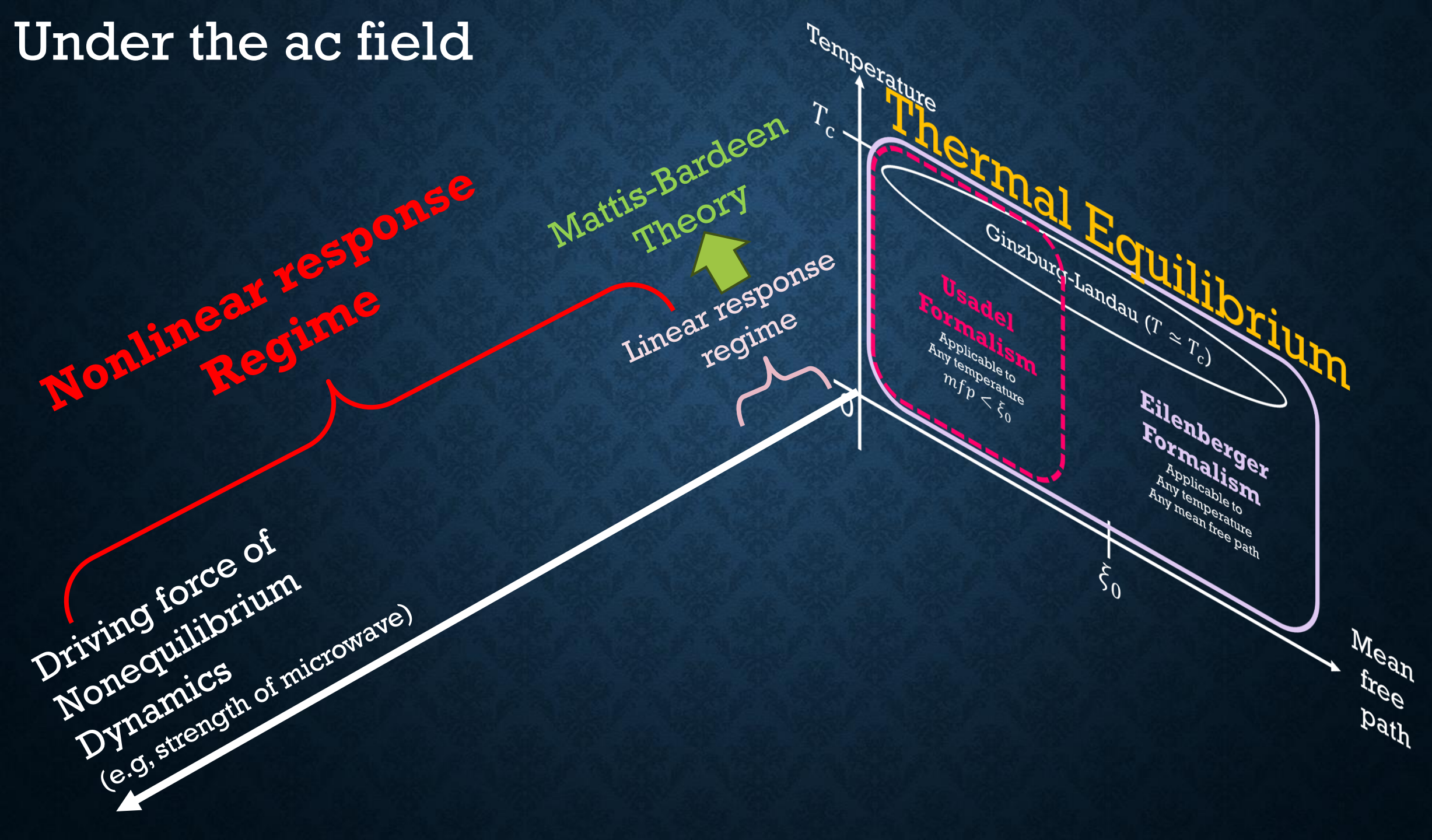
Under the ac field



Under the ac field



Under the ac field



Under the ac field

Unfortunately, this is
the **SRF** accelerator regime

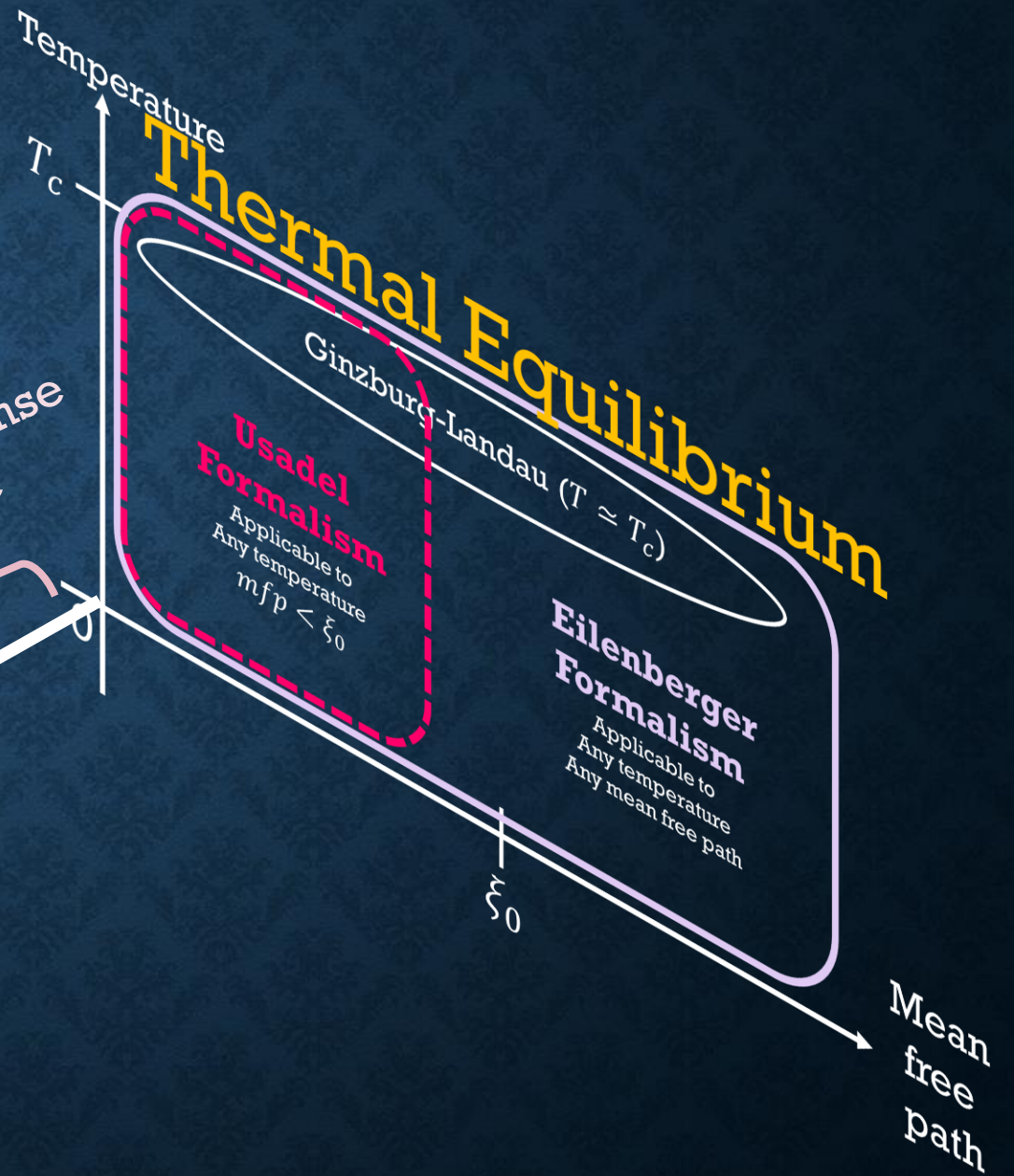


**Nonlinear response
Regime**

Driving force of
Nonequilibrium
Dynamics
(e.g, strength of microwave)

Mattis-Bardeen
Theory

Linear response
regime



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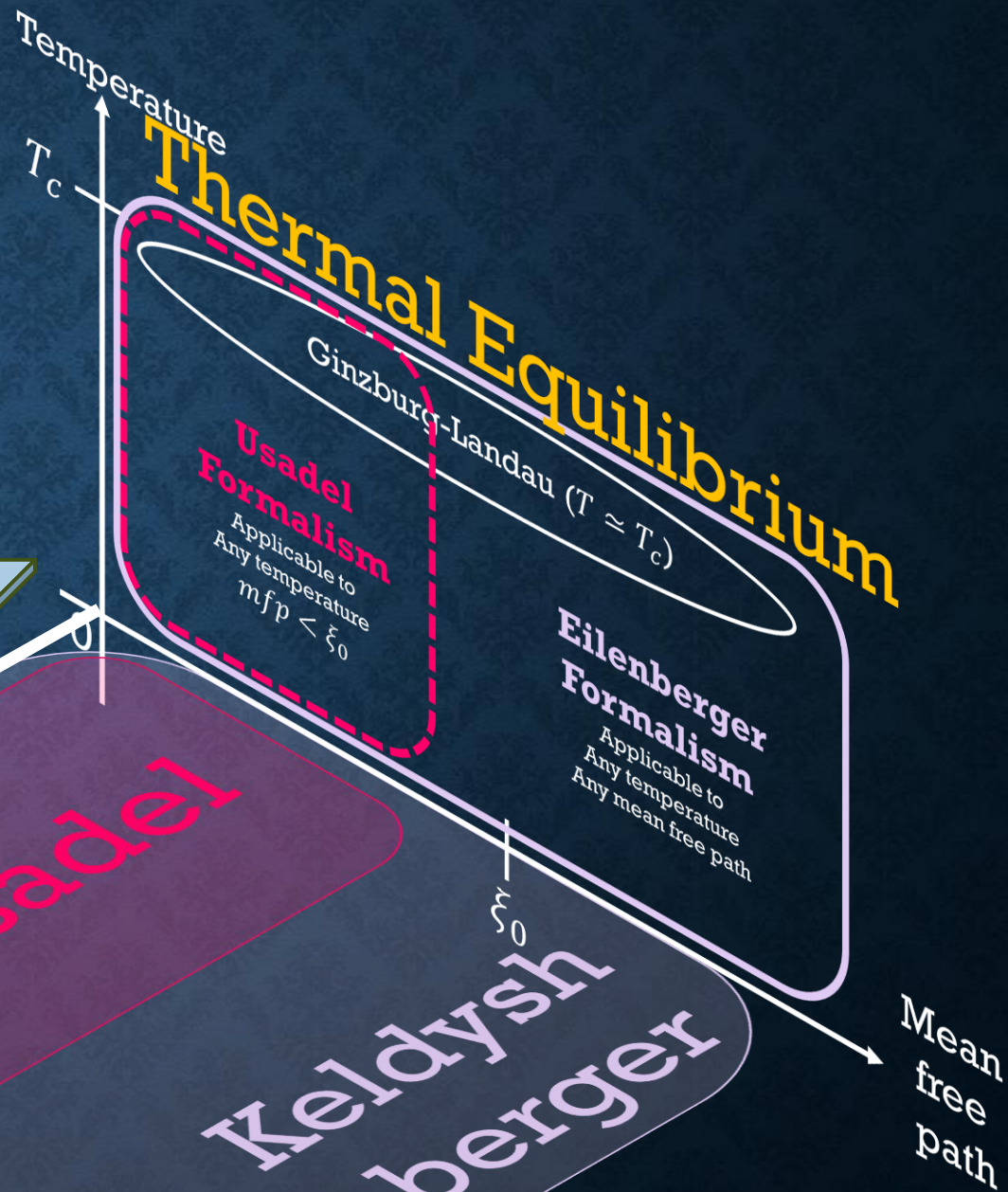
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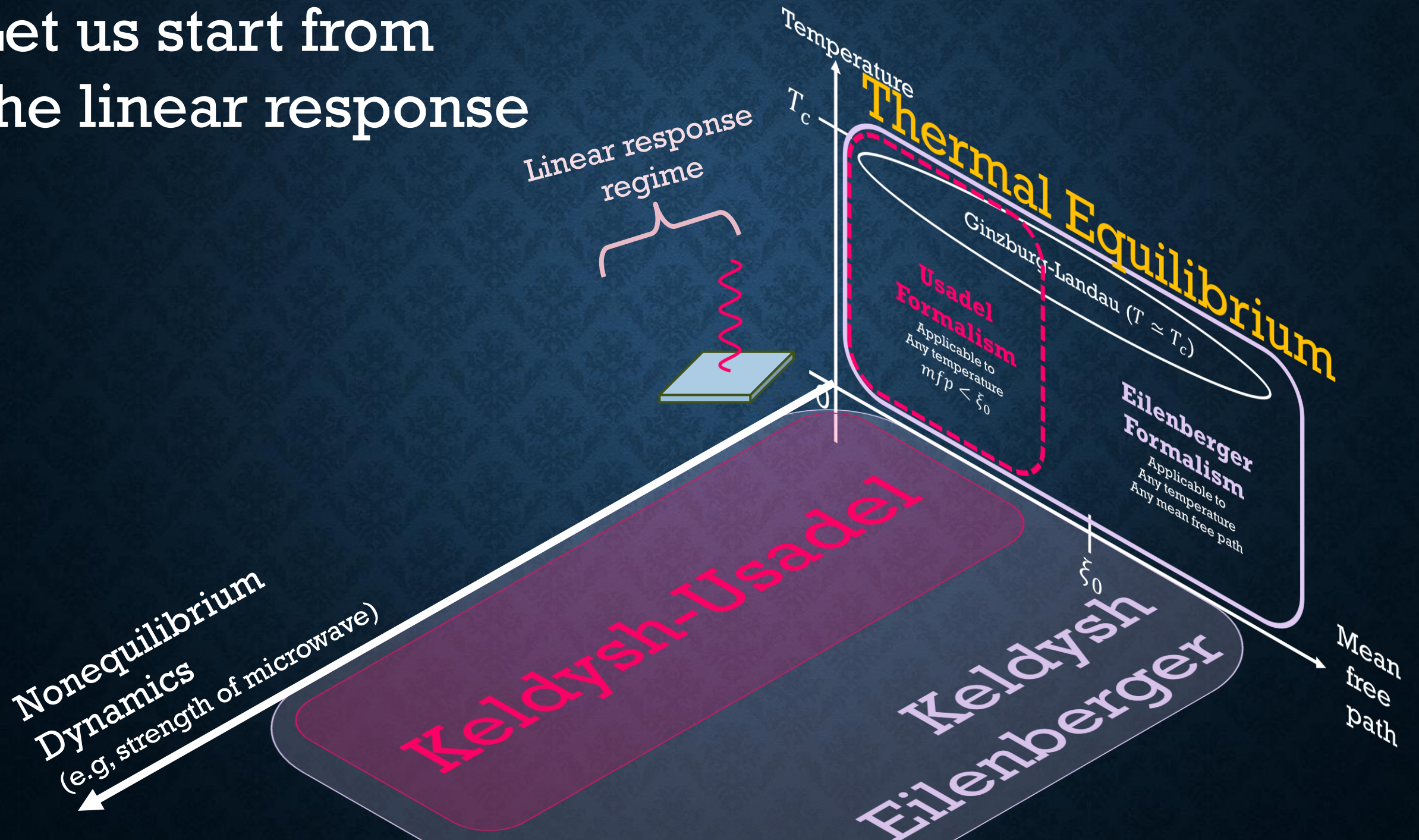
Nonequilibrium
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Keldysh-Usadel

Keldysh
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Let us start from the linear response



Start from the Keldysh-Eilenberger equations (a set of equations for Green's functions and pair potential Δ)

D. Rainer and J. A. Sauls,
Strong coupling theory of superconductivity, in Superconductivity:
From Basic Physics to the Latest Developments, edited by P. N.
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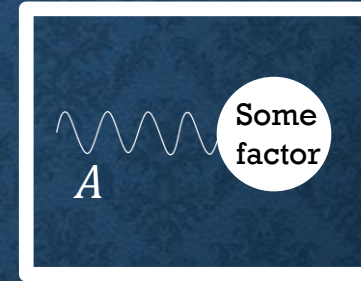
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We apply a small ac electromagnetic field and calculate the first-order perturbation to the current.

$$J \sim A \times \bullet \sim$$



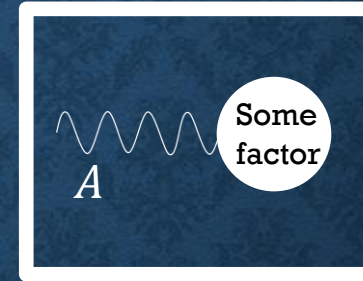
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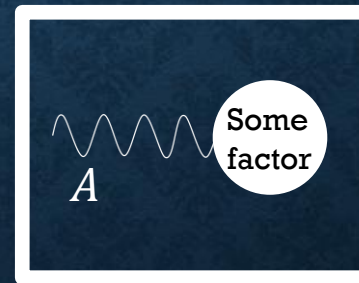
We apply a small ac electromagnetic field and calculate the first-order perturbation to the current.

$$J \sim A \times \bullet \sim$$



Then,
the complex conductivity is given by

$$\sigma = \frac{J}{E} \sim \frac{\boxed{\text{wavy line } A \text{ --- Some factor}}}{A}$$



Complex conductivity formula for arbitrary MFP (for $\lambda \gg \xi$)

$$\begin{aligned}\sigma(\ell, T, \omega) &= \sigma_1 + i\sigma_2 \\ &= \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon [\kappa\mathcal{T}_- + \kappa^*\mathcal{T}_+ + \kappa^a(\mathcal{T}_+ - \mathcal{T}_-)], \\ \kappa &= \frac{1}{3} \frac{g_+g_- + f_+f_- - 1}{d_+ + d_-}, \\ \kappa^a &= \frac{1}{3} \frac{-g_+g_-^* - f_+f_-^* - 1}{d_+ - d_-^*}.\end{aligned}$$

$\tau = \ell/v_f$ is the electron relaxation time. We define $\epsilon_{\pm} := \epsilon \pm \hbar\omega/2$ and set $g_{\pm} := g(\epsilon_{\pm})$, $f_{\pm} := f(\epsilon_{\pm})$, $d_{\pm} := d(\epsilon_{\pm})$, and $\mathcal{T}_{\pm} := \mathcal{T}(\epsilon_{\pm})$, where $g(\epsilon) = -i(\epsilon + i0)/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $f(\epsilon) = -i\Delta/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $d(\epsilon) = i\gamma + i\sqrt{\Delta^2 - (\epsilon + i0)^2}$, and $\mathcal{T}(\epsilon) := \tanh(\epsilon/2k_B T) = 1 - 2f_{\text{FD}}(\epsilon)$, with $f_{\text{FD}}(\epsilon)$ the

- W. Zimmermann, E. Brandt, M. Bauer, E. Seider, and L. Genzel, Optical conductivity of BCS superconductors with arbitrary purity, *Physica C* **183**, 99 (1991).
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**In the dirty limit
($\text{mfp} \ll \xi_0$)**



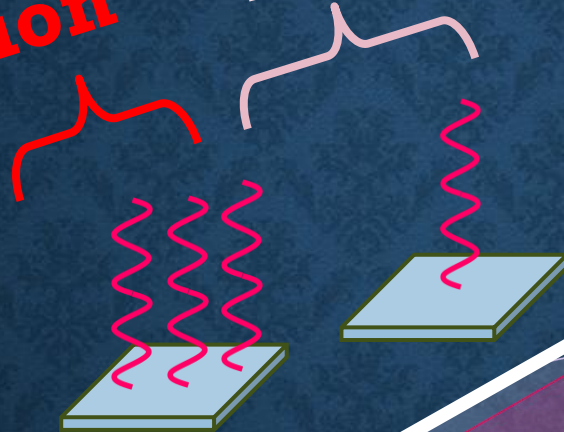
$$\begin{aligned}\sigma_1(T, \omega) &= \frac{\sigma_n}{\hbar\omega} \int_{-\infty}^{\infty} d\epsilon [f_{\text{FD}}(\epsilon) - f_{\text{FD}}(\epsilon')] M_1(\epsilon, \omega), \\ M_1(\epsilon, \omega) &= \text{Re}g(\epsilon)\text{Re}g(\epsilon') + \text{Re}f(\epsilon)\text{Re}f(\epsilon'), \\ \sigma_2(T, \omega) &= \frac{\sigma_n}{\hbar\omega} \int_{-\infty}^{\infty} d\epsilon \tanh \frac{\epsilon}{2k_B T} M_2(\epsilon, \omega), \\ M_2(\epsilon, \omega) &= \text{Re}g(\epsilon)\text{Im}g(\epsilon') + \text{Re}f(\epsilon)\text{Im}f(\epsilon').\end{aligned}$$

$$\epsilon' := \epsilon + \hbar\omega$$

The well-known
dirty-limit Mattis-Bardeen formula

**Leading order
nonlinear
correction**

Linear response
regime



Nonequilibrium
Dynamics
(e.g, strength of microwave)

Keldysh-Usadel

**Keldysh
Eilenberger**

Temperature
 T_c
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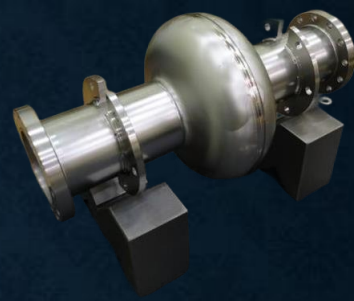
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Keldysh–Usadel Equation for SC under a strong ac field



arXiv:2509.09766
To be published in
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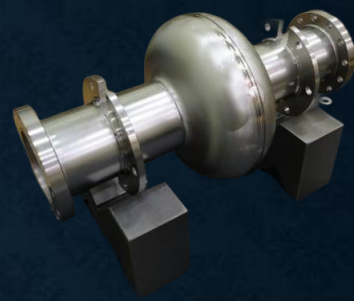
$$\begin{aligned}
 & -i\frac{s}{8} \sum_{\eta=\pm\omega_{ac}} \sum_{\eta'=\pm\omega_{ac}} 2\pi\delta(\omega + \eta + \eta') \times \\
 & \left[\hat{\tau}_3 \hat{g}_e^r \left(\epsilon - \frac{\hbar\eta}{2} + \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \hat{g}_e^r \left(\epsilon + \frac{\hbar\eta}{2} + \frac{\hbar\eta'}{2} \right) \right. \\
 & \left. - \hat{g}_e^r \left(\epsilon - \frac{\hbar\eta}{2} - \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \hat{g}_e^r \left(\epsilon + \frac{\hbar\eta}{2} - \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \right] \\
 & = \left(\epsilon + \frac{\hbar\omega}{2} \right) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \left(\epsilon - \frac{\hbar\omega}{2} \right) \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \delta \hat{\Delta}(\omega) \hat{g}_e^r \left(\epsilon - \frac{\hbar\omega}{2} \right) - \hat{g}_e^r \left(\epsilon + \frac{\hbar\omega}{2} \right) \delta \hat{\Delta}(\omega) \\
 & + [\hat{\Delta}_e, \delta \hat{g}^r(\epsilon, \omega)]. \quad (11)
 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (15)$$

$$\begin{aligned}
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 & + \hat{\tau}_3 \hat{g}_e^K \left(\epsilon - \frac{\hbar\eta}{2} + \frac{\hbar\eta'}{2} \right) \hat{\tau}_3 \hat{g}_e^A \left(\epsilon + \frac{\hbar\eta}{2} + \frac{\hbar\eta'}{2} \right) \\
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We solve these equations to obtain the **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

To obtain the current response,
we substitute the solutions
($\delta \hat{g}^{R,A,K}$, $\delta \Delta$) into

$$\begin{aligned}
 \mathbf{J}(t) &= \mathbf{J}^{(1)}(t) + \mathbf{J}^{(3)}(t) \\
 &= \frac{\sigma_n}{8e} \int \left\{ \mathbf{S}^{(1)}(\epsilon, t) + \mathbf{S}^{(3)}(\epsilon, t) \right\} d\epsilon, \quad (49)
 \end{aligned}$$

$$\mathbf{S}^{(1)}(\epsilon, t) = \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right] \quad (50)$$

$$\begin{aligned}
 \mathbf{S}^{(3)}(\epsilon, t) &= \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \delta \hat{g}^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \delta \hat{g}^A) \right. \right. \\
 & \left. \left. + \delta \hat{g}^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \delta \hat{g}^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right]. \quad (51)
 \end{aligned}$$

← Linear response

← **Nonlinear** correction
of $\mathcal{O}(A^3)$

Schematically, the results can be illustrated as

$$J \sim Agg + Ag\delta g$$

Linear
response

Nonlinear corrections



$$\begin{aligned} \mathbf{J}(t) &= \mathbf{J}^{(1)}(t) + \mathbf{J}^{(3)}(t) \\ &= \frac{\sigma_n}{8e} \int \left\{ \mathbf{S}^{(1)}(\epsilon, t) + \mathbf{S}^{(3)}(\epsilon, t) \right\} d\epsilon, \end{aligned} \quad (49)$$

$$\mathbf{S}^{(1)}(\epsilon, t) = \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right] \quad (50)$$

$$\begin{aligned} \mathbf{S}^{(3)}(\epsilon, t) &= \text{Tr} \left[\hat{\tau}_3 \left\{ \hat{g}_e^R \circ (\hat{\partial} \circ \delta \hat{g}^K) + \hat{g}_e^K \circ (\hat{\partial} \circ \delta \hat{g}^A) \right. \right. \\ &\quad \left. \left. + \delta \hat{g}^R \circ (\hat{\partial} \circ \hat{g}_e^K) + \delta \hat{g}^K \circ (\hat{\partial} \circ \hat{g}_e^A) \right\} \right]. \end{aligned} \quad (51)$$

← Linear response

← **Nonlinear** correction
of $\mathcal{O}(A^3)$

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$$J \sim Agg + Ag\delta g$$

Linear
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← Linear response

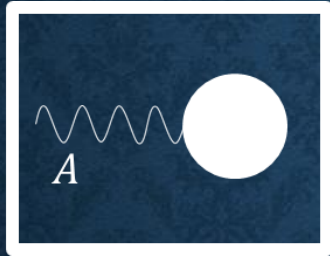
← **Nonlinear** correction
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Schematically, the results can be illustrated as

$$J \sim \boxed{Agg} + Ag\delta g$$

Linear
response

Nonlinear corrections

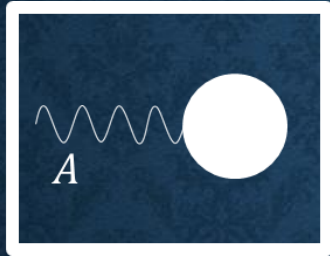


Schematically, the results can be illustrated as

$$J \sim \boxed{A g g} + \boxed{A g \delta g} \quad \delta g \sim A^2 + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi A^2$$

Linear
response

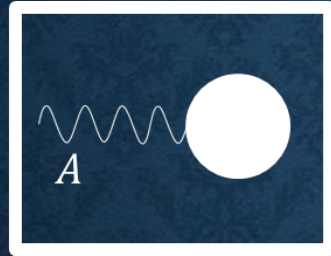
Nonlinear corrections



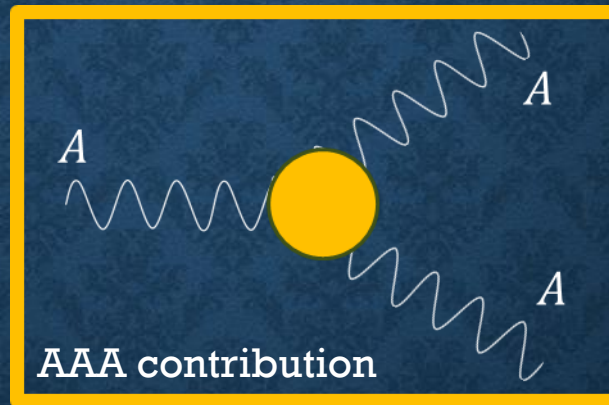
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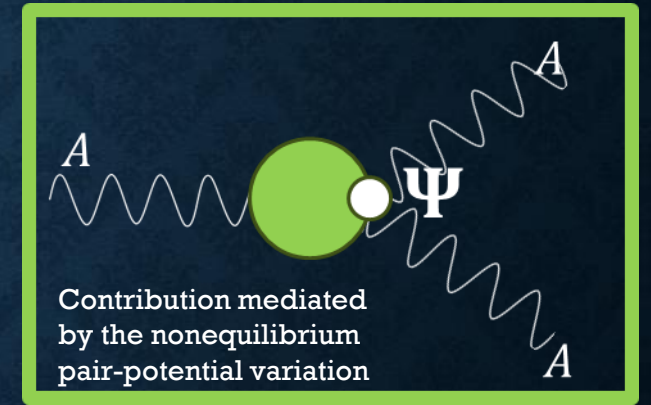
Linear
response



Nonlinear corrections



+

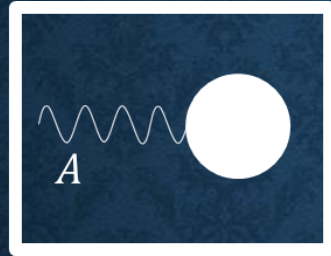


Contribution mediated
by the nonequilibrium
pair-potential variation

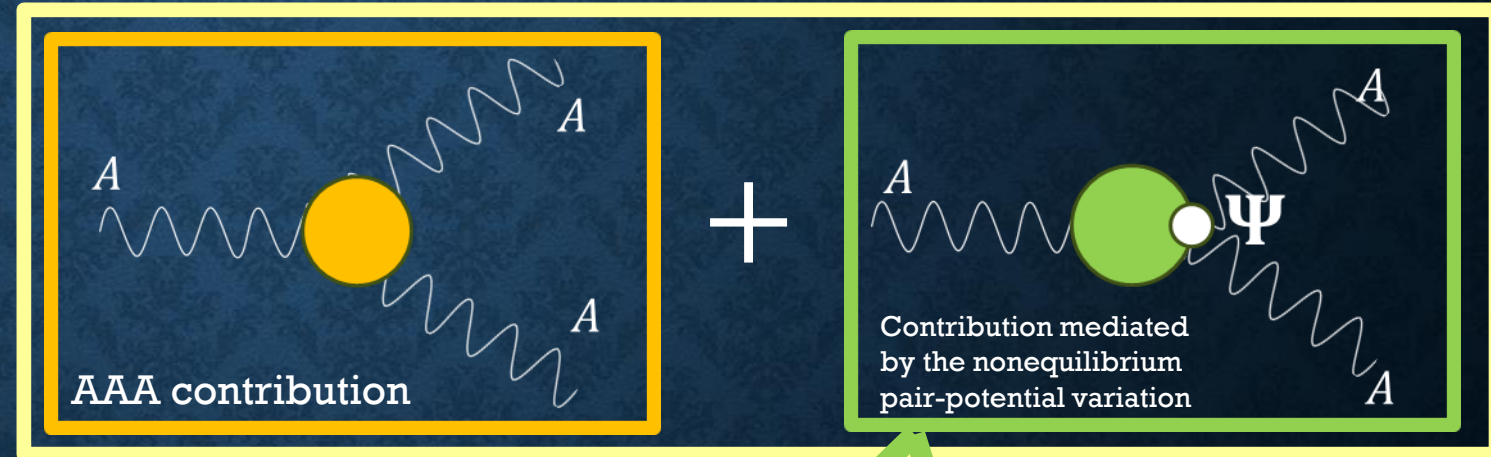
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Linear
response



Nonlinear corrections



Note: This theory automatically includes the **Eliashberg effect** (microwave-induced gap enhancement) as well as the **Higgs-mode** contribution.

Time averaged $\delta\Delta$:
Eliashberg effect

Time dependent $\delta\Delta$:
Higgs

Eliashberg effect?

Higgs?

Eliashberg effect?

Higgs?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardeen theory because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

$$\delta\Delta \propto A_0^2 \cos^2 \omega t$$

Eliashberg effect? Higgs?

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$$\delta\Delta \propto A_0^2 \cos^2 \omega t \propto 1 + \cos 2\omega t$$

Eliashberg effect?

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Time-averaged nonequilibrium gap variation

Eliashberg effect?

Higgs?

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Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation

Eliashberg effect?

Higgs?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardeen theory because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

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Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation

quasiparticles



Eliashberg effect?

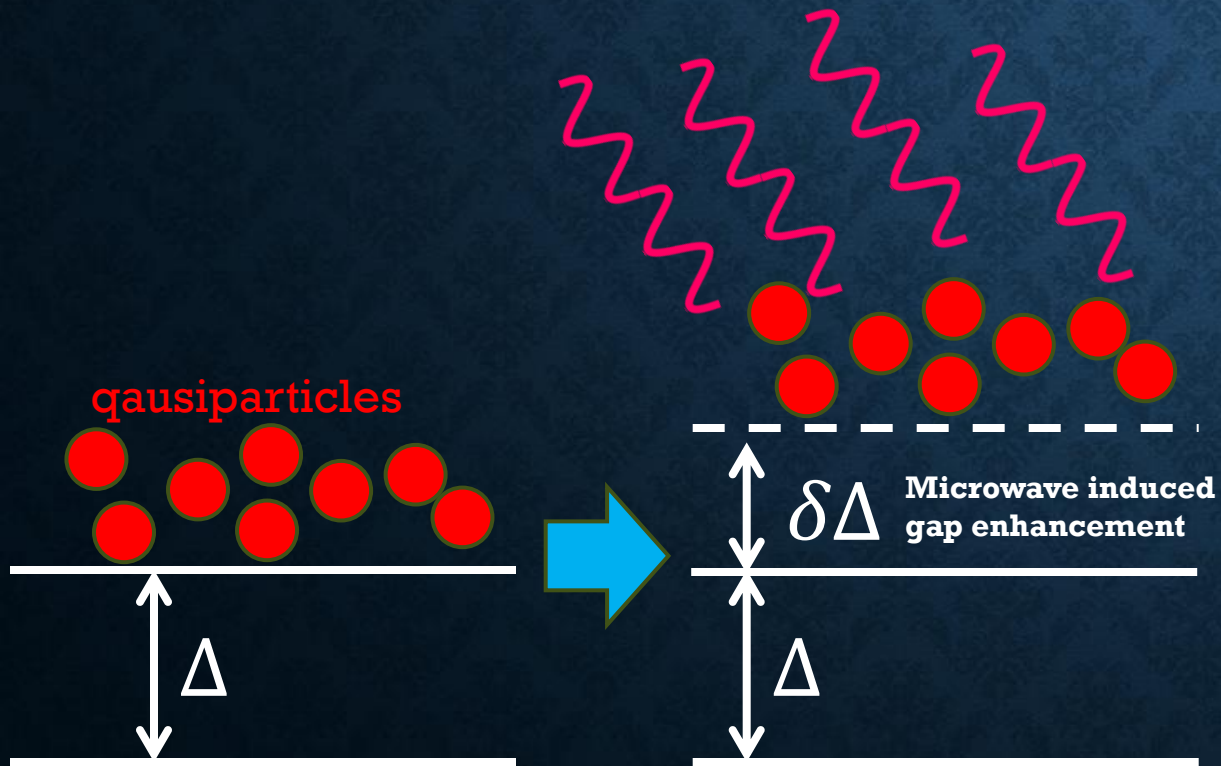
Higgs?

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Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation



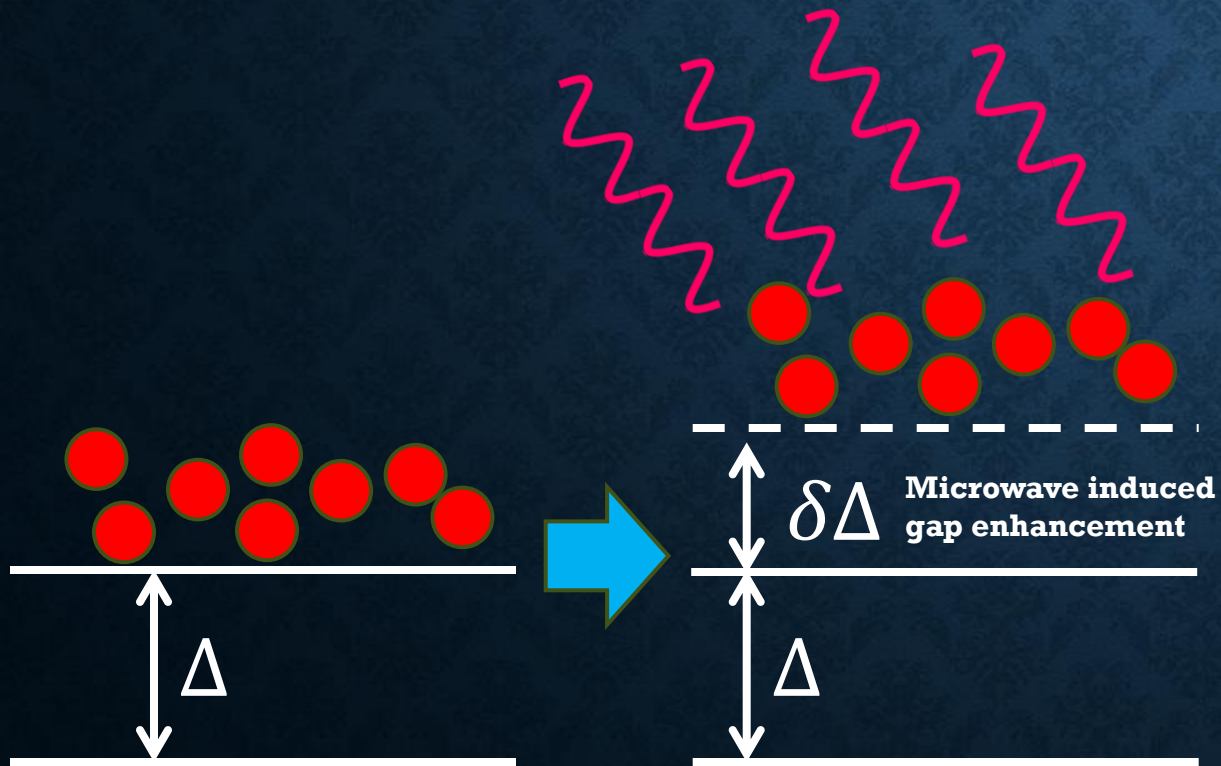
Eliashberg effect?

Higgs?

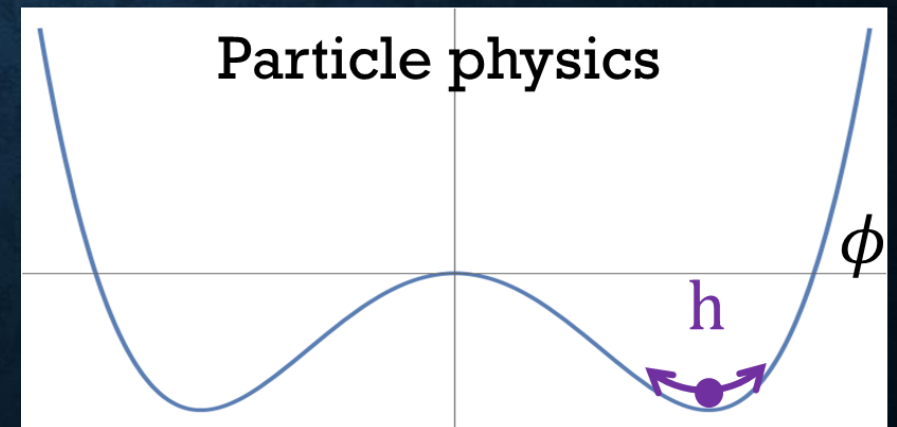
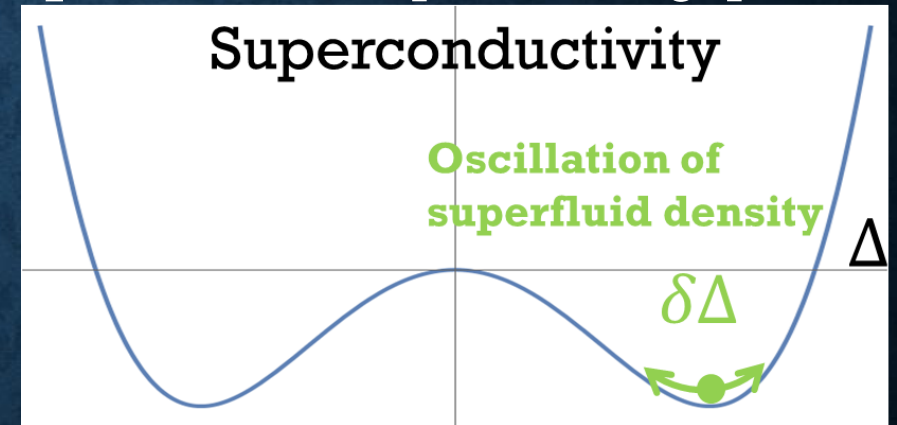
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Time-averaged nonequilibrium gap variation



Time dependent nonequilibrium gap variation



Nonlinear correction

arXiv:2509.09766

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<https://doi.org/10.1103/8fh8-nknj>

$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\text{Higgs}} + \delta\sigma_1^{\text{Eliash}}$$

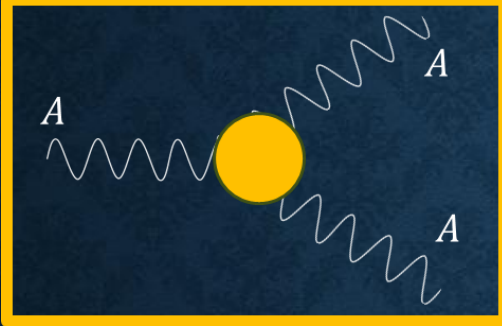
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$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\text{Higgs}} + \delta\sigma_1^{\text{Eliash}}$$



$$\delta\sigma_1^{qqq} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1\text{H}}^{qqq}] \frac{s}{\hbar\omega_{\text{ac}}} \quad I_{1\text{H}}^{\text{qqq}} = \frac{-1}{16\sqrt{\pi}} \int K_{1\text{H}} d\epsilon,$$

$$K_{1\text{H}}(\epsilon, \omega_{\text{ac}}) = \sum_{i=1}^6 K_{1\text{H},i}, \quad (61)$$

$$K_{1\text{H},1} = i \frac{(F_1 - F_{-3})F_{-1} + (G_1 - G_{-3})G_{-1}}{4\hbar\omega_{\text{ac}}(F_1 + F_{-3})} \times \left[\left\{ (F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^* \right\} (\mathcal{T}_{-1} - \mathcal{T}_{-3}) + \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1} \right], \quad (62)$$

$$K_{1\text{H},2} = i \frac{(G_3 - G_{-1})G_1 + (F_3 - F_{-1})F_1}{4\hbar\omega_{\text{ac}}(F_3 + F_{-1})} \times \left\{ (F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1 \right\} \mathcal{T}_{-1}, \quad (63)$$

$$K_{1\text{H},3} = -i \frac{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1}{4\hbar\omega_{\text{ac}}(F_3 - F_{-1}^*)} \times \left[\left\{ (G_3 + G_{-1}^*)G_1^* + (F_3 + F_{-1}^*)F_1^* \right\} (\mathcal{T}_3 - \mathcal{T}_1) + \left\{ (G_3 + G_{-1}^*)G_1 + (F_3 + F_{-1}^*)F_1 \right\} (\mathcal{T}_{-1} - \mathcal{T}_1) \right], \quad (64)$$

$$K_{1\text{H},4} = -i \frac{F_1 G_{-1} + G_1 F_{-1}}{2(\epsilon_{-1} G_{-1} - F_{-1} \Delta)} \times \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1}, \quad (65)$$

$$K_{1\text{H},5} = -i \frac{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1}{2(\epsilon_1 G_1 - F_1 \Delta)} \times \left\{ (G_1 F_{-1}^* + F_1 G_{-1}^*) (\mathcal{T}_{-1} - \mathcal{T}_1) + (G_1 F_{-1} + G_{-1} F_1) \mathcal{T}_{-1} \right\}, \quad (66)$$

$$K_{1\text{H},6} = -\frac{G_{-1}^* \text{Im} F_1 + F_{-1}^* \text{Im} G_1}{\epsilon_1 \text{Im} G_1 - \text{Im} F_1 \Delta} \times \left\{ (\text{Re} G_3 \text{Im} F_1 + \text{Im} G_1 \text{Re} F_3) (\mathcal{T}_3 - \mathcal{T}_1) + (\text{Re} G_{-1} \text{Im} F_1 + \text{Im} G_1 \text{Re} F_{-1}) (\mathcal{T}_{-1} - \mathcal{T}_1) \right\}, \quad (67)$$

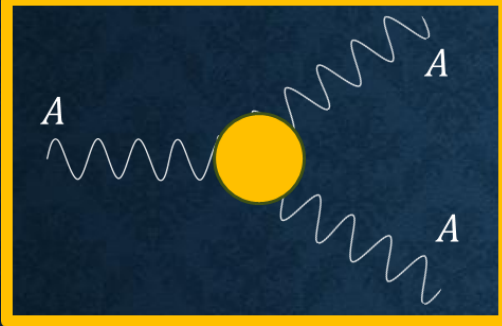
Nonlinear correction

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$$\delta\sigma_1^{qqq} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{qqq}] \frac{s}{\hbar\omega_{ac}} \quad I_{1H}^{qqq} = \frac{-1}{16\sqrt{\pi}} \int K_{1H} d\epsilon,$$

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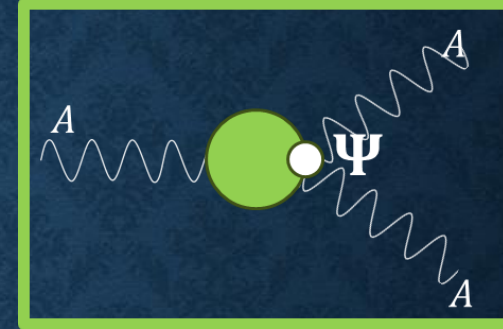
$$K_{1H,2} = i \frac{(G_3 - G_{-1})G_1 + (F_3 - F_{-1})F_1}{4\hbar\omega_{ac}(F_3 + F_{-1})} \times \left\{ (F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1 \right\} \mathcal{T}_{-1}, \quad (63)$$

$$K_{1H,3} = -i \frac{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1}{4\hbar\omega_{ac}(F_3 - F_{-1}^*)} \times \left[\left\{ (G_3 + G_{-1}^*)G_1^* + (F_3 + F_{-1}^*)F_1^* \right\} (\mathcal{T}_3 - \mathcal{T}_1) + \left\{ (G_3 + G_{-1}^*)G_1 + (F_3 + F_{-1}^*)F_1 \right\} (\mathcal{T}_{-1} - \mathcal{T}_1) \right], \quad (64)$$

$$K_{1H,4} = -i \frac{F_1 G_{-1} + G_1 F_{-1}}{2(\epsilon_{-1} G_{-1} - F_{-1} \Delta)} \times \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1}, \quad (65)$$

$$K_{1H,5} = -i \frac{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1}{2(\epsilon_1 G_1 - F_1 \Delta)} \times \left\{ (G_1 F_{-1}^* + F_1 G_{-1}^*) (\mathcal{T}_{-1} - \mathcal{T}_1) + (G_1 F_{-1} + G_{-1} F_1) \mathcal{T}_{-1} \right\}, \quad (66)$$

$$K_{1H,6} = -\frac{G_{-1}^* \text{Im} F_1 + F_{-1}^* \text{Im} G_1}{\epsilon_1 \text{Im} G_1 - \text{Im} F_1 \Delta} \times \left\{ (\text{Re} G_{-1} \text{Im} F_1 + \text{Im} G_1 \text{Re} F_3) (\mathcal{T}_3 - \mathcal{T}_1) + (\text{Re} G_{-1} \text{Im} F_1 + \text{Im} G_1 \text{Re} F_{-1}) (\mathcal{T}_{-1} - \mathcal{T}_1) \right\}, \quad (67)$$



Higgs
Contribution:

Time dependent $\delta\Delta$

$$\delta\sigma_1^{\text{Higgs}} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{\text{Higgs}}] \frac{s}{\hbar\omega_{ac}}$$

$$I_{1H}^{\text{Higgs}} = \frac{-1}{16\sqrt{\pi}} \Psi_{2H} \int Z_{1H}^{\text{Higgs}} d\epsilon$$

$$Z_{1H}^{\text{Higgs}}(\epsilon, \omega_{ac}) = \sum_{i=1,2,3} Z_{1H,i}^{\text{Higgs}}, \quad (71)$$

$$Z_{1H,1}^{\text{Higgs}} = -\frac{2(F_1 - F_{-3})}{\hbar\omega_{ac}(F_1 + F_{-3})} \times \left[\left\{ (F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^* \right\} (\mathcal{T}_{-1} - \mathcal{T}_{-3}) + \left\{ (F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1} \right\} \mathcal{T}_{-1} \right], \quad (72)$$

$$Z_{1H,2}^{\text{Higgs}} = -\frac{2(F_3 - F_{-1})}{\hbar\omega_{ac}(F_3 + F_{-1})} \times \left\{ (F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1 \right\} \mathcal{T}_{-1}, \quad (73)$$

$$Z_{1H,3}^{\text{Higgs}} = -\frac{2(F_3 + F_{-1}^*)}{\hbar\omega_{ac}(F_3 - F_{-1}^*)} \times \left\{ (F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1 \right\} (\mathcal{T}_3 - \mathcal{T}_{-1}) \quad (74)$$

Eliashberg
Contribution:

Time averaged $\delta\Delta$

$$\delta\sigma_1^{\text{Eliash}} = 2\sqrt{\pi}\sigma_n \text{Re}[I_{1H}^{\text{Eliash}}] \frac{s}{\hbar\omega_{ac}}$$

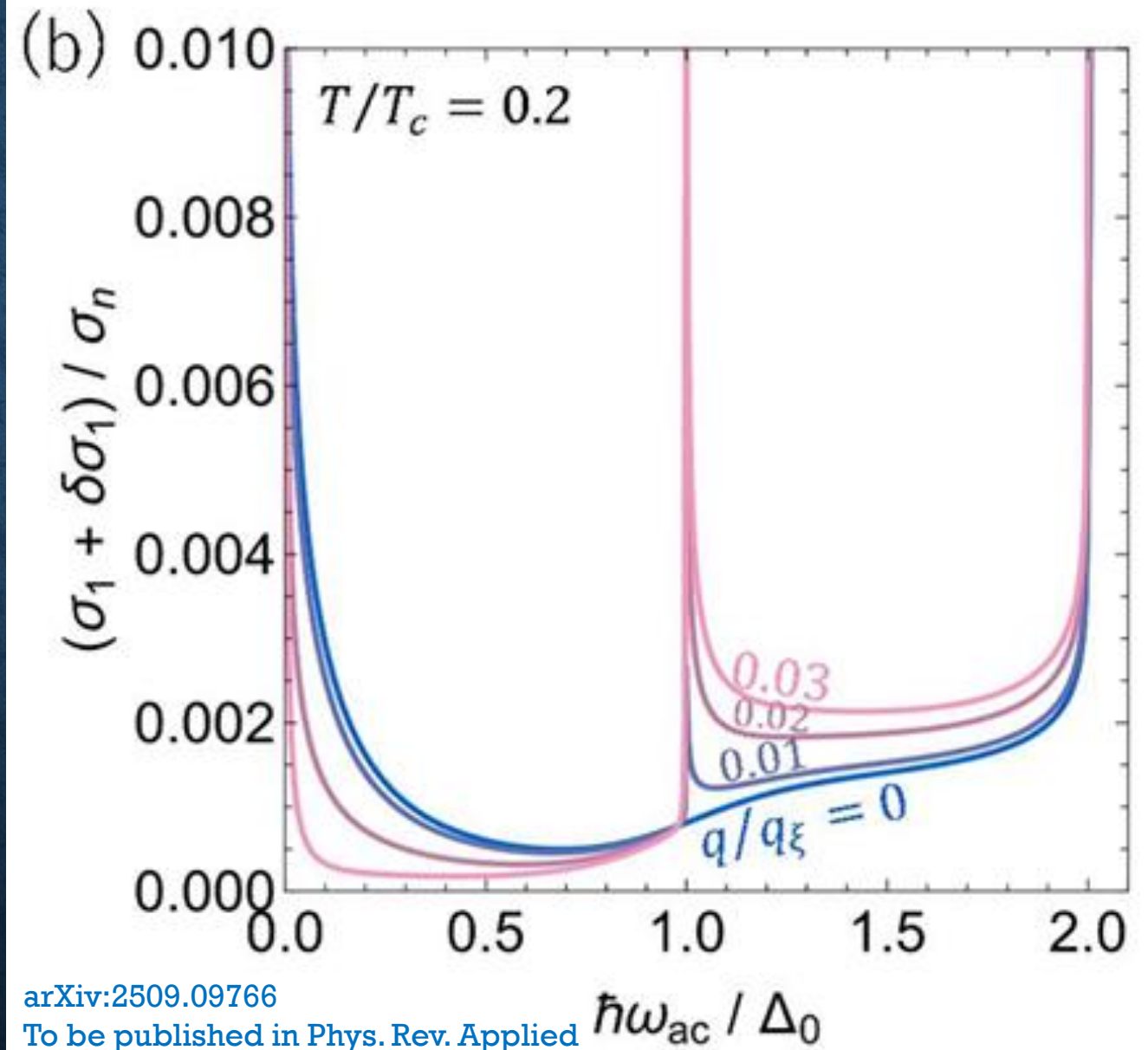
$$I_{1H}^{\text{Eliash}} = \frac{-1}{16\sqrt{\pi}} \Psi_{0H} \int Z_{1H}^{\text{Eliash}} d\epsilon$$

$$Z_{1H}^{\text{Eliash}}(\epsilon, \omega_{ac}) = \sum_{i=1,2} Z_{1H,i}^{\text{Eliash}}, \quad (68)$$

$$Z_{1H,1}^{\text{Eliash}} = \frac{4G_{-1}(G_1 F_{-1} + F_1 G_{-1})}{\epsilon_{-1} G_{-1} - F_{-1} \Delta} \mathcal{T}_{-1}, \quad (69)$$

$$Z_{1H,2}^{\text{Eliash}} = \frac{4G_1}{\epsilon_1 G_1 - F_1 \Delta} \left\{ (G_1 F_{-1} + G_{-1} F_1) \mathcal{T}_{-1} + (G_1 F_{-1}^* + F_1 G_{-1}^*) (\mathcal{T}_{-1} - \mathcal{T}_1) \right\}, \quad (70)$$

Results



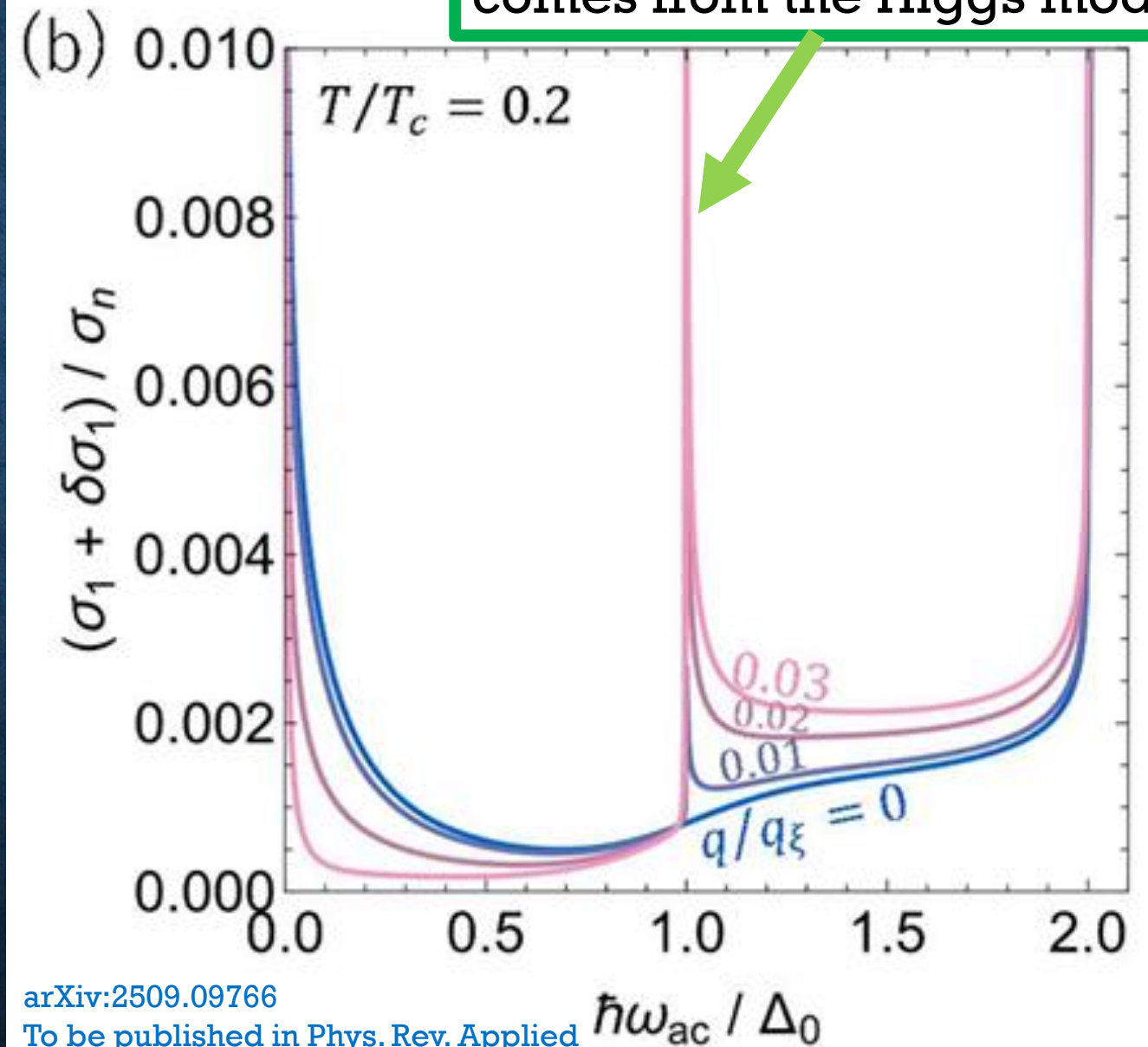
[arXiv:2509.09766](https://arxiv.org/abs/2509.09766)

To be published in Phys. Rev. Applied

<https://doi.org/10.1103/8fh8-nknj>

Results

The resonance peak at Δ comes from the Higgs mode



arXiv:2509.09766

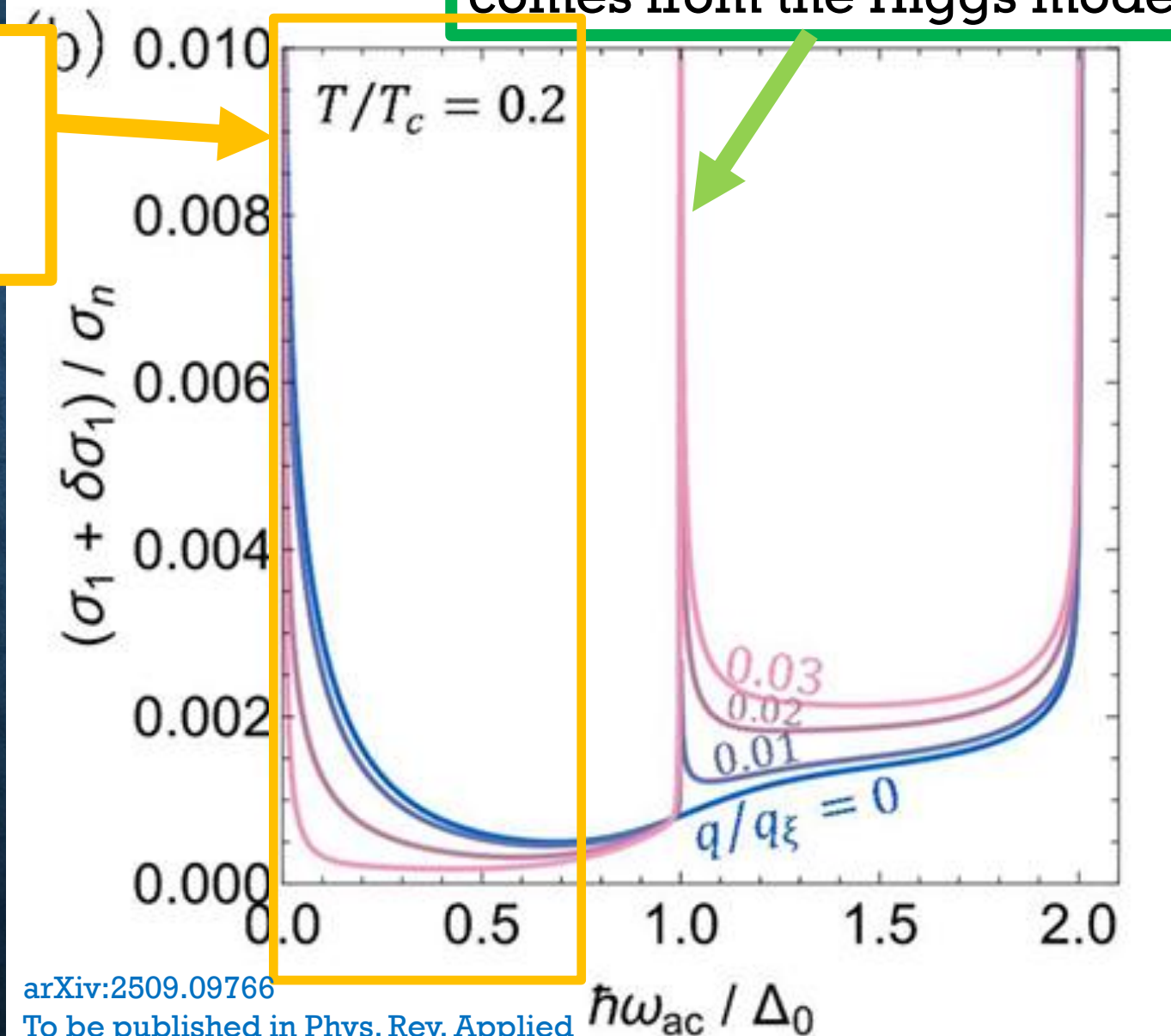
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<https://doi.org/10.1103/8fh8-nknj>

Results

At lower frequencies, $\sigma_1 + \delta\sigma_1$ can be a decreasing function of the ac amplitude (q).

The resonance peak at Δ comes from the Higgs mode



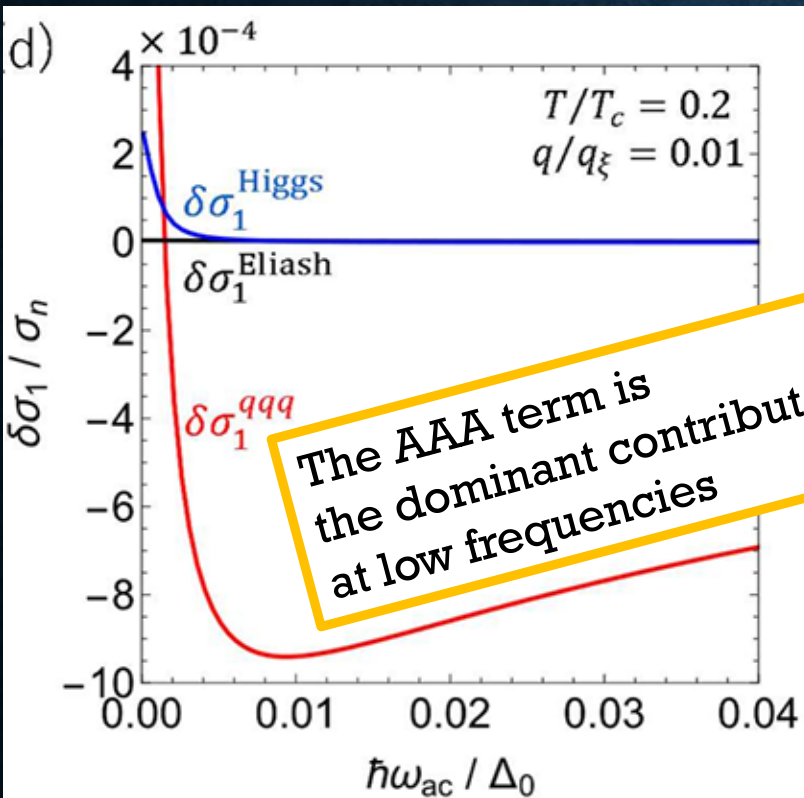
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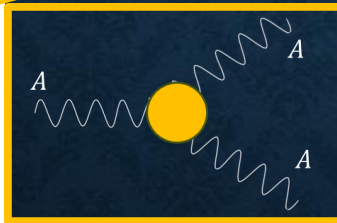
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Results

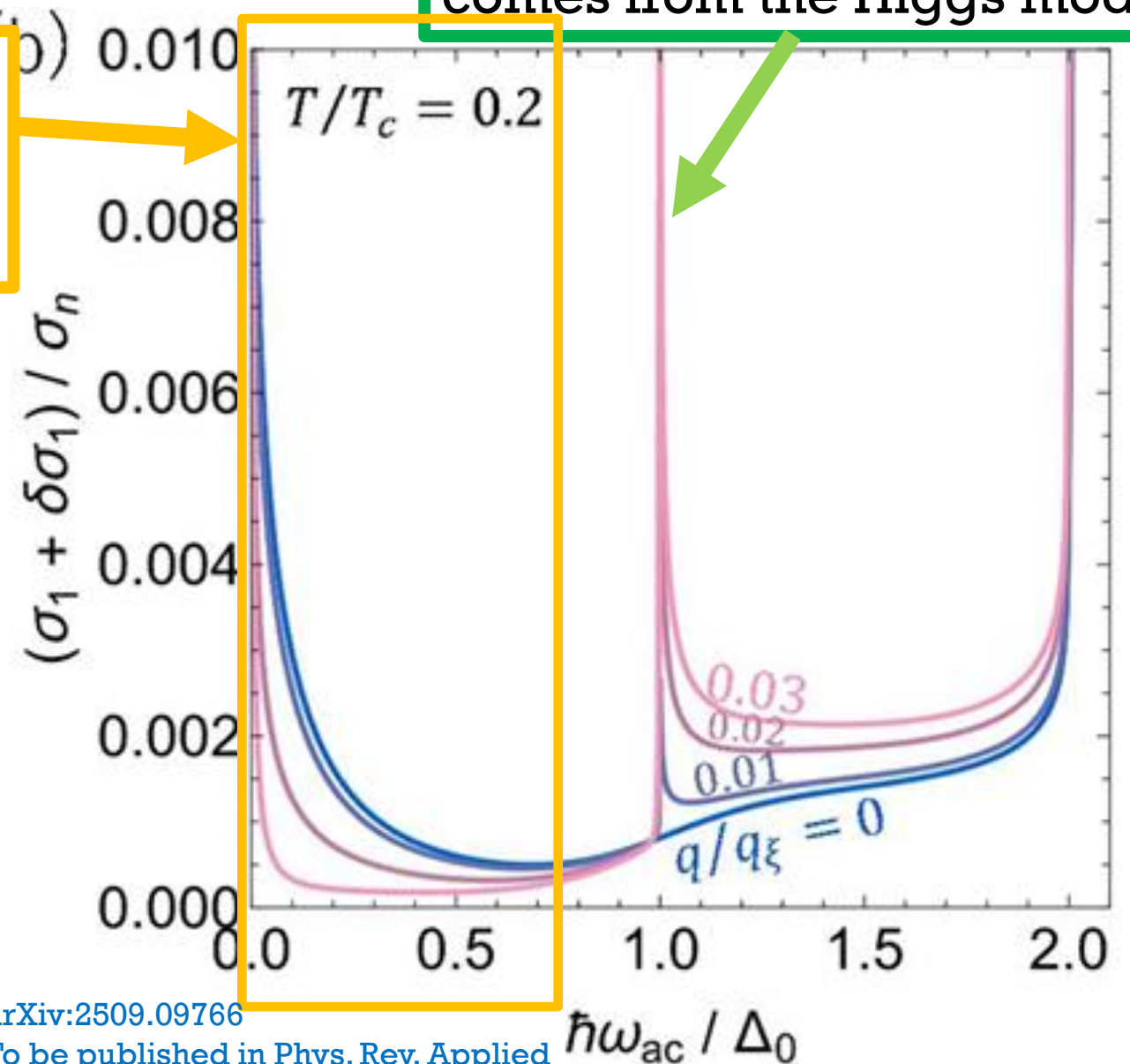
At lower frequencies, $\sigma_1 + \delta\sigma_1$ can be a decreasing function of the ac amplitude (q).



The AAA term is the dominant contributor at low frequencies



The resonance peak at Δ comes from the Higgs mode

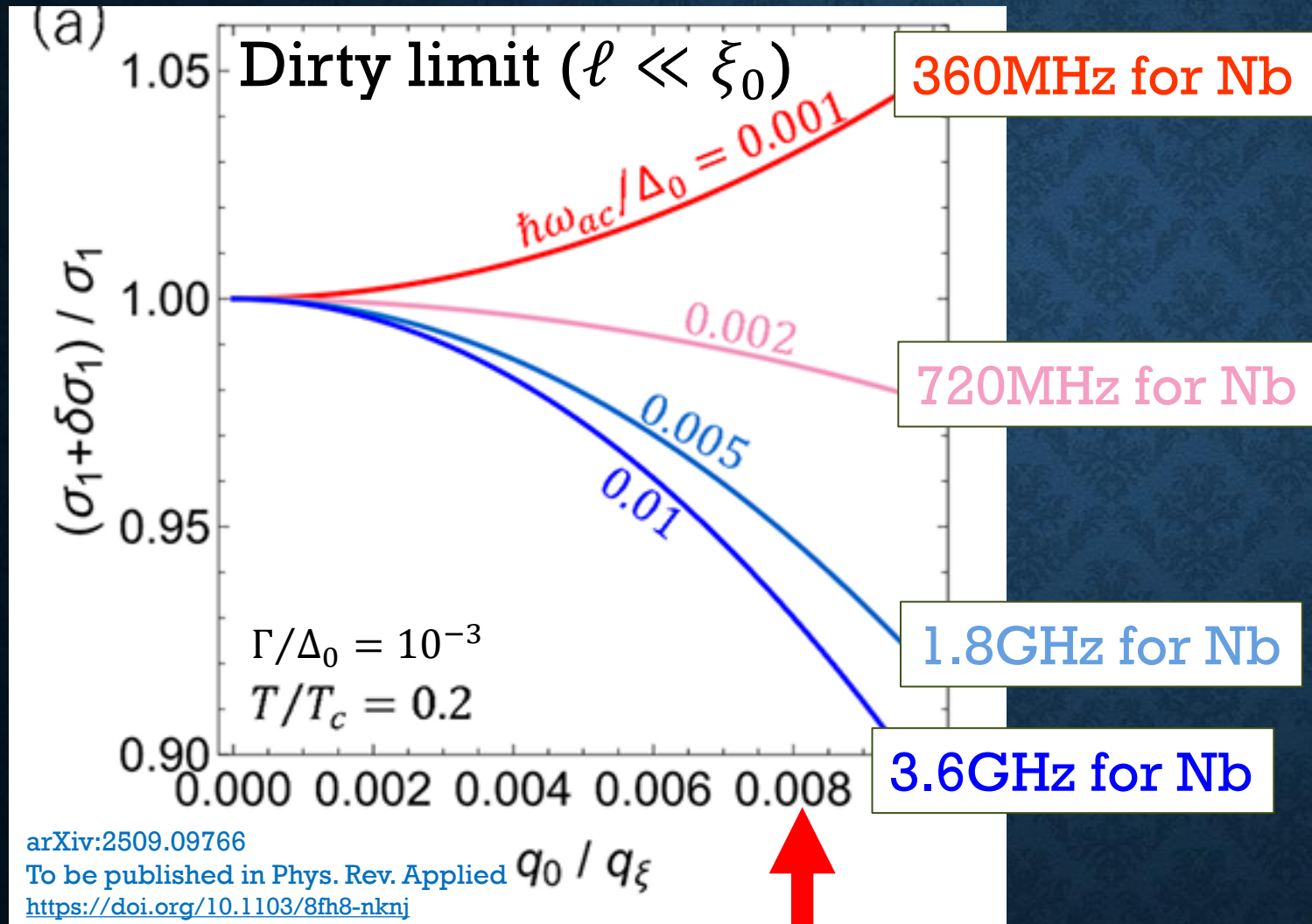


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ac amplitude dependent dissipative conductivity



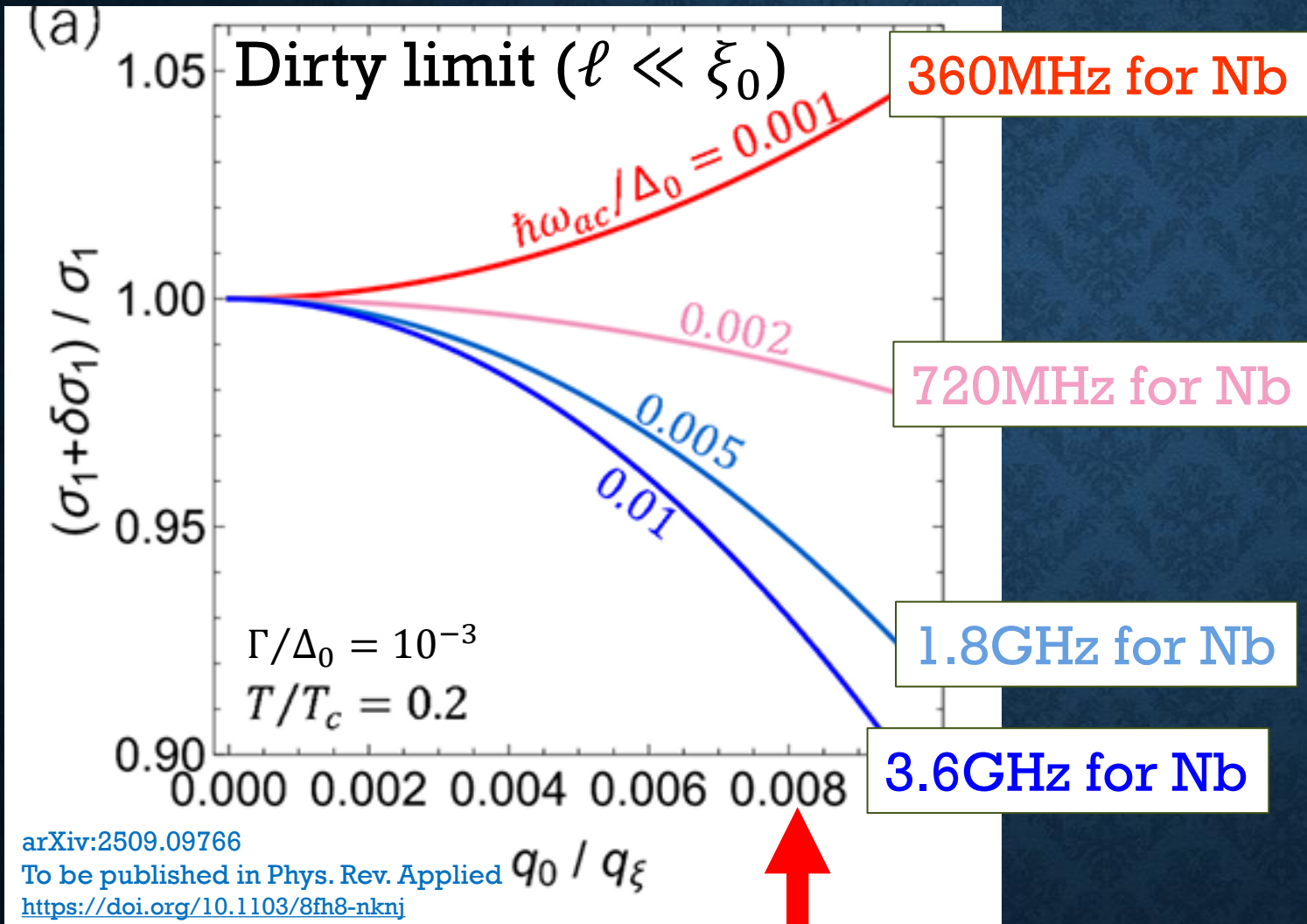
Note: this corresponds to a few (mT)

arXiv:2509.09766

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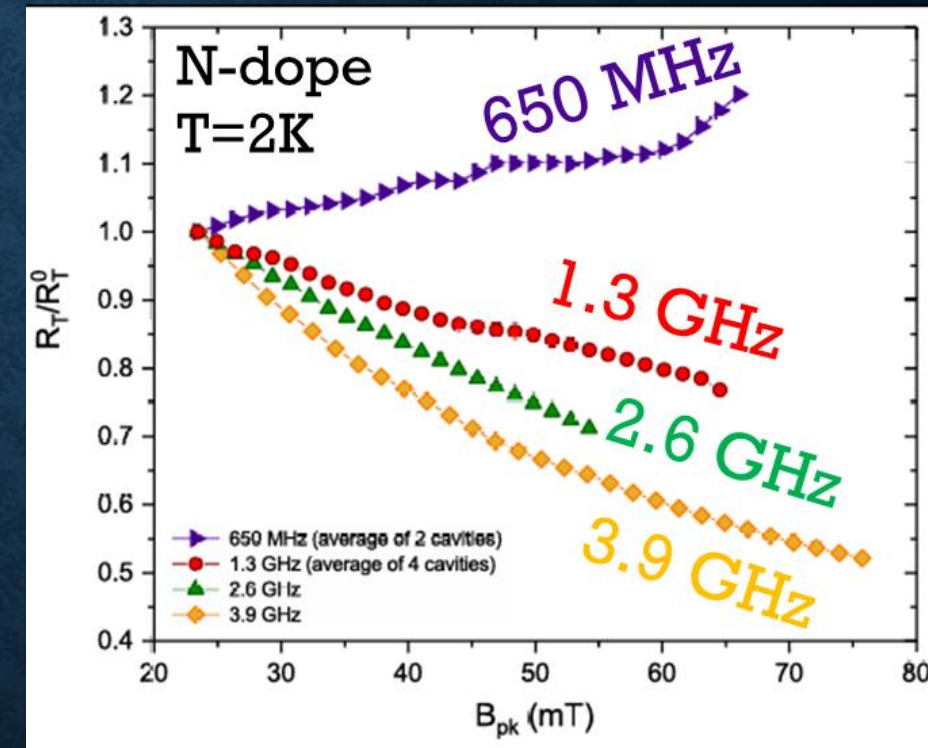
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ac amplitude dependent dissipative conductivity



Experiments: Moderately dirty ($\ell \sim \xi_0$)

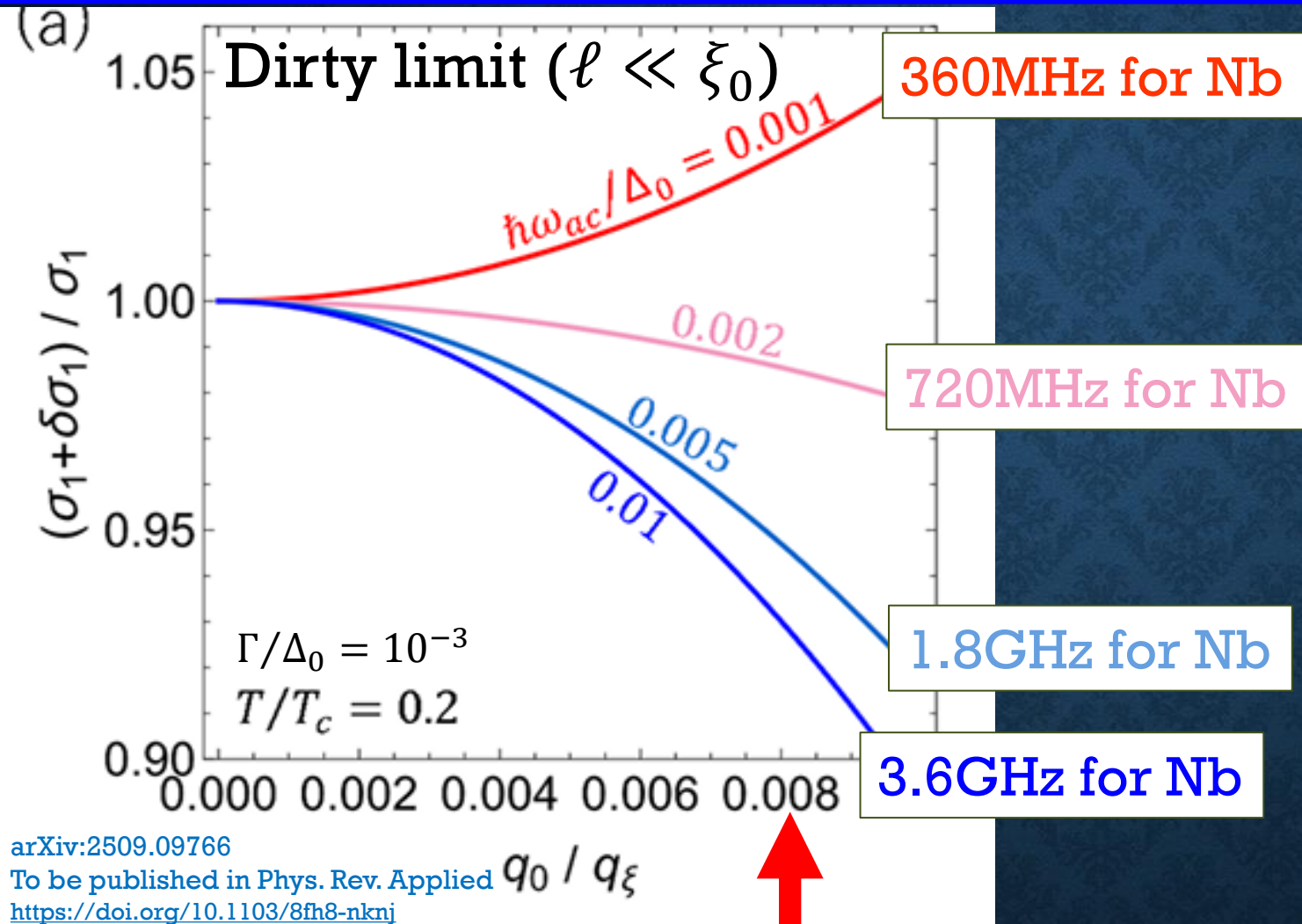
- M. Martinello et al., PRL **121**, 224801 (2018)
- P. Dhakal et al., PRAB **27**, 062001 (2024)



M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: <https://doi.org/10.1103/PhysRevLett.121.224801>

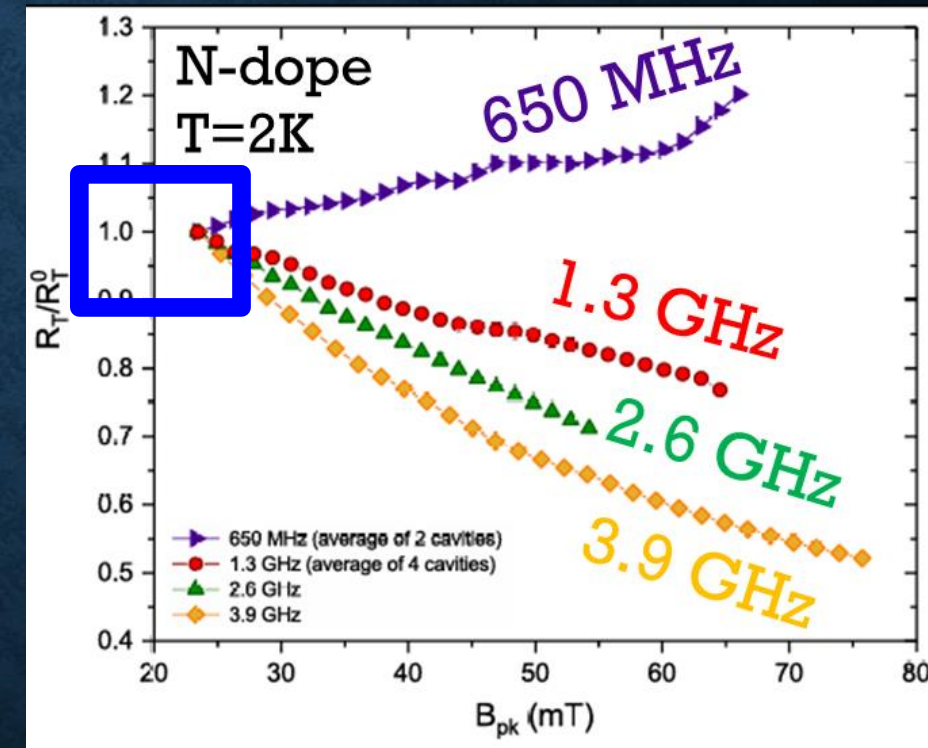
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Experiments: Moderately dirty ($\ell \sim \xi_0$)

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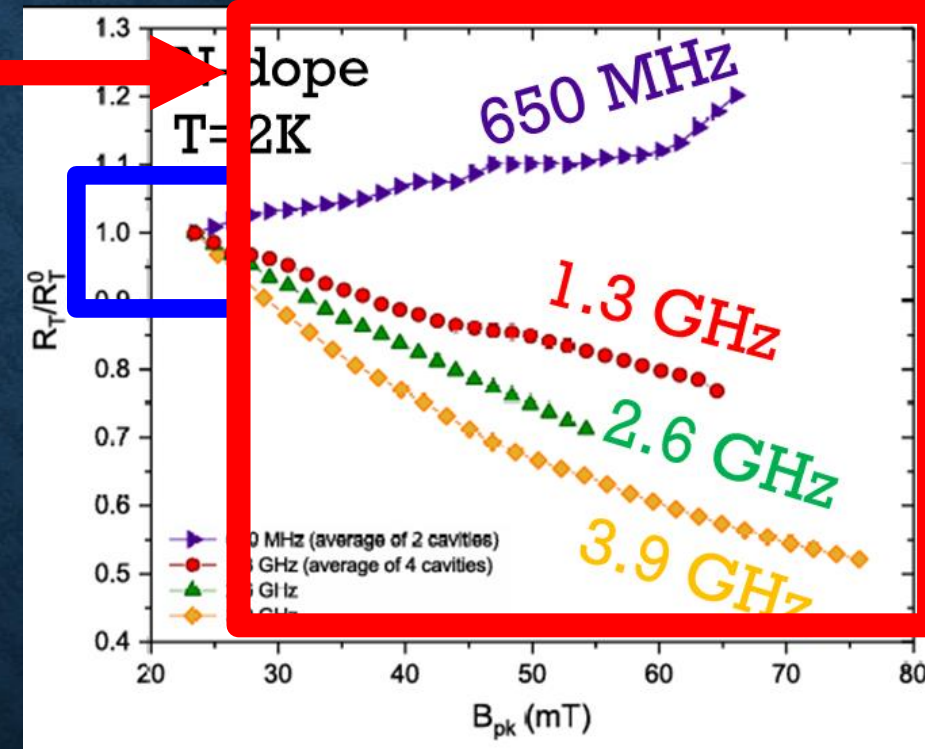
Experiments:

Moderately dirty ($\ell \sim \xi_0$)

- M. Martinello et al., PRL **121**, 224801 (2018)
- P. Dhakal et al., PRAB **27**, 062001 (2024)

To deal with **the red-framed region**, the theory must be extended nonperturbatively to capture the quasiparticle spectrum modified by strong AC currents (**DOS smearing**).

Gurevich, PRL **113**, 087001 (2014)
Kubo-Gurevich, PRB **100**, 064522 (2019)

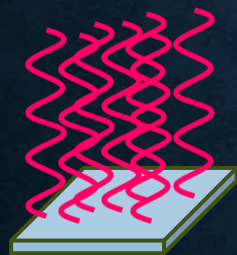


M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
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T. Kubo, arXiv:2509.09766
To be published in Phys. Rev. Applied
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**Leading order
nonlinear
correction**

Linear
response
regime



**Nonequilibrium
Dynamics**
(e.g, strength of microwave)

Keldysh-Usadel

**Keldysh
Eilenberger**

Thermal Equilibrium

Ginzburg-Landau ($T \approx T_c$)

**Usadel
Formalism**
Applicable to
Any temperature
 $mfp < \xi_0$

**Eilenberger
Formalism**
Applicable to
Any temperature
Any mean free path

ξ_0

Mean
free
path

T. Kubo, arXiv:2509.09766
To be published in Phys. Rev. Applied
<https://doi.org/10.1103/8fh8-nkni>

Leading order
nonlinear
correction

Linear
response
regime

The SRF theory
(Deep Nonlinear Regime)

Nonequilibrium
Dynamics
(e.g, strength of microwave)

Keldysh-Usadel

Keldysh
Eilenberger

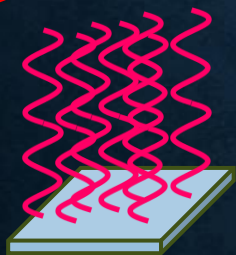
Temperature
 T_c
Thermal Equilibrium
Ginzburg-Landau ($T \approx T_c$)

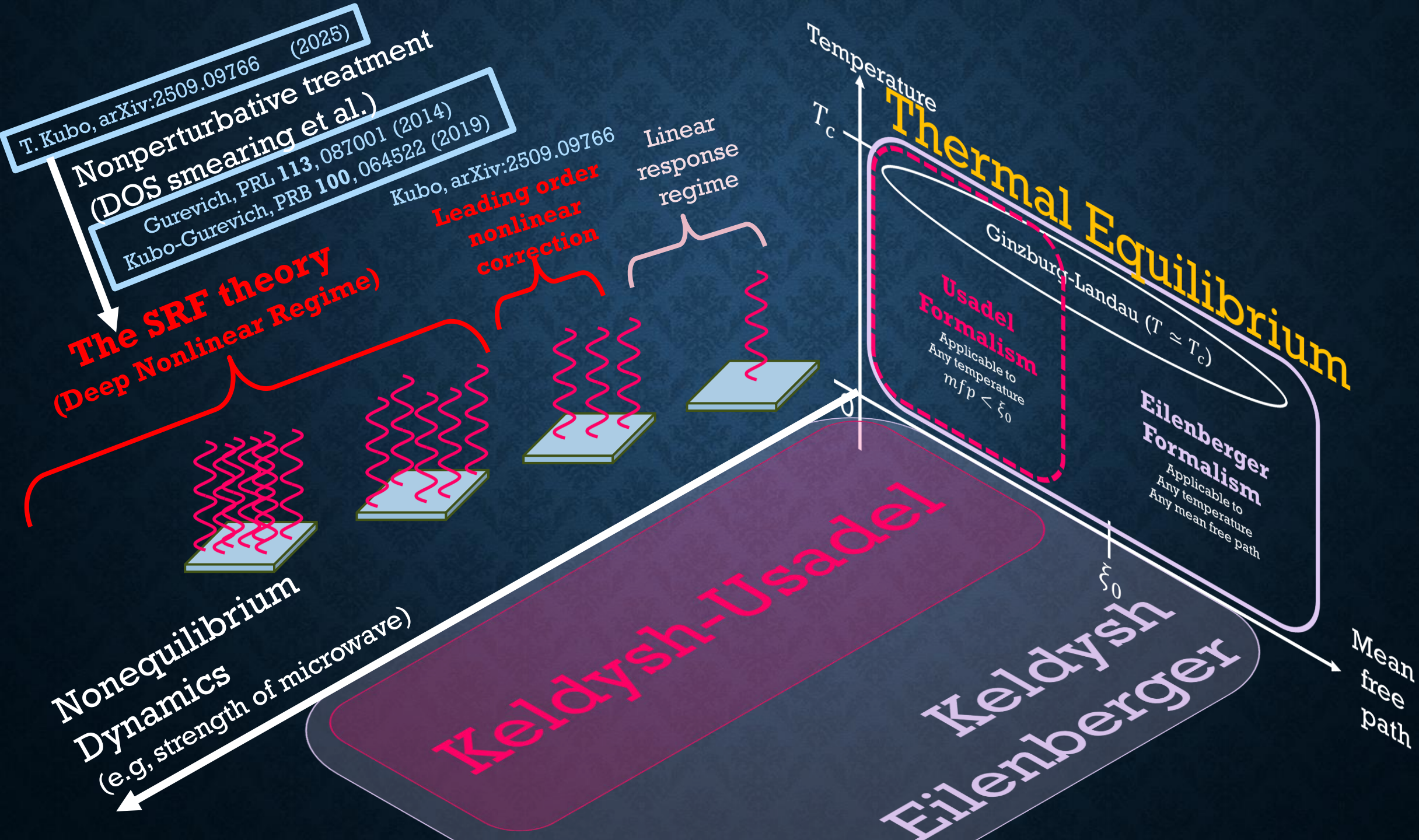
Usadel Formalism
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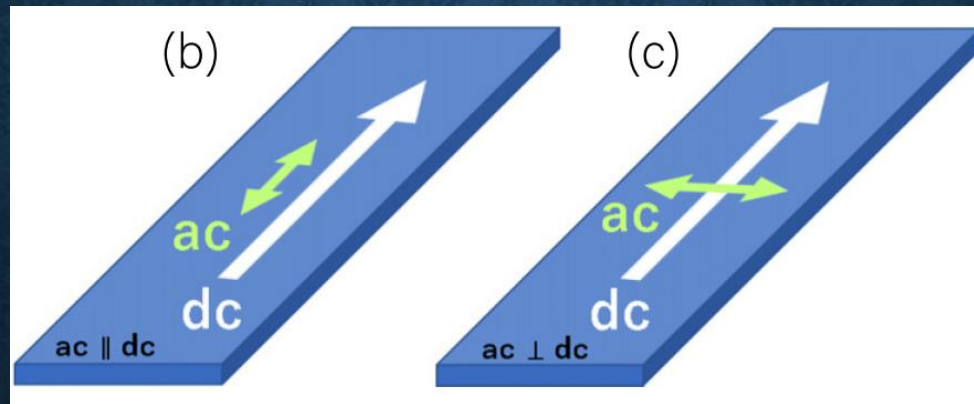
ξ_0

Mean
free
path



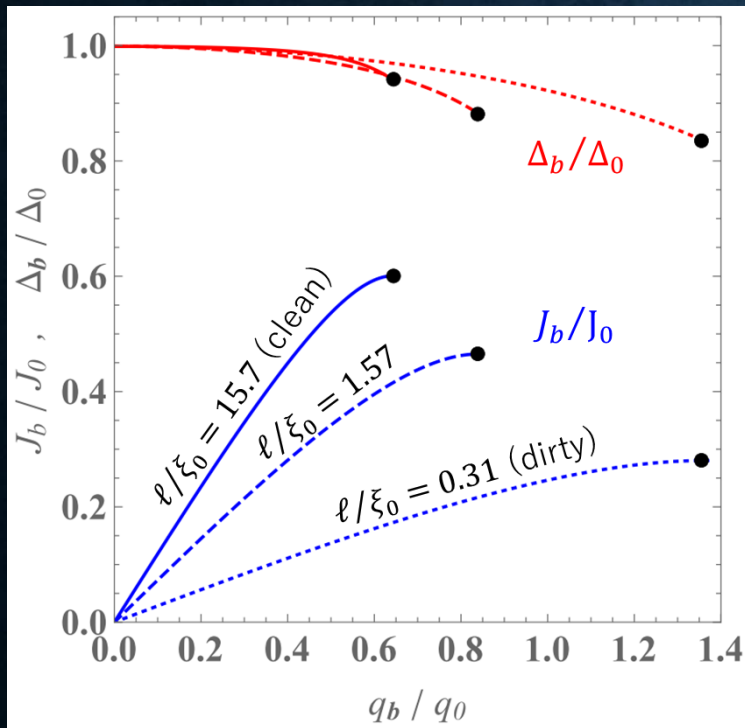


dc + ac





If the AC field is absent, this reduces to a problem of equilibrium superconductivity.
That case was already well understood decades ago.



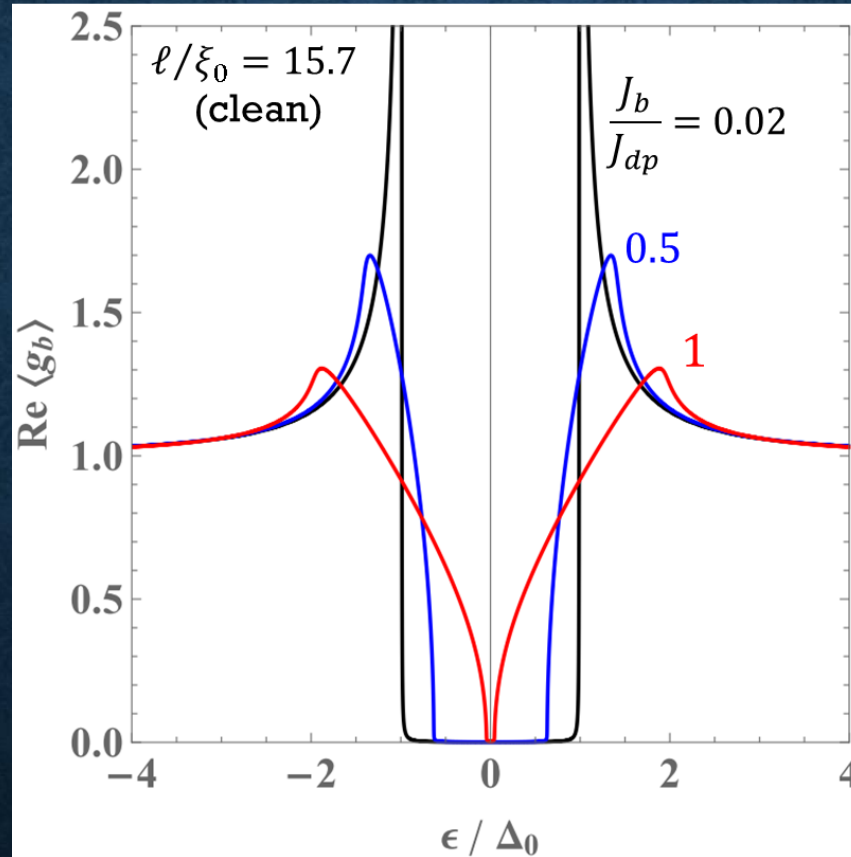
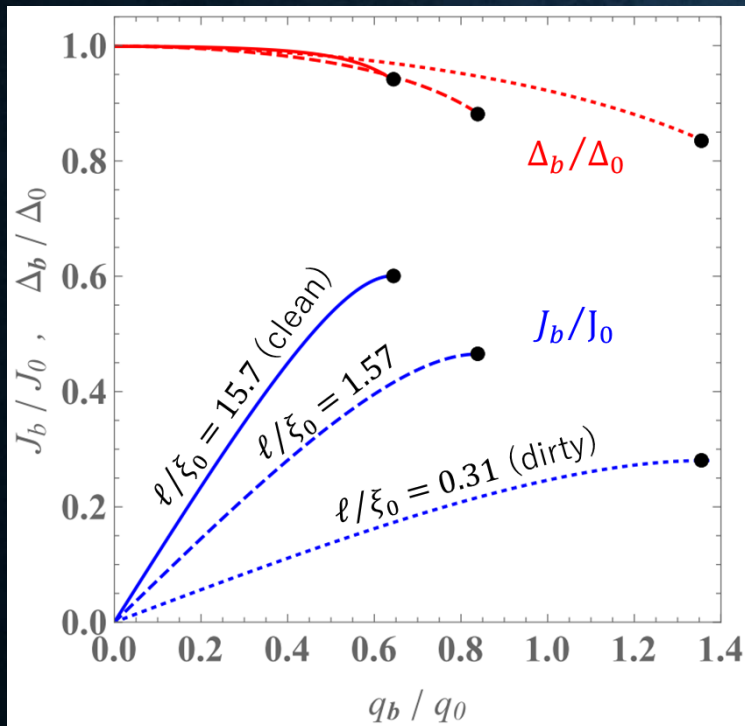
T. Kubo, Phys. Rev. Applied 22, 044042 (2024)

DOI: <https://doi.org/10.1103/PhysRevApplied.22.044042>

See also, T. Kubo, Phys. Rev. Research 2, 033203 (2020); F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012)



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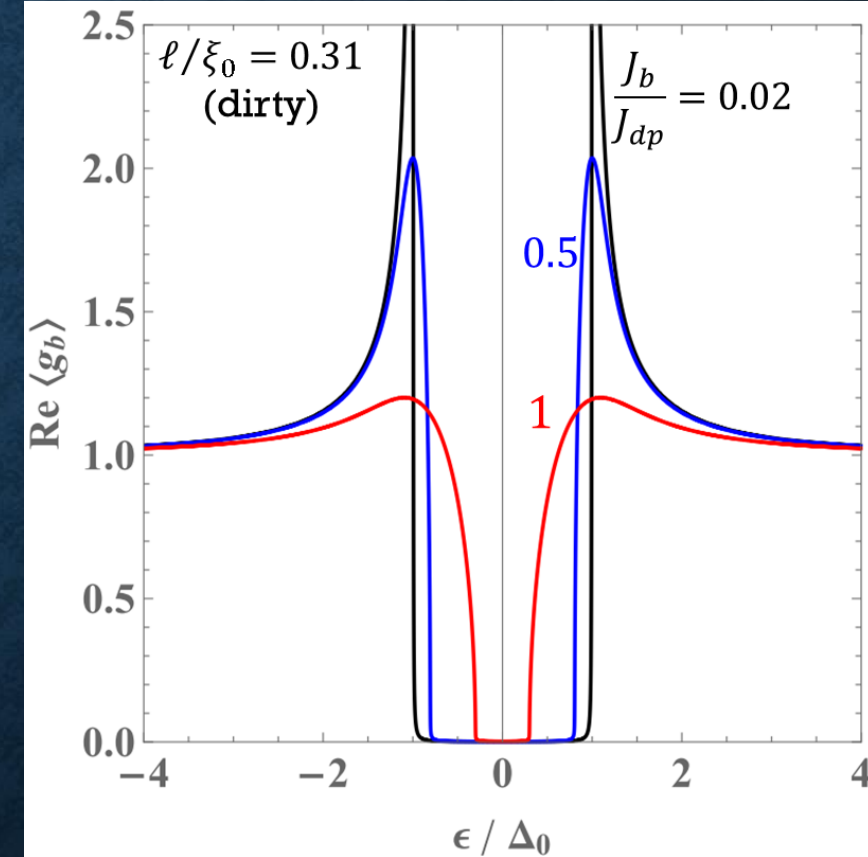
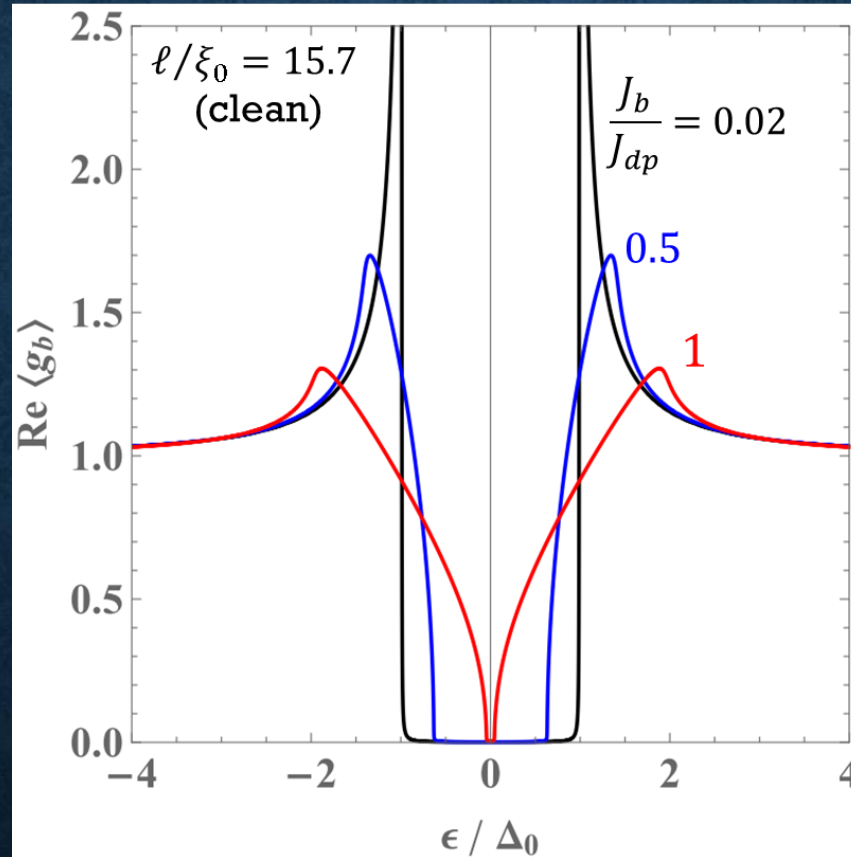
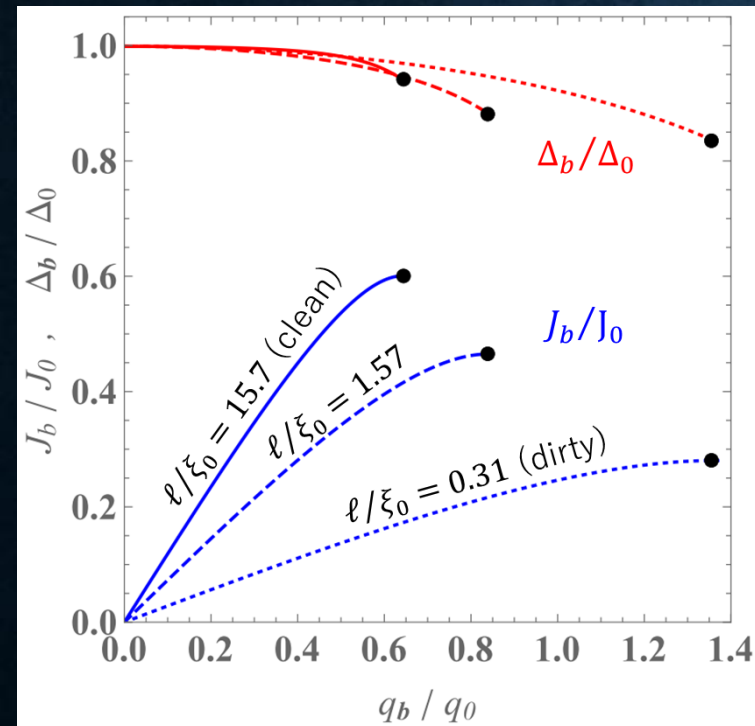
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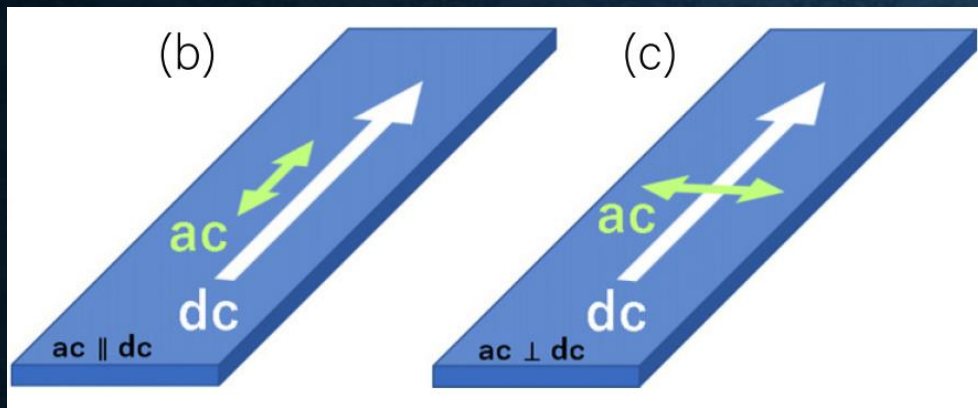
If the AC field is absent, this reduces to a problem of equilibrium superconductivity.
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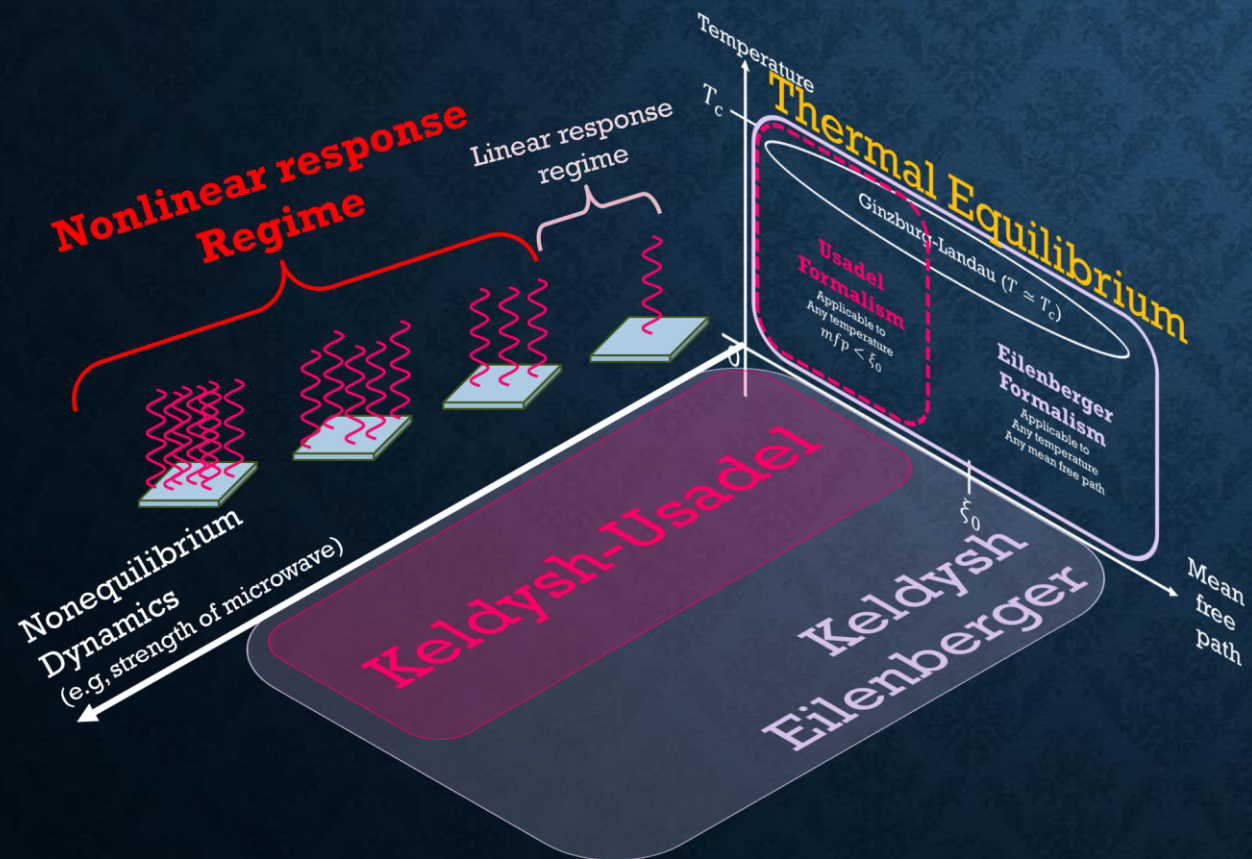
T. Kubo, Phys. Rev. Applied 22, 044042 (2024)

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See also, T. Kubo, Phys. Rev. Research 2, 033203 (2020); F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012)



However, once an AC field is present, the problem becomes much more complicated. In that case, we need to use the theory of **nonequilibrium** superconductivity.



A. Moor et al., Phys. Rev. Lett. **118**, 047001 (2017)
 T. Jujo, J. Phys. Soc. Jpn. **91**, 074711 (2022)
 T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)
 T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

- **The Keldysh-Eilenberger theory** is a microscopic theory of **nonequilibrium** superconductivity. It is applicable at any temperature ($0 \leq T \leq T_c$) and for arbitrary mean free path.
- **The Keldysh-Usadel theory** represents the dirty-limit reduction of the Keldysh-Eilenberger theory of nonequilibrium superconductivity, applicable at any T ($0 \leq T \leq T_c$).

Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: <https://doi.org/10.1103/PhysRevApplied.22.044042>

T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

$$\begin{aligned}
 & -i(s/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & -i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^R(\epsilon_-) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_+) \hat{\tau}_3] \\
 & = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
 \end{aligned}$$

$$\begin{aligned}
 & -i(s/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
 & -i(\delta W/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 + \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
 & - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \} = \epsilon_+ \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \epsilon_- \\
 & + [\hat{\Delta}_b, \delta \hat{g}^r(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^r(\epsilon_-) - \hat{g}_b^r(\epsilon_+) \delta \hat{\Delta}(\omega), \quad (13)
 \end{aligned}$$

$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr}[-i\tau_2 \delta \hat{g}^K(\epsilon, \omega)]. \quad (17)$$

Higgs
↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: <https://doi.org/10.1103/PhysRevApplied.22.044042>

T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

$$\begin{aligned}
 & -i(s/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3] \\
 & -i(\delta W/2) [\hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_+) \hat{\tau}_3] \\
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 & + [\hat{\Delta}_b, \delta \hat{g}^K(\epsilon, \omega)] + \delta \hat{\Delta}(\omega) \hat{g}_b^K(\epsilon_-) - \hat{g}_b^K(\epsilon_+) \delta \hat{\Delta}(\omega). \quad (15)
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$$\begin{aligned}
 & -i(s/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) - \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \\
 & + \hat{\tau}_3 \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) - \delta \hat{g}^r(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \hat{\tau}_3 \} \\
 & -i(\delta W/2) \{ \hat{\tau}_3 \hat{g}_b^r(\epsilon_+) \hat{\tau}_3 \hat{g}_b^r(\epsilon_-) \\
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$$\delta \Delta(\omega) = -\frac{\mathcal{G}}{8} \int d\epsilon \text{Tr} [(-i\tau_2) \delta \hat{g}^K(\epsilon, \omega)]. \quad (17)$$

Higgs
↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta \hat{g}^{R,A,K}$ and $\delta \Delta$).

To obtain the ac response,
we substitute the solutions ($\delta \hat{g}^{R,A,K}$, $\delta \Delta$) into

$$\delta \mathbf{J}(\omega) = -i \frac{\sigma_n}{e} \int d\epsilon \delta \mathbf{S}(\epsilon, \omega), \quad (18)$$

$$\begin{aligned}
 \delta \mathbf{S}(\epsilon, \omega) = & (i/16) \text{Tr} [i \mathbf{q}_b \\
 & \times \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \\
 & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \} \\
 & + i \delta \mathbf{q}_\omega \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \}]. \quad (19)
 \end{aligned}$$

Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim Agg + A_{dc}g\delta g$$

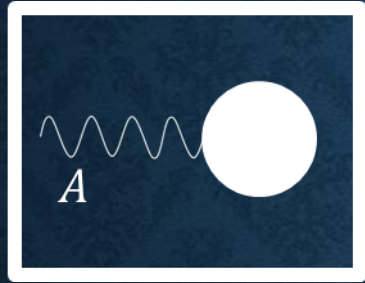


$$\delta \mathbf{J}(\omega) = -i \frac{\sigma_n}{e} \int d\epsilon \delta \mathbf{S}(\epsilon, \omega), \quad (18)$$

$$\begin{aligned} \delta \mathbf{S}(\epsilon, \omega) = & (i/16) \text{Tr} [i \mathbf{q}_b \\ & \times \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^R(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \\ & + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \delta \hat{g}^A(\epsilon, \omega) + \hat{\tau}_3 \delta \hat{g}^K(\epsilon, \omega) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \} \\ & + i \delta \mathbf{q}_\omega \{ \hat{\tau}_3 \hat{g}_b^R(\epsilon_+) \hat{\tau}_3 \hat{g}_b^K(\epsilon_-) + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+) \hat{\tau}_3 \hat{g}_b^A(\epsilon_-) \}]. \end{aligned} \quad (19)$$

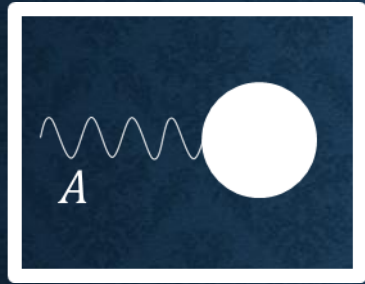
Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + A_{dc} g \delta g$$



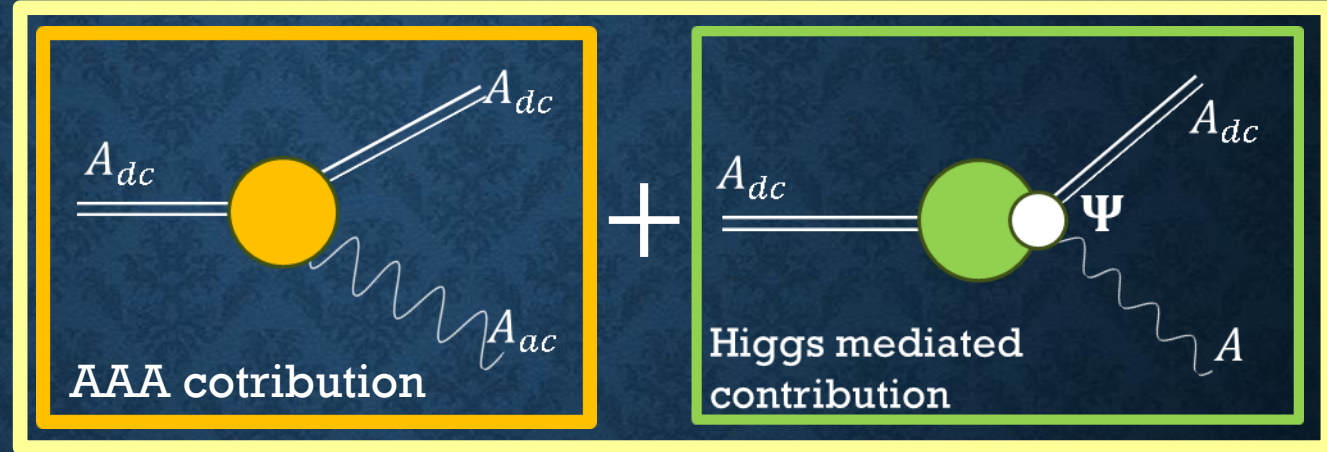
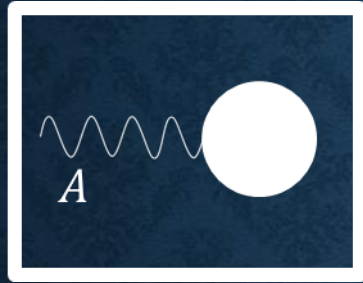
Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + A_{dc} g \delta g \quad \delta g \sim \mathbf{A}_{dc} \cdot \mathbf{A} + \delta \Delta \quad \text{and} \quad \delta \Delta \sim \Psi \mathbf{A}_{dc} \cdot \mathbf{A}$$



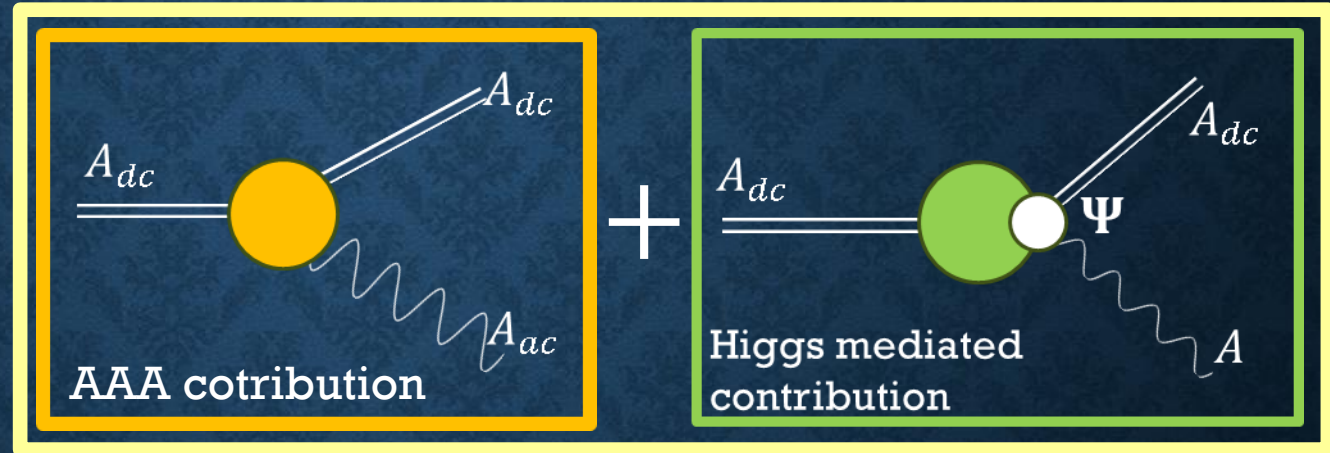
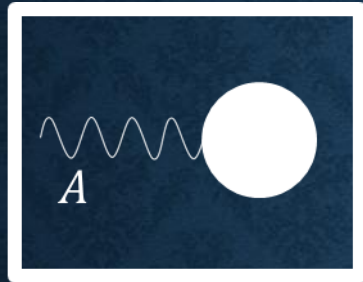
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$$J \sim \boxed{A g g} + \boxed{A_{dc} g \delta g} \quad \delta g \sim \mathbf{A}_{dc} \cdot \mathbf{A} + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi \mathbf{A}_{dc} \cdot \mathbf{A}$$



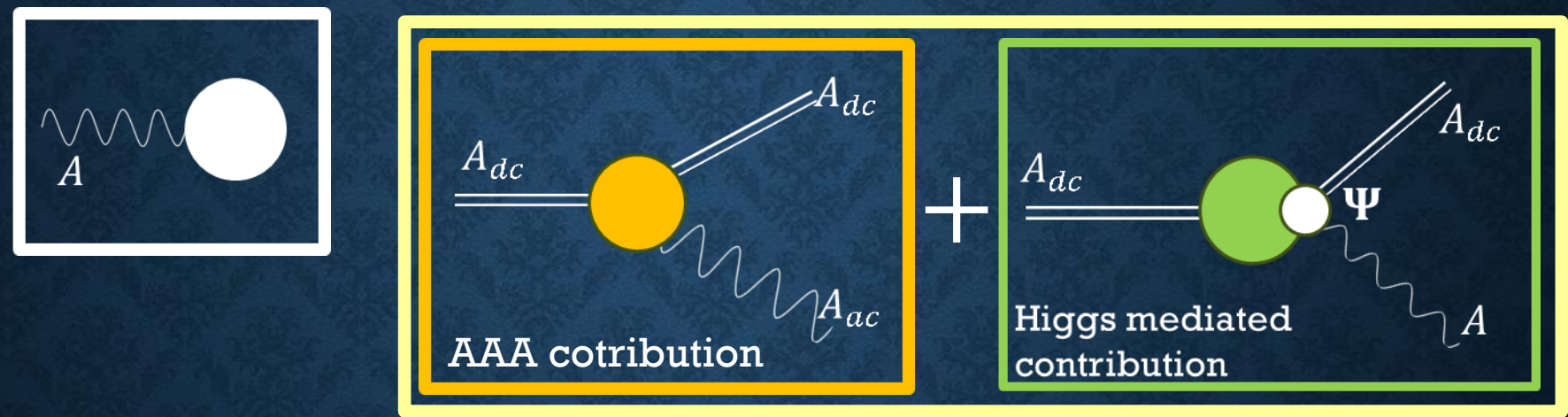
$$\propto \mathbf{A}_{dc} \cdot \mathbf{A}$$

$$\propto \mathbf{A}_{dc} \cdot \mathbf{A}$$

Nonequilibrium corrections due to the
Doppler fluctuation of flow $\propto \mathbf{A}_{dc} \cdot \mathbf{A}$

Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{A g g} + \boxed{A_{dc} g \delta g} \quad \delta g \sim \textcolor{brown}{A}_{dc} \cdot \textcolor{brown}{A} + \textcolor{green}{\delta\Delta} \quad \text{and} \quad \delta\Delta \sim \Psi \textcolor{green}{A}_{dc} \cdot \textcolor{green}{A}$$



Then, the complex conductivity is given by

$$\sigma = \frac{J}{E} \sim \frac{\boxed{\text{diagram 1}} + \boxed{\text{diagram 2}} + \boxed{\text{diagram 3}}}{A}$$

Complex conductivity formula

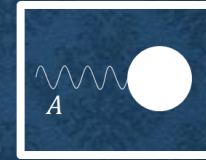
T. Kubo, Phys. Rev. Applied 22, 044042 (2024)

T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

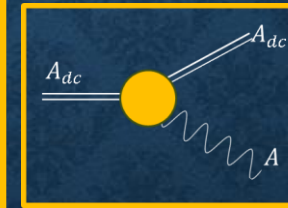
$ac \parallel dc$ case

$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

$$\begin{aligned} \frac{\sigma^{(0)}}{\sigma_n} &= \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Re } G'_b + \text{Re } F_b \text{Re } F'_b) (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Im } G'_b + \text{Re } F_b \text{Im } F'_b) (2f_{\text{FD}} - 1), \end{aligned} \quad (41)$$

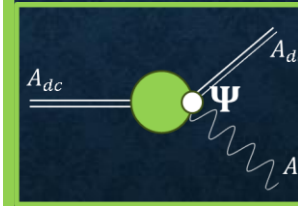


$$\begin{aligned} \frac{\sigma^{(1)}}{\sigma_n} &= \frac{8s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \text{Re } F_b \text{Im } F_b \text{Re } G'_b (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \frac{2s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} [2 \text{Re } F_b \text{Im } F_b \text{Im} \{G_b + G'_b\} + \{(\text{Re } F'_b)^2 \\ &- (\text{Re } F_b)^2 + (\text{Im } F_b)^2 - (\text{Im } F'_b)^2\} \text{Re } G_b] (2f_{\text{FD}} - 1), \end{aligned} \quad (42)$$



$$\propto A_{dc} \cdot A$$

$$\begin{aligned} \frac{\sigma^{(2)}}{\sigma_n} &= \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \\ &\times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im}(F_b - F'_b) \\ &+ \text{Re } F_b \text{Im}(G_b + G'_b) \} (2f_{\text{FD}} - 1). \end{aligned} \quad (43)$$



$$\propto A_{dc} \cdot A$$

Nonequilibrium corrections due to the Doppler fluctuation of flow
 $\propto A_{dc} \cdot A$

AAA
contribution

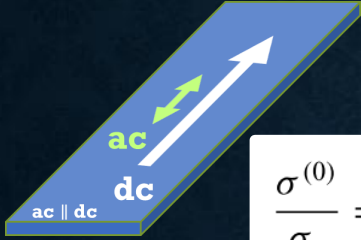
Higgs
mediated
contribution

Complex conductivity formula

T. Kubo, Phys. Rev. Applied 22, 044042 (2024)

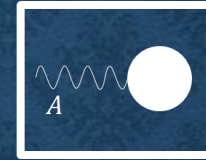
T. Kubo, Phys. Rev. Applied 23, 054091 (2025)

$ac \parallel dc$ case

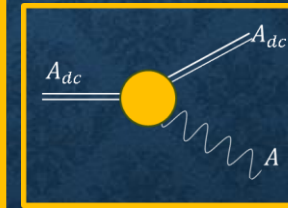


$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

$$\begin{aligned} \frac{\sigma^{(0)}}{\sigma_n} &= \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Re } G'_b + \text{Re } F_b \text{Re } F'_b) (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Im } G'_b + \text{Re } F_b \text{Im } F'_b) (2f_{\text{FD}} - 1), \end{aligned} \quad (41)$$

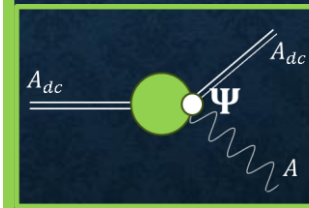


$$\begin{aligned} \frac{\sigma^{(1)}}{\sigma_n} &= \frac{8s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \text{Re } F_b \text{Im } F_b \text{Re } G'_b (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \frac{2s}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} [2 \text{Re } F_b \text{Im } F_b \text{Im } \{G_b + G'_b\} + \{(\text{Re } F'_b)^2 \\ &- (\text{Re } F_b)^2 + (\text{Im } F_b)^2 - (\text{Im } F'_b)^2\} \text{Re } G_b] (2f_{\text{FD}} - 1), \end{aligned} \quad (42)$$



$$\propto A_{dc} \cdot A$$

$$\begin{aligned} \frac{\sigma^{(2)}}{\sigma_n} &= \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} (\text{Re } F_b \text{Re } G'_b - \text{Re } G_b \text{Re } F'_b) \\ &\times (f_{\text{FD}} - f'_{\text{FD}}) + i \frac{2s\Psi}{\hbar\omega} \int \frac{d\epsilon}{\hbar\omega} \{ \text{Re } G_b \text{Im } (F_b - F'_b) \\ &+ \text{Re } F_b \text{Im } (G_b + G'_b) \} (2f_{\text{FD}} - 1). \end{aligned} \quad (43)$$



$$\propto A_{dc} \cdot A$$

AAA
contribution

Higgs
mediated
contribution

$ac \perp dc$ case



$$\sigma = \sigma^{(0)}$$

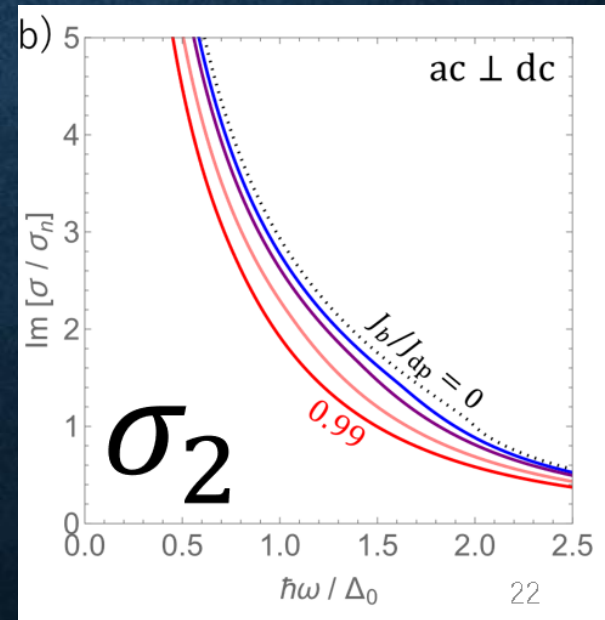
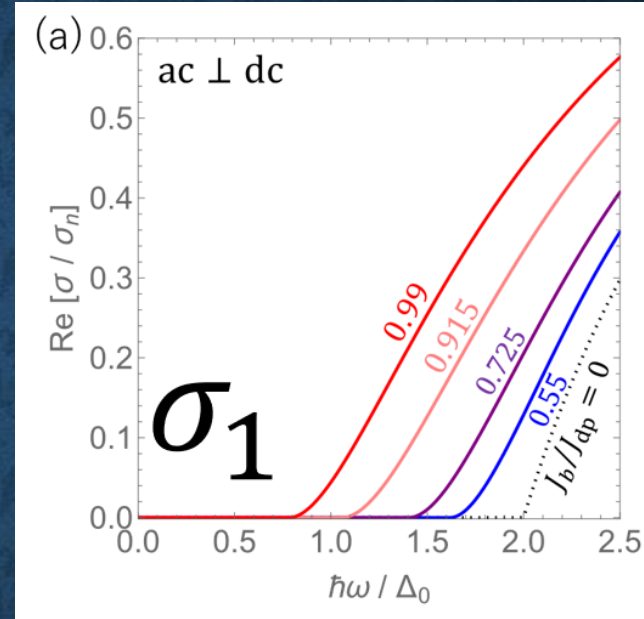
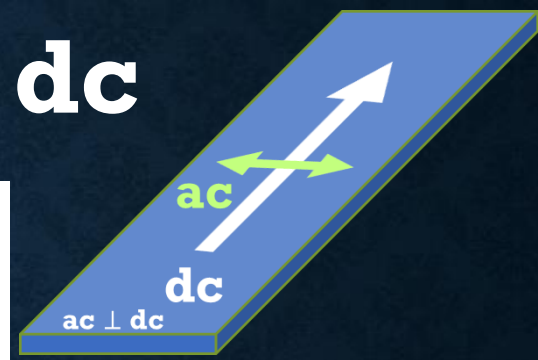


$$\begin{aligned} \frac{\sigma^{(0)}}{\sigma_n} &= \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Re } G'_b + \text{Re } F_b \text{Re } F'_b) (f_{\text{FD}} - f'_{\text{FD}}) \\ &+ i \int \frac{d\epsilon}{\hbar\omega} (\text{Re } G_b \text{Im } G'_b + \text{Re } F_b \text{Im } F'_b) (2f_{\text{FD}} - 1), \end{aligned} \quad (41)$$

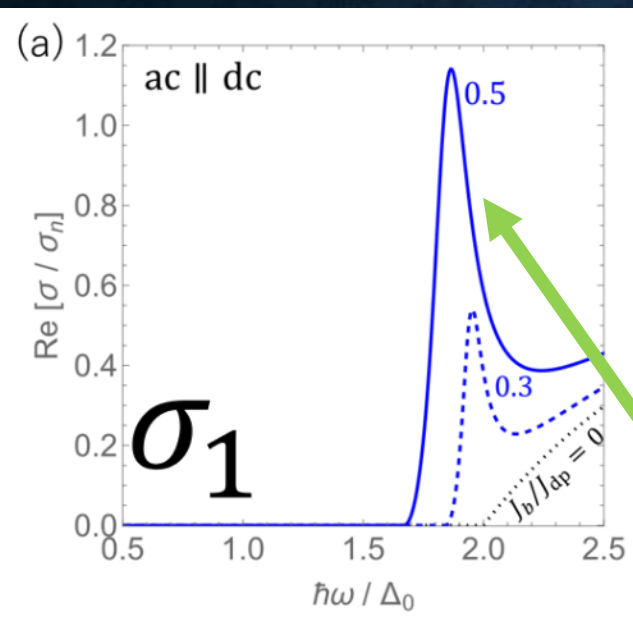
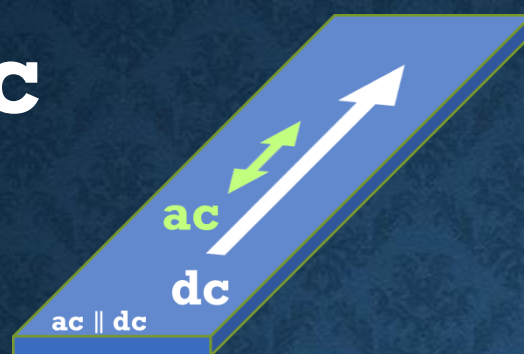
No Doppler fluctuation

Nonequilibrium corrections due
to the Doppler fluctuation of flow
 $\propto A_{dc} \cdot A$

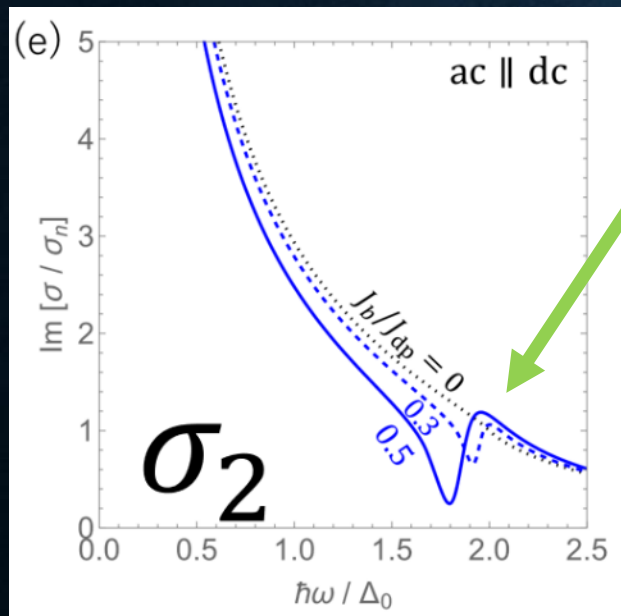
ac $\perp$ dc



ac || dc

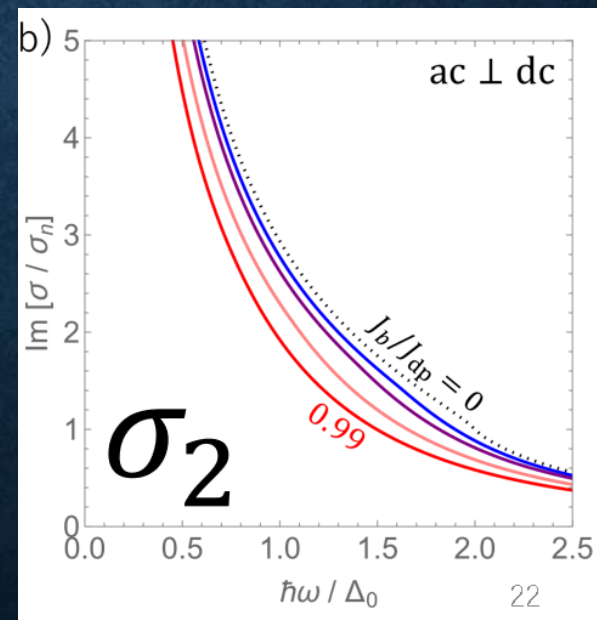
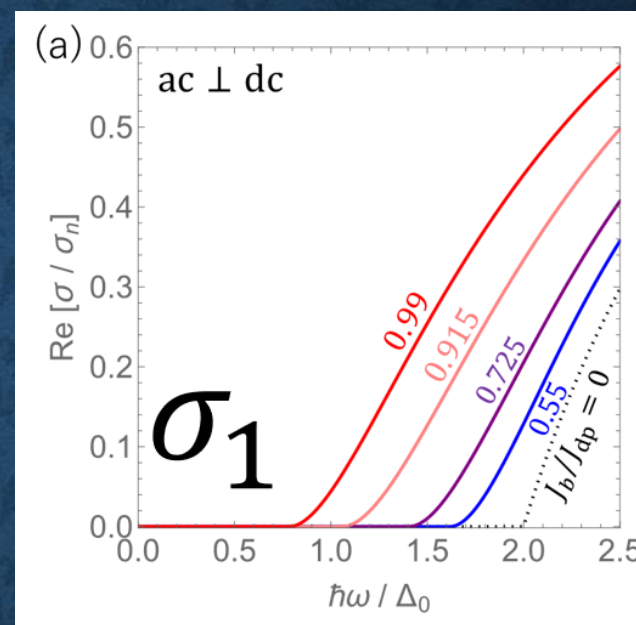
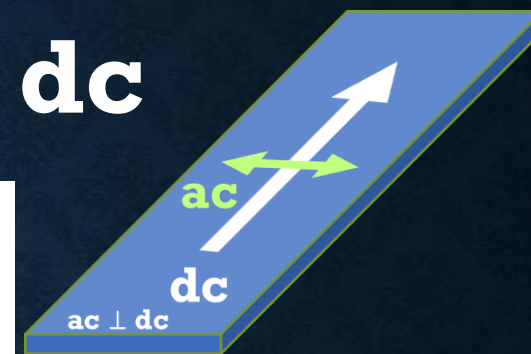


Resonance peak due to the **Higgs** mode.



Already observed in experiments!
S. Nakamura et al.,
PRL **122**, 257001 (2019)

ac ⊥ dc



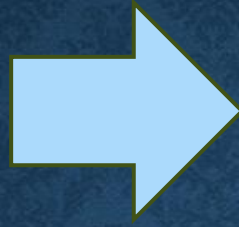
Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

$$L_k = \frac{E}{\dot{J}_s}$$

$$J_s \propto e^{-i\omega t}$$

$$L_k = \frac{1}{\omega \sigma_2}$$



$$L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

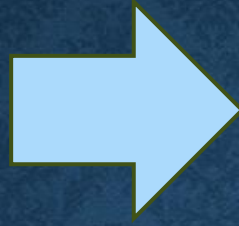
Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

$$L_k = \frac{E}{\dot{J}_s}$$

$$J_s \propto e^{-i\omega t}$$

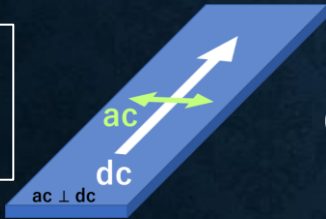
$$L_k = \frac{1}{\omega \sigma_2}$$



$$L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

$ac \perp dc$
case



$$C = C^{(0)} \simeq 0.136$$

Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

$$L_k = \frac{E}{\dot{J}_s}$$

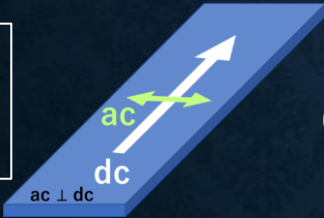
$$J_s \propto e^{-i\omega t}$$

$$L_k = \frac{1}{\omega \sigma_2}$$

$$L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

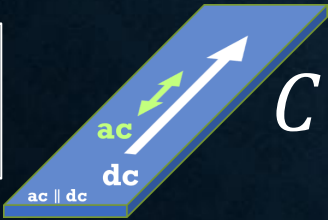
For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

$ac \perp dc$
case



$$C = C^{(0)} \simeq 0.136$$

$ac \parallel dc$
case



$$C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$$

$$\simeq 0.136$$

$$\simeq 0.0956 \text{ (AAA)} \\ (23\%)$$

$$\simeq 0.177 \text{ Higgs} \\ (43\%)$$

Insight for lower frequencies To calculate the kinetic inductance, we just go back to its definition.

$$L_k = \frac{E}{j_s} \xrightarrow{J_s \propto e^{-i\omega t}} L_k = \frac{1}{\omega \sigma_2} \xrightarrow{\quad} L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

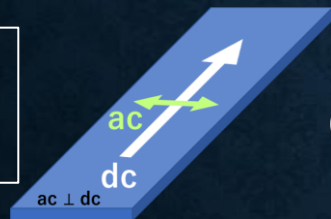
For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

Microscopic theory of nonequilibrium superconductivity

T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

**$ac \perp dc$
case**



$$C = C^{(0)} \simeq 0.136$$

**$ac \parallel dc$
case**



$$C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$$

$\simeq 0.136$

$\simeq 0.0956$
AAA
(23%)

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Higgs
(43%)

Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

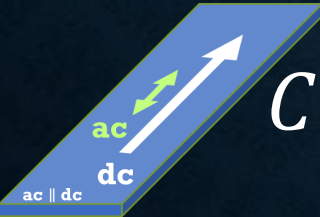
$$L_k = \frac{E}{j_s}$$

$J_s \propto e^{-i\omega t}$

$$L_k = \frac{1}{\omega \sigma_2}$$

$$L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

Microscopic theory of nonequilibrium superconductivity	Semi-phenomenological
T. Kubo, Phys. Rev. Applied 22, 044042 (2024) T. Kubo, Phys. Rev. Applied 23, 054091 (2025)	J. R. Clem and V. G. Kogan, Phys. Rev. B 86, 174521 (2012). T. Kubo, Physical Review Research 2, 033203 (2020).
$ac \perp dc$ case  $C = C^{(0)} \simeq 0.136$	
$ac \parallel dc$ case  $C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$ $\simeq 0.136$ $\simeq 0.0956$ AAA (23%) $\simeq 0.177$ Higgs (43%)	

Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

$$L_k = \frac{E}{j_s}$$

$$J_s \propto e^{-i\omega t}$$

$$L_k = \frac{1}{\omega \sigma_2}$$

$$L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

Microscopic theory of nonequilibrium superconductivity	Semi-phenomenological
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ac ⊥ dc case  $C = C^{(0)} \simeq 0.136$	Fast experiment (Frozen n_s) $C \simeq 0.136$
ac ∥ dc case  $C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$ $\simeq 0.136$ $\simeq 0.0956$ AAA (23%) $\simeq 0.177$ Higgs (43%)	

Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

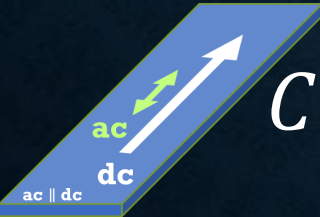
$$L_k = \frac{E}{j_s}$$

$J_s \propto e^{-i\omega t}$

$$L_k = \frac{1}{\omega \sigma_2}$$

$$L_k(J) = L_k(0) \left\{ 1 + \boxed{C} \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

For $(T, \omega) \rightarrow (0, 0)$, we can analytically calculate the coefficient C

Microscopic theory of nonequilibrium superconductivity	Semi-phenomenological
T. Kubo, Phys. Rev. Applied 22, 044042 (2024) T. Kubo, Phys. Rev. Applied 23, 054091 (2025)	J. R. Clem and V. G. Kogan, Phys. Rev. B 86, 174521 (2012). T. Kubo, Physical Review Research 2, 033203 (2020).
ac ⊥ dc case  $C = C^{(0)} \simeq 0.136$	Fast experiment (Frozen n_s) $C \simeq 0.136$
ac ∥ dc case  $C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$ $\simeq 0.136$ $\simeq 0.0956$ AAA (23%) $\simeq 0.177$ Higgs (43%)	Slow experiment (Oscillating n_s) $C \simeq 0.409$

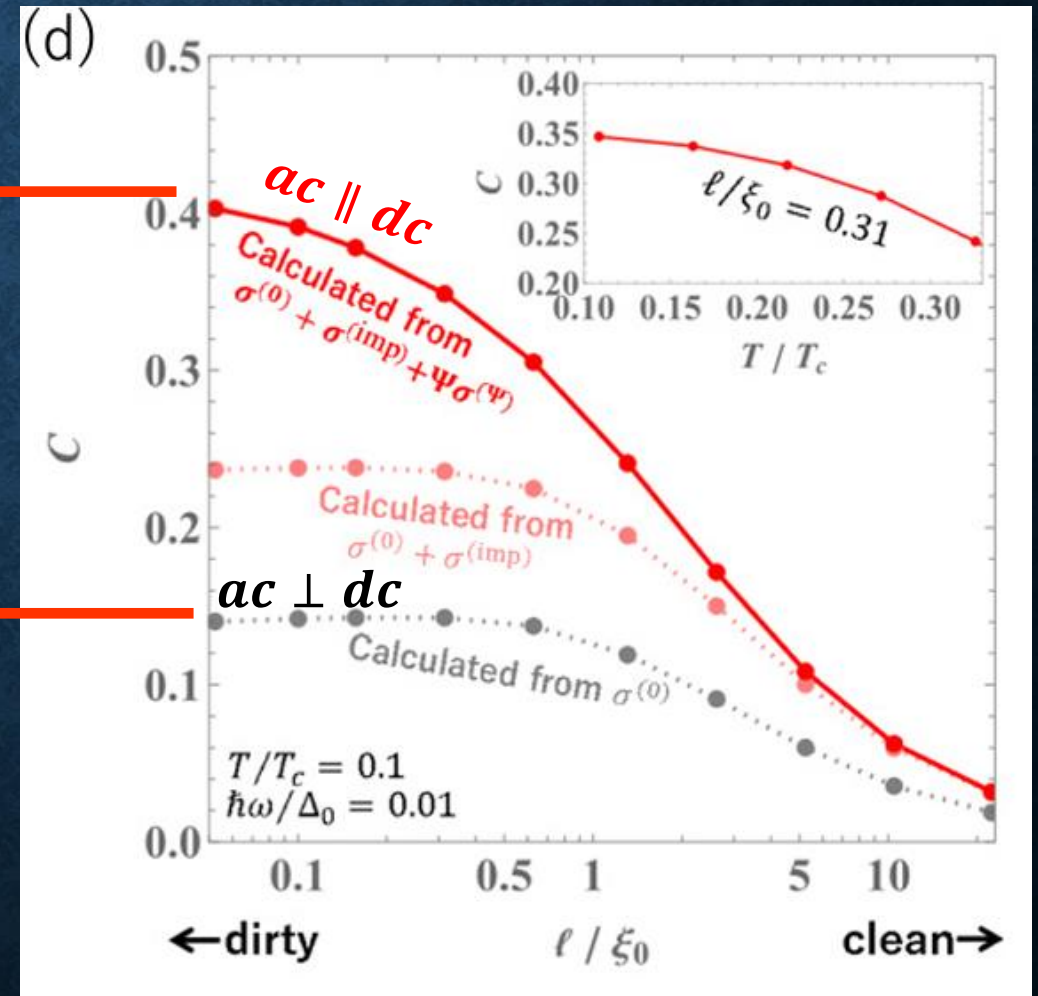
“C” from the Keldysh-Eilenberger theory, which is a microscopic theory of **nonequilibrium** superconductivity and is applicable at any temperature ($0 \leq T \leq T_c$) and for arbitrary mean free path.

$$L_k(J_b) = L_k(0) \left\{ 1 + \textcolor{red}{C} \left(\frac{J_b}{J_{dp}} \right)^2 + \dots \right\}$$

$$C \simeq 0.4$$

Consistent with the dirty limit result obtained from the Keldysh-Usadel

$$C \simeq 0.14$$



We have finally come to understand the *true meaning* of the semi-phenomenological approach: frozen and oscillating n_s

[When Does it happen?]

What has long been believed

Frozen n_s

The ac frequency is so fast that
the superfluid density cannot follow it.
(Fast experiment)

We have finally come to understand the *true meaning* of the semi-phenomenological approach: frozen and oscillating n_s

[When Does it happen?]

What has long been believed

Frozen n_s

The ac frequency is so fast that
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(Fast experiment)

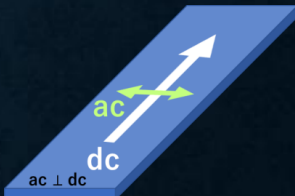
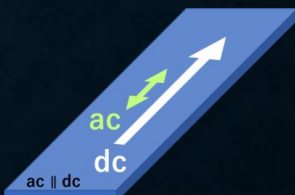
Oscillating n_s

The ac frequency is so slow that the
superfluid density can follow it.
(Slow experiment)

We have finally come to understand the *true meaning* of the semi-phenomenological approach: frozen and oscillating n_s

	[When Does it happen?] What has long been believed	[When Does it happen?] What has been revealed by the microscopic theory of nonequilibrium superconductivity
Frozen n_s	The ac frequency is so <u>fast</u> that the superfluid density cannot follow it. (Fast experiment)	
Oscillating n_s	The ac frequency is so <u>slow</u> that the superfluid density can follow it. (Slow experiment)	

We have finally come to understand the *true meaning* of the semi-phenomenological approach: frozen and oscillating n_s

	[When Does it happen?] What has long been believed	[When Does it happen?] What has been revealed by the microscopic theory of nonequilibrium superconductivity
Frozen n_s	The ac frequency is so <u>fast</u> that the superfluid density cannot follow it. (Fast experiment)	$ac \perp dc$ 
Oscillating n_s	The ac frequency is so <u>slow</u> that the superfluid density can follow it. <i>wrong</i> (Slow experiment)	$ac \parallel dc$  (AAA and Higgs contribution)

The microscopic theory of current dependent kinetic inductance is established.

$$L_k(J) = L_k(0) \left\{ 1 + C \left(\frac{J}{J_{dp}} \right)^2 + \dots \right\}$$

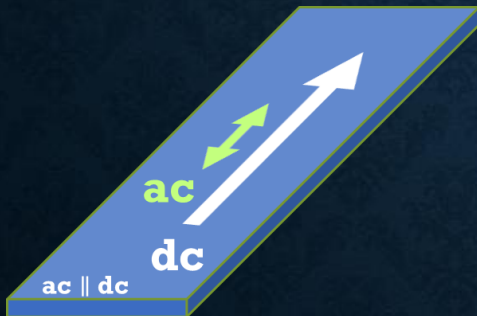
T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)



$ac \perp dc$
case

$$C = C^{(0)} \simeq 0.136$$



$ac \parallel dc$
case

$$C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$$

$$\begin{array}{l} \simeq 0.136 \\ \simeq 0.0956 \text{ AAA (23\%)} \\ \simeq 0.177 \text{ Higgs (43\%)} \end{array}$$

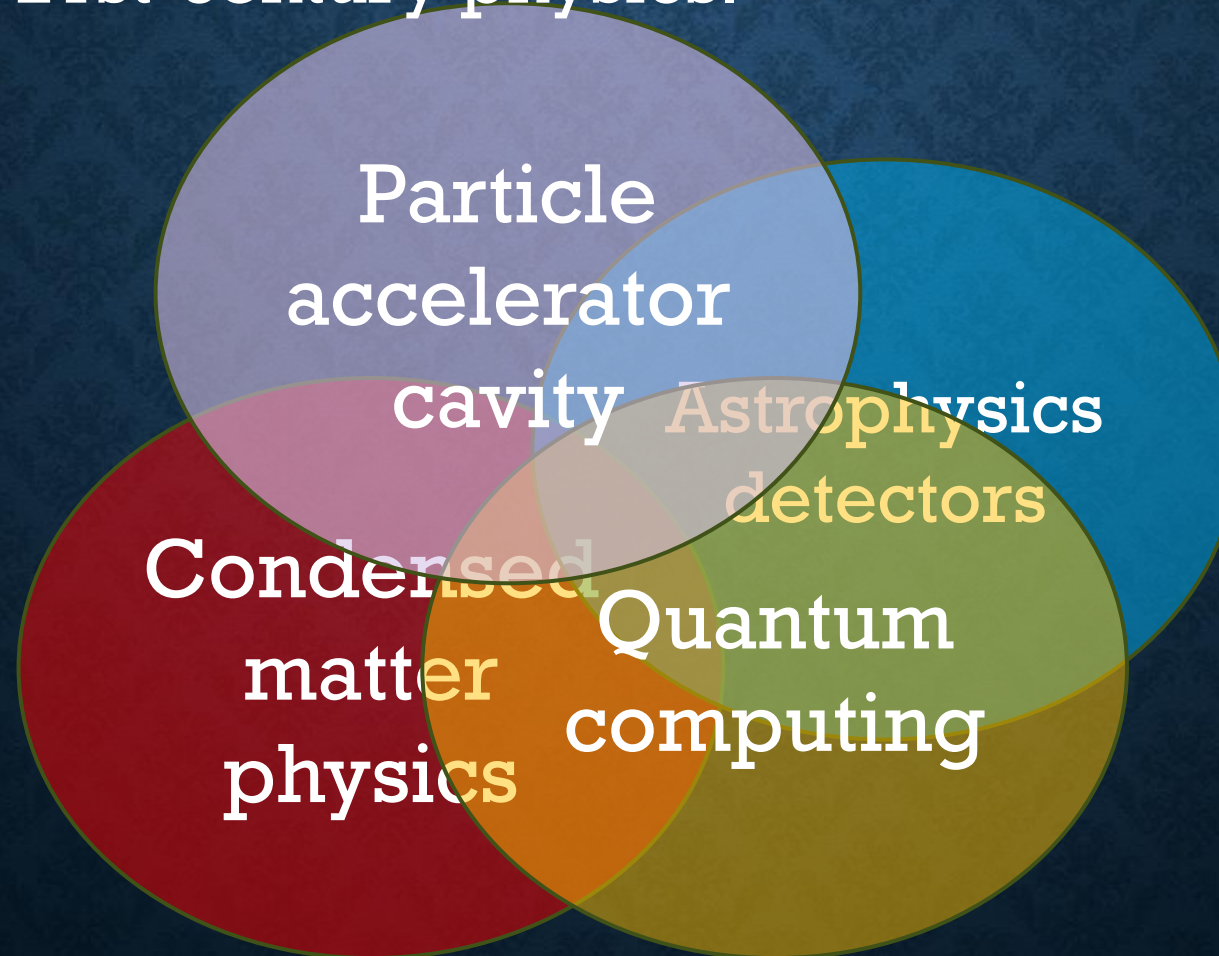
The Higgs mode is ubiquitous in superconducting devices.
The current dependence of the kinetic inductance is a representative example.

Summary

Research and development of superconducting resonators, which lie at the intersection of various fields, represent one of the most important efforts to pioneer 21st-century physics.



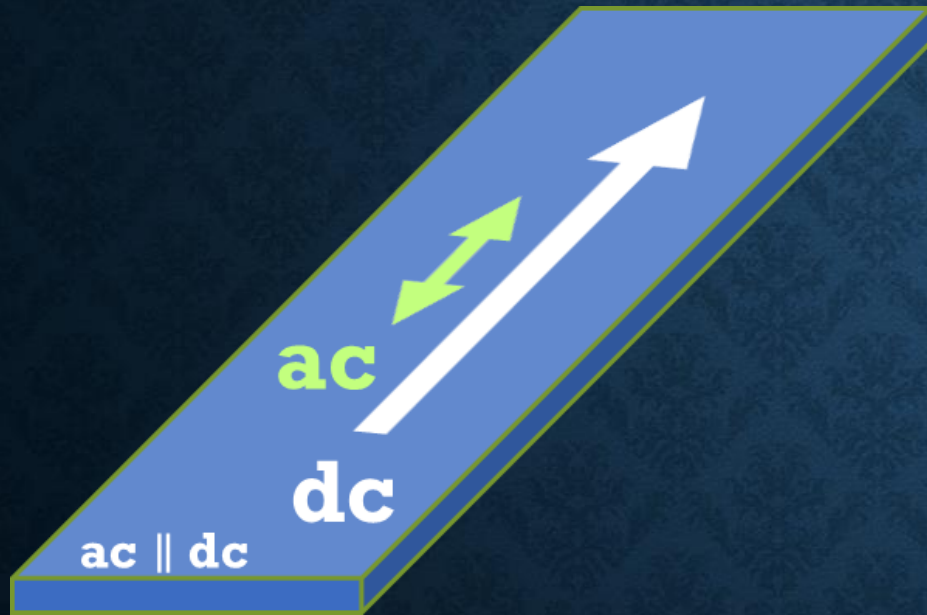
T. Kubo (2013)



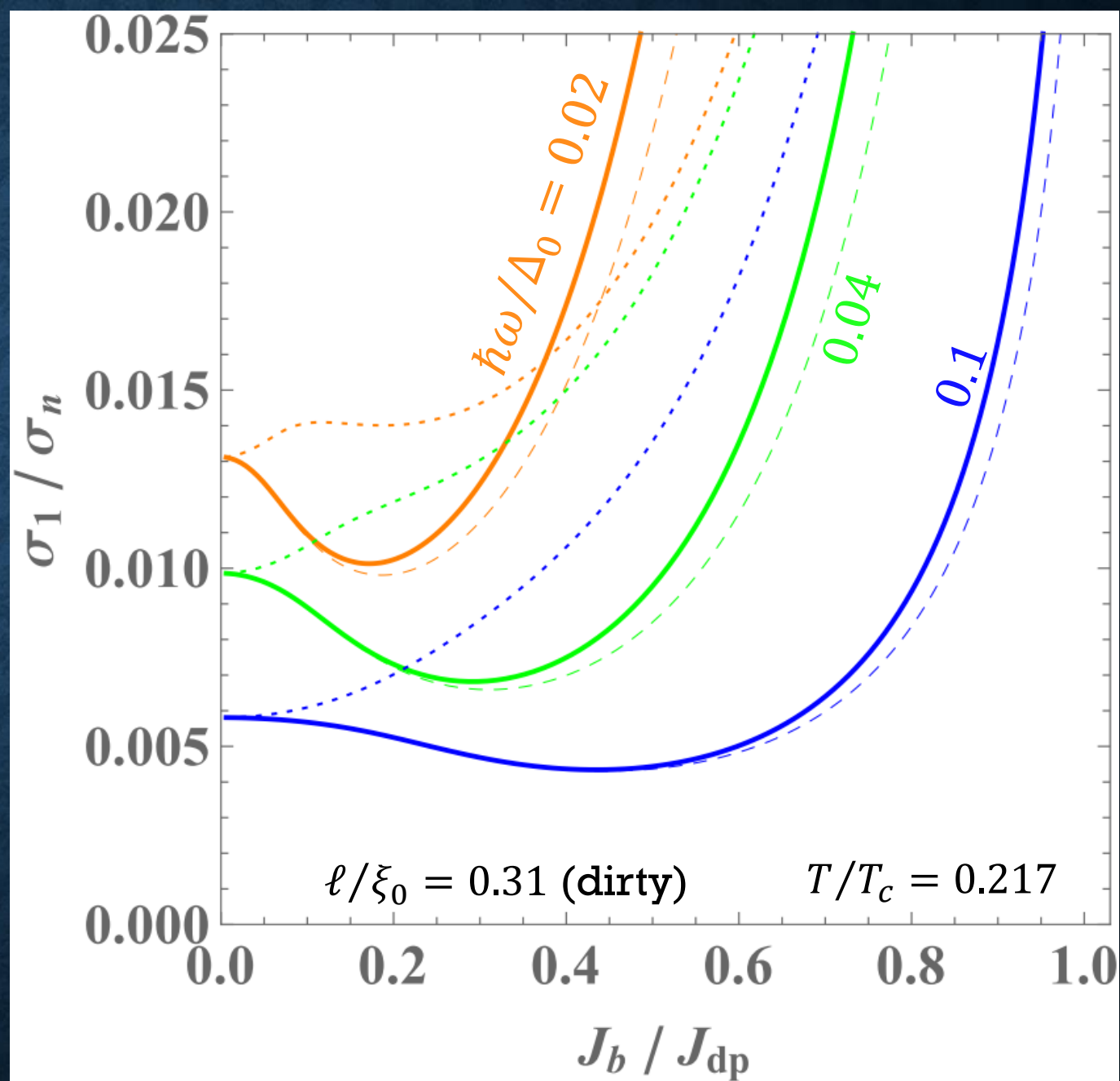
T. Kubo (2024)

Appendix (1)

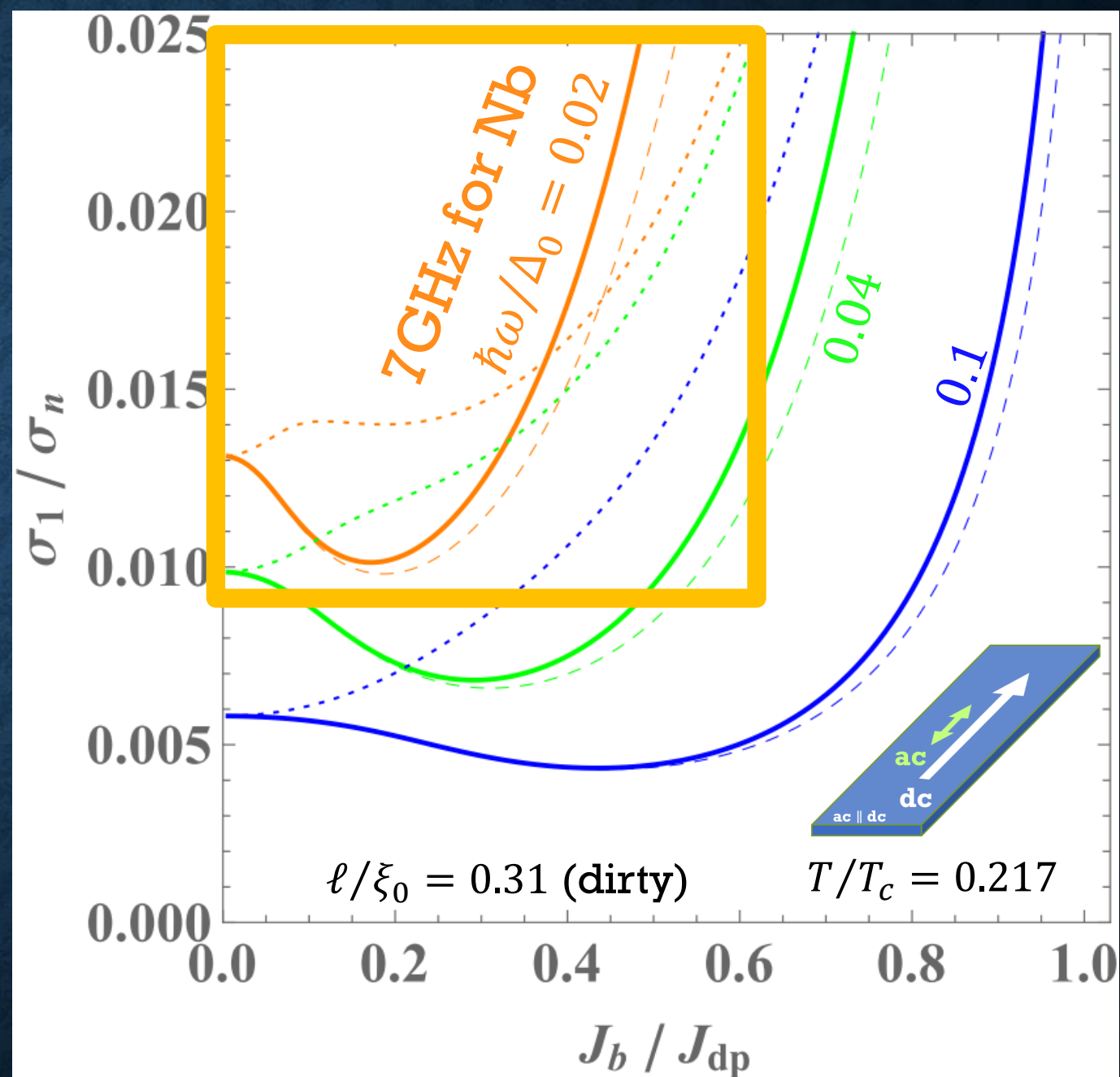
Current dependent $\sigma_1(J_b)$ curve in ac+dc system



dc: arbitrary strength
ac: perturbation



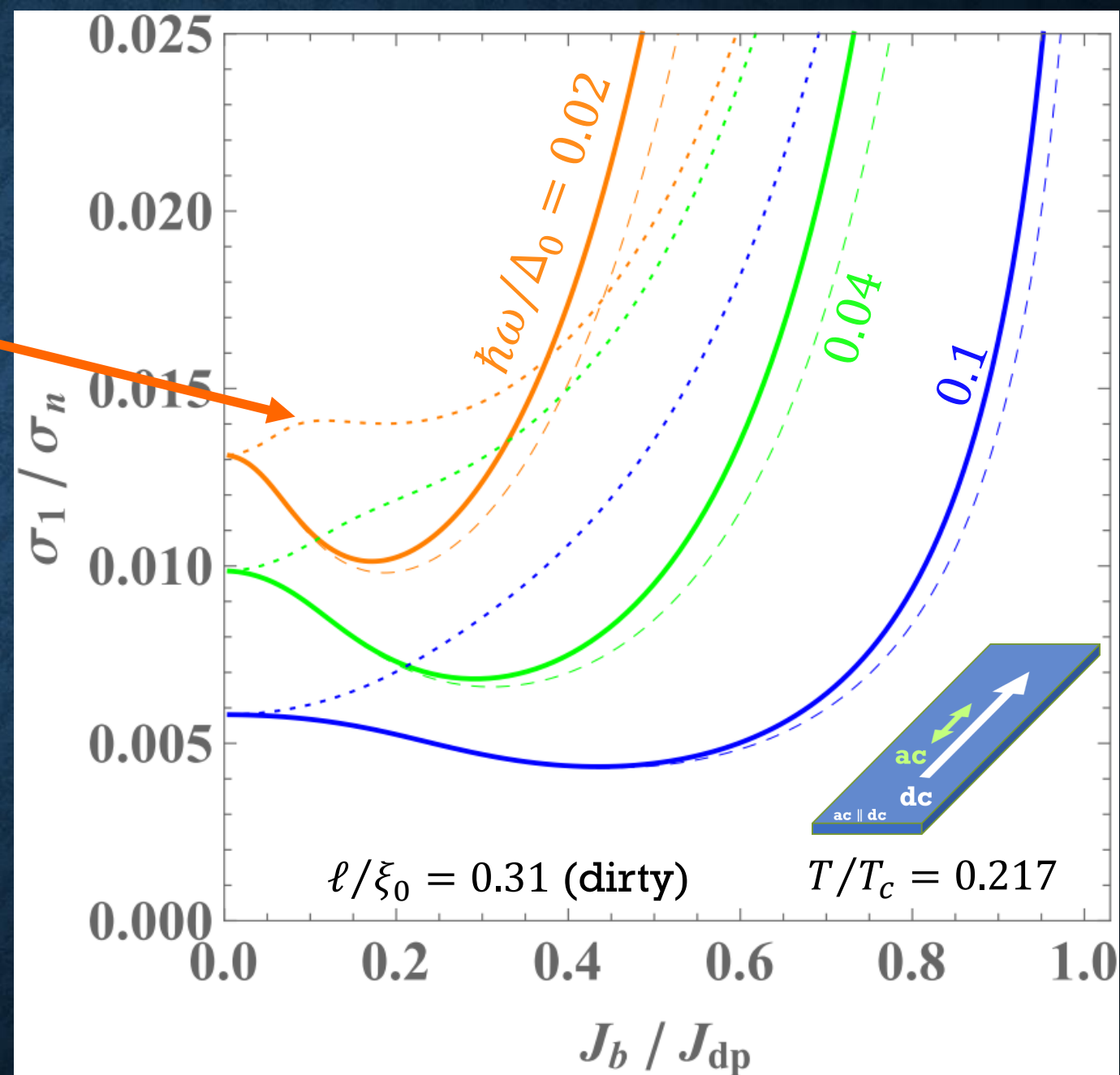
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$), which correspond to 7GHz for Nb.



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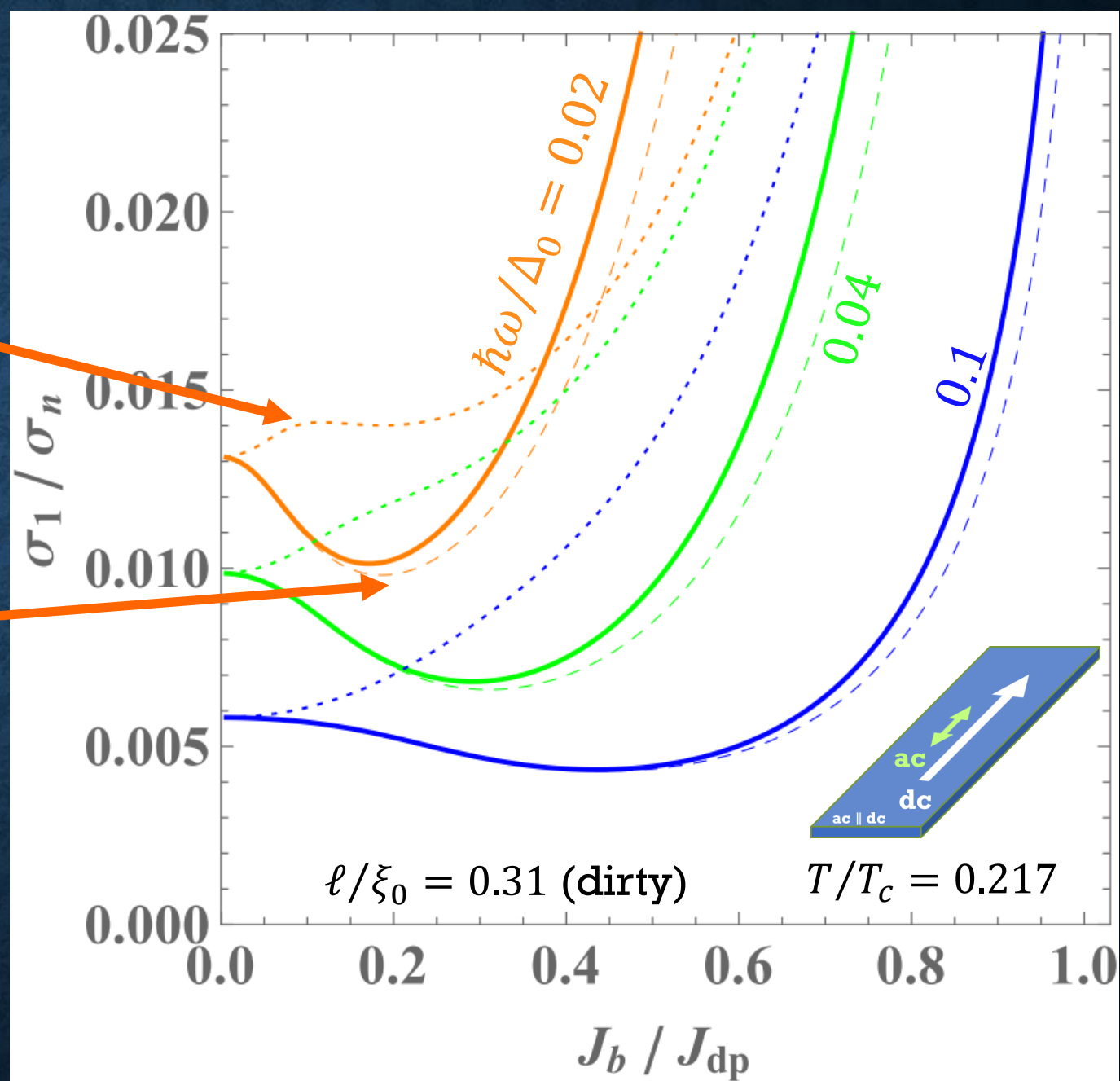
Dotted curve:
 $\text{Re}[\sigma^{(0)}]$



Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$), which correspond to 7GHz for Nb.

Dotted curve:
 $\text{Re}[\sigma^{(0)}]$

Dashed curve:
 $\text{Re}[\sigma^{(0)} + \sigma^{(1)}]$



Significant contribution from the direct AAA action term $\sigma^{(1)}$

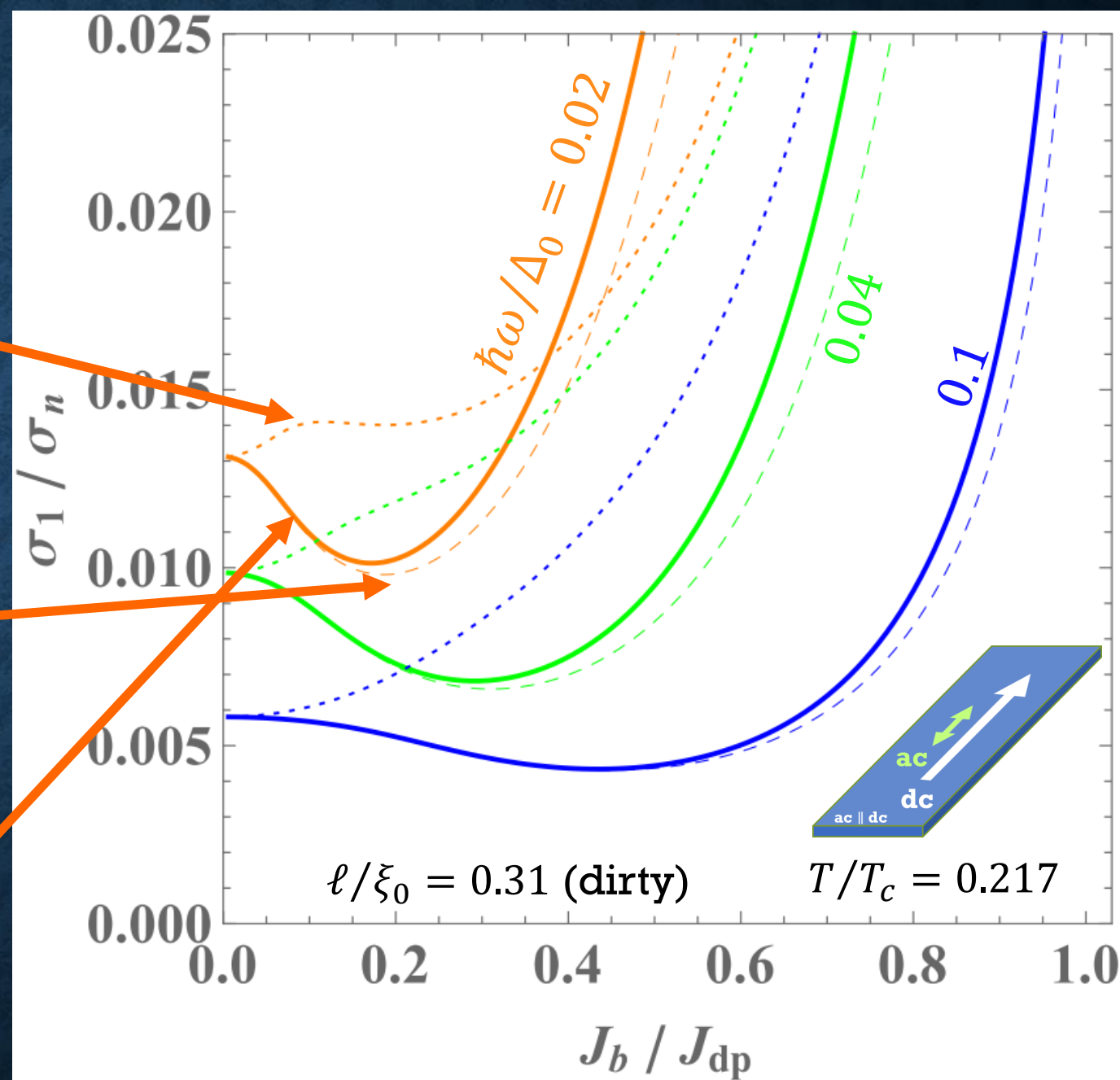
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$), which correspond to 7GHz for Nb.

Dotted curve:
 $\text{Re}[\sigma^{(0)}]$

Dashed curve:
 $\text{Re}[\sigma^{(0)} + \sigma^{(1)}]$

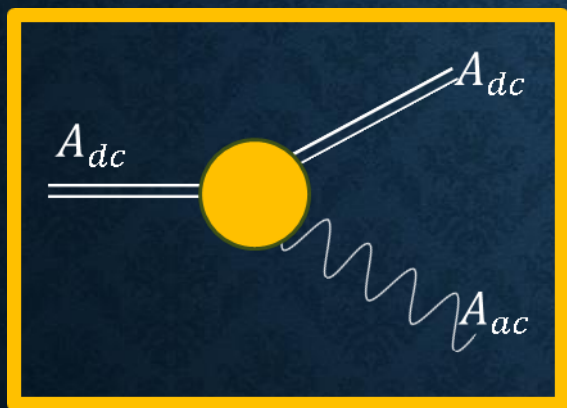
Solid curve:
 $\text{Re}[\sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}]$

Significant contribution from the direct AAA action term $\sigma^{(1)}$



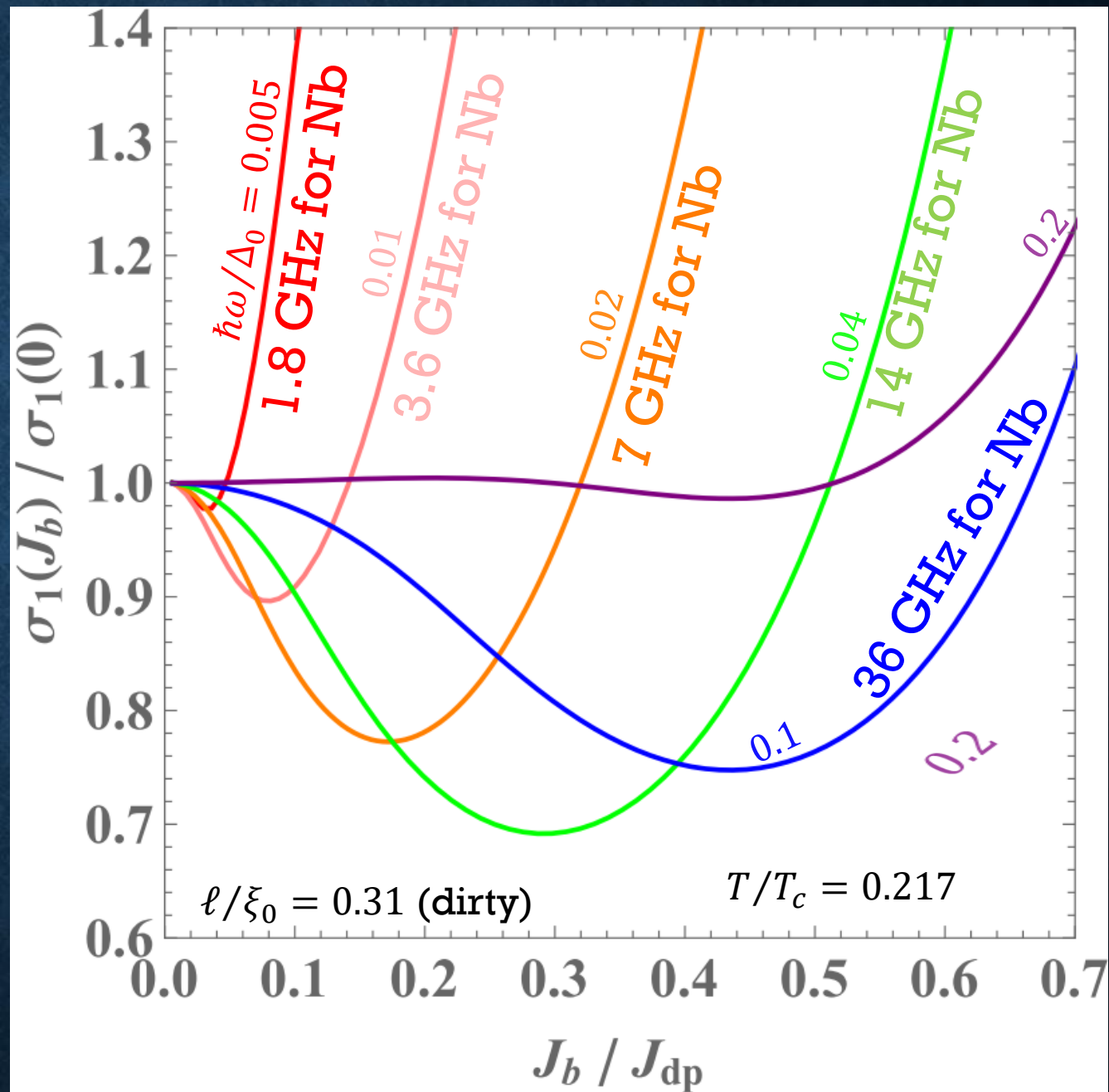
The current-induced reduction of $\sigma_1(J_b)$ is pronounced as the frequency increases

The key player is the direct AAA photon action



$$\sigma = \sigma^{(0)} + \boxed{\sigma^{(1)}} + \sigma^{(2)}$$

Big clue to understand the pronounced anti Q-slope with increasing frequency



Appendix (2)

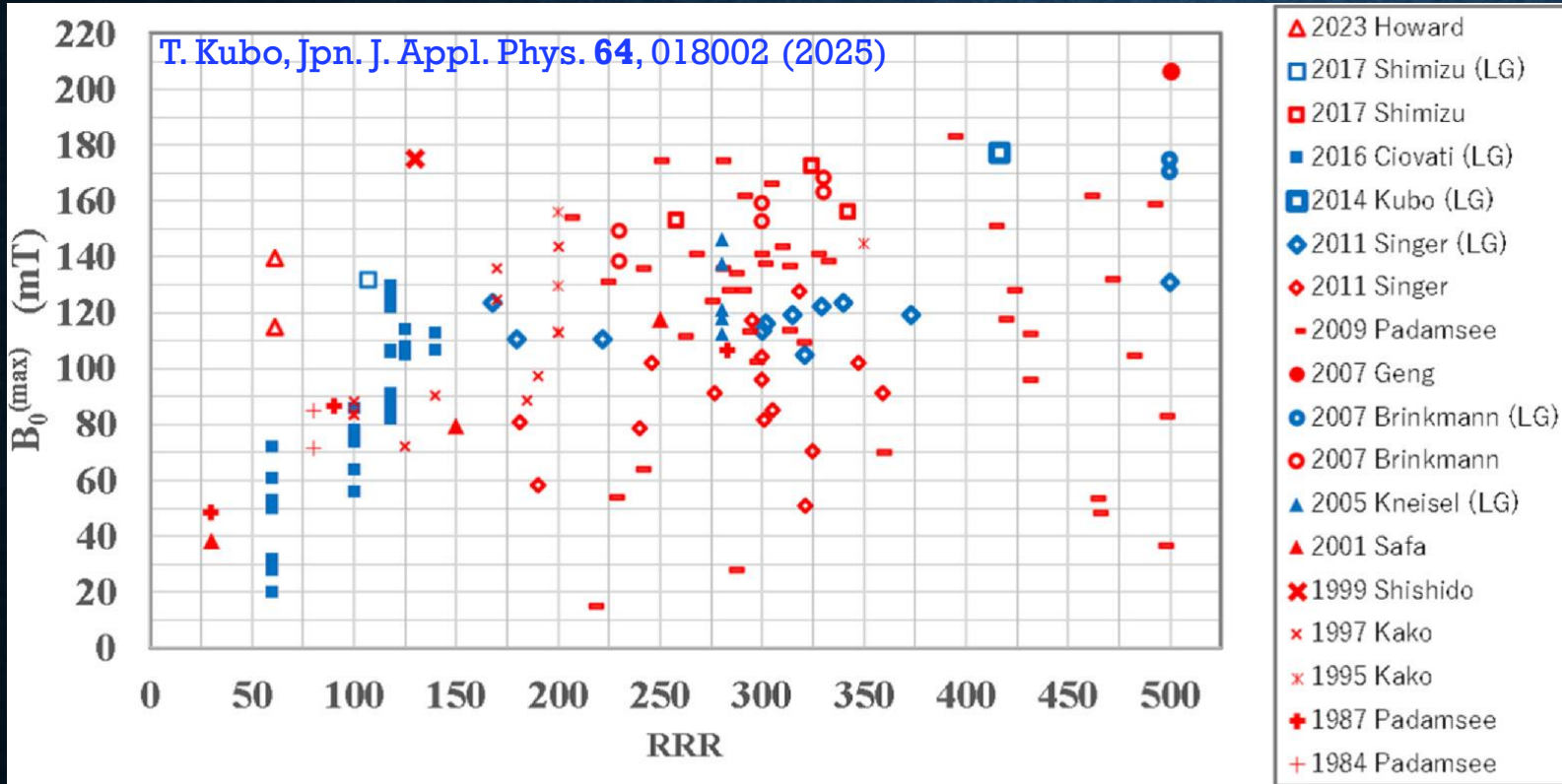
Summary of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to $T \ll T_c$)
$\frac{\lambda_0}{\xi_0} \gg 1$ (e.g., Nb ₃ Sn, NbN, NbTiN etc.)	$\ell \gg \xi_0$ (clean)	$H_{sh}(0) = 0.84H_c(0)$ $H_{sh}(T)$ for $0 \leq T \leq T_c$ Analytical result for T=0. V. P. Galaiko, Sov. Phys. JETP 23 , 475 (1966) G. Catelani and J. P. Sethna, Phys. Rev. B 78, 224509 (2008) F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012)
	any ℓ (from clean limit to dirty limit)	$H_{sh}(T)$ for $0 \leq T \leq T_c$ F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85 , 054513 (2012)
	$\ell \ll \xi_0$ (dirty)	$H_{sh}(T)$ for $0 \leq T \leq T_c$ $H_{sh}(0) = 0.795H_c(0)$ F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B 85, 054513 (2012) T. Kubo, Phys. Rev. Research 2, 033203 (2020)
$\frac{\lambda_0}{\xi_0} \lesssim 1$ (e.g., Nb)		Analytical result for T=0. T. Kubo, Phys. Rev. Research 2, 033203 (2020) T. Kubo, Supercond.Sci. Technol. 34, 045006 (2021)
	any ℓ (from clean limit to dirty limit)	N/A

Appendix (3)

Maximum field achieved so far

Already close to the theoretical limit?

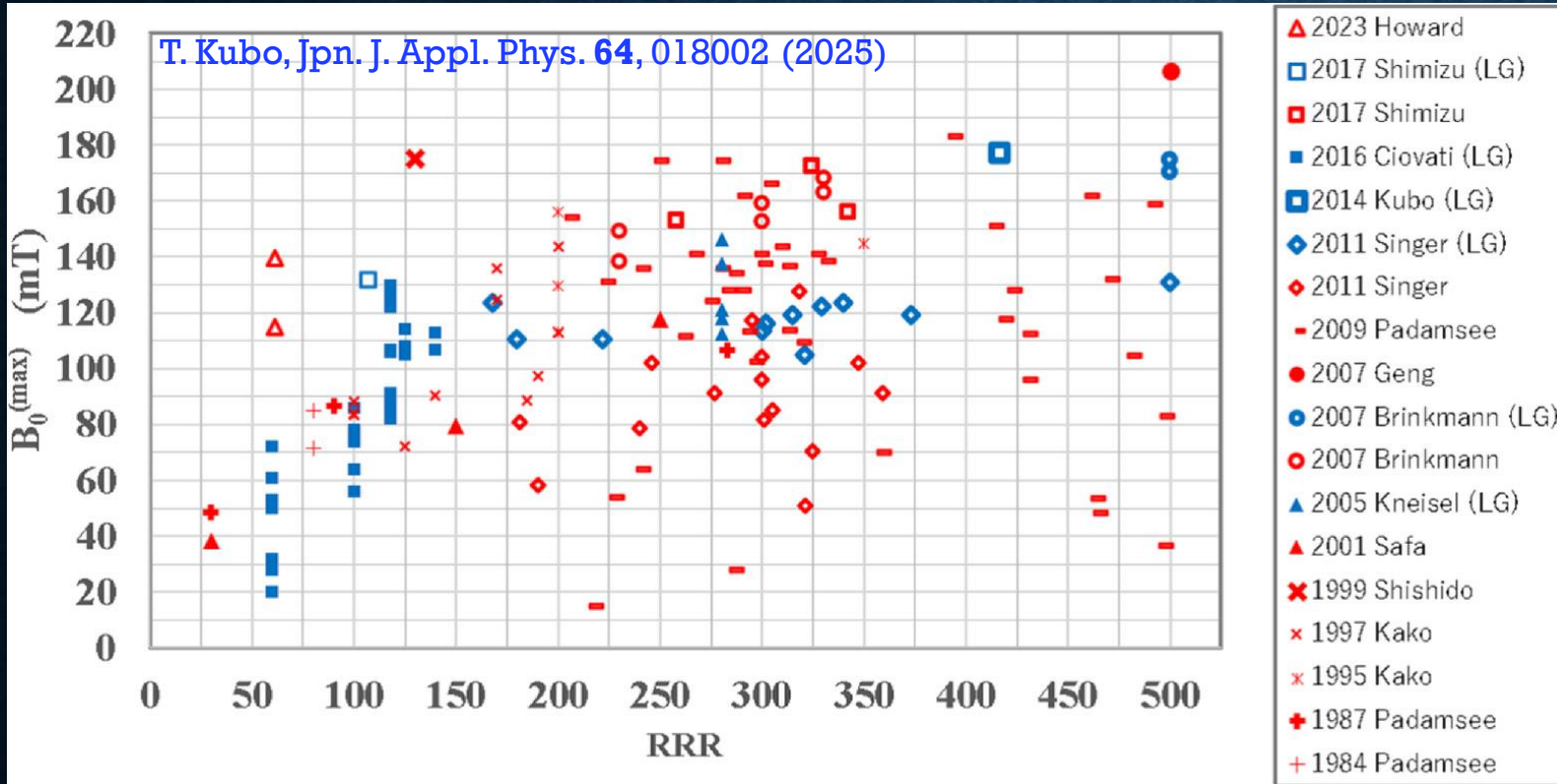


The answer: we currently have no reliable estimate, because there is **no theory of H_{sh} applicable to clean Nb at $T \ll T_c$** .

Fig. 1. Maximum surface magnetic field $B_0^{\max}$ for 1.3 and 1.5 GHz cavities as a function of RRR. The 149 data points are sourced from Refs. 8–22, where RRR values were explicitly reported. These cavities were processed using chemical polishing and/or electropolishing, with or without low-temperature baking. Some high-performing cavities, such as those in Refs. 29, 30 achieving 190 mT and 210 mT, are not included due to the absence of reported RRR data. Nitrogen-doped cavities are excluded as doping significantly alters the surface RRR, a topic discussed in the final paragraph of this Brief Note. Similarly, cavities subjected to heat treatments above 1000 °C, which can modify the RRR value, are not included. Tests affected by multipacting-induced quenches (e.g. Test numbers 1, 2, 3, 7, and 13 of cavity F1F2 in Ref. 20) are omitted, as are cavities whose performance was limited by field emission (e.g. cavity HF-5 in Ref. 21) In cases where field emission is not explicitly mentioned or may have been overlooked, data potentially affected by it might still be included in the figure. However, such instances do not significantly impact the overall trend, particularly the primary focus of this study: the maximum fields achieved across different RRR values.

Maximum field achieved so far

Already close to the theoretical limit?



The answer: we currently have no reliable estimate, because there is **no theory of H_{sh} applicable to clean Nb at $T \ll T_c$** .

Ginzburg–Landau (GL) theory, valid near $T \approx T_c$, gives

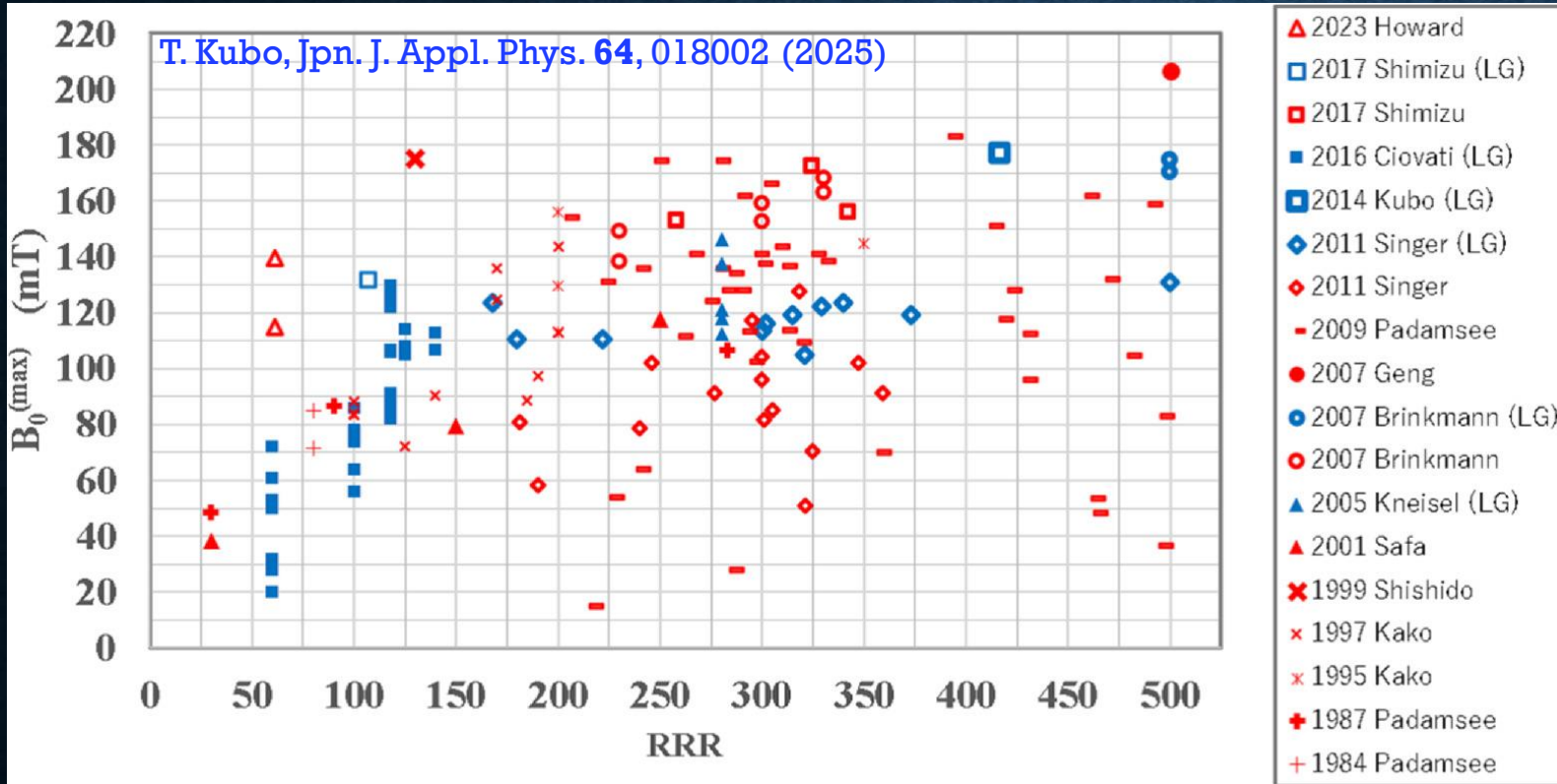
- A. J. Dolgert et al., Phys. Rev. B **53**, 5650 (1996)
- M. K. Transtrum et al., Phys. Rev. B **83**, 094505 (2011)

$$B_{sh}(T) \Big|_{T \approx T_c} = 1.26 B_c(T) \Big|_{T \approx T_c} \left(\kappa = \frac{\lambda}{\xi} = 1 \right)$$

Fig. 1. Maximum surface magnetic field $B_0^{\max}$ for 1.3 and 1.5 GHz cavities as a function of RRR. The 149 data points are sourced from Refs. 8–22, where RRR values were explicitly reported. These cavities were processed using chemical polishing and/or electropolishing, with or without low-temperature baking. Some high-performing cavities, such as those in Refs. 29, 30 achieving 190 mT and 210 mT, are not included due to the absence of reported RRR data. Nitrogen-doped cavities are excluded as doping significantly alters the surface RRR, a topic discussed in the final paragraph of this Brief Note. Similarly, cavities subjected to heat treatments above 1000 °C, which can modify the RRR value, are not included. Tests affected by multipacting-induced quenches (e.g. Test numbers 1, 2, 3, 7, and 13 of cavity F1F2 in Ref. 20) are omitted, as are cavities whose performance was limited by field emission (e.g. cavity HF-5 in Ref. 21) In cases where field emission is not explicitly mentioned or may have been overlooked, data potentially affected by it might still be included in the figure. However, such instances do not significantly impact the overall trend, particularly the primary focus of this study: the maximum fields achieved across different RRR values.

Maximum field achieved so far

Already close to the theoretical limit?



The answer: we currently have no reliable estimate, because there is **no theory of H_{sh} applicable to clean Nb at $T \ll T_c$** .

Ginzburg–Landau (GL) theory, valid near $T \approx T_c$, gives

- A. J. Dolgert et al., Phys. Rev. B **53**, 5650 (1996)
- M. K. Transtrum et al., Phys. Rev. B **83**, 094505 (2011)

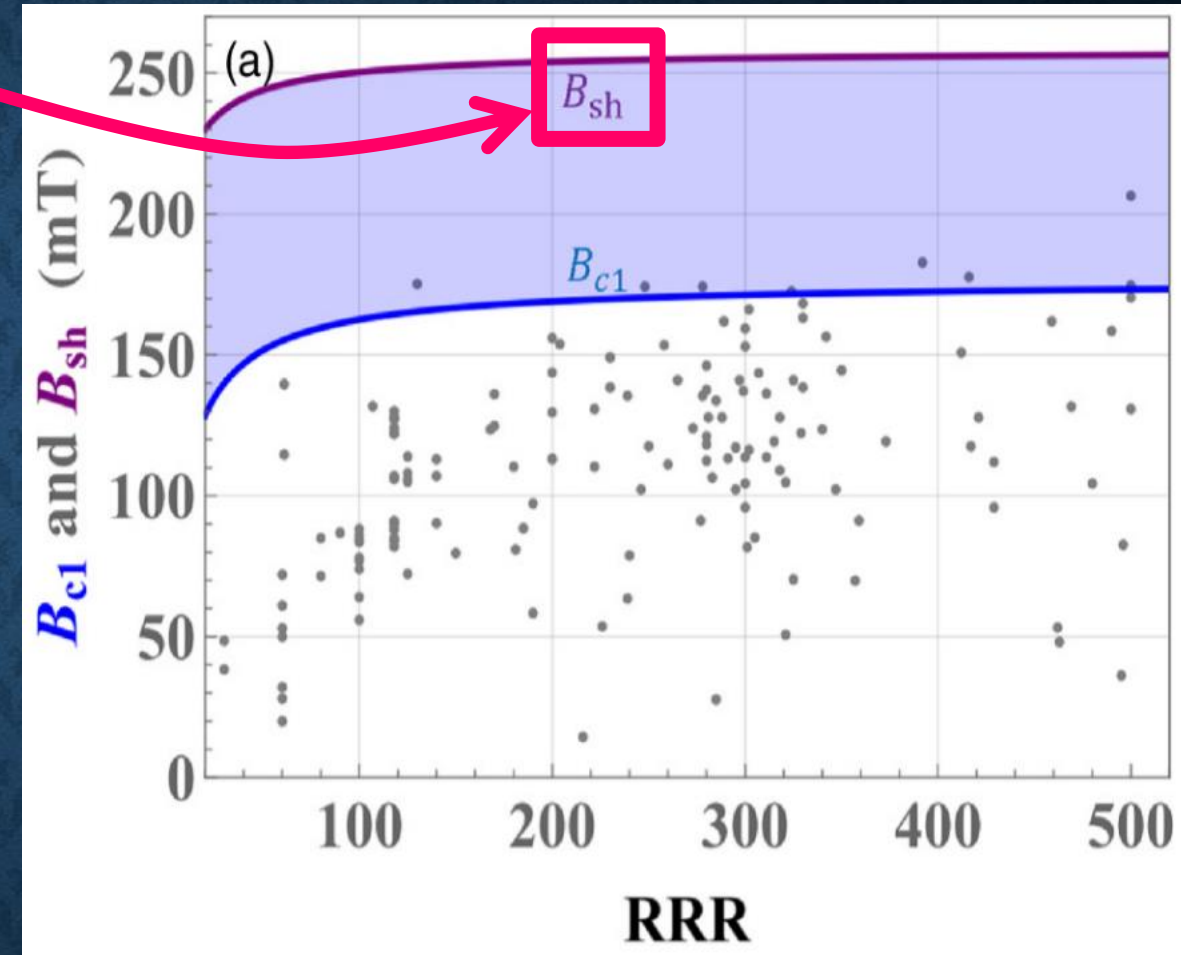
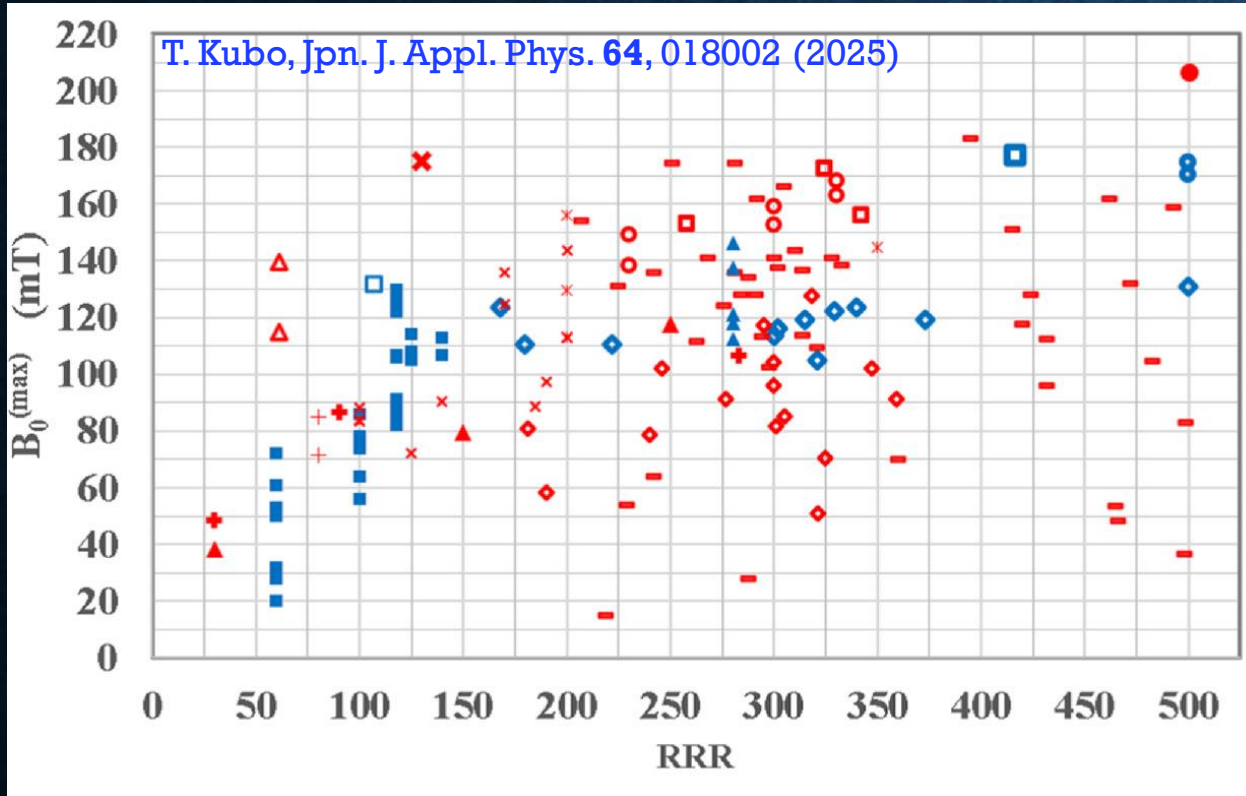
$$B_{sh}(T) \Big|_{T \approx T_c} = 1.26 B_c(T) \Big|_{T \approx T_c} \left(\kappa = \frac{\lambda}{\xi} = 1 \right)$$

If we naively extrapolate this to low temperatures — not theoretically justified — we obtain the rough estimate

$$\begin{aligned} B_{sh}(0) &= 1.26 B_c(0) \text{ ?} \\ &= 250 \text{ mT ?} \end{aligned}$$

Fig. 1. Maximum surface magnetic field $B_0^{\max}$ for 1.3 and 1.5 GHz cavities as a function of RRR. The 149 data points are sourced from Refs. 8–22, where RRR values were explicitly reported. These cavities were processed using chemical polishing and/or electropolishing, with or without low-temperature baking. Some high-performing cavities, such as those in Refs. 29, 30 achieving 190 mT and 210 mT, are not included due to the absence of reported RRR data. Nitrogen-doped cavities are excluded as doping significantly alters the surface RRR, a topic discussed in the final paragraph of this Brief Note. Similarly, cavities subjected to heat treatments above 1000 °C, which can modify the RRR value, are not included. Tests affected by multipacting-induced quenches (e.g. Test numbers 1, 2, 3, 7, and 13 of cavity F1F2 in Ref. 20) are omitted, as are cavities whose performance was limited by field emission (e.g. cavity HF-5 in Ref. 21) In cases where field emission is not explicitly mentioned or may have been overlooked, data potentially affected by it might still be included in the figure. However, such instances do not significantly impact the overall trend, particularly the primary focus of this study: the maximum fields achieved across different RRR values.

If we believe this rough estimate of B_{sh} based on GL, we get

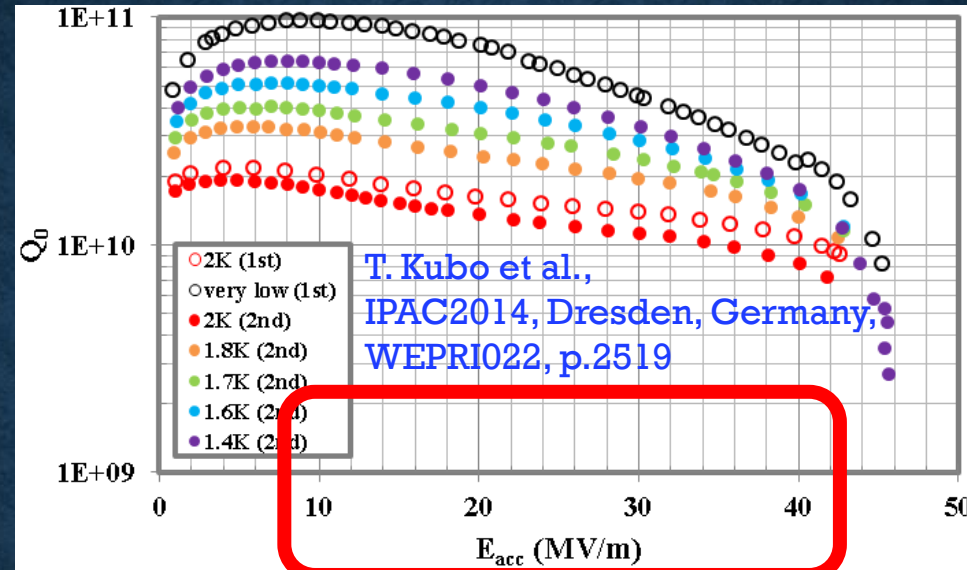
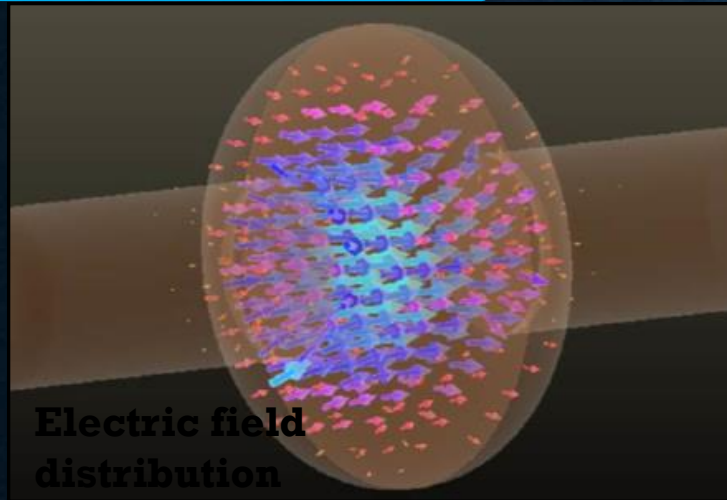


Note: To achieve B_{sh} , it is necessary to eliminate all surface defects. **However, this alone is not sufficient** to reach B_{sh} ; it is also essential to ensure that the thermal runaway field B_{run} exceeds B_{sh} . This condition is not always satisfied, particularly at 2 K. It is better to test the cavity at lower temperatures (e.g., 1.8K or 1.6K) to ensure that the condition $B_{run} > B_{sh}$ is satisfied.

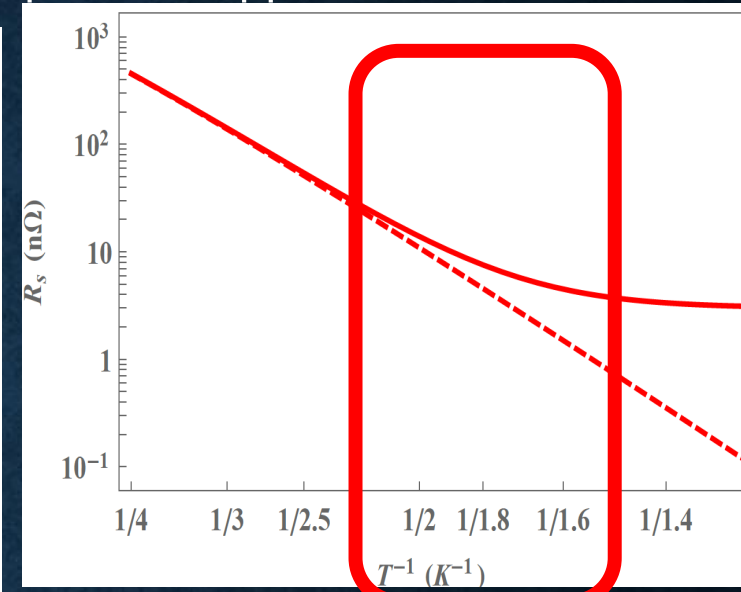
Appendix (4)

The biggest differences between cavities for accelerators and those for quantum applications

Accelerators



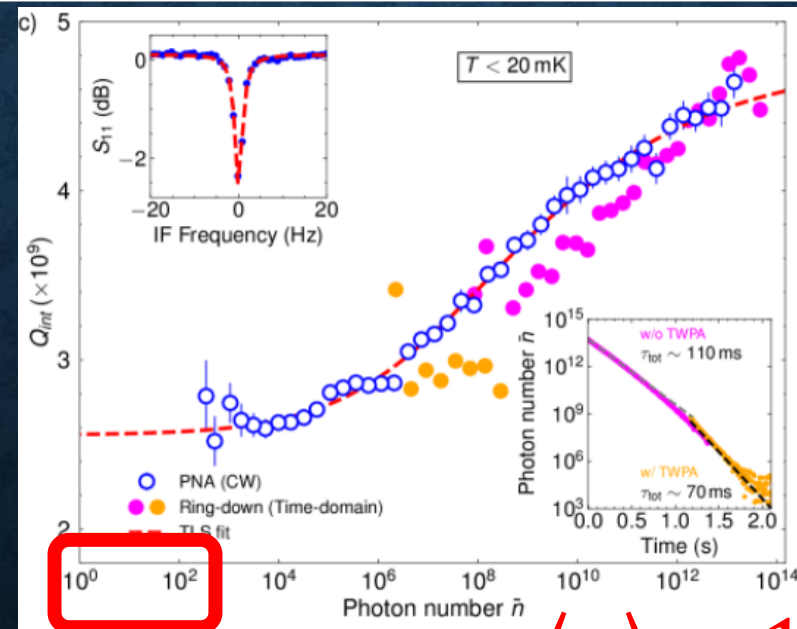
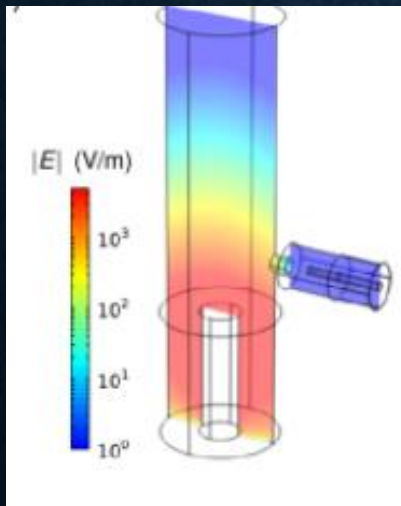
photon number $\langle n \rangle \sim 10^{25}$



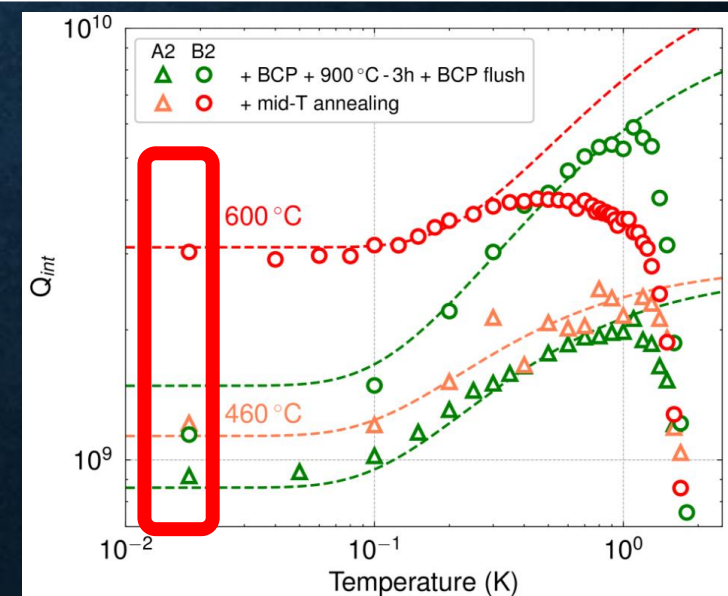
T ≈ 2 K

Quantum computing

T. Takenaka, T. Kubo et al.,
arXiv:2510.01819.



photon number $\langle n \rangle \sim 1$



T ≈ 20 mK